A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference

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Abstract

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Chapter 1

Introduction

1.1 Background of the Study

A functional legal system requires strict consistency so that citizens can clearly understand their rights and obligations [74]. However, the sheer volume of modern legislation has grown so rapidly that human practitioners can no longer track every active rule [181, 33]. As governing bodies continuously pass new regulations to address emerging issues, they frequently enact laws that inadvertently contradict older ones [191, 114]. Because the total number of documents is overwhelming, identifying these hidden conflicts through manual review is computationally impossible for legal staff. This issue is particularly severe in local government units, where historical ordinances are often scattered across fragmented files rather than maintained in a structured digital database [43].

Within the Philippine legal framework, local governments like Davao City operate under delegated authority and can only enact laws permitted by the national government [84]. Due to this structural limitation, the Magtajas v. Pryce Doctrine strictly prohibits any local ordinance from contradicting established national statutes [208]. Consequently, in the local context, city councils must rigorously audit every newly proposed draft ordinance to ensure it logically aligns with existing laws before final approval. If reviewers overlook a normative conflict, such as a local ordinance prohibiting an activity that a national law explicitly permits, the resulting local legislation is legally void. Passing invalid laws creates severe jurisdictional disputes, wastes administrative resources, and ultimately compromises the constitutional rights of the public.

To address these legal risks, the legislative process must utilize an ex-ante evaluation: a term derived from Latin meaning 'before the event.' Unlike an 'ex-post' review, which happens after a law is already in effect and often after legal damage has occurred, an ex-ante audit acts as a first line of defense. By checking a proposed ordinance for conflicts while it is still in the drafting phase, the Sangguniang Panlungsod can proactively align local rules with national statutes. This preventative approach is consistent with modern Philippine standards like Republic Act No. 11032 or the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which encourages government agencies to perform regulatory assessments to ensure that new rules do not create unnecessary or conflicting burdens on the public.

The decentralized drafting process of local ordinances further complicates manual review because the resulting texts are often highly unstructured [186]. City councils frequently splice and amend different drafts, producing verbose documents that blend formal legal English with regional dialects like Cebuano and administrative ”Taglish” [65]. Furthermore, because local archives heavily rely on digitized PDF scans, the text suffers from severe Optical Character Recognition (OCR) errors. While OCR is typically 98–99 percent accurate for clean English text, studies demonstrate that recognition accuracy degrades substantially for low-resource languages [104], with rapid lexicon extraction for Cebuano specifically encountering word error rates of 6.9 percent [157]. These errors cause an “error cascade” that shatters standard vocabulary, making it difficult for traditional search engines to semantically differentiate between a statutory obligation and a strict prohibition.

In light of these challenges, the proponents seek to develop a specialized application capable of retrieving relevant legal documents and comparing them against proposed ordinances to detect potential conflicts.

1.2 Problem Statement

This study is conducted to investigate the use of a multi-stage architecture to automate the manual auditing of proposed local legislation in Davao City.

Specifically, the study aims to answer the following research questions:

- What are the specific linguistic parameters and structural markers that define a ”semantic conflict” between a proposed local ordinance and existing local or national laws?
- How can a Coarse-to-Fine system be designed to efficiently retrieve relevant national and local statutes before performing deep logical inference on constrained LGU hardware?
- Which combination of a hybrid Information Retrieval (IR) algorithms and Natural Language Inference (NLI) models provides the highest accuracy when processing the specific hierarchy of Philippine legal texts?
- What is the empirical performance of the proposed pipeline when evaluated against a manually curated Ground Truth dataset of local and national conflict pairs?
- How usable and viable is the final system interface for key stakeholders within the Davao City Sangguniang Panlungsod?

1.3 Objectives of the Study

The primary goal of this research is to develop a computationally efficient, two-stage semantic conflict detection system for proposed Davao City ordinances.

Specifically, this study aims to:

- Identify the specific linguistic characteristics and structural constraints of Davao City's legislative archive that contribute to semantic ambiguity and retrieval failure;
- Design a Coarse-to-Fine computational pipeline that decouples document retrieval from logical inference to optimize resource usage for local deployment;
- Implement a hybrid Information Retrieval (IR) algorithm and Natural Language Inference (NLI) system specifically calibrated for the Philippine legal hierarchy;
- Evaluate the system's reliability by comparing its automated detections against a verified set of known legal conflicts to measure how accurately it identifies contradictions; and
- Develop a user-friendly web interface for the Davao City Sangguniang Panlungsod designed to accept uploaded draft documents and provide real-time conflict alerts.

1.4 Significance of the Study

The Davao City Sangguniang Panlungsod stands as the primary institutional beneficiary of this automated consistency checking system. By transitioning from manual statutory review to computational auditing, the tool saves legislative staff hundreds of hours of manual cross-referencing across multi-volume legislative digests. This computational intervention actively prevents the city from enacting legally vulnerable measures that invite judicial invalidation, public injunctions, or administrative deadlocks under the doctrine of statutory preemption. It equips the local legislative council with the operational capacity to draft regulatory measures with a heightened degree of legal certainty and procedural speed. Ultimately, the system modernizes the local legislative lifecycle by embedding rigorous logical safeguards directly into the pre-plenary committee review and Second Reading preparation phase, equipping drafters to resolve statutory conflicts during the floor amendment period.

The study provides substantial operational value to legal researchers, committee secretaries, and legislative bill drafters. In contemporary local legislative practice, legal researchers must examine extensive regulatory provisions across disparate national codes without centralized semantic indexing tools. The proposed system provides an automated cross-statutory audit trail, pinpointing candidate national statutes and fellow city ordinances that govern the same regulatory sphere. By surfacing relevant provisions and generating token-level self-attention attribution heatmaps, the architecture substantially mitigates cognitive fatigue and reading oversight during high-volume legislative sessions. Consequently, legal officers can conduct evidence-grounded preliminary reviews and focus their analytical expertise on high-level statutory interpretation.

Beyond the immediate city council, this research establishes an edge-deployable computational blueprint for Local Government Units (LGUs) nationwide and oversight agencies such as the Department of the Interior and Local Government (DILG). In compliance with the Ease of Doing Business and Efficient Government Service Delivery Act of 2018 (Republic Act No. 11032), local governments are mandated to perform systematic Regulatory Impact Assessments to ensure local regulations do not create redundant or conflicting business burdens. Demonstrating that an automated consistency checking pipeline operates efficiently on standard consumer-grade office desktop computers proves that advanced artificial intelligence can be deployed within budget-constrained local government environments without recurring cloud subscriptions or specialized computing clusters.

The research also contributes foundational advancements to the Philippine Natural Language Processing and legal informatics community. Prior domestic legal technology research has concentrated predominantly on national appellate court case summarization and judicial outcome forecasting, leaving local statutory conflict detection largely unaddressed. This investigation establishes an empirical baseline for processing low-resource, localized Philippine statutory corpora characterized by multi-generational scanning noise, administrative Taglish, and specialized legislative syntax. Furthermore, the empirical findings demonstrate that lightweight, fine-tuned cross-encoders can effectively overcome affirmative entailment bias and resolve directional statutory contradictions on resource-constrained local hardware.

Finally, the general public and the local business community benefit directly through the promulgation of coherent, constitutionally sound public policy. When local ordinances clash with superior national statutes, citizens and local enterprises bear the brunt of legal uncertainty, facing conflicting compliance directives or unauthorized penal sanctions. By serving as an ex-ante procedural safety net, the system helps ensure that enacted city ordinances harmonize with national standards and existing local legislation. This statutory coherence fosters institutional transparency, strengthens regulatory predictability, and upholds the integrity of the local rule of law.

1.5 Scope and Limitations

The scope of this study focuses on developing an automated semantic conflict detection system to evaluate draft ordinances from Davao City against a dual statutory search space. This collection encompasses national statutory enactments compiled to the extent accessible from public legal repositories, paired alongside the historical archive of enacted Davao City local ordinances collected to the fullest extent possible from local legislative records. Operationally, the system audits draft measures across two complementary dimensions: vertical statutory preemption, ensuring local provisions do not contradict superior national statutes under the *Magtajas* doctrine and Republic Act No. 7160; and horizontal consistency, verifying that proposed clauses do not contradict, duplicate, or inadvertently cause implied repeals of active local ordinances.

To establish an administratively grounded and computationally feasible engineering boundary, several explicit delimitations govern this research. The statutory knowledge base is strictly bounded to codified national legislative instruments and enacted local ordinances. To preserve retrieval index compactness and prevent ambiguous litigation records from distorting similarity rankings, judicial case law from Supreme Court decisions and departmental administrative circulars are excluded from the primary index. Furthermore, the local ordinance corpus reflects the practical realities of local legislative record-keeping, where recent council terms benefit from digital database records while earlier decades rely on curated landmark ordinances due to physical archive gaps. The system evaluates drafts strictly against the combined corpus gathered through public portals and direct institutional coordination with the Sangguniang Panlungsod, leaving the comprehensive optical character recovery of older paper files as an ongoing archival task.

The linguistic scope of the system operates exclusively on English-language enactments, following Section 20, Chapter 5, Book I of Executive Order No. 292, which establishes that the English text controls in formal statutory interpretation. While local drafts occasionally incorporate regional Cebuano phrases or administrative Taglish, restricting the primary reasoning pipeline to English prevents semantic drift and avoids translation distortion. At the inference tier, the system evaluates proposed ordinance sections against retrieved candidate provisions using asymmetric span-level pairing (1-vs-N). The pipeline deliberately excludes multi-hop reasoning across multiple independent statutes simultaneously. Detecting intricate legal contradictions that emerge only when synthesizing two or more disparate national statutes exceeds the computational capacity of standard workstation hardware and is designated as a recommendation for future research.

The computational architecture is intentionally restricted to an open-weight transformer encoder envelope capable of local deployment on standard Local Government Unit (LGU) desktop computers, specifically targeting sub-600M parameter models requiring less than 2 GB of inference VRAM. Massive autoregressive large language models and proprietary cloud APIs are excluded to maintain institutional data privacy, avoid recurring cloud API expenditures, and guarantee reliable offline operation. Within this architecture, the system operates strictly as a diagnostic decision-support tool providing transparent, white-box explainability, such as attention heatmaps and calibrated confidence scores, without rewriting or generating statutory text. Finally, the operational scope of the inference engine is delimited to direct normative and modal contradictions arising from explicit textual collisions, such as local prohibitions of federally authorized activities, unauthorized procedural preconditions, or penalty ceiling infractions. The framework deliberately does not model longitudinal legislative histories, historical tax rate compounding, or multi-year financial calculations across unprovided statutory baselines, as observed in complex judicial tax disputes [212]. Modeling such multi-hop temporal construction and arithmetic accounting exceeds the ex-ante diagnostic scope of cross-encoder architectures and is preserved as an avenue for specialized fiscal analytics.

1.6 Definition of Terms

Bi-Encoder. A dual-tower transformer architecture that maps queries and candidate statutory provisions independently into a shared dense vector embedding space, enabling rapid semantic similarity search via vector dot products or cosine distance.

Coarse-to-Fine Architecture. A two-stage decoupled computational framework that pairs a high-throughput, broad candidate retrieval module (coarse stage) with a compute-intensive pairwise sequence classification model (fine stage) to manage extensive textual collections on bounded hardware.

Contradiction. In Natural Language Inference, a directional semantic relationship where the truth of the hypothesis is incompatible with the regulatory mandate of the premise, indicating a direct normative conflict or statutory collision.

Cross-Encoder. A single-tower transformer architecture that ingests concatenated premise-hypothesis pairs into a unified self-attention matrix, enabling full bidirectional cross-attention across all tokens for granular semantic classification.

Entailment. In Natural Language Inference, a relationship where the substantive directive of the hypothesis naturally follows from, conforms with, or is explicitly authorized by the governing statutory premise.

Ex-Ante Legislative Review. The proactive legal and technical evaluation of proposed legislative drafts, ordinances, or administrative rules prior to their formal enactment, designed to detect statutory collisions and constitutional defects while the text remains amenable to revision.

Explainable Artificial Intelligence (XAI). Computational methods that provide human-interpretable rationales for machine learning decisions; in this system, operationalized through token-level self-attention attribution heatmaps highlighting specific conflicting text spans.

Fleiss' Kappa (κ). A statistical measure of inter-annotator agreement among three or more raters assessing categorical data, utilized to validate consensus reliability across the fifteen Sangguniang Panlungsod legal researchers.

Horizontal Inconsistency. A normative clash, redundancy, or unintended implied repeal occurring between a newly drafted local ordinance and an existing, fellow local ordinance enacted within the same legislative jurisdiction.

Information Retrieval (IR). The computational discipline and algorithmic process of searching, filtering, and ranking relevant candidate documents or legal provisions from extensive unstructured text collections in response to a query.

Local Ordinances. Enacted legislative measures promulgated by the Davao City Sangguniang Panlungsod under devolved legislative powers that carry the force and effect of law within the territorial jurisdiction of the city.

Magtajas v. Pryce Doctrine. The foundational Philippine constitutional principle establishing that local government ordinances are strictly subordinate to national statutes, rendering any ordinance that permits what an Act of Congress forbids, or forbids what an Act of Congress permits, *ultra vires* and void.

Natural Language Inference (NLI). A core natural language processing task that evaluates the directional semantic relationship between a premise text and a hypothesis text, categorizing the pair into Entailment, Contradiction, or Neutral.

Neutral. In Natural Language Inference, a relationship where the truth or enforcement of the hypothesis neither follows from nor contradicts the regulatory command of the statutory premise, indicating thematic or regulatory independence.

Normative Conflict. In legal theory and formal deontic logic, a condition where two or more legally valid enactments impose mutually incompatible obligations, prohibitions, or permissions upon the same subject under identical circumstances.

Optical Character Recognition (OCR) Noise. Textual distortions, typographical errors, broken character tokens, and structural formatting anomalies introduced when physical paper documents are scanned and converted into machine-readable text.

Reciprocal Rank Fusion (RRF). An unsupervised rank aggregation algorithm that merges ranked candidate lists from disparate retrieval systems (such as lexical BM25 and dense bi-encoders) based on reciprocal rank positions without requiring calibrated score normalization.

Sangguniang Panlungsod (SP). The unicameral legislative council of Davao City, vested by Republic Act No. 7160 with the corporate and police authority to enact local ordinances and approve resolutions.

Semantic Conflict Detection. The computational formulation and automated identification of directional normative contradictions between legislative provisions using continuous vector representations, lexical indexing, and natural language inference models.

Statutory Preemption. The constitutional and administrative principle under which superior national legislation invalidates, overrides, or limits the regulatory scope of local enactments governing the same subject matter.

Vision-Language Model (VLM). A multimodal deep learning architecture that combines visual processing backbones with transformer language decoders to transcribe document page images into structured text while preserving layout and alphanumeric precision.

Chapter 2

Literature Review

2.1 Semantic Conflict Detection

In jurisprudence and legal philosophy, the fundamental requirement for a functional legal order is internal consistency, where positive enactments provide unambiguous guidance without subjecting citizens to contradictory commands [74]. In legal theory and formal deontic logic, this phenomenon is strictly defined as a *normative conflict*—a structural clash between legal norms wherein an action is simultaneously commanded, prohibited, or permitted across distinct enactments, violating the principle of non-contradiction [129, 3]. Specifically, normative conflict occurs when two or more valid laws impose incompatible legal duties on the same subject under the same conditions, as when one law mandates an act that another law forbids [3, 105]. When statutory systems generate conflicting obligations, citizens cannot conform their conduct to the law without incurring liability under an opposing mandate, destabilizing legal predictability and public trust. Consequently, establishing systematic mechanisms to identify and resolve normative collisions represents a vital safeguard for the integrity of any rule-of-law framework.

However, in real-world legislative governance and local government archives, statutory enactments do not exist as pre-formalized formulas in symbolic logic. Rather, statutes are drafted and promulgated as unstructured, multi-generational natural language texts characterized by high syntactic variance, legal jargon, and implicit exceptions [33]. Consequently, from a computational perspective, this research formulates and operationalizes the challenge as *Semantic Conflict Detection*. By translating abstract deontic clashes into a computable natural language processing problem, semantic conflict detection leverages continuous vector representation spaces and cross-encoder Natural Language Inference (NLI) to identify directional contradictions between local drafts and governing statutory provisions without relying on brittle, hand-crafted legal ontologies [196, 105]. As Ruhl and Katz [181] established, the volume of positive law expands far faster than human auditing capacity, making automated semantic auditing an operational necessity for modern legislative bodies.

2.1.1 Unstructured Legal Corpora

Managing unstructured legal text—text not stored in structured databases—represents a fundamental maintenance challenge for statutory systems. Practical automated auditing requires addressing rapid textual accumulation alongside the linguistic complexity of legislative drafting. These structural realities drive the research transition toward neural retrieval and inference tools for statutory verification.

2.1.1.1 Legal Inflation

The term *legal inflation* describes the continuous, exponential expansion in the sheer volume and interconnected complexity of statutory enactments over time. Ruhl and Katz [181] observe that as modern societies grow more complex, legislative bodies enact new laws at a pace that vastly outpaces the capacity of human legal staff to manually review them for consistency. Within legal informatics and administrative science, this rapid accumulation generates *regulatory debt*—formally defined as the compounding institutional burden of uncoordinated, obsolete, or overlapping legal enactments that accumulate when new legislative measures are promulgated without systematically reconciling or harmonizing them against pre-existing statutory codes [181]. In local government administration, this phenomenon is particularly acute because municipal ordinances are rarely compiled into centralized, cross-referenced statutory codes [43]. Conchas and Salapa [43] emphasize that transforming historical municipal enactments into machine-readable data presents severe engineering hurdles because local records are scattered across disparate administrative formats, ranging from formal codified enactments to transitory executive orders.

Consequently, the statutory search space for identifying normative conflicts constitutes a fragmented, multi-generational archive rather than a structured relational database. Zhong et al. [250] characterize this challenge as a dynamic consistency problem, wherein governing facts and regulatory obligations continually shift across successive council terms. Because local statutory data lacks centralized indexing, municipal agencies are forced into a reactive posture, discovering statutory preemption defects only after an invalid measure is challenged in court. This systemic backlog motivates the transition in legal informatics from brittle, rule-based systems toward neural language models capable of learning semantic representations directly from unstructured text [33]. Such automated systems are required to process extensive legislative corpora that exceed the working memory and analytical throughput of human researchers.

2.1.1.2 Limits of Manual Review

In contemporary legal informatics, digitizing physical documents into electronic archives no longer represents the primary operational bottleneck; rather, the central hurdle is auditing whether newly introduced provisions logically align with governing positive law. As legislative throughput accelerates, manual review workflows become overwhelmed by the cognitive burden of cross-referencing multi-volume statutory digests. A primary challenge for human reviewers is distinguishing between beneficial regulatory overlap—where multiple enactments legitimately address adjacent dimensions of a public policy issue—and destructive statutory conflict, where an enactment inadvertently contradicts a superior mandate. Recent surveys confirm that conventional keyword-based search platforms consistently fail to capture these subtle legal distinctions [10]. Consequently, latent statutory contradictions frequently remain undetected throughout the drafting lifecycle, shifting the burden of conflict discovery from legislative staff during pre-enactment review to judicial authorities during adversarial litigation [223].

2.1.1.3 Semantic Ambiguity

Traditional search systems rely on exact character matching, operating on the premise that legal concepts can be retrieved by searching for specific vocabulary strings. However, natural legal language exhibits pervasive *semantic ambiguity* driven by two primary linguistic phenomena: *synonymy* (where different words express the identical legal concept, such as *prohibited* versus *unlawful*) and *polysemy* (where the same word assumes drastically different meanings depending on context, such as a *party* referring to a litigant rather than a social gathering) [33, 87]. In statutory drafting, normative directives are governed by deontic modal verbs—such as *shall*, *must*, or *may*—whose binding legal force cannot be determined by simple string matching. Chalkidis et al. [33] demonstrate that this syntactic ambiguity causes exact-match techniques to fail when processing complex legal tasks. For example, a local ordinance regulating "motor vehicles" and a national statute governing "two-wheeled conveyances" might address the exact same civic activity, yet a Boolean query for "automobile" will fail to retrieve either provision due to vocabulary divergence.

Furthermore, legal consistency frequently depends on implied normative boundaries rather than explicit textual declarations. Rigano et al. [175] establish that identifying these subtle contradictions requires computational systems that evaluate logical relationships and contextual semantics rather than surface character overlap. Transformer-based language models address this limitation by projecting text sequences into high-dimensional vector spaces that capture deep contextual semantics [249]. However, deploying neural architectures introduces an explainability trade-off. As Sovrano et al. [201] argue, while deep neural models achieve high classification accuracy, their uninspected internal activations function as opaque black boxes that fail to articulate the underlying legal reasoning behind a flagged conflict. In administrative and legislative settings, articulating the verifiable *ratio decidendi* behind an alert is just as vital as the detection itself [193].

2.1.2 Local Conflict Detection

While international legal informatics benchmarks such as LexGLUE advance automated conflict detection across Western legal corpora, corresponding research within post-colonial, developing jurisdictions like the Philippines remains in its nascent stages [33]. Since the passage of the Local Government Code of 1991, Philippine local government units have enacted thousands of local ordinances that remain largely unindexed in machine-readable formats [84]. Although theoretical algorithms for conflict detection are established in mainstream artificial intelligence literature, their empirical application to localized Philippine statutory text remains largely unaddressed.

2.1.2.1 Philippine Legal NLP

Within domestic computer science, research applying Natural Language Processing to Philippine law has historically focused on administrative classification and text summarization rather than deductive logical reasoning. Early investigations utilized basic supervised classifiers to forecast appellate outcomes in Supreme Court decisions [232] or applied extractive algorithms to condense extensive jurisprudence [183]. More recently, domestic legal informatics has explored transformer-based dense embedding architectures, which map text sequences into continuous mathematical vectors where geometric distance corresponds to topical relatedness [178]. While dense embeddings significantly outperform traditional keyword searches in retrieving domain-specific statutes (such as tax regulations), they remain fundamentally correlational: they identify that two provisions address the same subject, but lack the deductive reasoning layers required to determine whether the two provisions contradict each other.

Analyzing these research patterns reveals a pronounced structural gap in local literature. The Philippine NLP community has successfully automated the categorization, summarization, and retrieval of national statutory text, while also proving that inference models can resolve semantic contradictions in local dialects for general-domain tasks such as online fact-checking [232, 183, 178, 155]. However, extending these advanced inference models to detect normative conflicts within the decentralized, multi-generational ordinances of a Highly Urbanized City like Davao remains entirely unexplored.

2.1.2.2 Logic and Conflict Research

Within Philippine computational linguistics, the modeling techniques required for automated conflict detection—specifically Natural Language Inference (NLI)—have been successfully deployed across adjacent domains, most notably in detecting misinformation and rumors in low-resource vernacular text. Nuñez et al. [155] demonstrated that transformer architectures effectively resolve directional logical contradictions across code-switched Tagalog and administrative Taglish.

However, when domestic researchers analyze local government records, methodology routinely regresses to legacy information retrieval frameworks. Document indexing across Philippine local government units relies predominantly on Boolean keyword matching and rigid rule-based heuristics. As Naguio et al. [148] demonstrate, these traditional systems suffer from severe *lexical fragility*: they fail when synonymous terms are used across enactments (such as “firecrackers” versus “pyrotechnic devices”) and collapse entirely when encountering optical character recognition (OCR) noise in historical scans. Relying on manual rules to isolate statutory contradictions across thousands of localized city ordinances is mathematically unviable for human researchers and rule-based software alike.

2.1.2.3 Research Gaps

While the Philippine AI community has successfully automated the search and summarization of national laws, a significant gap remains in using deep logic models to audit local ordinances. Research indicates this gap persists because local legislation is substantially more challenging for automated systems to process than national statutory compilations.

The primary structural barrier to local legislative auditing is the critical dataset gap, characterized by the complete absence of curated ground-truth datasets for Philippine local enactments. Unlike national statutes that benefit from centralized repositories, local ordinances represent low-resource textual environments [168]. Standard legal NLP models are trained on massive Western corpora like LexGLUE, but there are no equivalent standardized benchmarks providing annotated pairs of conflicting local ordinances [75, 151]. Without a standardized benchmark to measure progress, local legal informatics research has historically struggled to move beyond superficial keyword search to deductive logical reasoning [75].

Compounding the lack of training data is a pronounced computational resource gap within Local Government Units (LGUs). Deductive reasoning models, particularly deep cross-encoders, demand substantial compute and memory resources that local government offices typically do not possess. Studies on Philippine public administration indicate that local digital adoption remains constrained by limited funding, scarce machine learning personnel, and legacy office desktop hardware [75]. In many local legislative councils, fewer than twenty percent of legislative personnel have received formal ICT training, with most offices still relying on paper records and basic text editors rather than structured databases [75]. This lack of digital infrastructure makes it difficult to justify complex models that would be too computationally demanding to execute on actual LGU workstations [101].

This computational disparity is directly reflected in the hardware profiles of municipal administrative offices. Under the Government Accounting Manual and Commission on Audit guidelines (COA Circular No. 2022-004), the estimated useful life of government information and communications technology equipment is fixed at five years, resulting in municipal workstation fleets that are routinely four to seven years old before replacement cycles occur [42, 75]. Furthermore, technical procurement specifications established by the Department of Information and Communications Technology (DICT) and the Procurement Service of the Department of Budget and Management (DBM-PS) for common-use clerical and legislative office computers specify mid-range quad-core or hexa-core processors (such as Intel Core i3 or i5 and AMD Ryzen 3 or 5 series), an 8 GB system memory ceiling, integrated graphics sharing system RAM without discrete CUDA acceleration, and standard solid-state drives [56]. Because standard local government computers lack dedicated tensor-processing hardware, deploying multi-billion-parameter autoregressive large language models locally is mathematically and financially unviable, while cloud-hosted commercial APIs introduce recurring subscription costs that exceed municipal operating budgets. Consequently, developing practical ex-ante conflict detection for local governance requires engineering lightweight, retrieve-then-entail pipelines capable of executing completely offline on consumer-grade 8 GB CPU-only workstations.

At the textual level, local legislative conflict detection faces a complex sociolinguistic gap. Local ordinances exhibit higher linguistic heterogeneity than national statutes because local drafting frequently blends formal legal English with regional dialects like Cebuano and administrative Taglish [65]. Standard NLP tokenizers designed for monolingual English often break down when encountering localized legal idioms or archaic Spanish loanwords embedded in Philippine local codes [142]. While recent studies confirm that neural inference models can parse regional dialects in informal social media text, extending this capacity to dense, multi-generational local enactments requires specialized domain adaptation that remains largely unexplored in local literature [37, 87].

Finally, the slow adoption of computational consistency checking is driven by an institutional trust gap across the legal profession. As emphasized by Supreme Court Senior Associate Justice Marvic Leonen, the legal community remains cautious regarding artificial intelligence because conventional deep learning models function as opaque black boxes that cannot articulate their underlying reasoning [126]. Legal researchers and legislative counsel legitimately fear that automated tools could introduce subtle statutory errors, leading to the enactment of void legislation [43]. Consequently, academic research has historically favored low-risk document indexing and retrieval systems, leaving deep semantic conflict detection unaddressed in local governance [12].

2.2 Philippine Legislative Domain

The Philippine legal system operates under a unitary constitutional framework, wherein Local Government Units (LGUs) possess purely delegated legislative powers conferred by Congress [84]. This jurisdictional structure establishes a strict hierarchical subordination that bounds municipal lawmaking beneath national legislative authority. Any computational system engineered to audit municipal enactments must operationalize these jurisdictional constraints. This section analyzes the statutory and structural conditions governing Philippine local law. First, it details the constitutional and statutory hierarchy subordinating local ordinances beneath national laws. Second, it traces the historical and stylistic roots of Philippine legal syntax. Third, it examines the parliamentary mechanics of the Three Readings legislative procedure during plenary sessions and the challenge of bilingual translation. Fourth, it contrasts ex-ante legislative auditing with ex-post judicial review, and finally reviews existing consistency checking methods to demonstrate the necessity of automated semantic auditing.

2.2.1 Hierarchical Constraints

The legal system in the Philippines depends on the fundamental rule that national laws always override local enactments. To understand the legal constraints governing local legislative drafting, automated conflict detection systems must reflect the structural principles that subordinate local ordinances beneath national statutes.

The Philippine legal system operates under a unitary constitutional framework, establishing a rigid hierarchy where national laws strictly control subordinate enactments [84]. At the apex stands the 1987 Constitution, followed by national statutes enacted by Congress, international treaties, presidential executive issuances, and local ordinances. Under this constitutional structure, no Local Government Unit can enact measures that contradict an Act of Congress. The doctrine of national preemption establishes an absolute boundary: whenever the national legislature enacts comprehensive statutory standards governing a given regulatory domain, local governments are legally precluded from enacting conflicting or diluting local rules [84].

The primary organic charter governing local lawmaking is the Local Government Code of 1991 (Republic Act No. 7160). Section 16 of the Code, known as the General Welfare Clause, delegates police powers to local legislative councils to enact ordinances promoting the health, safety, and general prosperity of their constituents, provided such enactments remain strictly within statutory boundaries [84]. Because Davao City is classified as a Highly Urbanized City (HUC), it operates autonomously from provincial administrative supervision, meaning its ordinances are not reviewed by a provincial board. This administrative independence places the entire burden of ex-ante consistency auditing directly upon the Davao City Sangguniang Panlungsod before measures proceed to final enactment [84].

The definitive judicial test governing the validity of local legislation is established by the Supreme Court in the landmark ruling *Magtajas v. Pryce Properties Corporation, Inc.* [208]. To withstand judicial scrutiny, an ordinance must satisfy six foundational substantive requirements: it must not contravene the Constitution or any national statute, it must not be unfair or oppressive, it must not be partial or discriminatory, it must not prohibit but may regulate trade, it must be general and consistent with public policy, and it must not be unreasonable. Because local legislative bodies possess purely delegated authority, local councils cannot prohibit activities that national legislation has expressly authorized or regulated [84]. This principle of statutory subordination forms the primary legal benchmark against which automated conflict detection systems must evaluate candidate local drafts.

When a local ordinance impermissibly enters an area governed by national legislation, Philippine courts consistently apply the *Magtajas* doctrine to declare the local measure invalid. In *Magtajas v. Pryce Properties Corporation, Inc.* [208], the City Council of Cagayan de Oro enacted ordinances prohibiting casino operations to protect public morals. The Supreme Court struck down the local enactments as *ultra vires*, ruling that a local council cannot prohibit commercial gambling activities that a national statute expressly authorized through the Philippine Amusement and Gaming Corporation (PAGCOR) charter. Similar preemption disputes have directly affected Davao City legislation. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [211], the Supreme Court invalidated Davao City Ordinance No. 0309-07, which imposed a local ban on aerial pesticide spraying across agricultural plantations, holding that the ordinance exceeded local police powers and collided with the regulatory authority granted to the national Fertilizer and Pesticide Authority under Presidential Decree No. 1144. In *City of Manila v. Laguio, Jr.* [210], the Court likewise struck down an ordinance prohibiting motels and massage parlors, holding that local police powers cannot arbitrarily prohibit lawful commercial activities under the guise of zoning regulation.

In contrast to vertical statutory preemption, local lawmaking also involves horizontal consistency, where newly proposed ordinances must align with existing, fellow local enactments. When drafting local legislation, the Sangguniang Panlungsod routinely uses the Three-Reading legislative process, particularly the committee debates and line-by-line amendments during the Second Reading, to harmonize new proposals with older local enactments [186]. Unlike vertical conflicts that represent fatal legal defects, inconsistencies between a new draft and an older local ordinance frequently reflect intentional legislative updates, such as revising local tax rates or adjusting traffic routes. To manage these horizontal modifications systematically, legislative drafters rely on standardized legislative saving devices recommended by the Department of the Interior and Local Government (DILG) [57]. Every formal ordinance incorporates an express or general repealing clause, declaring that all prior local enactments inconsistent with the newly approved measure are amended or repealed accordingly. Furthermore, ordinances feature a separability clause, acting as a structural legal safeguard ensuring that if any single section is declared invalid by judicial authorities, the unoffending sections remain in full legal force [57]. However, because local legal researchers must review hundreds of active local enactments under strict legislative session schedules, subtle inconsistencies often go unnoticed during the Second Reading, allowing problematic clauses to proceed to final enactment.

A fundamental design decision in legal informatics is delineating which positive legal instruments establish binding preemption standards. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III, Title I, Chapter 2, Section 2 of the Administrative Code of 1987 (Executive Order No. 292) [171], Executive Orders (EOs) are acts of the President providing rules of a general or permanent character in implementation or execution of constitutional or statutory mandates. Unlike Supreme Court decisions—which function as *ex-post* judicial dispute resolutions settling adversarial controversies under the doctrine of *stare decisis* (Article 8, Civil Code)—Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force and effect of law. Furthermore, during both the Martial Law period (1972–1986) and the Revolutionary Government under the Freedom Constitution (Proclamation No. 3, 1986–1987), the President exercised direct legislative powers through Executive Orders and Presidential Decrees, creating substantive national enactments that remain operative law today. Modern Executive Orders frequently establish binding nationwide regulatory standards—such as Executive Order No. 26 (s. 2017) establishing nationwide public smoking prohibitions, or Executive Order No. 168 (s. 2014) creating the Inter-Agency Task Force for the Management of Emerging Infectious Diseases (IATF)—that legally preempt conflicting local legislation. Excluding Executive Orders would therefore omit critical nationwide regulatory mandates that local ordinances are legally bound to respect.

2.2.2 Historical and Stylistic Heritage

The structural complexity of Philippine legislative text stems directly from the nation's historical experience as a mixed legal jurisdiction [4, 49]. Philippine jurisprudence combines continental civil law traditions with Anglo-American statutory common law conventions. Understanding why Philippine statutes and local ordinances are drafted in their characteristic syntax requires examining how these colonial and post-colonial traditions fused over more than a century of codification.

During more than three centuries of Spanish rule, the Philippine legal system was rooted firmly in the Roman-Germanic civil law tradition [4]. This continental tradition emphasized systematic codification, deductive legal reasoning, and broad normative principles organized into structured books, titles, and chapters. Foundational enactments, such as the Spanish Civil Code of 1889 and the Penal Code of 1870, relied on generalized legal mandates rather than granular factual definitions. Furthermore, this civilian heritage embedded a rich vocabulary of Latin doctrinal maxims into Philippine jurisprudence, including *in pari materia* (construing related laws together), *generalia specialibus non derogant* (a general law does not override a specific one), and *ultra vires* (acts beyond legal authority) [4]. Modern Philippine codes retain this civilian structure, expecting adjudicators to derive specific legal duties from comprehensive normative commands.

Following the Treaty of Paris and the establishment of the American Insular Government (1900–1935), American legal administrators introduced common law statutory drafting habits without fully displacing the civilian codification base [49]. Enacted by the Philippine Commission and later the Philippine Legislature, early statutes introduced formal drafting conventions borrowed directly from United States federal and state legislation. These conventions included explicit statutory title preambles, exhaustive definitions sections, severability clauses, and general repealing provisions [4]. Significantly, American administrators introduced the practice of drafting extensive conditional provisos (introduced by the traditional formula "*Provided, that...*" and "*Provided, further, that...*"), designed to restrict, qualify, or create exceptions to preceding operative clauses within the same section. The Revised Administrative Code of 1917 (Act No. 2711) and the contemporary Administrative Code of 1987 (Executive Order No. 292) solidified this mixed tradition, confirming English as the controlling statutory language while maintaining civil law institutional categories [171].

This dual heritage produces a legislative drafting style marked by high syntactic density and multi-clause complexity. Unlike standard conversational English, a single statutory provision frequently spans one hundred to two hundred words contained within a solitary grammatical sentence [20]. These sentences embed multiple dependent clauses, qualifying prepositional phrases, and recursive proviso exceptions. At the local level, legislative staff adopt this formal national drafting template while adding procedural boilerplate, resulting in verbose ordinances where the operative regulatory command is separated from its governing subject by dozens of intervening words [57, 186]. From an analytical and document processing perspective, this hybrid syntax presents substantial structural hurdles: archaic legal terms resist straightforward vocabulary normalization, long-distance dependencies obscure the core legal directive, and qualifying provisos invert the practical meaning of preceding clauses unless the text is examined in its full structural context.

2.2.3 Plenary Legislative Procedure and the Three Readings Framework

Besides the hierarchy of laws, the actual parliamentary mechanism through which laws are drafted, debated, and approved determines how statutory text evolves over time. In the Philippine legal architecture, positive law enactment is governed by a mandatory three-reading procedure conducted during formal plenary sessions of the legislative body. At the national level, Article VI, Section 26, Paragraph 2 of the 1987 Philippine Constitution explicitly dictates that no bill passed by either the House of Representatives or the Senate shall become law unless it has passed three readings on separate days, with printed copies in final form distributed to members at least three days prior to passage [45, 99]. At the local government level, Sections 53 and 54 of the Local Government Code of 1991 (Republic Act No. 7160) [84], the Local Legislative Drafting Manual of the Department of the Interior and Local Government (DILG) [57], and the Internal Rules of Procedure of the 20th Sangguniang Panlungsod of Davao City (Resolution No. 001-22) [186] replicate this parliamentary doctrine. Under these statutory rules, every proposed municipal or city ordinance must undergo three distinct readings before the full assembly in plenary session. This parliamentary mechanism is designed to guarantee institutional transparency, deliberate consideration, and democratic accountability before any regulatory constraint attains the force of law.

The legislative progression begins on the plenary floor with the First Reading. In both national congressional chambers and the Davao City Sangguniang Panlungsod, the

First Reading is strictly ministerial and introductory: the presiding officer directs the Secretary to read solely the assigned measure number, title, and the name of the author or proponent before the assembled plenary body [99, 186]. Crucially, parliamentary law forbids any substantive debate, interpellation, or floor amendment during First Reading. Instead, the presiding officer refers the proposed measure immediately to the appropriate standing committee—such as the Committee on Rules, Laws, and Ordinances—initiating the inter-session committee review phase. In this committee stage, technical working groups, legal counsel, and committee researchers examine the draft, hold public hearings, and prepare the formal committee report. Because local ordinance drafting is decentralized—with individual council members authoring distinct proposals rather than relying on a centralized drafting bureau [84, 186]—drafting conventions and legal terminology vary widely. Legal researchers bear the heavy burden of manually cross-referencing proposed clauses against thousands of national statutes and prior city enactments under strict legislative session schedules. When staff lack automated semantic search tools, potential preemption defects and horizontal contradictions frequently survive committee scrutiny undetected.

Following committee review, the proposed ordinance returns to the plenary floor for the Second Reading, which represents the primary substantive and deliberative phase of the legislative lifecycle. The Second Reading commences with a formal sponsorship speech delivered by the committee chair or measure proponent, summarizing the policy objectives and legal basis of the enactment. Immediately following sponsorship, the plenary assembly enters the Period of Interpellation and General Debate. During this period, colleagues interrogate the sponsor, probe statutory ambiguities, question fiscal or regulatory feasibility, and raise substantive legal objections regarding whether the measure contravenes superior national legislation. Once general debate is formally terminated, the body transitions into the Period of Amendments, where committee amendments and individual floor amendments are presented, debated, and voted upon section by section [99, 186]. This period of amendments is the decisive procedural window where proposed clauses are actively reshaped, modified, or deleted to align with governing law.

The Period of Interpellation and Amendments on Second Reading constitutes the exact procedural nexus where an automated semantic conflict detection system delivers its highest institutional utility. Because the Second Reading is the sole phase on the plenary floor where substantive textual alterations, statutory reconciliations, and corrective rewrites can be introduced, legal researchers and sponsors require granular, evidence-backed conflict diagnostics before and during floor deliberations. If a draft ordinance clause encroaches upon a national statutory domain—violating the *Magtajas* doctrine by attempting to prohibit what national law permits, or by exceeding statutory penalty caps under Section 458 of Republic Act No. 7160—the automated system pinpoints the conflicting statutory premise, flags the contradictory deontic modalities, and provides an explainable legal audit trail. Equipping Sangguniang Panlungsod legal researchers with these automated findings allows them to draft precise, legally sound floor amendments during the Period of Amendments, rectifying latent preemption flaws before the measure advances past the amendable phase.

After the assembly adopts all floor amendments and finalizes the draft on Second

Reading, the measure is engrossed, printed in its final form, and scheduled for Third Reading in plenary session. In stark contrast to the dynamic debates of Second Reading, the Third Reading is procedurally rigid and unmodifiable. Under Article VI, Section 26, Paragraph 2 of the 1987 Constitution and corresponding local Sanggunian rules, no amendment whatsoever is permitted upon the Third Reading of a bill or ordinance [45, 99, 186]. Plenary action is confined strictly to a nominal roll-call vote, where each member registers an affirmative, negative, or abstaining vote, accompanied by a brief oral explanation. This procedural finality creates severe institutional risk: if a statutory collision is not discovered and rectified prior to or during the Second Reading, the flawed provision is permanently locked into positive law upon Third Reading approval. The enacted ordinance becomes vulnerable to immediate judicial nullification under the *Magtajas* doctrine, administrative disputes before national executive departments, or citizen litigation. This structural irreversibility underscores why ex-ante computational conflict detection is essential during committee review and Second Reading preparation.

Compounding the procedural complexities of the Three Readings is the statutory mandate requiring local legislation to be accessible across languages. Under Section 469 of the Local Government Code, the Secretary to the Sanggunian is mandated to translate approved ordinances into local dialects, such as Cebuano for Davao City [84, 186, 65]. However, this creates legal friction because the Administrative Code of 1987 establishes that the English statutory text strictly controls in cases of conflict [171]. If an error occurs in regional translation, a citizen relying on the Cebuano text may inadvertently violate the authoritative English version, conflicting with the public notice doctrine in *Tañada v. Tuvera* [206]. Despite policy initiatives such as the *Batas sa Sariling Wika Act* [169], subtle dialect differences often produce semantic drift where the exact legal obligation in English is distorted in regional vernacular text [142]. Consequently, consistency checking systems must operate primarily on the authoritative English statutory provisions while accounting for regional linguistic artifacts generated during local committee debates and drafting iterations.

2.2.4 Ex-Ante Legislative Review

The term "ex-ante" derives from Latin meaning "before the event." In legislative informatics and administrative jurisprudence, it denotes the rigorous evaluation of a proposed law or local regulation prior to its official enactment [72]. This prospective auditing process aims to identify and remediate potential statutory collisions, constitutional infirmities, and unintended regulatory burdens while the draft remains amenable to revision.

In the Philippines, ex-ante regulatory review has achieved heightened institutional prominence following the enactment of Republic Act No. 11032, known as the Ease of Doing Business and Efficient Government Service Delivery Act of 2018 [85]. Under this statutory framework, the Anti-Red Tape Authority (ARTA) alongside the Department of the Interior and Local Government (DILG) mandates that Local Government Units conduct a formal Regulatory Impact Assessment (RIA) for proposed regulations impacting commerce and public welfare [85]. The objective is to verify that proposed ordinances possess valid statutory authority and do not impose duplicative or contradictory requirements on citizens and businesses.

The fundamental distinction between ex-ante and ex-post evaluation lies in timing, methodology, and legal impact across the hierarchy of legal instruments. Ex-ante evaluation focuses strictly on prospective, forward-looking legislative texts—such as draft local ordinances, pending congressional bills, and proposed executive issuances—prior to their passage into positive law. In contrast, ex-post evaluation operates retrospectively through judicial dispute resolution, encompassing adversarial litigation, appellate court rulings, and Supreme Court jurisprudence governed by *stare decisis*. While ex-post judicial adjudication rectifies constitutional injuries after harm has occurred, it incurs substantial societal and litigation expenses, as repealing an active ordinance requires protracted court proceedings [225]. Ex-ante auditing is therefore substantially more efficient, protecting local governments from administrative deadlock and judicial invalidation.

Deploying automated computational tools for ex-ante conflict detection empowers local legislative researchers to identify normative collisions during the drafting stage. Because legislative bodies generate extensive statutory volumes, manual review frequently fails to catch subtle inconsistencies between new drafts and existing laws [181]. By operating as an automated diagnostic triage tool, computational systems flag potential preemption conflicts while the measure remains in committee deliberation. This facilitates alignment with superior national statutes, mitigating the risk of enacting local measures that are legally void *ab initio* [208].

2.2.5 Legislative Consistency Checking

Because local governments hold subordinate legislative authority within a decentralized drafting environment, systematic consistency checking represents an essential institutional safeguard. However, verifying that an ordinance does not contradict superior statutes requires far more than title matching; it demands deep comprehension of legal semantics and subtle jurisdictional boundaries. Current consistency checking methods remain encumbered by syntactic ambiguity and reliance on basic tracking platforms.

Discourse analysis of Philippine statutory language reveals pervasive ambiguity in how normative keywords and modalities are deployed [207]. In particular, the modal auxiliary “shall” introduces frequent interpretive friction, as it may denote an absolute command, an administrative directive, or a directory permission depending on statutory context. Empirical studies confirm that such syntactic ambiguities complicate legal interpretation even when enactments adhere to formal drafting guidelines [20].

Philippine jurisprudence recognizes that statutory conflicts rarely present themselves through explicit textual declarations [49]. A local ordinance virtually never declares an overt intent to contradict an Act of Congress; rather, conflicts typically arise as implied preemption. This occurs when a national statute occupies a regulatory field, and a local council introduces unauthorized preconditions or local fees that restrict activities already licensed by the state [207]. In such cases, the Supreme Court has consistently held that the local measure impermissibly modifies national law under the guise of police power regulation. For manual reviewers and keyword-based search tools, detecting implied preemption is exceptionally difficult because the conflicting enactments employ disparate vocabularies, necessitating deep semantic analysis to uncover the contradictory regulatory effect.

In the local context, the Davao City Sangguniang Panlungsod processes extensive legislative volume, passing nearly 400 ordinances and over 2,000 resolutions within a single three-year council term [159]. To manage this throughput, the city deploys administrative platforms: the Resolution Ordinance Tracking System (ROTS) manages document routing and metadata, while the Legislative Information Support System Program (LISSP) enables council members to access digitized enactments [37]. However, while these digital systems streamline document logistics, substantive consistency auditing across provisions remains entirely manual. Furthermore, archival scans frequently contain optical character recognition (OCR) noise, causing keyword search queries to miss critical governing provisions [133].

2.3 Digital Legislative Archives

Building a functional Semantic Conflict Detection system requires access to comprehensive and machine-readable legal corpora. In the Philippines, the digital landscape of legislative data is highly fragmented, split between national statutory repositories and localized city archives. This structural division presents significant technical hurdles for statutory data ingestion. Before deep logical inference can occur, automated auditing pipelines must first secure reliable data streams from these disparate legal environments. Consequently, understanding the specific characteristics of these sources informs the design of statutory retrieval and preprocessing methodologies.

2.3.1 National Sources

National legal repositories represent the authoritative apex of the Philippine legislative hierarchy, housing laws that universally preempt local ordinances. These databases are maintained by centralized government agencies and contain highly formalized texts such as Republic Acts, Commonwealth Acts, and Batas Pambansa. They serve as the primary knowledge base for external legal context, providing the authoritative baseline rules against which local drafts are evaluated. Because national legislation undergoes a rigorous, standardized drafting process, the vocabulary within these repositories is generally consistent and adheres to strict legal drafting standards. Accessing these national databases provides computational systems with the necessary macroscopic view of state-level mandates and prohibitions.

The Official Gazette serves as the constitutionally mandated repository for Philippine statutes, rendering it the primary journal of record for national legislative text [86]. It acts as the official publisher for all newly enacted national laws, presidential executive orders, and administrative issuances, as required by Republic Act No. 453 [162] and Commonwealth Act No. 638 [150]. Because it represents the definitive, legally binding version of a statute, any computational system evaluating local statutory preemption must anchor its baseline rules to this repository. This archive effectively acts as the authoritative historical ledger for state-level mandates, preserving the exact phrasing approved by the legislative and executive branches. Consequently, it remains the most critical target for establishing an authoritative statutory knowledge base.

Complementing the official journal, developers utilize the Open Congress API provided by the House of Representatives Legislative Information System (LEGIS) to access structured, machine-readable metadata regarding legislative routing and statutory amendments [100]. This programmatic interface tracks a statute's formal lifecycle, charting its progression from initial committee filing to final executive approval. By providing standardized output formats, the API allows computational pipelines to establish relational dependencies between different legislative acts without manually parsing unstructured text.

Finally, the Lawphil Project functions as a massive, privately maintained digital repository of statutory records and jurisprudential decisions [9]. Unlike official government channels that frequently change web templates, this repository provides a comprehensive, pre-digitized plaintext corpus of historical and contemporary Philippine law. This accessibility makes it mathematically ideal for the unsupervised pre-training and vectorization of foundational language models tailored to the local legal domain. By ingesting this vast corpus of clean text, NLP architectures learn the highly specific syntactic structures and deontic operators inherent in Philippine jurisprudence, establishing the core vocabulary required for transformer models to process specialized statutory drafting [249].

2.3.2 Local Sources

In contrast to centralized national databases, local legal repositories are maintained independently by individual Local Government Units (LGUs). These archives house local ordinances, resolutions, and executive orders that govern specific geographic jurisdictions. The legislative output at this tier is entirely decentralized, resulting in highly variable archiving practices depending on the technological capacity of the specific city council. These local databases form the immediate target corpus for local conflict detection, containing the actual regulatory constraints imposed on city residents. Consequently, they represent a high-volume, dynamic data environment that continuously expands as the local council holds regular drafting sessions.

The locality's primary corpus consists of the Davao City Sangguniang Panlungsod website, which serves as the official digital repository for local legislative outputs [38]. This portal acts as the primary historical record for city regulations, resolutions, and zoning policies enacted within the city's jurisdiction. It represents a dynamic and continuously expanding text environment that directly reflects the decentralized nature of local lawmaking. The texts housed here encapsulate the specific regulatory constraints and policy directions of the city government, providing the exact ordinance text required to evaluate new drafts for logical consistency against the existing legislative framework.

To manage the extensive volume of local legislation, the city introduced the Legislative Information Support System Program (LISSP) as its primary administrative backbone [37]. This platform serves as a digital tracking system designed to maintain metadata regarding ordinance routing, committee assignments, and plenary approval statuses. The system provides transparency for monitoring ordinances and resolutions in real time [160]. By digitizing the legislative workflow, LISSP effectively functions as a centralized database of ordinances, providing the structured tracking and contextual data necessary for evaluating the lifecycle of local laws, while addressing the interoperability challenges identified in recent regional e-governance assessments [43].

2.3.3 Data Extraction Challenges

While the aforementioned repositories contain the necessary legal knowledge, extracting the needed data for computational analysis presents severe technical challenges. Scraping and formatting these diverse sources exposes the pipeline to significant data degradation and structural inconsistencies across multiple dimensions.

2.3.3.1 Archival Scanning and OCR Degradation

A primary barrier to automated statutory auditing is the lack of native, machine-readable text across archival collections. Both the Official Gazette and the Davao City Sangguniang Panlungsod public archive predominantly host scanned, image-based PDFs rather than selectable digital documents. During manual inspection of these collections, enactments were observed to contain varying degrees of visual noise, including slanted scans, faded ink, low contrast, and compression artifacts. As Lopresti [133] documents, such noisy visual data results in severe Optical Character Recognition (OCR) degradation, where standard extraction tools produce broken alphanumeric strings and nonsensical tokens. These irregularities suggest that a direct ingestion pipeline suffers from semantic fragmentation, requiring an additional high-precision extraction layer and extensive text-cleaning heuristics to ensure data viability.

The resulting character and word-level corruptions severely hinder automated semantic analysis. In statutory conflict detection, a single corrupted character can alter the legal interpretation of a provision, such as transforming a negative operator into an affirmative assertion or scrambling a penalty amount. Standard tokenizers struggle to process these degraded sequences, fragmenting broken words into out-of-vocabulary sub-tokens that dilute contextual representations [182]. Furthermore, scanning artifacts frequently obscure section demarcations and legislative numbering, preventing standard parsers from isolating self-contained statutory units. Consequently, automated preemption pipelines must incorporate noise-tolerant document normalization to restore structural integrity before downstream retrieval.

2.3.3.2 Repository Scarcity and Digital Isolation

Another critical challenge is archival fragmentation, wherein legislative repositories suffer from systemic gaps in their digital records. This poses a significant problem for high-fidelity auditing because the absence of even a single related ordinance can lead to false negatives in conflict detection, where a system declares a draft valid simply because it cannot access the contradicting record. A prominent example is the Davao City Sangguniang Panlungsod public archive, which features intermittently missing ordinance files and unorganized chronological listings that hinder systematic data collection [188]. Even relatively complete national databases like Lawphil can lack specific Republic Acts or historical amendments. As Westermann et al. [92] observe, fragmented legal databases force developers to move beyond simple scraping toward fault-tolerant data engineering to ensure an exhaustive corpus.

A further obstacle in statutory data acquisition is the discrepancy between the theoretical and actual accessibility of local legislative databases. For example, while administrative systems like Davao City's LISSP are promoted in institutional reports as public transparency portals [37, 160], empirical verification reveals strict authentication barriers requiring authorized credentials. Technical inspection indicates that local legislative portals often remain computationally isolated behind authentication interfaces, restricting automated extraction by public web scrapers. Correspondence with local legislative administrators confirms that full-text enactments require administrative clearance and manual provisioning. As Conchas and Salapa [43] observe, isolated administrative data often exists as digitally orphaned artifacts, where metadata about an enactment is disconnected from its full-text searchable content, requiring specialized ingestion pipelines to reconcile sparse public metadata with offline full-text documents.

2.4 Exact and Symbolic Methods

Before the adoption of neural networks, computational legal analysis relied heavily on deterministic systems to process text. Deterministic systems operate on predefined rule matrices, wherein a given input produces reproducible outputs without probabilistic inference or statistical estimation. Traditional methods for searching and evaluating legal information assumed that legislative ideas could be cleanly mapped either to specific vocabulary strings or to rigid, hand-coded logic formulas. While highly transparent and easy to debug, these foundational approaches frequently fail when forced to detect hidden conflicts in complex, unstructured legal texts. The core issue is that human language is inherently messy, and rigid systems break down when faced with the ambiguous phrasing typical of local ordinances.

2.4.1 Boolean Information Retrieval

Evaluating why legacy legal repositories struggle with semantic conflict detection requires examining how deterministic retrieval architectures process text at the machine level. Boolean Information Retrieval (BIR) represents the foundational search architecture underlying traditional legal databases. BIR operates on the bag-of-words assumption, abstracting statutory enactments into unordered multisets of terms while discarding grammatical syntax, clause order, and relational context. At its core, the architecture relies on an *inverted index*—a mapping that pairs every distinct vocabulary term with a postings list recording document identifiers and term frequencies. Because an inverted index registers solely the discrete presence or absence of literal character strings, the retrieval engine cannot infer latent semantic relationships, grammatical qualifications, or underlying regulatory intent.

At its core, the retrieval mechanism in BIR is strictly based on set theory and formal Boolean logic. A user constructs a query using logical operators like `AND` ($\wedge$), `OR` ($\vee$), and `NOT` ($\neg$) to create intersecting sets of documents. For example, a search for a specific ordinance regarding explosives is formulated mathematically as:

Boolean Query Set Intersection Formulation:

$$Q = (\text{fireworks} \vee \text{pyrotechnics}) \wedge \neg\text{exempt} \tag{2.1}$$

Where the compound query Q combines disjunctive keyword sets under conjunction with a negated exemption condition.

The system then retrieves documents by calculating the exact mathematical intersection of these sets, pulling only the files that satisfy every condition simultaneously. If a document contains the specific string of characters requested, it is retrieved and scored as a 1; if it misses even a single required term, it is completely ignored and scored as a 0. It evaluates the raw presence of strings, entirely ignoring the conceptual meaning or legal intent behind them [23].

2.4.1.1 Challenges with BIR

The primary limitation of BIR in legal auditing is its absolute, uncompromising reliance on exact string matching. This rigidity creates a severe "vocabulary mismatch" problem when processing natural language. Human language naturally features synonymy, where entirely different words possess the exact same meaning, and polysemy, where the identical word takes on drastically different meanings depending on the context. In statutory drafting, one author might use "prohibited" while another uses "unlawful," referring to the exact same legal concept. Because BIR operates on character-level matching rather than conceptual mapping, it fundamentally cannot connect these two terms, leading to massive blind spots in the search results.

A landmark empirical evaluation of these systems demonstrated this vocabulary gap in a highly practical setting. Legal researchers conducting queries using standard BIR systems confidently estimated they were finding over 75 percent of the relevant documents for a given case. They experienced an illusion of effectiveness because the documents they did retrieve looked highly accurate, masking the massive volume of relevant files they missed. In reality, the system retrieved less than 20 percent of the actual relevant files contained within the database [23]. The researchers missed the remaining 80 percent simply because the authors of those hidden documents used slightly different, completely valid phrasing that did not perfectly align with the strict search query [227].

In a local government context like Davao City, this specific margin of error is administratively unacceptable. Missing even one existing ordinance during an ex-ante review due to a simple vocabulary mismatch could directly result in the passage of an invalid, conflicting law. Furthermore, legislative vocabulary inherently evolves over time as drafting conventions change. An ordinance passed in the 1990s and a modern draft written in 2026 might strictly regulate the exact same local regulatory subject but share absolutely zero common keywords. Recent studies emphasize that resolving this requires the implementation of "Transformer" models, which abandon exact matching entirely in favor of continuous vector spaces that recognize when different vocabulary points to the same underlying concept [116].

2.4.2 Symbolic AI

Because BIR fundamentally fails to capture underlying meaning, early artificial intelligence researchers attempted to automate legal reasoning by directly modeling the rules of law themselves. This approach, widely known as Symbolic AI or "Good Old-Fashioned AI" (GOFAI), operates on the premise that human cognition can be replicated by manipulating formal symbols [230]. Symbolic AI formalizes legal reasoning by modeling statutory constraints into explicit, hand-coded production rules. Rather than searching unstructured text strings, Symbolic AI translates legislative commands into formal deontic and first-order logic representations, allowing computational engines to execute deductive reasoning over regulatory relationships.

To execute this logic, these systems specifically utilize propositional or first-order logic combined with deontic operators. Deontic logic is a specialized branch of mathematical philosophy that formally evaluates normative modalities governing conduct: obligations ($\mathcal{O}$, representing mandatory actions that must be performed), permissions ($\mathcal{P}$, representing discretionary actions that may be undertaken), and prohibitions ($\mathcal{F}$, representing unlawful actions that must not be performed). For example, a system represents a local zoning prohibition mathematically as:

First-Order Deontic Prohibition Formulation:

$$\forall x(\text{Business}(x) \wedge \text{ResidentialZone}(x) \rightarrow \text{Prohibited}(x)) \tag{2.2}$$

Where the universal quantifier ($\forall$) asserts that for any entity x, if it is classified as a business ($\wedge$) and located in a residential zone, it is strictly prohibited ($\rightarrow$).

By converting ordinances into these rigid formulas, a system can theoretically use deductive reasoning to spot a conflict. If Rule A dictates $\mathcal{O}(X)$ (an absolute obligation to do X) and Rule B dictates $\mathcal{F}(X)$ (an absolute prohibition against doing X), the system successfully flags a formal logical contradiction ($\perp$) [13].

2.4.2.1 Challenges with Symbolic AI

While Symbolic AI provides perfectly transparent and mathematically sound reasoning, its absolute reliance on manual programming makes it practically impossible to scale for real-world governance. The primary architectural barrier is the classical "knowledge acquisition barrier" in expert systems: human domain experts and software engineers must manually read, interpret, and translate thousands of unstructured city ordinances into rigid computer rules. This manual translation process is unacceptably slow, extremely expensive, and highly prone to human error during the coding phase [137]. Furthermore, every time the city council amends an existing law, the underlying symbolic formula must be manually rewritten, making the system incredibly difficult to maintain over time.

Beyond the logistical challenge of manual coding, the rigid binary architecture of Symbolic AI fundamentally clashes with what legal philosopher H.L.A. Hart famously termed the *open texture of law* [94]. As Hart demonstrated in *The Concept of Law*, statutory language cannot anticipate every future factual scenario; consequently, lawmakers deliberately employ open-ended, discretionary standards—such as mandating “reasonable care,” “due diligence,” or “good moral conduct”—to allow adjudicators flexibility across changing societal contexts. These subjective legal standards cannot be reduced to binary IF-THEN rules because their meaning shifts depending on circumstance and judicial interpretation [12]. While symbolic systems achieve deterministic precision in tightly constrained, domain-restricted formal micro-environments, they fail when exposed to the linguistic variance, unformalized exceptions, and semantic ambiguities characteristic of municipal legislative archives [250].

2.5 Graph-Based Architectures

Because exact word matching can be unreliable, computational legal studies frequently explore graph-based architectures to model relationships between statutory provisions. Instead of treating legal texts as isolated character sequences, graph-based frameworks represent statutes as interconnected nodes within a structural network. By analyzing how enactments cite, amend, and conceptually reference one another, these architectures capture latent legal context that bag-of-words models invariably miss.

2.5.1 Knowledge Graphs and GNNs

Legal Knowledge Graphs (LKGs) model statutory ecosystems by representing legal entities, provisions, and administrative bodies as structured vertices linked by directional, typed edges. In formal graph theory, a Legal Knowledge Graph models a statutory corpus as a directed, multi-relational graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, wherein vertices $v \in \mathcal{V}$ represent discrete statutory enactments, administrative agencies, or substantive regulatory provisions, while edges $(u, r, v) \in \mathcal{E}$ designate formal juridical relationships $r \in \mathcal{R}$ such as statutory delegation, cross-reference citation, explicit amendment, or regulatory enforcement. Unlike flat text indices, these relational graphs enable Graph Neural Networks (GNNs) to learn node embeddings by aggregating feature representations across local structural neighborhoods [21]. Researchers have developed automated frameworks like GNN-ERE to extract legal entities and relations directly from unstructured statutory text, establishing structured taxonomies without manual relational coding [139]. Furthermore, empirical evaluations demonstrate that graph-based representations improve legal document retrieval by mapping semantically related court judgments even in the absence of verbatim keyword overlap [93].

2.5.1.1 Graph Modeling and Multi-Hop Reasoning

The foundational strength of Graph Neural Networks lies in their ability to propagate structural context across multi-hop relational paths. In statutory interpretation, normative conflicts rarely occur through isolated direct collisions; rather, they frequently manifest indirectly across chains of administrative definitions and delegated authorities [244]. Formally, Graph Neural Networks execute iterative neighborhood aggregation (message passing), wherein the representation $\mathbf{h}_v^{(l)}$ of node v at layer l is recursively updated by aggregating the latent feature vectors of its adjacent 1-hop topological neighbors. This inductive propagation allows the network to incorporate multi-hop structural context, evaluating whether an ordinance provision conflicts with a superior statutory mandate through intermediate administrative definitions or delegated authority chains. This topological awareness enables systems to distinguish between different legislative functions, such as separating procedural requirements from substantive prohibitions [21]. Consequently, graph neural architectures provide a principled mechanism for structural reasoning across complex, multi-tiered statutory hierarchies.

Furthermore, graph representations capture institutional relationships that remain hidden during sequential text reading. In decentralized governance frameworks like the Philippines, local legislative acts operate within a matrix of national agencies, provincial guidelines, and local zoning boundaries. Graph neural models learn relational node embeddings that reflect these jurisdictional hierarchies, allowing algorithms to evaluate legislative preemption within its broader administrative network [139]. By mapping shared regulatory targets across different enactments, the graph structure guides inference models toward relevant statutory clusters that standard search engines fail to connect. This relational capacity makes graph modeling an appealing conceptual paradigm for analyzing legislative coherence across decentralized jurisdictions.

2.5.1.2 Structural Vulnerabilities in Unstructured Archives

While Graph Neural Networks offer substantial theoretical utility, their practical deployment depends strictly on the completeness and fidelity of the underlying graph representation. A primary vulnerability is their reliance on explicit statutory cross-references to establish graph edges [103]. In Philippine local archives, local ordinances frequently omit formal citations to the governing Republic Acts they inadvertently contradict, relying instead on localized colloquial descriptions. When explicit relational links are missing from the initial graph construction, the GNN remains fundamentally blind to the underlying normative collision, unable to propagate message vectors across disconnected nodes.

This limitation is exacerbated by systemic disconnection and physical degradation across local government records. City archives like Davao City's LISSP store enactments as isolated scanned documents rather than interconnected, machine-readable data nodes [43]. When automated relation extraction pipelines encounter noisy optical character recognition outputs, they generate hallucinated edges and misattributed entity linkages [133]. These erroneous connections propagate through successive graph convolution layers, contaminating node representations and introducing false conflict signals throughout the network. Consequently, without expensive, manual human curation to verify every edge, graph-based systems prove too brittle for noisy, real-world local archives.

2.5.2 Abstract Meaning Representation

While Graph Neural Networks capture macro-level relationships between distinct legal documents, Abstract Meaning Representation (AMR) evaluates the semantic and logical structure inside individual sentences. AMR represents the meaning of a sentence as a rooted, directed acyclic graph where nodes correspond to semantic concepts and edges represent semantic roles [15]. In legal informatics, AMR provides a formal representation of predicate-argument semantics, abstracting away superficial syntactic variations such as active versus passive voice [233]. For example, whether a statute declares that "The City Mayor shall enforce the penalty" or that "The penalty shall be enforced by the City Mayor," AMR generates the identical semantic rooted graph, identifying the City Mayor as the agent (`:ARG0`) and the penalty as the patient (`:ARG1`). This semantic grounding allows computational systems to evaluate legal statements based on their core logical assertions rather than surface grammar.

2.5.2.1 Semantic Graph Representations and Role Labeling

A principal advantage of Abstract Meaning Representation in statutory analysis is its ability to isolate predicate-argument structures and deontic operators within complex legislative commands. Statutory provisions rely heavily on deontic modalities—such as obligations, permissions, and prohibitions—to govern citizen behavior and administrative authority [220]. Standard grammar parsers often conflate these modal verbs with general descriptive predicates, obscuring the primary regulatory command. In contrast, AMR maps deontic operators directly to core predicates, explicitly identifying whether an action is mandatory, permissible, or prohibited [233]. This structural clarity allows systems to identify subtle policy collisions, such as when a local ordinance authorizes a commercial practice that a national statute expressly forbids.

Furthermore, AMR graphs facilitate semantic normalization across diverse legislative drafting styles. Because local ordinances are drafted by different local committees across multiple decades, identical regulatory concepts are routinely phrased using disparate vocabulary and sentence structures. AMR normalizes these surface differences into standardized semantic frames, allowing different legislative phrasings to map to identical concept graphs [220]. Recent legal NLP benchmarks demonstrate that evaluating semantic graph similarity outperforms raw token comparisons when identifying core regulatory requirements [233]. This structural consistency positions AMR as a valuable framework for extracting precise semantic representations from dense legislative text.

2.5.2.2 Sentence Boundary Constraints and Parser Degradation

Despite its structural precision, Abstract Meaning Representation faces severe operational limitations when applied to Philippine local legislation. The primary architectural constraint is AMR's restriction to single-sentence parsing [220]. In statutory drafting, preemption conflicts rarely manifest within a single isolated sentence; rather, normative contradictions typically emerge across distinct sections, such as a general mandate in one clause negated by an unprovided exception several paragraphs later. Because standard AMR parsers evaluate individual sentences independently, they cannot perform cross-sentence coreference resolution or integrate distant provisos, creating significant logical blindspots in statutory conflict detection.

Compounding this sentence-level restriction is AMR's high sensitivity to complex statutory syntax and optical character recognition noise. Philippine statutory provisions, such as sections within Republic Act No. 7160, frequently exceed one hundred words, featuring deeply nested conditional clauses and multiple provisos within a single period [222].

When processing these complex syntactic structures alongside scanning artifacts or localized administrative phrasing, standard AMR parsers frequently fail, producing fragmented or malformed logic graphs [226]. Without extensive domain-specific parser retraining tailored to Philippine legal drafting, AMR cannot provide a dependable, production-ready foundation for ex-ante local statutory auditing.

2.6 Deep Learning and Transformers

The application of Natural Language Processing (NLP) in the legal domain has shifted from rigid, rule-based systems and basic word-matching toward Deep Learning [113]. Legacy algorithms treat legal texts as simple character strings, which frequently fail to capture the dense and interconnected nature of statutory drafting. Deep Learning, conversely, utilizes multi-layer artificial neural networks to map out the actual semantic meaning of legal language. This allows systems to process complex syntax and perform the deductive reasoning required to spot conflicts. This section breaks down the Transformer architecture, explaining its underlying mechanics, computational challenges, and why it remains the most practical framework for auditing unstructured local archives.

2.6.1 Transformers and Self-Attention

A major breakthrough was the Transformer architecture, introduced by Vaswani et al. [229] in 2017. Prior neural sequence models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed text sequentially from left to right, one token at a time. This sequential dependency created severe computational bottlenecks during training and suffered from vanishing gradient dynamics over long sequences, wherein contextual information from earlier tokens attenuated as sequence length expanded, hindering the model's capacity to resolve dependencies across multi-clause statutory provisions.

Transformers eliminate this sequential bottleneck entirely through the *Self-Attention* mechanism. Rather than reading words one by one, self-attention evaluates all tokens in an input sequence simultaneously in parallel. For every word in a provision, the mechanism computes a set of attention weights measuring how much every other word in the text contributes to its contextual meaning, regardless of their physical distance in the sentence. For example, in a provision stating that "Any licensee who operates without a valid health clearance shall be penalized," self-attention directly connects "licensee" to "penalized" across the intervening relative clause in a single computational step. Because it processes data in parallel, Transformers train substantially faster on modern hardware, directly enabling foundational language models. Modern iterations have introduced refinements like Disentangled Attention to represent token content and relative positional offsets separately, alongside Windowed Attention to process long-context legal documents efficiently. However, while generative Large Language Models (LLMs) excel at text summarization, they are prone to "hallucinations"—fabricating plausible but non-existent statutory citations or misinterpreting statutory cross-references [58]. To ensure absolute legal reliability, consistency auditing requires a discriminative approach: specifically classifying whether two statutory provisions stand in a relationship of Entailment, Neutrality, or Contradiction, rather than probabilistically generating unverified text.

2.6.2 Transformer Challenges

Despite their architectural versatility, deploying transformer models within the computational constraints of Philippine Local Government Units (LGUs) introduces substantial operational hurdles.

The foremost engineering bottleneck arises from quadratic memory scaling and context length limits. The standard self-attention mechanism scales quadratically ($O(n^2)$) with input sequence length n. Because computing pairwise attention scores requires constructing an explicit $n \times n$ attention matrix across all query-key token pairs, memory consumption scales quadratically with sequence length, such that doubling sequence length quadruples the memory footprint required for attention computation. Consequently, standard foundational architectures like BERT are constrained to a rigid 512-token context window [61], whereas Philippine ordinances frequently span multiple pages. While practitioners employ sliding-window techniques to segment documents into overlapping spans [8, 97], this segmentation severs the semantic dependency between general regulatory mandates and distant provisos [193]. Consequently, sequence truncation creates structural blind spots that can lead to false conflict classifications.

A secondary challenge involves morphological noise and domain adaptation limits. Pretrained transformers require massive corpora of clean text, and while Western legal models like Legal-BERT perform well on foreign statutory benchmarks [31], no dedicated foundation model exists for Philippine local legislation. Applying general tokenizers to local archives—saturated with scanning artifacts, archaic statutory phrases, and administrative Taglish—triggers severe token fragmentation, where domain-specific terms are broken into meaningless sub-tokens that degrade contextual comprehension.

Finally, edge deployment is heavily restricted by local government hardware realities. Executing deep logical inference across thousands of document pairs demands substantial GPU compute and memory bandwidth [170]. Because most local government units operate standard desktop workstations with modest hardware specifications, deploying unoptimized neural models leads to system latency spikes or memory exhaustion, necessitating decoupled, multi-stage architectures optimized for local inference.

2.6.3 Suitability of Transformers

Notwithstanding these hardware limitations, Transformer models remain the most viable approach for Semantic Conflict Detection in local ordinances. General-domain language models typically struggle with legal text because common words frequently take on entirely different legal definitions. For instance, “party” refers to a litigant rather than a celebration, and “suit” signifies a legal action rather than clothing. Greco et al. [87] emphasize that this high degree of polysemy is exactly why domain-specific models are required. While older static systems assign a single, fixed meaning to a word [87], a Transformer dynamically adjusts its reading of a word based on the surrounding sentence context [31]. Chalkidis et al. [32] empirically showed that this contextual awareness is what allows a model to distinguish between amending a law and violating it—a distinction that simple keyword searches invariably miss.

The transition to Deep Learning necessitates explicitly moving away from legacy methodologies. Boolean keyword searches fundamentally fail because legislative drafting vocabulary constantly evolves over the decades. Conversely, manually mapping thousands of unlinked Republic Acts and local ordinances into deterministic graph structures creates an intractable administrative burden. As Li et al. [128] observed, hand-crafting these legal knowledge graphs simply does not scale for volatile, unstructured local records. Ultimately, Transformers represent the ideal method: they possess the capacity to map the deep, latent semantic relationships in unstructured legal text without requiring an intractable amount of manual data entry.

2.7 Single-Stage and Multi-Stage Architectures

In mainstream Natural Language Processing (NLP), end-to-end, single-stage neural architectures serve as the conventional baseline for document classification, sentiment detection, and question answering. These systems funnel raw input sequences directly into a single neural network that outputs the final classification without intermediate filtering. However, in legal informatics, detecting statutory conflicts requires a level of deductive precision across vast corpora that a monolithic model structurally cannot provide. Forcing a single architecture to simultaneously scan tens of thousands of statutory records and evaluate fine-grained deontic logic creates severe mathematical and operational trade-offs.

2.7.1 Single-Stage Monolithic Pipelines

The appeal of a single-stage neural architecture is its conceptual simplicity: a single model handles the entire input-output mapping without requiring intermediate data handoffs. In standard NLP tasks operating over small document collections, this approach minimizes engineering complexity and eliminates pipeline latency bottlenecks. However, deploying a monolithic model to audit draft local legislation against expansive national repositories introduces severe operational barriers, inevitably sacrificing either candidate recall or deductive precision.

One common monolithic approach involves prompting generative Large Language Models to directly declare legal preemption. However, autoregressive language models operate as probabilistic token predictors rather than deterministic logical discriminators, rendering them vulnerable to legal hallucinations where non-existent statutory citations are fabricated [131]. Furthermore, their massive parameter sizes make offline deployment on standard local government hardware infeasible without heavy quantization, which severely degrades deductive reasoning fidelity in specialized statutory domains [59].

Conversely, relying exclusively on single-stage bi-encoders introduces severe semantic compression limits. In a bi-encoder (or dual-tower) architecture, the search query and the candidate statute are processed through separate neural networks into independent vector coordinates. Because the two texts never interact directly during encoding, the model can pre-compute and store embeddings for all 27,000+ laws in advance, enabling sub-second nearest-neighbor search across extensive archives [170, 35]. However, compressing an entire multi-clause provision into a single fixed-dimensional vector inevitably discards granular syntactic nuance [111, 218]. This architecture optimizes for broad topical similarity rather than logical consistency, causing bi-encoders to conflate permission and prohibition provisions that share identical topical vocabulary [116].

Furthermore, executing a single-stage cross-encoder across an entire statutory repository is computationally prohibitive. In contrast to bi-encoders, a cross-encoder concatenates the draft query and the candidate statute into a single unified sequence, allowing every token in the draft to interact directly with every token in the statute through full bidirectional cross-attention. While this joint comparison achieves high logical sensitivity, full cross-attention scales quadratically ($O(n^2)$) with input length. Comparing a newly proposed local draft against 25,432 national statutes simultaneously would require hundreds of thousands of complex forward passes, instantly exhausting workstation memory and causing intolerable system latency [18]. Forcing the model to process fragmented sliding windows across thousands of enactments severs inter-clause dependencies. This computational ceiling confirms that isolating a compact candidate shortlist via coarse retrieval is essential prior to executing fine-grained sequence classification.

2.7.2 Two-Stage Retrieve-then-Entail Architectures

To resolve the mathematical and computational limitations of monolithic architectures, contemporary legal informatics adopts decoupled, two-stage systems. In the retrieve-then-entail paradigm, the workflow is split into two specialized stages: an initial coarse Information Retrieval (IR) phase that scans the entire statutory corpus to extract a top-k candidate shortlist, followed by a fine Natural Language Inference (NLI) phase that performs deep cross-attention on the candidate pairs [240]. Williams et al. [240] demonstrate that this two-tier design reduces manual auditing overhead by up to seventy percent while preserving high inferential accuracy.

The primary engineering advantage of this decoupled architecture is its alignment with edge computing realities. By offloading broad corpus search to pre-computed vector indices that require minimal runtime memory, the computationally intensive $O(n^2)$ matrix operations of the cross-encoder are restricted exclusively to the top-k candidate pairs. Khattab and Zaharia [115] established that this decoupled structure enables advanced neural language processing to execute locally on consumer-grade hardware without requiring distributed GPU clusters. Furthermore, decoupling retrieval from inference provides a pragmatic resolution to the vector compression trade-off. The initial retrieval stage accepts the loss of granular syntactic nuance to maximize search throughput across extensive statutory archives. In contrast, the subsequent inference stage bypasses vector pooling entirely, performing joint token-level cross-attention over raw, uncompressed text spans to achieve high-precision deontic reasoning [102].

2.7.3 Comparative Architectural Trade-offs

Comparing single-stage and dual-stage architectures highlights a fundamental division of labor in statutory preemption analysis. A single-stage retriever maximizes search speed but remains blind to subtle modal negators, whereas a single-stage cross-encoder possesses high logical sensitivity but cannot scale beyond a few dozen provisions without memory failure. By chaining both mechanisms in a coarse-to-fine hierarchy, the system leverages the complementary strengths of each model family: Stage One absorbs corpus volume and context length, while Stage Two resolves logical nuance and morphological noise [135].

The empirical superiority of this decoupled design is reinforced by over a decade of findings from the Competition on Legal Information Extraction and Entailment (COLIEE). As Goebel et al. [78] document, COLIEE divides legal reasoning into statutory information retrieval (Task 3) and statutory textual entailment (Task 4). Year after year, top-performing teams reject single-stage monoliths in favor of pipelined approaches [6]. While early benchmarks in COLIEE 2021 established this separation, modern iterations through COLIEE 2025 [79] and COLIEE 2026 [80] confirm that decoupled retrieval and inference remain the gold standard. For statutory retrieval, hybrid dense-sparse systems achieve near-perfect candidate recall [152], while cross-encoder entailment systems reach over 95% accuracy by isolating granular reasoning from broad corpus retrieval [5, 221]. This sustained global validation confirms that separating Information Retrieval from Natural Language Inference is the established operational blueprint for resolving semantic conflict across statutory corpora.

2.8 Information Retrieval

Information Retrieval (IR) is the process of extracting relevant documents from massive, unstructured datasets based on a specific query. In legal informatics, IR acts as the foundational filtering mechanism. It serves as the "coarse" stage of a dual-pipeline system, isolating candidate statutes from thousands of irrelevant local records before any intensive logical analysis takes place.

2.8.1 Information Retrieval Challenges in Statutory Corpora

Applying information retrieval techniques to statutory collections introduces domain-specific hurdles that general-domain search engines are not engineered to handle. A primary challenge is severe lexical divergence and vocabulary mismatch across legislative eras [23]. In statutory law, statutes governing identical civic activities may be enacted decades apart, using drastically different terminology, archaic phrasing, or specialized administrative acronyms. When newly drafted local ordinances employ modern regulatory expressions, exact keyword matching fails to retrieve governing historical statutes, creating dangerous blindspots in preemption review [227].

Furthermore, statutory corpora exhibit extreme length asymmetry and structural complexity that disrupt standard retrieval scoring. Statutory enactments frequently contain verbose, multi-clause provisions exceeding one thousand tokens, where general commands are qualified by distant provisos and statutory exceptions [33]. Standard passage retrievers struggle to balance term saturation across these extended provisions, frequently penalizing lengthy, comprehensive enactments. Compounding this structural challenge is administrative hierarchy noise: in national legal archives, subordinate issuances (such as executive orders, circulars, and departmental memos) outnumber primary legislative acts, frequently diluting candidate shortlists unless protected by statutory priority safeguards [84].

2.8.2 Relevance in Retrieval

To understand how computational systems extract documents, we must first examine the human cognitive process that defines relevance in the physical world. In cognitive science, relevance is not a static property but a dynamic relationship between information and a specific task. According to Relevance Theory, humans naturally filter information by seeking the greatest cognitive benefit for the least mental effort [202]. When a legal researcher checks for conflicts, they do not simply count matching words; they instinctively evaluate the “intent” and “applicability” of a rule based on their background knowledge of the legal system. This human-centric view suggests that a document is relevant only if it provides a new piece of information that changes the researcher’s understanding of the legal landscape [190].

Beyond the cognitive baseline, the theoretical definition of relevance in legal information retrieval is anchored in the philosophical requirements of a functional legal system. According to the jurisprudence established by Lon Fuller in *The Morality of Law*, a primary requirement for the internal morality of law is consistency; a system that issues contradictory rules fails to be a legal system at all [74]. Within this philosophical framework, relevance is not merely about finding documents that talk about the same subject matter, but about identifying provisions that share a “normative overlap.” This means that a retrieved document is relevant if it occupies the same jurisdictional or behavioral space as a draft local ordinance, potentially creating a situation where a citizen cannot obey one law without violating another. Grounding the retrieval stage in this quest for consistency treats relevance as the systematic identification of potential normative collisions and threats to the rule of law [74].

However, traditional computational methods struggle to replicate this human judgment because they rely on a rigid, mathematical definition of relevance. In standard deterministic systems, relevance is treated as a binary mathematical property based entirely on set intersection [23]. If we denote a candidate document as D and a user query as Q, this lexical relevance is formally evaluated as a binary function $R(D, Q) \in \{0, 1\}$. Under this strict lexical definition, a document is deemed fully relevant (i.e. 1) if and only if it contains the exact sequence of characters requested by the user [227]. Consequently, this mechanism establishes a purely topical baseline, effectively grouping documents together simply because they share a surface-level vocabulary [23]. While this mathematical simplicity allows for rapid indexing across massive databases, it completely strips away the legal context required for complex legislative analysis.

This stark discrepancy between superficial lexical matching and actual legal consequence represents a major stumbling block for legacy search architectures. In statutory law, two documents discussing the exact same topic might be lexically relevant to one another, but they are only *legally* relevant if one provision actively modifies, permits, restricts, or preempts the other [227]. For instance, a retrieved document might use identical vocabulary but apply to a completely different jurisdiction or regulatory scope, rendering it completely useless for an ex-ante conflict check. Conversely, a document that governs the exact same civic behavior but uses entirely different phrasing—such as substituting "motor vehicle" for "automobile"—will be scored as mathematically irrelevant (i.e. 0) by a strict Boolean system [23]. Therefore, to accurately audit local ordinances against superior law, statutory retrieval systems must move beyond binary, keyword-driven matching in favor of dense semantic representations capable of capturing actual jurisprudential relationships [116].

To resolve these limitations, modern computational pipelines fundamentally redefine relevance spatially rather than lexically. Instead of scanning a query as a raw string of text, dense retrieval architectures utilize Bi-Encoder models to compress entire legal provisions into high-dimensional mathematical coordinates, commonly referred to as dense vectors [170]. Under this advanced semantic framework, relevance is no longer determined by counting shared characters, but by calculating the geometric distance—such as cosine similarity—between these coordinate points [116]. If a newly drafted local ordinance and an existing national statute govern the same underlying civic behavior, the neural network learns to map their respective vectors closely together in the mathematical space, completely bypassing the vocabulary mismatch problem [241]. Consequently, the retrieval mechanism evaluates relevance as a continuous proximity score rather than a rigid binary output, successfully surfacing conceptually intertwined laws that basic keyword searches otherwise miss.

2.8.3 Long-Context Retrieval

Relying on standard BERT-based architectures introduces an artificial constraint on legal analysis: the 512-token limit. Because standard Self-Attention mechanisms scale quadratically with sequence length, processing long texts requires severe truncation. To get around this, researchers like Chalkidis et al. [33] routinely use Sliding Window techniques to segment documents into overlapping chunks. However, applying this to Philippine ordinances—which are historically wordy and often bury exceptions deep within 1,000-token provisions—is structurally destructive. It severs the substantive semantic link between a primary rule and its distant exception.

Consequently, the model operates with a logical blindspot. Beltagy et al. [18] note that this causes logical hallucinations, where a rule might be flagged as an absolute prohibition simply because its exempting clause was pushed into a separate, unconnected processing window. Furthermore, these systems must navigate the trade-off between Sparse Retrieval (e.g., BM25) for exact keyword matching, and Dense Retrieval for capturing conceptual meaning even without direct word overlap.

Resolving this fragmentation means moving beyond short sliding-window chunking where possible. The architectural solution requires an initial retrieval mechanism capable of native long-context retention. Recent Transformer models incorporate advanced positional encoding, such as Rotary Positional Embeddings (RoPE), alongside sparse or optimized attention matrices. Warner et al. [238] demonstrated that these upgrades extend context windows up to 8,192 tokens without triggering memory failures. Deploying this class of architecture allows the system to read entire local ordinances as cohesive semantic units, preserving the law's internal structure and eliminating the Sliding Window fallacy.

2.8.4 Retrieval and Fusion Strategies

In computational legal informatics, isolating candidate enactments from extensive statutory collections requires understanding the theoretical foundations, failure modes, and complementary trade-offs of diverse retrieval strategies. Recent findings from the COLIEE benchmarks and legislative consistency literature identify five core methodological dimensions governing coarse statutory search.

The foundational retrieval strategies in computational law encompass sparse probabilistic scoring and dense semantic embeddings. Each relevance scoring metric is to be computed as follows:

Okapi BM25 Probabilistic Relevance Scoring Function:

$$S_{\text{BM25}}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \tag{2.3}$$

Where $f(q, D)$ denotes term frequency of query token q in document D, $|D|/\text{avgdl}$ normalizes for provision length, and parameters $k_1 = 1.5$ and $b = 0.75$ govern term saturation and length penalization within Robertson and Zaragoza's probabilistic relevance framework [176].

The Okapi BM25 formulation balances three core retrieval properties within a unified probabilistic relevance framework. First, Inverse Document Frequency ($\text{IDF}(q)$) assigns higher numerical weight to rare, discriminative terms (such as specific statutory numbers, formal administrative acronyms, or penal sanctions) while suppressing ubiquitous statutory stopwords that appear across all enactments. Second, Term Frequency Saturation ($k_1 = 1.5$) enforces sub-linear term weighting, reflecting diminishing marginal relevance so that repeating a keyword multiple times does not disproportionately inflate provision relevance over texts with concise occurrences. Third, Document Length Normalization ($b = 0.75$) penalizes exceptionally verbose provisions, ensuring that lengthy multi-page legislative digests do not dominate search rankings simply because their word volume increases the incidental probability of keyword matches. In statutory retrieval competitions such as COLIEE Task 3, BM25 serves as an indispensable baseline (as demonstrated by Team AIIR [241]) because legislative drafting routinely depends on exact statutory citations, formal administrative acronyms, and verbatim penal monetary amounts. However, exact lexical matching suffers from the vocabulary mismatch problem when local ordinances paraphrase statutory commands [23, 140].

Dense Semantic Cosine Proximity Relevance Function:

$$S_{\text{dense}}(D, Q) = \frac{\mathbf{e}_Q \cdot \mathbf{e}_D}{\|\mathbf{e}_Q\| \|\mathbf{e}_D\|} \tag{2.4}$$

Where $\mathbf{e}_Q$ and $\mathbf{e}_D$ represent pooled contextual embeddings generated by dense bi-encoders such as `all-MiniLM-L6-v2` [170]. Rather than evaluating token overlap, dense retrievers score relevance via cosine proximity in latent semantic space, capturing conceptual legal intent and retrieving governing statutes even when local drafts employ disparate phrasing.

In heterogeneous statutory databases spanning tens of thousands of enactments, researchers frequently investigate topic classification or domain partitioning to reduce search space complexity. However, empirical literature highlights the severe danger of hard categorical filtering in legal domains. Foundational statutory instruments, such as the Philippine Local Government Code of 1991 (RA 7160) or comprehensive penal codes, function as omnibus legislation that simultaneously govern local administration, taxation, public safety, and environmental sanitation. Hard filtering by a single predicted regulatory category discards the governing omnibus code whenever a query touches an adjacent regulatory area, causing catastrophic candidate recall collapse. In contrast, information retrieval theory establishes soft domain priors [176]. Instead of binary candidate exclusion, soft priors apply a calibrated multiplicative score boost (e.g., +20%) to candidate documents matching the predicted topic or latent semantic cluster, preserving access to multi-domain statutory codes while gently elevating topically congruent provisions.

Because lexical and semantic retrievers exhibit orthogonal failure modes, leading computational law frameworks combine both streams through hybrid fusion. In score-level linear combination, popularized by Team JNLP in the COLIEE competitions [78], weighted interpolation combines normalized sparse and dense scores according to $S_{\text{hybrid}}(D, Q) = \alpha \hat{S}_{\text{dense}}(D, Q) + (1 - \alpha)\hat{S}_{\text{BM25}}(D, Q)$, where $\alpha \in [0, 1]$ balances semantic and lexical bias. However, because BM25 scores are unbounded while cosine similarities span $[-1, 1]$, linear combination requires min-max score normalization, introducing sensitivity across heterogeneous query lengths. To eliminate score normalization artifacts, Cormack et al. [46] formulated Reciprocal Rank Fusion (RRF), defined as $\text{RRF}(D) = \sum_{m \in \mathcal{M}} (k_{\text{rrf}} + \text{rank}_m(D))^{-1}$, where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ and $k_{\text{rrf}} = 60$.

The mathematical advantage of Reciprocal Rank Fusion lies in its scale-invariant rank aggregation. Because BM25 produces unbounded positive relevance scores while dense bi-encoders output bounded cosine similarities within $[-1, 1]$, score-level linear combination requires sensitive calibration and min-max normalization. In contrast, RRF abstracts away disparate score distributions entirely by operating strictly on the discrete ordinal rank positions assigned to candidate provisions within each individual retrieval stream. The smoothing constant $k_{\text{rrf}} = 60$ stabilizes reciprocal scoring against extreme rank disparities while systematically rewarding candidates that achieve high concordance across both lexical keyword matching and dense vector projections.

Finally, statutory retrieval systems must respect constitutional hierarchies of authority. Anchored in the doctrine of legislative supremacy and the hierarchy of legal norms [114, 84], statutory systems operate under strict formal ordering. In national digital archives, subordinate administrative issuances (such as executive orders, administrative circulars, and departmental memos) frequently constitute nearly half of all indexed records. Because administrative issuances share broad governmental vocabulary with local drafts, unconstrained retrieval often allows routine circulars to crowd out superior national statutes. Literature in legal informatics emphasizes structural safeguards that prioritize primary legislative acts (Republic Acts) over subordinate executive issuances. Implementing a hierarchical statutory priority safeguard shields governing statutes from administrative noise, preventing dilution of candidate shortlists and significantly improving Mean Reciprocal Rank (MRR).

2.8.5 Retrieval Benchmarks

Selecting effective retrieval architectures from the extensive landscape of open-weight models requires objective empirical benchmarking. In contemporary legal information retrieval, candidate models are evaluated against standardized empirical leaderboards that assess zero-shot generalization across dense statutory text. The Benchmarking Information Retrieval (BEIR) suite evaluates a dense retriever's capacity for out-of-domain transfer across diverse, specialized datasets [218]. Because Philippine local legislation lacks extensive annotated training corpora, candidate retrievers must possess strong zero-shot generalization. Dominant performance on the BEIR leaderboard serves as a primary indicator of a model's capacity to map novel, localized legal terminology into a structured latent space without prior fine-tuning [218].

Expanding on BEIR, the Massive Text Embedding Benchmark (MTEB) establishes an international standard for ranking embedding systems across more than one hundred datasets and diverse linguistic tasks [147]. In information retrieval literature, candidate architectures are frequently filtered using MTEB's retrieval and clustering leaderboards. Models excelling on these benchmarks demonstrate an empirical capacity to cluster conceptually related doctrines and retrieve relevant provisions based on continuous semantic proximity. Complementing general embedding benchmarks, the annual COLIEE Task 3 evaluation provides a specialized testbed for statute law retrieval [79, 80]. Systems evaluated on Task 3 must extract precise governing articles from civil code collections in response to legal queries. Success in this competition requires balancing broad candidate recall with computational throughput, providing a proven testing foundation for leading statutory retrieval models [241, 78].

2.8.6 Candidate Retrieval Models

Surveying leading open-weight architectures documented across recent retrieval literature and standardized benchmarks reveals eight prominent bi-encoder models relevant to statutory candidate discovery. Among distilled edge baselines, Reimers and Gurevych [170] established the Sentence-BERT framework, mapping text into fixed-dimensional dense vector spaces optimized for cosine proximity. The `all-MiniLM-L6-v2` model compresses a 6-layer BERT architecture down to 22.7 million parameters through knowledge distillation, producing 384-dimensional dense vectors. Despite its ultra-compact footprint, empirical benchmarks demonstrate that it retains strong semantic clustering capabilities with sub-two-millisecond latency on consumer CPUs. Complementing this, Xiao et al. [243] introduced `bge-small-en-v1.5` (33.4M parameters), which optimizes contrastive representation learning for high rank sensitivity on MTEB. In public-sector edge deployment, these distilled bi-encoders provide high-throughput candidate filtering without requiring dedicated GPU accelerators.

Scaling bi-encoder capacity to standard 110M-parameter base dimensions enables more expressive semantic representations across general retrieval benchmarks. Xiao et al. [243] developed `bge-base-en-v1.5` (109M parameters, 768 dimensions), utilizing retrofitted contrastive pretraining and instruction-tuned query prefixes to excel across multi-domain retrieval leaderboards. Similarly, Wang et al. [236] introduced `e5-base-v2` (109M parameters), trained on over one billion weakly-supervised text pairs via contrastive objectives, demonstrating high generalization on unseen technical and legal search tasks. Furthermore, Song et al. [200] developed MPNet, combining masked language modeling with permuted language pretraining to eliminate positional dependence, which serves as the foundational architecture for `all-mpnet-base-v2` (109M parameters).

Beyond general-domain models, legal-domain adapted encoders capture specialized statutory taxonomies. Chalkidis et al. [31] demonstrated that pretraining bidirectional encoders on specialized legal repositories enables dense representations to capture statutory taxonomy and regulatory jargon that general-domain embedders conflate. Mean-pooled sentence embeddings derived from `legal-bert-base-uncased` (110M parameters) map statutory provisions into vector spaces that reflect legislative hierarchy and administrative codification rather than colloquial word associations. This domain-specific pretraining provides a structured baseline to evaluate whether legal pre-training improves candidate provision ranking over general embeddings.

To overcome the quadratic attention complexity of traditional 512-token encoders, modern architectures incorporate native long-context and multilingual designs. Warner et al. [238] developed `ModernBERT-base` (149M parameters), which integrates FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activations, and sequence unpadding to natively process up to 8,192 tokens. In statutory retrieval, this allows direct encoding of verbose multi-paragraph enactments without destructive sliding-window chunking. Expanding to the multilingual domain, Chen et al. [35] introduced `bge-m3` (568M parameters, 1,024 dimensions), supporting 8,192-token contexts and multi-functionality across dense, sparse, and multi-vector representations. In statutory consistency checking, its multi-vector representations provide resilience across code-switched local records and bilingual administrative issuances.

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Because lexical and semantic retrievers exhibit orthogonal failure modes, leading computational law frameworks combine both streams through hybrid fusion. In score-level linear combination, popularized by Team JNLP in the COLIEE competitions [78], weighted interpolation combines normalized sparse and dense scores according to $S_{\text{hybrid}}(D, Q) = \alpha \hat{S}_{\text{dense}}(D, Q) + (1 - \alpha)\hat{S}_{\text{BM25}}(D, Q)$, where $\alpha \in [0, 1]$ balances semantic and lexical bias. However, because BM25 scores are unbounded while cosine similarities span $[-1, 1]$, linear combination requires min-max score normalization, introducing sensitivity across heterogeneous query lengths. To eliminate score normalization artifacts, Cormack et al. [46] formulated Reciprocal Rank Fusion (RRF), defined as $\text{RRF}(D) = \sum_{m \in \mathcal{M}} (k_{\text{rrf}} + \text{rank}_m(D))^{-1}$, where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ and $k_{\text{rrf}} = 60$. Because RRF operates strictly on ordinal ranking positions, it provides scale invariance, eliminates distribution sensitivity, and ensures that provisions performing well across both lexical and semantic modalities are prioritized.

Finally, statutory retrieval systems must respect constitutional hierarchies of authority. Anchored in the doctrine of legislative supremacy and the hierarchy of legal norms [114, 84], statutory systems operate under strict formal ordering. In national digital archives, subordinate administrative issuances (such as executive orders, administrative circulars, and departmental memos) frequently constitute nearly half of all indexed records. Because administrative issuances share broad governmental vocabulary with local drafts, unconstrained retrieval often allows routine circulars to crowd out superior national statutes. Literature in legal informatics emphasizes structural safeguards that prioritize primary legislative acts (Republic Acts) over subordinate executive issuances. Implementing a hierarchical statutory priority safeguard shields governing statutes from administrative noise, preventing dilution of candidate shortlists and significantly improving Mean Reciprocal Rank (MRR).

2.9 Natural Language Processing

Natural Language Processing (NLP) provides the computational techniques needed to analyze and represent text at various linguistic levels, aiming for human-like comprehension [130]. As Jurafsky and Martin [108] explain, it acts as the bridge between human language and machine logic, allowing algorithms to parse and interpret unstructured text. In the legal field, NLP shifts document auditing from slow manual reviews to scalable, automated systems. However, applying standard NLP to law requires more than basic grammar checks; it demands systems that can navigate dense sentence structures, deontic operators (obligations, permissions, prohibitions), and the highly specific vocabulary found in statutory law.

Early Legal NLP relied heavily on Symbolic Artificial Intelligence (AI) [113]. These systems used manually built ontologies and strict Boolean logic to extract legal information. However, because local legislative drafting language changes constantly over decades, these exact-match frameworks proved far too brittle. Contemporary NLP architectures solve this by moving to continuous vector representations. By mapping textual provisions into high-dimensional semantic spaces, modern Transformers bypass superficial vocabulary mismatches to mathematically quantify the deeper concepts connecting different legal terms. This allows the system to act as an active diagnostic engine that structurally maps legal conflicts, rather than just a passive search index.

This section breaks down how NLP operates within legislative analysis. It covers the systemic challenges of processing localized legal text, the structural need for Natural Language Inference (NLI), the benchmarks used for model selection, and the specific architectures built for these deductive tasks.

2.9.1 NLP Challenges

Deploying NLP frameworks against local archives quickly exposes the mechanical limits of standard Language Models. A primary hurdle is the pronounced Global North bias in pre-training data and legal benchmarking. As Ranathunga et al. [168] point out, the vast majority of Legal NLP benchmarks, like LexGLUE [33] and CaseHOLD, are trained exclusively on pristine, digitally native Western legal text. These highly optimized models degrade when applied to Philippine Local Government Unit (LGU) archives, which are filled with scanned PDFs suffering from Optical Character Recognition (OCR) errors, archaic American colonial phrasing, untranslated Spanish legal terms, and local Taglish. Rust et al. [182] demonstrated that standard WordPiece tokenizers shatter unrecognized words into sub-tokens, destroying a model's ability to infer legal intent. Addressing localized morphological noise requires deploying models equipped with massive Byte-Pair Encoding (BPE) vocabularies that absorb linguistic variants without triggering out-of-vocabulary errors, paired with Disentangled Attention architectures that separate word content from corrupted positions [95].

Beyond geographic and morphological bias, traditional NLP methods suffer from structural shortcomings in lexical similarity. Early NLP systems excel at identifying topical overlap by measuring surface-level token intersection using algorithms like TF-IDF or shallow word embeddings. However, Bowman et al. [24] explain that pure lexical similarity is fundamentally unsuited for detecting legal contradictions because lexical metrics are entirely blind to deontic logic and syntactic negation. For example, an ordinance stating that the city permits an activity and a statute stating that the state strictly prohibits it share near-perfect lexical similarity scores. A standard similarity model clusters them as highly related, completely missing the absolute policy collision occurring beneath the shared vocabulary. Consequently, detecting normative conflicts requires advancing from lexical proximity to directional logical inference.

2.9.2 From NLP to NLI

Historically, checking legislative consistency relied on Symbolic NLP (hand-written rules) and, later, Statistical NLP. In Statistical NLP, models learned probabilistic weights for specific inputs, like Term Frequency-Inverse Document Frequency (TF-IDF) or N-grams [113].

These models depended heavily on human-engineered features to define grammar rules and categories. While they allowed for soft, probabilistic decisions, they consistently struggled with the deep lexical and semantic ambiguities inherent in legal drafting.

The field largely transitioned to Neural NLP models in the mid-2010s [165]. Unlike statistical systems that treat features as separate entities, neural networks operate end-to-end, mapping raw text directly to output predictions through hidden layers. These architectures capture deeper linguistic patterns than traditional statistical methods by learning complex relationships during optimization. However, while machine learning excels at finding patterns, it can overly mimic its training data, making domain-specific pre-training essential for high performance in rigorous legal tasks.

Within modern neural approaches, Natural Language Inference (NLI)—originally formalized under the Recognizing Textual Entailment (RTE) challenges initiated by Dagan et al. [51]—has emerged as the definitive benchmark for evaluating whether an algorithm can perform genuine logical deduction. Yang et al. [245] define NLI as determining the directional logical relationship between two structured text sequences: a governing *Premise* and a candidate *Hypothesis*. In statutory preemption auditing, this formulation maps directly to constitutional hierarchy: the Premise represents the established positive legal standard (the superior national statute), while the Hypothesis represents the newly proposed local claim (the draft city ordinance provision).

The neural network must classify the relationship into one of three mutually exclusive logical states. In an Entailment relationship, the draft ordinance directly carries out, derives its authority from, or compliantly adheres to the governing national statute. In a Contradiction relationship, the draft ordinance directly collides with the national statute (for example, permitting what national law prohibits, or exceeding statutory penalty ceilings under the *Magtajas* doctrine). Finally, in a Neutral relationship, the two provisions govern complementary or non-overlapping subject matter without normative conflict. This directional formulation exposes the fatal flaw of earlier statistical NLP, which mistakenly equated high word overlap with logical entailment [245]. Modern neural approaches fix this by capturing deep compositional semantics. As Le Hong et al. [125] point out, this allows modern models to resolve complex linguistic phenomena—like coreferences and modality—that rigid rule-based systems historically failed to grasp.

2.9.3 NLI Benchmarks

Identifying effective Cross-Encoder architectures for the inference stage requires isolating models with robust, pre-trained deductive logic. In natural language processing, research literature relies on established benchmarks to evaluate candidate reasoning models. Wang et al. introduced the General Language Understanding Evaluation (GLUE) [235] and its successor SuperGLUE [234] as foundational baselines for model reasoning. These frameworks evaluate a Transformer's ability to handle diverse linguistic challenges, such as Recognizing Textual Entailment (RTE) and resolving complex coreferences. While high performance on SuperGLUE confirms baseline deductive capacity, it does not independently guarantee success in the legal domain due to the specialized vocabulary shifts present in statutory drafting.

To evaluate specialized legal logic, legal NLP research relies on the Legal General Language Understanding Evaluation (LexGLUE) benchmark. Chalkidis et al. [33] designed LexGLUE specifically to test a model's ability to navigate legalese, where common words assume highly context-dependent meanings. High performance on LexGLUE entailment tasks indicates that an architecture possesses the foundational deductive capacity required to navigate complex statutory drafting and regulatory provisions.

Complementing domain-specific suites, contemporary Hugging Face NLI leaderboards evaluate models on diverse, adversarial textual entailment tasks aggregating MultiNLI (MNLI), FEVER, Adversarial NLI (ANLI R1–R3) [153], and WANLI [132]. Benchmarking candidate models across these suites tests their resilience against spurious lexical heuristics, premise-hypothesis artifacts, and negation traps. High-performing architectures on these leaderboards, such as the pre-aligned DeBERTa-v3 variants curated by Laurer et al. [124], demonstrate strong zero-shot deductive capacity, serving as leading candidate foundations for statutory conflict detection. Finally, within legal informatics, the definitive international competition benchmark for statutory inference is Task 4 of the Competition on Legal Information Extraction and Entailment (COLIEE) [166, 79, 80]. In Task 4, systems evaluate whether a natural language query is legally entailed by or contradicts a governing civil code article. Evaluating architectures against COLIEE Task 4 methodologies provides direct empirical evidence of a model's capacity to navigate statutory conditions, exceptions, and deontic modalities.

2.9.4 Candidate NLI Models

Evaluating deontic relationships across statutory provisions demands specialized inductive biases across candidate sequence encoders. Standard language models trained exclusively on general corpora frequently fail to capture the specialized semantics of legal drafting, where everyday terms carry specialized juridical meanings. To address this vocabulary divergence, researchers develop domain-adapted encoders initialized with custom legal vocabularies or continuously pretrained on expansive legal repositories. Chalkidis et al. [31] introduced `LEGAL-BERT`, pretrained on diverse English legislation, court judgments, and contracts, demonstrating superior term sensitivity across legal classification tasks. Henderson et al. [96] expanded this approach with `PoL-BERT` (Pile of Law), pretraining encoders on a 256 GB corpus encompassing administrative regulations, municipal codes, and statutory provisions. Complementary efforts by Paul et al. [161] (`InLegalBERT`) and Zheng et al. [249] (`CaseLaw-BERT`) investigate the transferability of common-law statutory pretraining, while Xiao et al. [242] evaluate long-document legal encoders. In literature, this family provides a critical testbed for assessing whether domain pretraining provides intrinsic protection against legal misinterpretation.

A pervasive vulnerability of unadapted sequence encoders on deontic tasks is *affirmative entailment bias*, where models exhibit an innate psychological tendency to default toward predicting Entailment or Neutral when reading complex clauses containing subtle prohibitions [5, 221]. In standard language corpora, true logical contradictions are exceedingly rare; text passages typically elaborate, confirm, or describe unrelated facts. Consequently, unadapted neural models learn a strong statistical prior that pairs generally agree. Devlin et al. [61] established `bert-base-uncased` (110M parameters) with masked language modeling, serving as the canonical supervised baseline against which improvements are measured. To overcome heuristic reliance on lexical overlap and break affirmative bias, contemporary literature explores models pre-aligned on multi-task natural language inference datasets (such as MNLI, FEVER, ANLI, and WANLI). Laurer et al. [124] demonstrate that initializing sequence classifiers from warm NLI checkpoints equips models with robust directional contradiction sensitivity before domain adaptation. Structurally, He et al. [95] developed `DeBERTa-v3`, introducing Disentangled Attention to represent word content and relative positions with separate vectors alongside adversarial replaced token generation. In parallel, Clark et al. [39] developed `ELECTRA` with Replaced Token Detection, optimizing contextual representations across all input positions. These architectures excel at distinguishing subtle deontic shifts that standard unidirectional or masked encoders routinely miss.

Statutory drafting is inherently verbose, characterized by complex preamble clauses, extensive definitions, and multi-sentence penal qualifications. Traditional transformer encoders are constrained by quadratic self-attention complexity, imposing a rigid 512-token ceiling that necessitates destructive document chunking or sliding window heuristics [33]. Early long-context architectures attempted to resolve this constraint using localized or sparse attention approximations, such as the sliding-window attention in `Longformer` [18] and the block-sparse random attention in `BigBird` [247]. However, empirical legal informatics evaluations note that sparse attention patterns can dilute or sever semantic dependencies between distant provisos and governing commands. To overcome these limitations, modern architectures integrate architectural optimizations that support extended contexts natively. Warner et al. [238] introduced `ModernBERT`, an 8,192-token encoder incorporating FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activation functions, and sequence unpadding. Complementing this, Nussbaum et al. [156] developed `Nomic-BERT` to extend context retention. These modern long-context designs enable joint encoding of complete multi-sentence statutory provisions without sparse approximation artifacts.

Unlike bi-encoders that process queries and documents independently into pooled vector coordinates, cross-encoders execute full bidirectional self-attention across the concatenated pair simultaneously. In retrieval and inference benchmarks, joint cross-attention captures deep inter-token interactions that vector similarity cannot represent. In COLIEE statute retrieval benchmarks, researchers deploy multilingual cross-encoder rerankers, notably `BGE-Reranker-v2-m3` [243], to score pairwise relevance across multi-thousand token contexts. Similarly, sequence classifiers adapted from MS-MARCO reranking frameworks employ deep cross-attentive scoring to evaluate pair compatibility. In multi-stage neural retrieval paradigms, Nogueira and Cho [154] demonstrated that candidate passages retrieved by coarse search can be evaluated in parallel batches via cross-attentive models (such as MonoBERT), aggregating candidate pairs into structured mini-batches to utilize hardware vectorization. In literature, native cross-encoders represent the conceptual bridge between Stage 1 candidate filtering and Stage 2 logical validation.

Evaluating candidate sequence pairs through multi-layer cross-encoders introduces computational throughput trade-offs during live document review. In standard transformer implementations, mini-batch tensors are padded with dummy zero tokens to match a static global sequence limit, such as 256 or 512 tokens [61]. Because self-attention complexity scales quadratically $(O(L^2))$ with input sequence length L, allocating static global padding causes significant floating-point operations to be expended on empty padding positions when input clauses are short. In neural language processing literature, dynamic sequence padding addresses this inefficiency by clamping batch tensor dimensions strictly to the length of the longest clause within each mini-batch, substantially reducing redundant self-attention calculations while producing mathematically identical hidden states. Furthermore, in multi-document and multi-clause document auditing, batch collation literature establishes that flattening candidate pairs across multiple queries into a single continuous tensor pass amortizes runtime call overhead and utilizes multi-threaded central processing unit vectorization.

In post-colonial statutory jurisdictions like the Philippines, statutory archives exhibit unique sociolinguistic complexities. While statutory enactments are drafted in legal English, local ordinances and administrative records frequently feature localized terminology, regional administrative phrasing, Latin legal maxims, and English-Tagalog code-switching. Conneau et al. [44] introduced `XLM-RoBERTa`, trained across 100 languages, demonstrating strong cross-lingual transfer on semantic benchmarks. He et al. [95] extended disentangled attention to the multilingual domain with `mDeBERTa-v3`, providing resilience to mixed-language syntax. Furthermore, Cruz and Cheng [50] developed Tagalog-adapted transformers (`roberta-tagalog-base`) to capture Philippine syntactic structures. In statutory consistency checking, multilingual encoders evaluate whether localized linguistic variances can be processed without catastrophic comprehension loss.

Deploying deep learning in public-sector local government environments requires strict adherence to available edge computing infrastructure. Because local government units typically operate without high-performance computing clusters or enterprise cloud APIs, running billion-parameter language models is computationally unfeasible. To establish realistic performance baselines under constrained hardware, the literature explores knowledge distillation and parameter sharing. Sanh et al. [189] introduced `DistilBERT`, compressing transformer depth to reduce model size by 40% while retaining 97% of BERT's language comprehension capabilities and running 60% faster. Lan et al. [120] introduced `ALBERT`, utilizing factorized embedding parameterization and cross-layer parameter sharing to reduce parameter size down to 12 million in its base configuration (an 89% reduction compared to BERT-base). In dense semantic representation, Reimers and Gurevych [170] established the Sentence-BERT (SBERT) siamese embedding framework, providing the structural foundation for compact distilled bi-encoders such as `all-MiniLM-L6-v2` (22.7M parameters) to deliver sub-two-millisecond latency on commodity CPU hardware. Evaluating distilled edge models allows researchers to quantify the exact accuracy penalty incurred when prioritizing sub-second CPU latency over multi-layer neural capacity.

2.10 Ground Truth

In supervised machine learning, a Ground Truth (or Gold Standard) dataset provides the verified target labels used to optimize objective functions and evaluate model parameters. Formally, the ground truth establishes the empirical reference standard against which supervised learning models optimize loss functions [81]. During training, the optimization algorithm computes empirical risk between model predictions and verified target labels, computing gradients via backpropagation to iteratively adjust model weights. In domain-specific legal NLP, expert-adjudicated ground truth is essential to prevent models from learning superficial statistical artifacts or spurious annotation heuristics that fail to generalize to actual statutory logic.

In model evaluation methodology, the ground truth provides the objective benchmark for measuring generalization capacity across held-out partitions. Datasets are partitioned into independent training, validation, and test subsets. The test partition remains strictly unobserved during hyperparameter tuning and model checkpoint selection, serving as a clean out-of-sample evaluation set. Evaluating the model on this held-out data enables objective computation of classification metrics—including precision, recall, Macro F_1, and class-specific F_2-scores—validating whether the architecture generalizes reliably to unseen legislative drafts [25].

2.10.1 Ground Truth in NLI

In Natural Language Inference (NLI), ground truth datasets define the logical relationship between two pieces of text. Annotators read a source text (the Premise) and a claim (the Hypothesis), then categorize their relationship into one of three distinct labels. If the premise guarantees the hypothesis is true, it is labeled as Entailment. If they contradict each other, it is labeled as Contradiction. If there is not enough information to prove either, the pair is labeled Neutral [245].

Most traditional NLI datasets, such as the widely used SNLI (Stanford Natural Language Inference) corpus, are built using crowdsourced workers who write simple variations of everyday sentences [25]. While this approach is cheap and easily generates hundreds of thousands of training pairs, the resulting datasets are incredibly basic and often embed unintended annotation artifacts [91]. They usually feature short, single-sentence premises with obvious logical connections. Training a model exclusively on these simple datasets leaves it entirely unprepared for the dense, multi-layered logic required in specialized fields.

2.10.2 Annotation and Reliability Metrics

Turning raw text into a useful NLI dataset requires strict guidelines to minimize human bias and prevent annotators from relying on subjective legal interpretations. In standard NLI, guidelines dictate simple factual agreement. However, legal conflict detection requires evaluating normative relationships based on deontic logic. Drawing from the frameworks established by Agotnes et al. [3] and Li et al. [129], the annotation guidelines strictly instruct experts to map the text into obligations (must do), permissions (may do), and prohibitions (cannot do). A “Contradiction” is only flagged if the local ordinance mandates an action the national statute prohibits, or prohibits an action the national statute explicitly permits. If a local ordinance merely adds a regulatory step (e.g., a local processing fee) to a national permit without banning the core activity, the guidelines strictly dictate a “Neutral” classification. This rigid deontic framework is designed to ensure that the dataset measures actual legal preemption rather than just topical friction.

To evaluate the effectiveness of these strict guidelines, researchers must utilize Inter-Rater Reliability (IRR) metrics [144] rather than mathematically flawed percentage agreements. Raw percentage agreement is structurally inadequate for evaluating multi-annotator consistency because it fails to correct for expected chance concordance (p_e). Under a three-class discrete annotation schema ($\mathcal{C} = 3$), independent stochastic labeling yields a baseline expected agreement of approximately 33.3% purely through chance alignment. Reporting unadjusted percentage concordance confounds random stochastic concordance with genuine evaluative consensus, producing an inflated assessment of inter-annotator reliability.

The two primary statistical formulas used to subtract this random chance are Cohen’s Kappa [40] and Fleiss’ Kappa [71]. Each inter-rater reliability metric is to be computed as follows:

Cohen’s Pairwise Kappa Reliability Metric (κ):

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2.5}$$

Where p_o represents the relative observed agreement between the two individual raters, and p_e represents the hypothetical probability of chance agreement. Cohen’s Kappa evaluates agreement between exactly two raters.

Fleiss’ Generalized Multi-Rater Kappa Reliability Metric (κ):

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \tag{2.6}$$

Where $\bar{P}$ represents the mean observed agreement across all pairs of raters, while $\bar{P}_e$ represents the expected chance agreement across the entire pool. Fleiss’ Kappa generalizes Cohen’s formula to accommodate three or more domain experts establishing a majority consensus.

The resulting κ score is interpreted using the universally accepted statistical matrix established by Landis and Koch [122]:

Table 2.1: Interpretation Ranges for Fleiss’ Kappa Reliability Metric

Kappa (κ) Score	**Interpretation**
< 0.00	Poor agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

Table 2.1 outlines the standard categorical intervals for evaluating inter-rater agreement across multiple independent judges. It consists of two columns mapping numerical κ ranges to their corresponding qualitative interpretations, providing an objective statistical benchmark for assessing annotation consensus. The table is included to ground the evaluation of the 15 Sangguniang Panlungsod legal researchers within an authoritative psychometric standard. As the score progresses from negative values toward unity, it reflects a transition from chance-level alignment to robust consensus across evaluators. For this research, the matrix establishes the operational target of $\kappa \geq 0.61$ ("Substantial Agreement"), ensuring that the resulting ground truth benchmark represents dependable legal consistency rather than subjective individual interpretation.

While objective tasks routinely achieve κ scores above 0.90, applying this expectation to the legal domain is mathematically unrealistic. Bowman et al. [25], in their foundational study for the Stanford Natural Language Inference (SNLI) corpus, established that evaluating semantic relationships is inherently subjective. Their benchmark study hit an approximate performance ceiling with an overall Fleiss' Kappa of 0.70. They explicitly demonstrated that achieving near-perfect agreement (> 0.80) in textual inference usually indicates overly simplistic or fabricated test data.

2.10.2.1 Sub-Threshold Reliability

While achieving a "Substantial" Inter-Rater Reliability score ($\kappa \geq 0.61$) validates the clarity of the annotation guidelines, failing to meet this threshold does not inherently invalidate the dataset or halt the experimental pipeline. Klebanov and Beigman [117] establish that a sub-threshold κ score typically indicates systematic ambiguity within the text itself—categorized as "hard" instances where seasoned legal professionals can reasonably disagree on statutory scope—rather than carelessness by the annotators. Their research demonstrates that it is still entirely possible to extract a reliable, low-noise subset from the instances where annotators do achieve consensus.

When researchers encounter this sub-threshold reliability, the mitigation strategy depends entirely on the size of the dataset. For large-scale studies, including the disaster-tweet classification by Ablazo [1], the standard procedure is data omission. Beigman et al. [117] explicitly advocate for filtering out difficult, borderline cases from the gold standard, arguing that evaluating algorithms on inherently ambiguous text introduces stochastic label noise that degrades model optimization. In massive general-domain corpora, items that fail to achieve majority consensus are omitted because the extensive volume of unambiguous data absorbs the sample reduction without compromising statistical power.

However, applying this standard omission strategy to a specialized, few-shot legal dataset mathematically destroys the test set's Minimum Detectable Effect. Furthermore, in local legislation, these "hard" cases often represent the exact jurisdictional friction the logic engine needs to learn. To preserve dataset volume without introducing benchmarking noise, researchers in specialized domains employ Expert Adjudication rather than omission. Gao et al. [77] establish that in these scenarios, a hierarchical adjudication procedure is used where a senior domain expert reviews the conflicting annotations. The senior expert resolves the ambiguity based on governing jurisprudential principles and manually establishes the Gold Standard label, treating the initial low κ metric as a measure of text difficulty rather than a rigid experimental blocker.

2.10.3 Ground Truth in Legal NLP

When applying NLI to the legal domain, the standard approach of comparing short, isolated sentences quickly breaks down. Statutory phrasing is notoriously complex, filled with conditional clauses, exceptions, and domain-specific vocabulary. Consequently, a hypothesis in a legal context rarely aligns perfectly with a single sentence in the premise. Instead, proving an entailment or contradiction often requires the model to connect evidence scattered across multiple paragraphs or pages.

To address this, researchers have developed specialized legal datasets that reflect real-world document structures. For instance, the Competition on Legal Information Extraction/Entailment (COLIEE) dataset constructs its ground truth by pairing civil code statutes with legal bar exam queries, requiring models to identify specific legal provisions before determining entailment [166]. Similarly, the ContractNLI dataset by Koreeda and Manning [118] engineers its gold standard around non-disclosure agreements. Rather than giving the model pre-cut sentence pairs, ContractNLI forces the system to evaluate hypotheses against entire legal documents, requiring annotators to both classify the logical relationship and extract the specific supporting text span.

Furthermore, unlike general-domain datasets crowdsourced through platforms like Amazon Mechanical Turk, legal ground truth relies heavily on domain experts. Constructing datasets like CaseHOLD [249], which evaluates legal holding statements, requires law students and legal professionals to ensure the labels accurately reflect actual jurisprudence. The push towards comprehensive legal benchmarks, such as LexGLUE [33], demonstrates that evaluating legal NLP models demands rigorously structured, expert-annotated data rather than automated or amateur labeling.

This difficulty is magnified when dealing with local legislation. While national or federal laws generally follow standardized drafting conventions, local ordinance texts are highly heterogeneous, exhibiting significant variations in terminology, structure, and overall organization across different jurisdictions [27]. Because of this localized variance, local legal documents have historically been neglected in NLP research, hampered primarily by a severe lack of annotated ground truth datasets [28]. Standard legal corpora fail to capture the specific legislative templates, localized governance terminology, or idiosyncratic cross-referencing styles used in local legislation, such as Philippine city ordinances. Without existing resources that reflect these structural and linguistic complexities, evaluating models on local legislation requires building highly specialized, expert-annotated datasets from the ground up.

A critical architectural decision in dataset construction is determining the exact granularity of the extracted legal premise. While the hypothesis (the local draft) is evaluated as a multi-sentence span to capture context, the premise (the national statute) must be strictly isolated at the sentence level. In Philippine statutory drafting, a single paragraph frequently contains a general rule, multiple conditional modifiers, and distinct exceptions. Holzenberger et al. [98] demonstrate that feeding an entire statutory paragraph into an NLI model as a premise dilutes the model's attention mechanism, causing it to lose track of specific variable constraints. By forcing human annotators to isolate the national rule down to its exact sentence boundary, the resulting Ground Truth prevents the candidate logic engines from getting distracted by surrounding, non-governing text, ensuring the classification mathematically targets the exact point of semantic collision.

The difficulty of dataset construction is magnified by the scarcity of legal domain experts available for annotation. As Zhong et al. [250] explicitly note, the complex terminology required for legal texts makes manual labeling highly time-consuming and difficult to scale, as it strictly requires domain experts rather than crowdsourced workers. Because constructing massive datasets for localized city law is practically unfeasible, researchers must rely on constrained, highly curated Gold Standards. To compensate for this limited data volume, the current NLP standard mandates treating legal conflict detection as a few-shot learning scenario. Gao et al. [76] validate this approach, proving that transformer models can be successfully fine-tuned on specialized datasets containing merely hundreds of examples, provided the evaluation framework rigorously controls for overfitting.

2.10.4 Multi-Sentence and Cross-Document Limits

Traditional Natural Language Inference (NLI) evaluates text using a strict 1-to-1 pairing, comparing a single premise sentence directly against a single hypothesis sentence. However, this highly granular approach fundamentally breaks down in the legal domain due to two specific edge cases. First, intra-document context: a primary rule in one section of an ordinance is frequently modified or negated by an exception buried several paragraphs later (the "1v2" or multi-sentence problem). Second, cross-document dependencies: a contradiction may only emerge when synthesizing provisions from two entirely separate statutes simultaneously.

To address the first edge case, modern legal frameworks utilize Asymmetric Span-Level Pairing. Yin et al. [246] critique traditional sentence-level datasets, proving that capturing contextual exceptions requires evaluating multi-sentence blocks rather than isolated lines. As demonstrated by Koreeda and Manning in the ContractNLI dataset [118], the premise remains a single, focused legal rule, while the corresponding hypothesis is extracted as a multi-sentence chunk. This asymmetric structure allows the model's attention mechanism to accurately evaluate a primary rule against a distant exception contained within the same local provision.

While asymmetric spans successfully capture intra-document context, resolving the second edge case—cross-document multi-hop reasoning—remains computationally prohibitive. Identifying semantic preemption that spans multiple independent laws represents the current research frontier in legal natural language processing and remains unverified for local governance applications. Recent frameworks have attempted to solve this, but they introduce severe architectural overhead. For instance, Lu et al. [136] address cross-document retrieval by programmatically mapping Entity Graphs to mine evidence paths between laws, a process that is highly resource-intensive and unscalable for unstructured LGU archives. Alternatively, Shen et al. [197] proposed the HopWeaver framework, which relies on massive generative Large Language Models (LLMs) to synthesize complex reasoning paths across documents.

Both of these state-of-the-art solutions immediately violate the strict hardware constraints of consumer-grade LGU workstations. Consequently, practical, edge-deployable systems must intentionally bound their theoretical scope to single-premise, asymmetric span evaluations, deliberately avoiding the quadratic memory scaling required for multi-document fusion.

2.10.5 Dataset Partitions

When evaluating NLI models on constrained datasets, researchers often default to k-fold cross-validation to maximize testing coverage. However, Vabalas et al. [224] mathematically demonstrated that applying cross-validation to datasets under 1,000 items introduces severe optimistic bias, causing models to perform significantly worse in real-world deployment than their test scores suggest. To rigorously control for overfitting in few-shot legal scenarios, contemporary evaluation frameworks must abandon k-fold approaches. Instead, they require strict, static Train/Validation/Test partitions (e.g., 70/15/15) to ensure the evaluation metrics remain entirely untainted by training exposure.

Beyond merely preventing optimistic bias, utilizing a static partition allows for rigorous validation monitoring during the fine-tuning phase. In datasets with limited volume, neural networks are highly susceptible to memorizing the training data rather than learning underlying logical patterns. Goodfellow et al. [81] establish that permanently isolating a validation set allows architectures to employ early stopping criteria, halting the training process the moment the model's accuracy on unseen data begins to degrade. Furthermore, as Zhang et al. [248] note regarding few-sample fine-tuning, this static separation ensures strict document-level boundaries, preventing clauses from a specific draft ordinance from accidentally bleeding across the training and testing sets to artificially inflate performance metrics.

2.10.6 Minimum Detectable Effect

A standard critique of static splits on small datasets is that the resulting test set lacks the statistical power to reliably compare candidate architectures. However, Card et al. [29] established that a test set's validity is fundamentally determined by its Minimum Detectable Effect (MDE) rather than arbitrary size thresholds.

Formally, statistical power $(1 - \beta)$ quantifies the probability of correctly rejecting the null hypothesis (H_0) when a true performance disparity exists between competing model architectures. In hypothesis testing, the required sample size N is inversely proportional to the square of the standardized effect size $(\delta = |\mu_1 - \mu_2|/\sigma)$. When comparing architectures with marginal performance differentials (e.g., 1%–2%), large evaluation corpora $(N > 10^4)$ are statistically required to detect the variation at a significance level of $\alpha = 0.05$. Conversely, when evaluating fine-tuned legal models against unadapted zero-shot baselines where the performance delta is substantial $(\delta > 0.80)$, a curated holdout partition of approximately 50 instances provides sufficient statistical power to confirm that observed improvements represent genuine domain adaptation rather than random sampling variance.

The practical application of MDE fundamentally shifts how model performance is evaluated in specialized domains. In general NLP research, massive test sets are required because researchers are often evaluating models against benchmarks like GLUE to prove marginal, incremental gains of one percent to two percent over previous state-of-the-art architectures [235]. However, applying a base foundation model—which possesses no prior knowledge of Philippine local governance for example—to a highly specialized task generally results in baseline performance barely above random chance [31]. Zheng et al. [249] demonstrate that in the legal domain, fine-tuning foundational architectures on expertly curated pairs triggers a massive, double-digit jump in classification accuracy compared to zero-shot baselines. Because this expected performance delta is substantial, legal NLP frameworks do not require thousands of annotated examples to demonstrate statistically valid improvements, proving that an evaluation set of approximately 50 curated pairs possesses sufficient statistical power to mathematically capture this distinct leap in model reasoning capacity.

2.11 Explainable AI

In legal informatics, a “black box” prediction is fundamentally unacceptable, regardless of nominal classification accuracy. Automated systems that flag normative conflicts between statutory enactments must provide verifiable reasoning (*ratio decidendi*) behind their classifications to genuinely assist legislative analysis [64]. Without transparent justification, legal practitioners and legislative staff cannot trust, audit, or formally defend algorithmic determinations. Consequently, contemporary legal NLP literature emphasizes Explainable AI (XAI) not as an optional secondary layer, but as an intrinsic structural requirement of high-stakes legal decision support, prioritizing faithful mathematical traces over ungrounded generative summaries.

2.11.1 Intrinsic Explainability

Within the explainability literature, researchers distinguish between Post-Hoc Surrogate Models and Intrinsic Explainability Mechanisms. Surrogate methods, such as Local Interpretable Model-agnostic Explanations (LIME) [173] and SHapley Additive exPlanations (SHAP) [138], attempt to approximate the behavior of complex deep networks by fitting a simplified, interpretable linear model around individual predictions. Formally, intrinsic explainability extracts feature attributions directly from the internal mathematical parameters and representations of the model during inference. In contrast, post-hoc surrogate methods treat the deep neural network as an unobservable black box, approximating local decision boundaries by iteratively perturbing input tokens thousands of times and fitting auxiliary linear models over the resulting output shifts [180]. In text classification benchmarks, these external explainers are computationally demanding and inherently prone to approximation errors, as they explain an external proxy rather than the underlying network itself [180]. In contrast, intrinsic approaches extract internal model representations—such as self-attention weights and contextualized hidden states—directly from the transformer layers during the forward pass. Because native attention extraction reflects the mathematical weights computed during inference without fitting secondary approximations, it offers an intrinsically faithful and computationally lightweight mechanism for structural transparency.

While utilizing attention weights as a proxy for feature importance has sparked significant debate, empirical findings demonstrate its utility for targeted textual inspection. Jain and Wallace [107] argued that raw attention distributions do not always correlate consistently with gradient-based feature importance. However, Wiegreffe and Pinter [239] provided critical counter-evidence, demonstrating that attention mechanisms provide faithful representations of internal prioritization under rigorous diagnostic testing. In statutory conflict detection and deontic reasoning, extracting token-level attention spans allows reviewers to identify the specific conflicting provisions (for example, isolating an "absolute prohibition" in a superior statute against an "unauthorized permit" in a local draft), establishing a direct visual link between textual evidence and classification decisions without incurring the computational overhead of auxiliary surrogate models.

Complementing structural attention heatmaps in modern explainability frameworks is Confidence Calibration. Deep learning models, particularly multi-layer transformers, frequently exhibit severe overconfidence in erroneous predictions when evaluated on out-of-domain or specialized text. As Guo et al. [89] demonstrated, modern neural networks trained with standard cross-entropy loss are systematically miscalibrated, exhibiting pronounced overconfidence where output probability vectors assign extreme values that exceed the true empirical likelihood of correct classification. In high-stakes regulatory auditing, relying on uncalibrated softmax scores generates distorted risk assessments. To restore probabilistic reliability, calibration literature advocates post-hoc temperature scaling applied directly to pre-softmax logit vectors. Temperature scaling introduces a learned strictly positive scalar parameter $\tau > 0$ that rescales logit magnitudes before softmax normalization without modifying the rank-order of class predictions. This formulation directly minimizes Expected Calibration Error (ECE), ensuring that predicted probability scores correspond to observed empirical accuracy rates across decision thresholds.

2.11.2 Edge Deployment Feasibility

In edge and on-premise computing environments, operational feasibility requires strict adherence to memory and latency boundaries. Post-hoc explainability methods operate by repeatedly perturbing input tokens, requiring hundreds or thousands of forward passes per document to approximate Shapley values or linear boundary slopes [173, 138]. Empirical evaluations by Kumar et al. [119] confirm that permutation-based explainers inflate per-document inference latency by several orders of magnitude, rendering them computationally prohibitive for lengthy legislative texts. Furthermore, Wang et al. [237] demonstrate that running complex post-hoc explainers on edge workstations severely degrades throughput, making interactive review impractical. In resource-constrained deployment settings, intrinsic extraction of native attention weights and calibrated logits provides a computationally viable pathway to transparency without requiring specialized high-performance computing clusters.

2.11.3 Explanation Quality Metrics

Evaluating explanation quality in natural language processing requires distinguishing between two fundamental evaluation criteria: Plausibility and Faithfulness. Formally, *plausibility* measures the extrinsic degree to which an explanation appears convincing, fluent, and domain-intuitive to human legal practitioners, whereas *faithfulness* measures the intrinsic fidelity with which an explanation reflects the actual internal computational mechanics and feature weights utilized by the model to produce its output [106, 64]. Jacovi and Goldberg [106] demonstrate that optimizing explanations purely for plausibility frequently produces deceptive explanations that mask internal reasoning failures and biases. In legal domains, legal scholarship emphasizes that faithfulness must take precedence, as an explanation that sounds plausible but misrepresents statutory text can lead to catastrophic compliance failures.

This trade-off is particularly evident in generative explainability approaches that deploy auto-regressive large language models to draft plain-language explanations of legal predictions. While generative summaries achieve high human plausibility scores, empirical legal informatics studies demonstrate that generative models frequently suffer from legal hallucinations, fabricating non-existent statutory citations or altering legal interpretations during text generation [131, 112]. Because generated natural language explanations are decoupled from the classifier's internal mathematical representations, they remain inherently unfaithful [106]. Consequently, legal informatics literature increasingly favors extractive and attention-grounded visualizations that display the exact mathematical spans influencing classification, avoiding generative distortions.

2.11.4 Explainability in Multi-Stage Pipelines

In multi-stage NLP pipelines, literature on Explainable AI emphasizes situating interpretability mechanisms at the fine-grained inference stage rather than the coarse retrieval phase. In retrieve-then-entail systems, the initial retrieval phase functions as a high-throughput candidate selector across extensive document repositories. As Anand et al. [8] observe, computing dense feature attributions during coarse document filtering provides negligible analytical utility while incurring prohibitive computational latency. In contrast, the subsequent cross-encoder inference stage conducts fine-grained semantic entailment across a constrained candidate set, making it the mathematically appropriate locus for interpretability analysis.

At this second stage, legal NLP frameworks utilize token-level saliency and attention visualizations to bridge model internals with human inspection [231]. By mapping cross-attention weights and calibrated class probabilities back to corresponding spans in the premise and hypothesis texts, systems provide structural verification of detected contradictions without introducing generative paraphrasing risks. This dual mechanism of intrinsic attention mapping and post-hoc confidence calibration allows legislative researchers to trace classification outputs directly back to source statutory phrases.

2.12 Theoretical Framework

As established in legal informatics literature, the Retrieve-then-Entail architecture represents the leading theoretical framework for statutory conflict detection and legislative consistency checking. Originating in the international benchmarks of the Competition on Legal Information Extraction and Entailment (COLIEE 2025 and 2026) [79, 80], this framework integrates two complementary theoretical pillars: a scalable Information Retrieval (IR) architecture to isolate relevant statutory context from expansive corpora, followed by a fine-grained Natural Language Inference (NLI) architecture for rapid deontic conflict detection. Specifically, this theoretical framework synthesizes the dense statutory retrieval methodology of Team AIIR Lab [241, 123], the multi-stage ensemble retrieval principles of Team JNLP [79, 78], and the encoder-based textual entailment framework of Team KIS [109].

2.12.1 Statutory Retrieval (Team AIIR)

In statute law retrieval, dense bi-encoder architectures offer an effective theoretical foundation for bridging the semantic gap between newly drafted provisions and expansive statutory corpora without relying on computationally demanding generative text decoding or manual taxonomy engineering [241, 123]. Drawing from the empirical methodology developed by the AIIR Lab in recent COLIEE benchmarks, this retrieval pipeline operates across four theoretical dimensions: domain-adapted data augmentation, semantic vectorization, corpus pre-computation, and nearest-neighbor distance evaluation. Figure 2.1 illustrates this dense bi-encoder workflow.

Figure 2.1 details the structural dataflow of the dense statutory retrieval architecture adapted from the AIIR Lab methodology. The diagram illustrates four primary operational stages: synthetic legal query augmentation, offline statutory embedding precomputation, dense vector indexing, and approximate nearest neighbor similarity matching. This visual is presented to demonstrate how modern legal search pipelines computationally decouple corpus-wide indexing from runtime query execution. As depicted, the incoming query provision is projected into a shared semantic coordinate space, allowing its vector representation to be compared directly against pre-computed statutory embeddings via cosine distance. This architectural separation ensures that high-volume statutory collections spanning tens of thousands of provisions can be searched in sub-second latency on consumer-grade hardware, providing the high-recall candidate shortlist necessary to feed the downstream entailment engine.

To operationalize this dense bi-encoder architecture, the pipeline executes across four theoretical dimensions. First, targeted data augmentation and domain adaptation expose dense retrieval models to synthetically augmented legal variants, training the neural network to recognize identical statutory concepts across archaic terminology and modern drafting syntax to enhance robustness against lexical divergence. Second, semantic vectorization via bi-encoders (such as `all-mpnet-base-v2`) projects statutory provisions into a dense, continuous vector space where queries and corpus documents are encoded independently, optimizing embedding coordinates such that logically relevant statutory pairs exhibit high geometric proximity under cosine distance. Third, offline corpus pre-

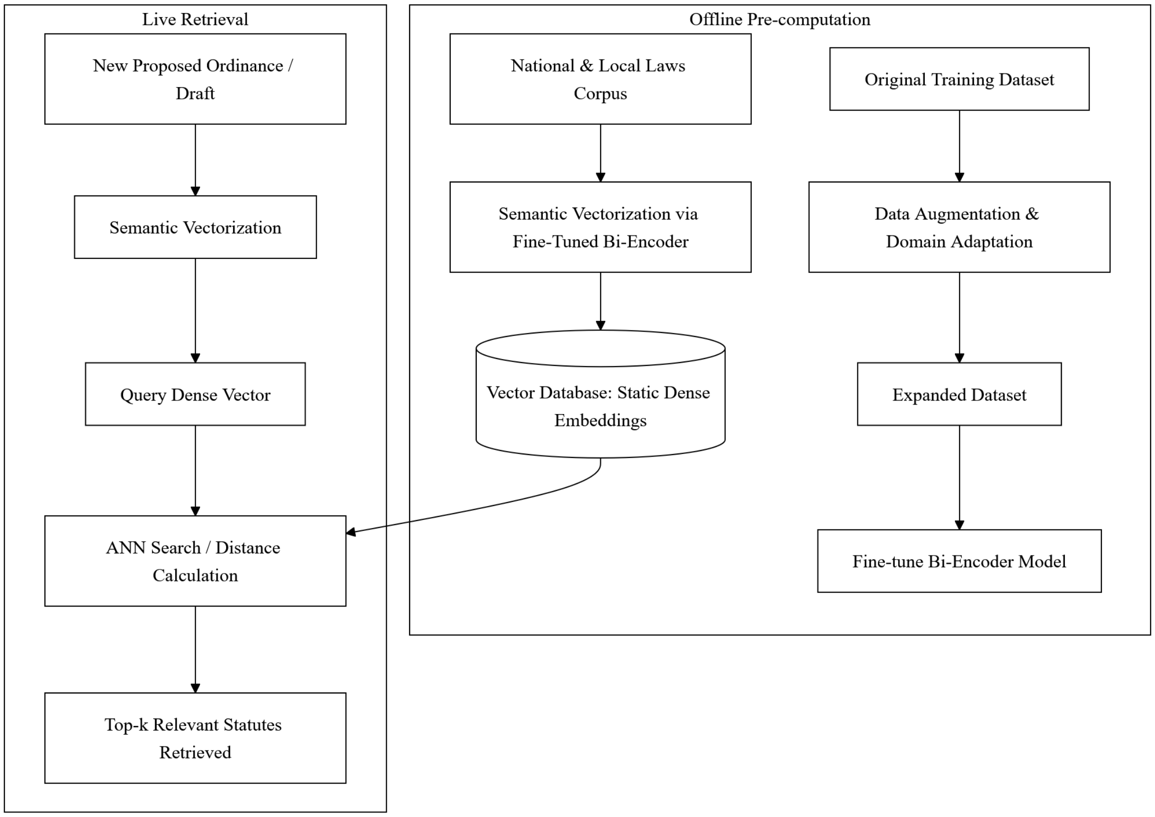

Live Retrieval
New Proposed Ordinance / Draft
Semantic Vectorization
Query Dense Vector
ANN Search / Distance Calculation
Top-k Relevant Statutes Retrieved
Offline Pre-computation
National & Local Laws Corpus
Semantic Vectorization via Fine-Tuned Bi-Encoder
Vector Database: Static Dense Embeddings
Original Training Dataset
Data Augmentation & Domain Adaptation
Expanded Dataset
Fine-tune Bi-Encoder Model

Figure 2.1: Statute Law Retrieval Pipeline Using Dense Bi-Encoder Architecture

computation and vector indexing pre-calculate static embeddings across the statutory repository within an indexed vector database, decoupling computationally intensive encoding from live runtime queries and allowing dense retrieval to scale across extensive archives without real-time GPU re-encoding. Fourth, live query vectorization and proximity scoring project incoming draft provisions into the shared embedding space, enabling approximate nearest-neighbor search to calculate continuous proximity scores in sub-second latency and delivering a high-recall candidate shortlist for downstream inference.

2.12.2 Multi-Stage Retrieval (Team JNLP)

Complementing pure bi-encoder retrieval, literature in computational law emphasizes multi-stage retrieval pipelines to balance candidate recall against ranking precision. Exemplified by Team JNLP's high-performing architecture in COLIEE benchmarks [79, 78], this multi-stage pipeline progressively filters statutory search spaces across three structured phases: Pre-Retrieval Candidate Pruning, Model Fine-Tuning, and Result Ensembling. Figure 2.2 illustrates this multi-stage workflow.

Figure 2.2 outlines the progressive candidate filtering architecture developed by Team JNLP. The diagram delineates the modular progression from initial corpus-level candidate pruning, through parameter-efficient neural re-ranking, to multi-model score

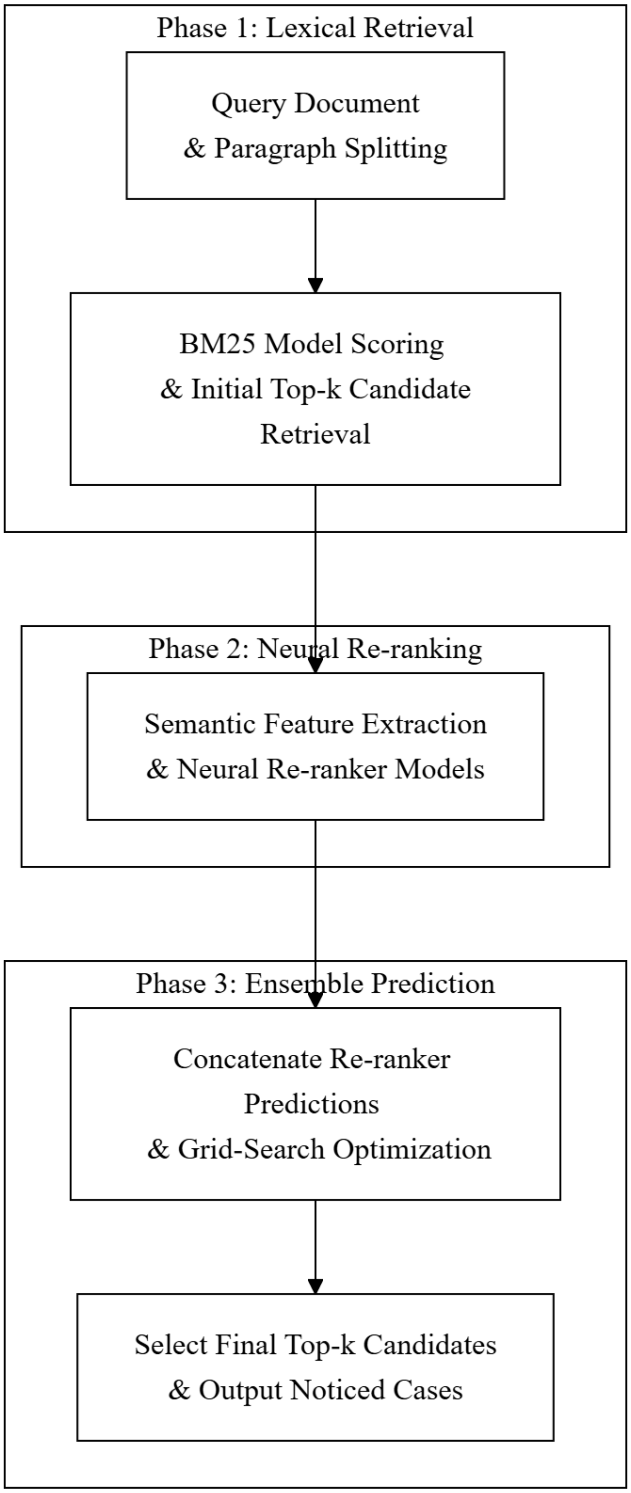

Phase 1: Lexical Retrieval
Query Document & Paragraph Splitting
BM25 Model Scoring & Initial Top-k Candidate Retrieval
Phase 2: Neural Re-ranking
Semantic Feature Extraction & Neural Re-ranker Models
Phase 3: Ensemble Prediction
Concatenate Re-ranker Predictions & Grid-Search Optimization
Select Final Top-k Candidates & Output Noticed Cases

Figure 2.2: Multi-Stage Statutory Retrieval Pipeline With Candidate Pruning and Ensembling

ensembling. This framework is included to provide theoretical justification for combining sparse lexical matching with dense semantic embeddings in statutory search. In the initial phase, heuristic distance thresholds and lexical filters rapidly eliminate irrelevant enactments, after which fine-tuned language models compute calibrated relevance scores across the remaining shortlist. The resulting predictions are synthesized through rank fusion, ensuring that the candidate set benefits from both keyword specificity and conceptual semantic coverage.

2.12.2.1 Pre-Retrieval

The primary objective of the pre-retrieval phase is to drastically reduce the expansive statutory search space while maintaining high recall across governing provisions. Team JNLP achieves this through a multi-stage candidate pruning procedure combining distance-based heuristic filtering with neural re-ranking [79, 78]. Dense embeddings for statutory queries and corpus provisions are extracted using high-capacity multilingual representations. Rather than relying solely on raw cosine similarity, the methodology calculates multi-dimensional distance metrics (such as L1 distance bins), training a gradient-boosted decision tree to predict relevance and counter severe sample imbalance. Following initial filtering, a specialized passage re-ranker evaluates candidate texts and eliminates non-relevant provisions, narrowing the candidate pool to top-ranked candidates while preserving recall.

2.12.2.2 Model Fine-Tuning

Once the candidate space is constrained, the framework applies targeted natural language models to perform fine-grained relevance scoring. Because fine-tuning large models across full parameter sets is computationally intensive, legal NLP frameworks employ Parameter-Efficient Fine-Tuning (PEFT) techniques, such as LoRA and QLoRA. These techniques enable foundational language models to adapt to specialized statutory semantics with minimal memory overhead. Furthermore, because statutory candidate sets exhibit severe class imbalance where only a small minority of retrieved provisions contain true preemption conflicts, sampling strategies and asymmetric loss functions are applied during training to preserve sensitivity to positive conflict signals.

2.12.2.3 Result Ensemble

To maximize retrieval robustness and mitigate individual model variance, multi-stage retrieval pipelines aggregate relevance scores across distinct retrieval architectures. Relevance predictions are aggregated through weighted score aggregation, combining calibrated probabilities from complementary lexical and neural models. The weighting coefficients assigned to individual scoring components are optimized via systematic hyperparameter search algorithms such as Optuna. This optimization ensures balanced contributions between lexical keyword precision and dense semantic recall, yielding a stable candidate shortlist for downstream reasoning.

2.12.3 Textual Entailment (Team KIS)

In the second phase of legal retrieve-then-entail architectures, the theoretical objective transitions from broad candidate discovery to fine-grained deontic reasoning. To ensure high computational throughput and deterministic classification, modern legal NLP frameworks deploy non-generative cross-encoder architectures, exemplified by the ModernBERT-based rationale extraction methodology developed by Team KIS [109]. Figure 2.3 shows this cross-encoder classification structure.

Figure 2.3 illustrates the sequence classification pipeline utilized for statutory textual entailment. The visual depicts the concatenation of the retrieved statutory premise and draft hypothesis into a unified input sequence demarcated by delimiter tokens, which is then processed through bidirectional self-attention layers to produce calibrated class probabilities. This model is included to illustrate how fine-grained deontic reasoning operates without the hallucination risks inherent in generative text decoders. By evaluating full cross-attention between premise and hypothesis tokens simultaneously, the cross-encoder identifies subtle modal contradictions that independent vector projections fail to capture. The calibrated classification head then maps the combined representation into deterministic Entailment, Neutral, or Contradiction outputs with formal statistical reliability.

2.12.4 Theoretical Synthesis

This synthesis integrates the foundational theoretical models reviewed throughout this chapter into a unified framework for ex-ante legislative conflict detection. While each individual methodology addresses an isolated phase of legal document processing, their conceptual integration establishes a principled retrieve-then-entail architecture. The theoretical foundation for statutory candidate discovery draws from the dense retrieval methodology established by the AIIR Lab [241, 123]. Their approach demonstrates that targeted domain augmentation mitigates OCR noise in archival legal text, while continuous semantic vectorization enables high-throughput candidate identification across large statutory collections without relying on rigid keyword indices.

To balance candidate recall against precision, the framework incorporates the multi-stage ranking principles demonstrated by Team JNLP [79, 78]. This methodology utilizes lexical filtering (BM25) to prune extraneous documents, followed by neural scoring and rank fusion to isolate authoritative candidate enactments on standard computing hardware. For granular deontic conflict evaluation, the framework adopts the Team KIS framework for textual entailment [109]. Their methodology formats premise-hypothesis pairs into a continuous sequence evaluated via lightweight cross-encoders, utilizing token-level separation boundaries to classify logical relationships without generative hallucination risks.

Synthesizing these foundational frameworks establishes an end-to-end theoretical foundation for automated preemption auditing. The retrieval stage narrows extensive statutory search spaces down to high-probability candidate provisions, while the textual entailment stage conducts deterministic normative conflict checks. The latest findings from the COLIEE 2026 proceedings [80, 152] reinforce this architectural consensus, vali-

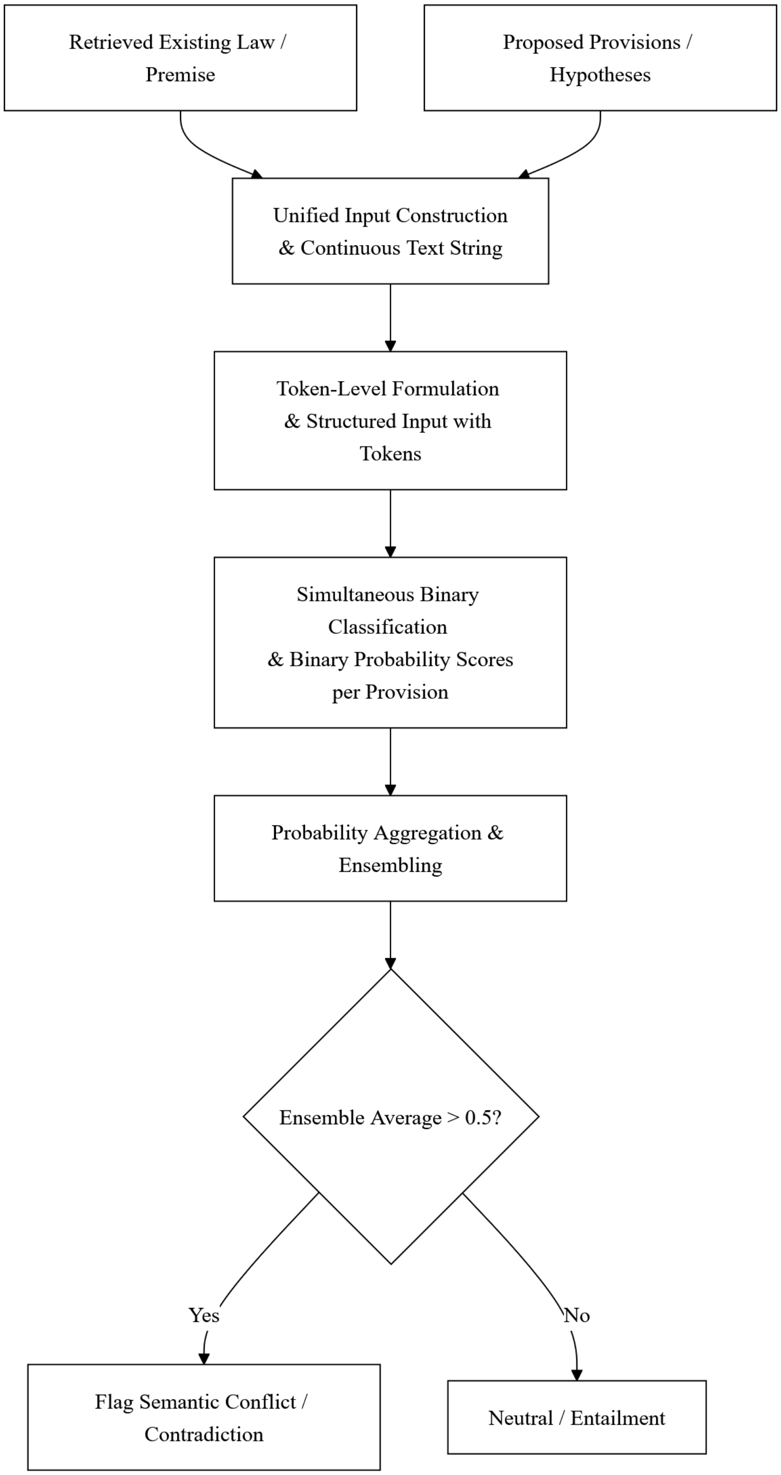

Retrieved Existing Law / Premise
Proposed Provisions / Hypotheses
Unified Input Construction & Continuous Text String
Token-Level Formulation & Structured Input with Tokens
Simultaneous Binary Classification & Binary Probability Scores per Provision
Probability Aggregation & Ensembling
Ensemble Average > 0.5?
Yes
No
Flag Semantic Conflict / Contradiction
Neutral / Entailment

Figure 2.3: Statutory Textual Entailment Pipeline Using Bidirectional Cross-Encoder

dating that pipelined retrieval and textual entailment provide the most reliable foundation for automated legal consistency analysis.

Chapter 3

Methodology

This chapter explains the methodology used to build and evaluate the Coarse-to-Fine Semantic Conflict Detection System. It describes how Philippine legal texts are collected and cleaned, and how the two-stage Retrieve-then-Entail architecture is organized. The chapter also details the creation of the ground truth dataset and the selection of candidate models. Finally, it outlines the evaluation metrics and hardware constraints, which are chosen to match the computing capacity of an average Local Government Unit (LGU).

3.1 Conceptual Framework

This section maps how data flows through the proposed system. The architecture follows an Input-Process-Output (IPO) design organized into a two-stage "Coarse-to-Fine" retrieve-then-entail pipeline (Figure 3.1). In information retrieval and natural language processing, a coarse-to-fine retrieve-then-entail architecture is a cascaded processing design that couples a fast, broad candidate search phase with a computationally intensive, fine-grained semantic classification phase. In this research, Stage 1 serves as the coarse retrieval stage: it rapidly scans the comprehensive statutory corpus of 27,096 enactments (176,421 provisions) using hybrid sparse-dense search to filter the vast document collection down to a compact shortlist of top-k ($k = 50$) candidate provisions. Stage 2 serves as the fine inference stage: it pairs the draft ordinance section with each retrieved candidate statute to perform deep bidirectional cross-attention, computing the calibrated probability of a substantive legal contradiction. By decoupling high-throughput statutory candidate retrieval from compute-intensive cross-encoder inference, this two-stage design circumvents the quadratic memory limits ($O(n^2)$) and runtime overhead that otherwise prohibit whole-corpus cross-attention, enabling deep semantic conflict detection to execute on standard local government office computers without requiring multi-million-peso cloud clusters or dedicated data centers. The workflow is divided into two operational phases: Phase 1 (Offline Foundation and Optimization) and Phase 2 (Live Online Retrieve-then-Entail Pipeline).

Figure 3.1 illustrates the complete end-to-end information flow of the conflict detection architecture, detailing the functional separation between offline foundation

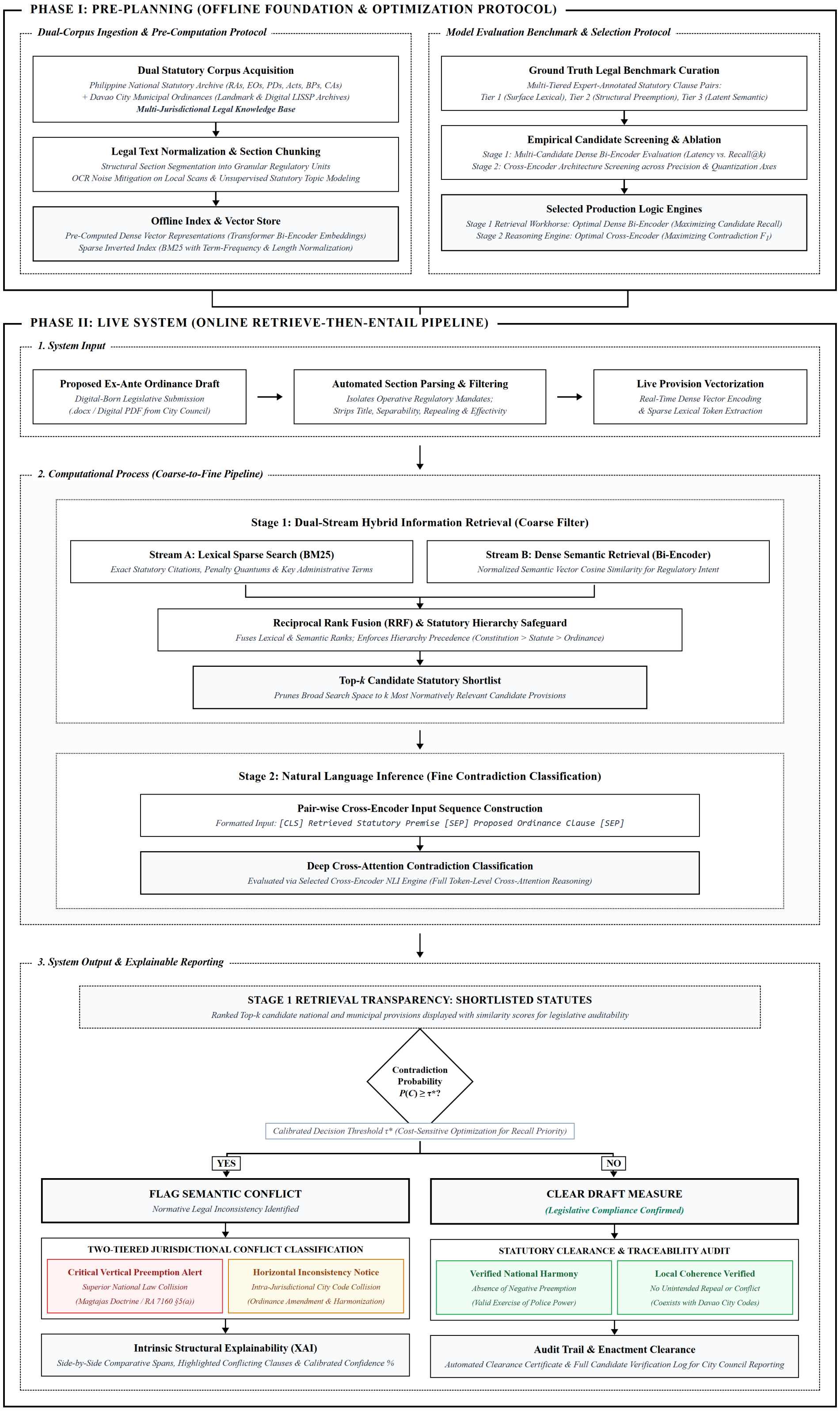

PHASE I: PRE-PLANNING (OFFLINE FOUNDATION & OPTIMIZATION PROTOCOL)
Dual-Corpus Ingestion & Pre-Computation Protocol
Dual Statutory Corpus Acquisition
Philippine National Statutory Archive (RAs, EOs, PDs, Acts, BPs, CAs)
+ Davao City Municipal Ordinances (Landmark & Digital LISSP Archives)
Multi-Jurisdictional Legal Knowledge Base
Legal Text Normalization & Section Chunking
Structural Section Segmentation into Granular Regulatory Units
OCR Noise Mitigation on Local Scans & Unsupervised Statutory Topic Modeling
Offline Index & Vector Store
Pre-Computed Dense Vector Representations (Transformer Bi-Encoder Embeddings)
Sparse Inverted Index (BM25 with Term-Frequency & Length Normalization)
Model Evaluation Benchmark & Selection Protocol
Ground Truth Legal Benchmark Curation
Multi-Tiered Expert-Annotated Statutory Clause Pairs:
Tier 1 (Surface Lexical), Tier 2 (Structural Preemption), Tier 3 (Latent Semantic)
Empirical Candidate Screening & Ablation
Stage 1: Multi-Candidate Dense Bi-Encoder Evaluation (Latency vs. Recall@k)
Stage 2: Cross-Encoder Architecture Screening across Precision & Quantization Axes
Selected Production Logic Engines
Stage 1 Retrieval Workhorse: Optimal Dense Bi-Encoder (Maximizing Candidate Recall)
Stage 2 Reasoning Engine: Optimal Cross-Encoder (Maximizing Contradiction F_1)
PHASE II: LIVE SYSTEM (ONLINE RETRIEVE-THEN-ENTAIL PIPELINE)
1. System Input
Proposed Ex-Ante Ordinance Draft
Digital-Born Legislative Submission
(.docx / Digital PDF from City Council)
Automated Section Parsing & Filtering
Isolates Operative Regulatory Mandates;
Strips Title, Separability, Repealing & Effectivity
Live Provision Vectorization
Real-Time Dense Vector Encoding
& Sparse Lexical Token Extraction
2. Computational Process (Coarse-to-Fine Pipeline)
Stage 1: Dual-Stream Hybrid Information Retrieval (Coarse Filter)
Stream A: Lexical Sparse Search (BM25)
Exact Statutory Citations, Penalty Quantums & Key Administrative Terms
Stream B: Dense Semantic Retrieval (Bi-Encoder)
Normalized Semantic Vector Cosine Similarity for Regulatory Intent
Reciprocal Rank Fusion (RRF) & Statutory Hierarchy Safeguard
Fuses Lexical & Semantic Ranks; Enforces Hierarchy Precedence (Constitution > Statute > Ordinance)
Top-k Candidate Statutory Shortlist
Prunes Broad Search Space to k Most Normatively Relevant Candidate Provisions
Stage 2: Natural Language Inference (Fine Contradiction Classification)
Pair-wise Cross-Encoder Input Sequence Construction
Formatted Input: [CLS] Retrieved Statutory Premise [SEP] Proposed Ordinance Clause [SEP]
Deep Cross-Attention Contradiction Classification
Evaluated via Selected Cross-Encoder NLI Engine (Full Token-Level Cross-Attention Reasoning)
3. System Output & Explainable Reporting
STAGE 1 RETRIEVAL TRANSPARENCY: SHORTLISTED STATUTES
Ranked Top-k candidate national and municipal provisions displayed with similarity scores for legislative auditability
Contradiction Probability $P(C) \geq \tau^*$?
Calibrated Decision Threshold τ^* (Cost-Sensitive Optimization for Recall Priority)
YES
NO
FLAG SEMANTIC CONFLICT
Normative Legal Inconsistency Identified
CLEAR DRAFT MEASURE
(Legislative Compliance Confirmed)
TWO-TIERED JURISDICTIONAL CONFLICT CLASSIFICATION
Critical Vertical Preemption Alert
Superior National Law Collision
(Magtajas Doctrine / RA 7160 §5(a))
Horizontal Inconsistency Notice
Intra-Jurisdictional City Code Collision
(Ordinance Amendment & Harmonization)
STATUTORY CLEARANCE & TRACEABILITY AUDIT
Verified National Harmony
Absence of Negative Preemption
(Valid Exercise of Police Power)
Local Coherence Verified
No Unintended Repeal or Conflict
(Coexists with Davao City Codes)
Intrinsic Structural Explainability (XAI)
Side-by-Side Comparative Spans, Highlighted Conflicting Clauses & Calibrated Confidence %
Audit Trail & Enactment Clearance
Automated Clearance Certificate & Full Candidate Verification Log for City Council Reporting

Figure 3.1: Detailed Information Flow of the Coarse-to-Fine Semantic Conflict Detection Architecture.

pre-computation (Phase 1) and the live pre-enactment review pipeline (Phase 2). The flowchart depicts an Input-Process-Output workflow organized into modular processing layers, highlighting how heterogeneous statutory archives are ingested, normalized, indexed, and logically verified under local edge compute constraints. In Phase 1, the system compiles the dual statutory knowledge base (combining 25,432 national statutes and 1,664 local ordinances), builds pre-computed BM25 and dense embedding indexes, and screens candidate neural architectures against an adviser-reviewed 350-pair ground truth benchmark. In Phase 2, an ex-ante ordinance draft undergoes structural boundary parsing and procedural boilerplate filtering before being routed through Stage 1 hybrid retrieval (fusing lexical BM25 and dense bi-encoder streams via Reciprocal Rank Fusion) and Stage 2 Cross-Encoder inference. This coarse-to-fine design restricts compute-intensive cross-attention to a focused candidate shortlist ($k = 50$), allowing deep semantic conflict detection to execute efficiently on standard local government desktop workstations while producing transparent, explainable audit reports.

3.1.1 Phase 1: Offline Foundation and Optimization

This preliminary phase establishes the empirical foundation of the system by compiling the statutory corpus, building pre-computed retrieval indexes, and selecting candidate transformer logic engines through a structured benchmark screening process.

During dual-corpus ingestion and pre-computation, the system compiles a comprehensive dual-jurisdiction statutory archive combining the Philippine national statutory collection (Republic Acts, Executive Orders, Presidential Decrees, Acts, Batas Pambansa, and Commonwealth Acts) with Davao City local legislation (historical Landmark Ordinances and digital LISSP ordinance records). The compiled corpus undergoes legal text normalization, optical character recognition (OCR) noise mitigation on historical scans, and structural section segmentation into granular, self-contained regulatory units ($\sim$ 200–300 tokens). Cleaned statutory provisions are then indexed offline: dense semantic representations are pre-computed using a lightweight transformer bi-encoder, while sparse inverted lexical indexes are constructed using BM25 ($k_1 = 1.5, b = 0.75$) alongside unsupervised statutory topic modeling.

Following corpus indexing, model benchmarking and selection evaluates candidate neural architectures against a curated Ground Truth Legal Benchmark stratified across three granular difficulty tiers: Tier 1 (Surface Lexical), Tier 2 (Structural Preemption), and Tier 3 (Latent Semantic). A multi-stage empirical screening and ablation process evaluates candidate bi-encoders by balancing inference latency against candidate Recall@k, alongside candidate cross-encoders evaluated across precision, latency, and quantization axes. Based on these empirical evaluations, the highest-performing models are selected as the production engines: an optimal dense bi-encoder for Stage 1 candidate retrieval, and an optimal fine-tuned cross-encoder for Stage 2 contradiction reasoning.

3.1.2 Phase 2: Live System Workflow

The live system executes the automated conflict detection pipeline when a newly drafted local ordinance is submitted for pre-enactment review, operating through an Input-Process-

Output structure.

The workflow begins with system input ingestion, where a legislative researcher uploads a newly proposed ex-ante ordinance draft. Because draft measures originate within local government offices as digital-born documents (.docx or digital PDF), they are free from optical scanning noise. The ingestion pipeline segments the draft into individual substantive provisions, filtering out non-regulatory procedural boilerplate such as the Title, Separability, Repealing, and Effectivity clauses. The isolated substantive provisions are then vectorized in real time via the selected dense bi-encoder alongside sparse lexical token extraction.

The computational process core then executes two complementary stages: hybrid candidate retrieval followed by natural language inference. In Stage 1 (Dual-Stream Hybrid Information Retrieval), the system narrows the comprehensive statutory search space down to a focused candidate pool. Stream A executes lexical sparse search via BM25 to match exact statutory citations, numerical penalty amounts, and administrative agency terms, while Stream B executes dense semantic search via the selected bi-encoder, measuring cosine similarity between the draft provision and statutory embeddings to capture abstract regulatory intent. The ranked outputs are combined using Reciprocal Rank Fusion (RRF) integrated with a Statutory Hierarchy Safeguard prioritizing higher-ranking national statutes over local provisions, extracting the Top-k candidate provisions without overburdening local LGU compute resources.

In Stage 2 (Natural Language Inference), following the standard sequence-pair classification formulation established by Devlin et al. [61] and adapted for transformer cross-encoders by Reimers and Gurevych [170], each retrieved statutory candidate is paired with the draft ordinance provision into a single cross-encoder sequence:

Cross-Encoder Joint Sequence Input Representation:

$$\mathbf{x} = [\text{CLS}]\,\text{Statutory Premise}\,[\text{SEP}]\,\text{Ordinance Hypothesis}\,[\text{SEP}] \tag{3.1}$$

Where [CLS] marks the classification sequence start and [SEP] separates jurisdictional text boundaries.

Equation 3.1 combines the retrieved statutory section and the draft ordinance provision into a single token sequence separated by delimiter tokens. This allows every word in the statutory premise to interact directly with every word in the draft hypothesis through bidirectional cross-attention. This direct comparison is necessary because legal conflicts often depend on small details—such as specific conditions, penalty ceilings, or exceptions—that independent bi-encoder vectors can easily miss. The model's classification head then calculates calibrated probabilities for Entailment, Neutral, and Contradiction, outputting a conflict probability $P(\text{Contradiction})$.

Finally, the system presents an audit trail showing the top-k shortlisted laws for human review. It compares the contradiction probability against a calibrated decision threshold τ^*, tuned with an F_2-score weighting to prioritize catching true conflicts. When the probability meets or exceeds this threshold ($P(\text{Contradiction}) \geq \tau^*$), the system flags a potential legal conflict and classifies it into one of two alert types: a Vertical Preemption Alert for conflicts with superior national statutes under the *Magtajas* doctrine, or a Horizontal Inconsistency Notice for contradictions or overlaps with existing Davao City ordinances. To help legislative researchers interpret the result, the user interface displays side-by-side text comparisons, highlighted conflicting phrases, and model confidence scores. If the probability remains below the threshold ($P(\text{Contradiction}) < \tau^*$), the system reports that no direct conflicts were found, confirming consistency with both national statutes and prior local enactments.

3.2 Research Design

This study uses an applied research design to build a practical software tool for local governance. The researchers use an experimental systems development approach to solve the delays and difficulties of manual ordinance review through automated detection. This setup connects principles from legal informatics and natural language processing to address a real local need. By focusing on practical transformer-based models, the design connects theoretical computer science with everyday government office work. The experiment directly tests whether a retrieve-then-entail pipeline can operate reliably in a local legislative setting.

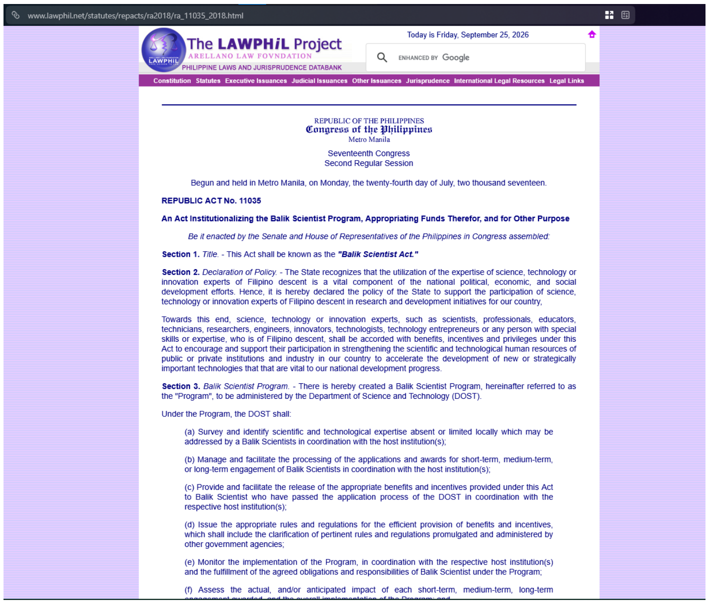
The experimental setup tests the retrieval and classification modules separately to measure search speed and accuracy independently. Building the system step-by-step allows the researchers to adjust the information retrieval and natural language inference stages using local legal texts. This engineering process ensures that the finished tool accounts for the writing styles found in both city ordinances and national statutes. In this way, applied research methods help turn modern NLP concepts into a usable tool for the Davao City Sangguniang Panlungsod. As a result, the findings establish a clear benchmark for automating pre-enactment legal checks on local legislation.

3.3 Corpus Acquisition and Exploratory Analysis

Building the legal database requires collecting text from different public repositories so that the system includes as many relevant laws as possible. These sources provide the national statutes and city ordinances needed to check draft legislation for consistency. To keep the language consistent and make vector processing simpler, the system gathers only the English-language versions of all legal texts. This choice matches Philippine statutory rules under Section 20, Chapter 5, Book I of Executive Order No. 292 (Administrative Code of 1987) [171], which explicitly mandates that in the interpretation of laws promulgated in official languages, the English text shall control unless otherwise specifically provided. National statutes and local ordinances are gathered using separate automated scripts designed for the format of each source.

3.3.1 Corpus Composition

The Lawphil Project (Arellano Law Foundation) serves as the primary digitized extraction source for national statutory text because its documents are provided in clean HTML, follow a consistent layout, and offer broad historical coverage [9]. Automated scraping scripts browse through chronological and alphabetical indexes to collect Republic Acts, Batas Pambansa, Commonwealth Acts, Acts, Executive Orders, and Presidential Decrees. Using a custom web extraction pipeline built by the proponents, statutory extraction was conducted as of **March 2026**, collecting 25,432 unique legal records enacted between 1900 and 2026. The Official Gazette of the Republic of the Philippines [86] and the Chan Robles Virtual Law Library were used as authoritative secondary cross-references to verify enactment dates, chronological numbering, and historical amendment status. All collected national laws are converted into standardized JSON records with metadata, such as dates of approval, titles, and section numbers. As illustrated in Figure 3.2, the Lawphil repository presents statutory text in clean, unstyled HTML typography with a predictable hierarchical URL schema (e.g., `/statutes/repacts/ra2018/ra_11035_2018.html` for Republic Act No. 11035). This standardized structure separates introductory legislative headers from operative provisions and eliminates the need for dynamic JavaScript browser rendering, facilitating rapid, fault-tolerant web scraping.


www.lawphil.net/statutes/repacts/ra2018/ra_11035_2018.html
The LAWPHiL Project
ARELLANO LAW FOVNDATION
PHILIPPINE LAWS AND JURISPRUDENCE DATABANK
LAWPHIL
Today is Friday, September 25, 2026
ENHANCED BY Google
Constitution
Statutes
Executive Issuances
Judicial Issuances
Other Issuances
Jurisprudence
International Legal Resources
Legal Links

REPUBLIC OF THE PHILIPPINES
Congress of the Philippines
Metro Manila

Seventeenth Congress
Second Regular Session

Begun and held in Metro Manila, on Monday, the twenty-fourth day of July, two thousand seventeen.

REPUBLIC ACT No. 11035

An Act Institutionalizing the Balik Scientist Program, Appropriating Funds Therefor, and for Other Purpose

Be it enacted by the Senate and House of Representatives of the Philippines in Congress assembled:

Section 1. *Title.* - This Act shall be known as the ***"Balik Scientist Act."***

Section 2. *Declaration of Policy.* - The State recognizes that the utilization of the expertise of science, technology or innovation experts of Filipino descent is a vital component of the national political, economic, and social development efforts. Hence, it is hereby declared the policy of the State to support the participation of science, technology or innovation experts of Filipino descent in research and development initiatives for our country,

Towards this end, science, technology or innovation experts, such as scientists, professionals, educators, technicians, researchers, engineers, innovators, technologists, technology entrepreneurs or any person with special skills or expertise, who is of Filipino descent, shall be accorded with benefits, incentives and privileges under this Act to encourage and support their participation in strengthening the scientific and technological human resources of public or private institutions and industry in our country to accelerate the development of new or strategically important technologies that that are vital to our national development progress.

Section 3. *Balik Scientist Program.* - There is hereby created a Balik Scientist Program, hereinafter referred to as the "Program", to be administered by the Department of Science and Technology (DOST).

Under the Program, the DOST shall:

(a) Survey and identify scientific and technological expertise absent or limited locally which may be addressed by a Balik Scientists in coordination with the host institution(s);

(b) Manage and facilitate the processing of the applications and awards for short-term, medium-term, or long-term engagement of Balik Scientists in coordination with the host institution(s);

(c) Provide and facilitate the release of the appropriate benefits and incentives provided under this Act to Balik Scientist who have passed the application process of the DOST in coordination with the respective host institution(s);

(d) Issue the appropriate rules and regulations for the efficient provision of benefits and incentives, which shall include the clarification of pertinent rules and regulations promulgated and administered by other government agencies;

(e) Monitor the implementation of the Program, in coordination with the respective host institution(s) and the fulfillment of the agreed obligations and responsibilities of Balik Scientist under the Program;

(f) Assess the actual, and/or anticipated impact of each short-term, medium-term, long-term

Figure 3.2: Sample Statutory Enactment Webpage from the Lawphil Project Repository Illustrating Clean Document Hierarchy and Structured Legislative Metadata (Republic Act No. 11035).

Figure 3.2 displays a representative national statutory webpage from the Lawphil Project repository, featuring the complete legislative record of Republic Act No. 11035 (*Balik Scientist Act*). The document interface exhibits a clean, unstyled HTML layout that cleanly isolates the formal congress heading, approval date, official title, and individual numbered provisions into predictable DOM elements. This visual structure is included to illustrate why the national statutory repository facilitates automated web crawling without requiring dynamic JavaScript execution or complex layout segmentation. In the displayed layout, the operative provisions appear in sequential chronological order, flanked by transparent URL parameters (`/statutes/repacts/ra2018/ra_11035_2018.html`) that enable automated validation against congressional records. By capturing these structured HTML pages directly, the ingestion preprocessor compiles over 25,000 national enactments into standardized text records while completely avoiding the visual noise and character degradation present in scanned archival paper.

Collecting national laws from Lawphil helps avoid the disorganization common to public legal websites. Although the Official Gazette and congressional websites also host statutory records, their web pages frequently change layout and often provide older laws only as image-only PDF scans. In contrast, Lawphil keeps a standard HTML structure across all historical eras, clearly separating introductory preambles from operative legal sections. Scraping from this repository ensured uniform text formatting across the national collection (over 25,000 enactments), preventing formatting differences from skewing vector embeddings during Stage 1 retrieval. Listing 3.1 illustrates the web scraping and provision extraction logic implemented to retrieve and structure the statutory archive, establishing a consistent text foundation for semantic analysis and conflict detection.

```python
def extract_statutory_corpus(base_url, start_year=1900, end_year=2026):
    """Extracts, sanitizes, and structures national statutes from the online archive."""
    statute_records = []
    for year in range(start_year, end_year + 1):
        year_index = fetch_page(f"{base_url}/year/{year}")
        statute_links = parse_index_links(year_index)
        for link in statute_links:
            # Fetch raw web page and parse HTML DOM tree
            html_content = fetch_page_with_retry(link, delay_seconds=1.0)
            soup = BeautifulSoup(html_content, "html.parser")

            # Extract statutory metadata and clean document body
            metadata = extract_header_metadata(soup)
            body_text = clean_document_body(soup)
            sections = segment_into_sections(body_text)

            statute_records.append({
                "statute_id": metadata.id,
                "title": metadata.title,
                "instrument": metadata.type,
                "date_enacted": metadata.date,
                "sections": sections
            })
    return statute_records

```

Listing 3.1: Code Snippet of the Statutory Corpus Acquisition and Provision Extraction Pipeline

Listing 3.1 details the programmatic execution logic of the statutory web extraction script, outlining the sequential functions for annual index traversal, DOM parsing, metadata extraction, and provision boundary segmentation. The script is structured to crawl legislative archives spanning from 1900 through 2026, parsing HTML document trees via BeautifulSoup while implementing rate-limiting delays to preserve source server availability. During execution, the pipeline extracts key statutory metadata—including enactment identifiers, approval dates, and legislative titles—before passing the document body to a regex-based section segmenter that isolates individual statutory provisions into distinct JSON fields. Methodologically, this automated extraction guarantees uniform data serialization across all 25,432 national enactments, eliminating web scraping artifacts and ensuring that downstream embedding models evaluate pure statutory syntax.

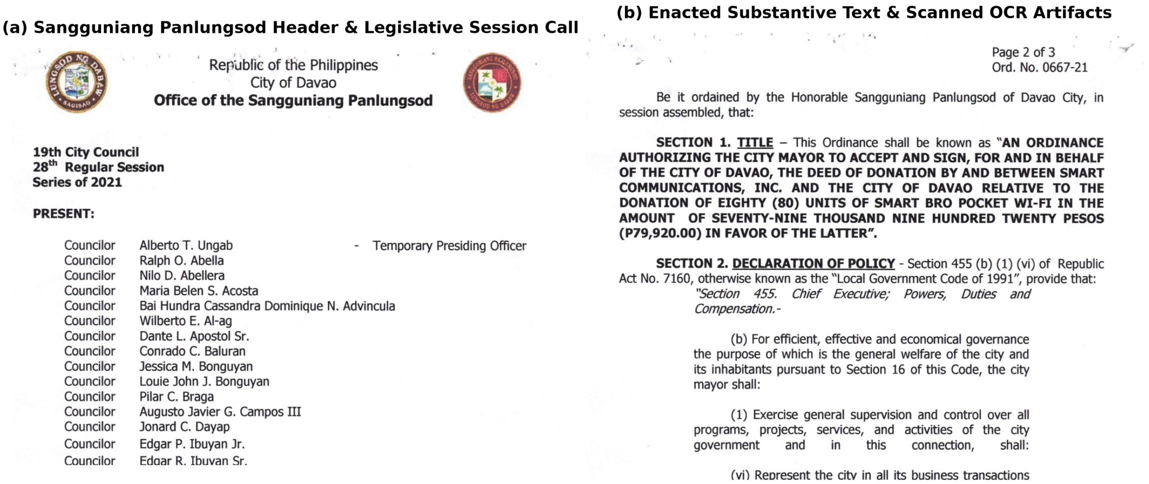
To capture the complete statutory search space, the national collection is paired with Davao City local legislation gathered across two complementary acquisition streams. The first channel consists of the official Davao City Sangguniang Panlungsod Public Web Portal, which provides selective historical landmark ordinances and city uploads spanning approximately fifteen years, capturing prominent regulatory codes in public safety, zoning, and health [188]. The second and primary channel comprises the local Legislative Information Support System Program (LISSP), acquired directly through formal administrative coordination with the Sangguniang Panlungsod Secretariat and Information Technology division [37]. Through this institutional coordination, local authorities granted the proponents direct access to digitized enactments via a shared repository, encompassing enactments passed primarily by the 19th and 20th City Councils between 2021 and 2025. Together, these two channels yielded an archive of exactly 1,664 local enactments spanning from 1950 to 2025. These 1,664 documents represent the complete accessible digital corpus gathered within the proponents' institutional capabilities. In practice, the true cumulative historical total of ordinances enacted since Davao City was chartered under Commonwealth Act No. 51 in 1936 remains unrecorded on the internet and unknown to local authorities. In institutional consultations, Sangguniang Panlungsod personnel confirmed that an official cumulative ledger does not exist because local legislation is numbered sequentially per calendar year and council term rather than in an unbroken cumulative sequence. Furthermore, the vast majority of historical enactments, particularly those prior to the 2021 LISSP digital initiative, exist exclusively as physical paper documents preserved in archive rooms and filing cabinets at the Sangguniang Panlungsod building on San Pedro Street, requiring manual page-by-page physical inventorying to tally. When combined with the 25,432 national statutes, the total statutory search space comprises **27,096 legal enactments** ($N_{\text{sections}} = 176,421$).

3.3.2 Corpus Characteristics and Inherent Biases

A close look at the collected national laws reveals a large imbalance across legal topics. Over 40% of the national statutes consist of routine administrative actions, such as converting local schools, renaming roads, adjusting municipal boundaries, or granting broadcast franchises. In contrast, major regulatory areas that interact with city powers, such as public health, land zoning, and local taxation, make up less than 15% of the total national collection. Searching through this raw distribution without filtering would cause search algorithms to return irrelevant administrative laws instead of actual regulatory rules. To keep search results useful, the system must separate these routine administrative laws from substantive regulatory provisions.

The local ordinance archive also shows an uneven time distribution and gaps in older records. For ordinances passed before 2021, city record-keeping mainly focused on digitizing major "Landmark Ordinances," leaving ordinary regulations from earlier decades in physical paper storage [188]. As a result, the pre-2021 local records represent a selective collection of prominent laws rather than a complete legislative record. While post-2021 records from the digital LISSP database are much more complete, some numerical gaps remain due to missing committee drafts and unindexed attachments. These temporal biases mean that the available local archive contains substantially more recent ordinances than older enactments.

A fundamental technical distinction must be maintained regarding the document media of the legislative corpus. All 1,664 archival ordinance documents received from the Sangguniang Panlungsod are historical scanned physical paper PDFs, rather than digital-born documents exported from word processing software. Older ordinances scanned at low resolution suffer from faded print, ink bleed-through, staple shadows, and rotational skew, as illustrated in Figure 3.3. The scanned copies also contain non-text items such as government stamps, official signatures, and handwritten marginal notes that confuse standard sentence splitting tools. Clean, digital-born text (documents exported directly from modern word processors or Google Docs without scanning noise) appears exclusively during Phase 2 live system testing when legislative staff submit newly drafted ex-ante measures for pre-enactment conflict review. At the language level, local ordinances sometimes include regional terms and administrative Taglish in their introductory sections [65]. These informal drafting habits differ noticeably from the formal, standardized English found in national laws.



(a) Sangguniang Panlungsod Header & Legislative Session Call

Republic of the Philippines
City of Davao
Office of the Sangguniang Panlungsod

19th City Council
28th Regular Session
Series of 2021

PRESENT:

Councilor	Alberto T. Ungab	- Temporary Presiding Officer
Councilor	Ralph O. Abella	
Councilor	Nilo D. Abellera	
Councilor	Maria Belen S. Acosta	
Councilor	Bai Hundra Cassandra Dominique N. Advincula	
Councilor	Wilberto E. Al-ag	
Councilor	Dante L. Apostol Sr.	
Councilor	Conrado C. Baluran	
Councilor	Jessica M. Bonguyan	
Councilor	Louie John J. Bonguyan	
Councilor	Pilar C. Braga	
Councilor	Augusto Javier G. Campos III	
Councilor	Jonard C. Dayap	
Councilor	Edgar P. Ibuyan Jr.	
Councilor	Edgar R. Ibuyan Sr.	

(b) Enacted Substantive Text & Scanned OCR Artifacts

Page 2 of 3
Ord. No. 0667-21

Be it ordained by the Honorable Sangguniang Panlungsod of Davao City, in session assembled, that:

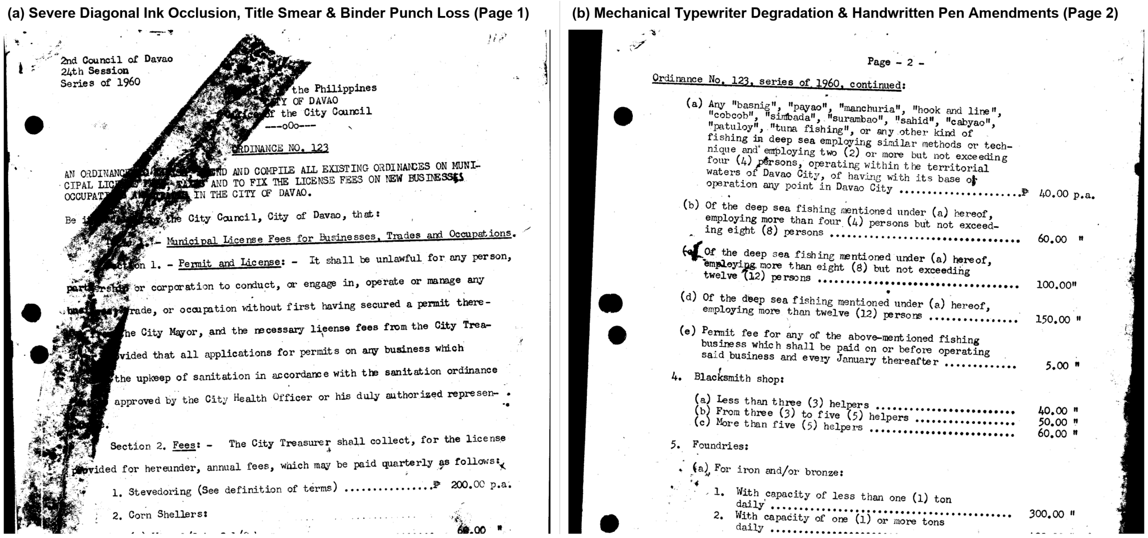
SECTION 1. TITLE – This Ordinance shall be known as **"AN ORDINANCE AUTHORIZING THE CITY MAYOR TO ACCEPT AND SIGN, FOR AND IN BEHALF OF THE CITY OF DAVAO, THE DEED OF DONATION BY AND BETWEEN SMART COMMUNICATIONS, INC. AND THE CITY OF DAVAO RELATIVE TO THE DONATION OF EIGHTY (80) UNITS OF SMART BRO POCKET WI-FI IN THE AMOUNT OF SEVENTY-NINE THOUSAND NINE HUNDRED TWENTY PESOS (P79,920.00) IN FAVOR OF THE LATTER".**

SECTION 2. DECLARATION OF POLICY - Section 455 (b) (1) (vi) of Republic Act No. 7160, otherwise known as the "Local Government Code of 1991", provide that:

> *"Section 455. Chief Executive; Powers, Duties and Compensation.-*
>
> (b) For efficient, effective and economical governance the purpose of which is the general welfare of the city and its inhabitants pursuant to Section 16 of this Code, the city mayor shall:
>
> (1) Exercise general supervision and control over all programs, projects, services, and activities of the city government and in this connection, shall:
>
> (vi) Represent the city in all its business transactions

Figure 3.3: Sample Archival Document from the Davao City Sangguniang Panlungsod Corpus Illustrating Physical Scanning Artifacts and Character Noise (Ordinance No. 0667-21, Series of 2021).

Figure 3.3 displays a representative physical document scan from the Davao City Sangguniang Panlungsod legislative archive (Ordinance No. 0667-21, enacted in 2021). Notably, this document represents one of the tamest and cleanest historical scans across the local collection, having been produced using modern electronic typesetting and printed on clean bond paper. Even on this mild scan, off-the-shelf optical character recognition tools still fail: standard OCR engines suffer from character corruptions (misreading "2021" as "0667-2L" and "TITLE" as "TITTE") and erroneously fold the non-regulatory council session attendance roster from Panel (a) into substantive regulatory commands. While contemporary enactments present manageable surface noise, older historical enactments in the repository exhibit far more severe physical degradation. Figure 3.4 illustrates an authentic historical scan from the pre-LGC archive (Ordinance No. 123, Series of 1960), representing one of the most intense and severely degraded documents preserved in the local collection.



(a) Severe Diagonal Ink Occlusion, Title Smear & Binder Punch Loss (Page 1)

2nd Council of Davao
24th Session
Series of 1960

[illegible] the Philippines
[illegible]Y OF DAVAO
[illegible] the City Council
---oOo---

[illegible]RDINANCE NO. 123

AN ORDINANC[illegible] [illegible]ND AND COMPILE ALL EXISTING ORDINANCES ON MUNICIPAL LIC[illegible] AND TO FIX THE LICENSE FEES ON NEW BUSINESSES OCCUPATI[illegible] IN THE CITY OF DAVAO.

Be i[illegible] the City Council, City of Davao, that:

[illegible] – Municipal License Fees for Businesses, Trades and Occupations.

[illegible]on 1. – Permit and License: – It shall be unlawful for any person, part[illegible]ership or corporation to conduct, or engage in, operate or manage any [illegible]rade, or occupation without first having secured a permit there- [illegible]he City Mayor, and the necessary license fees from the City Trea- [illegible]vided that all applications for permits on any business which [illegible] the upkeep of sanitation in accordance with the sanitation ordinance approved by the City Health Officer or his duly authorized represen- [illegible]

Section 2. Fees: – The City Treasurer shall collect, for the license provided for hereunder, annual fees, which may be paid quarterly as follows:

1. Stevedoring (See definition of terms) .... ₱ 200.00 p.a.
2. Corn Shellers:

[illegible] 60.00 "

(b) Mechanical Typewriter Degradation & Handwritten Pen Amendments (Page 2)

Page – 2 –

Ordinance No. 123, series of 1960, continued:

(a) Any "basnig", "payao", "manchuria", "hook and line", "cobcob", "simbada", "surambao", "sahid", "cabyao", "patuloy", "tuna fishing", or any other kind of fishing in deep sea employing similar methods or technique and employing two (2) or more but not exceeding four (4) persons, operating within the territorial waters of Davao City, of having with its base of operation any point in Davao City .... ₱ 40.00 p.a.

(b) Of the deep sea fishing mentioned under (a) hereof, employing more than four (4) persons but not exceeding eight (8) persons .... 60.00 "

(c) Of the deep sea fishing mentioned under (a) hereof, employing more than eight (8) but not exceeding twelve (12) persons .... 100.00"

(d) Of the deep sea fishing mentioned under (a) hereof, employing more than twelve (12) persons .... 150.00 "

(e) Permit fee for any of the above-mentioned fishing business which shall be paid on or before operating said business and every January thereafter .... 5.00 "

4. Blacksmith shop:

   (a) Less than three (3) helpers .... 40.00 "
   (b) From three (3) to five (5) helpers .... 50.00 "
   (c) More than five (5) helpers .... 60.00 "

5. Foundries:

   (a) For iron and/or bronze:

      1. With capacity of less than one (1) ton daily .... 300.00 "
      2. With capacity of one (1) or more tons daily ....

Figure 3.4: Severe Archival Document Degradation in Historical Davao City Local Ordinances Illustrating Heavy Carbon Smudging, Hole Punch Clipping, and Faded Mechanical Typewriting (Ordinance No. 123, Series of 1960).

Figure 3.4 displays the extreme physical degradation present in mid-century local legislation, examining two pages of Davao City Ordinance No. 123 from the 1960 legislative term. Panel (a) illustrates a severe diagonal carbon smudge and ink smear that obliterates nearly half of the opening legislative preamble, ordinance title, and Section 1 permit mandates, alongside three physical binder punch holes that cut directly through operative characters along the left margin. Panel (b) reveals mechanical typewriter degradation characterized by uneven ink ribbon transfer, misaligned baseline characters, and handwritten pen strike-throughs and amendments (such as "As amended by Ord. No. 131, S. 1961"). When standard optical character recognition is applied to documents with this level of physical deterioration, traditional rule-based pipelines suffer catastrophic failure, producing character error rates exceeding 40% and misinterpreting or completely dropping substantive penal provisions. This visual evidence justifies why the proponents implemented rigorous layout perimeter cropping, noise whitelisting, and local vision-language model (VLM) extraction to recover authoritative legal text from historical city archives.

In real-world legislation, direct contradictions between laws are very rare. Lawmakers generally try to write rules that work alongside existing higher laws rather than directly opposing them [74]. As a result, actual legal conflicts make up less than 1% of random pairs formed between local ordinances and national statutes. Taking random pairs of text from the unedited collections would lead to an overwhelming majority of unrelated pairs. A machine learning model trained on such unbalanced data could score 99% nominal accuracy simply by always predicting "Neutral," while remaining unable to detect real legal conflicts.

3.3.3 National Statutory Corpus Analysis

Across the five main types of legislation with sequential numbers, the collected corpus includes 19,683 statutes out of an estimated 20,246 historical laws, achieving a coverage rate of 97.22%. When combined with the 5,749 Executive Orders from fifteen presidential administrations, the compiled national archive holds 25,432 out of about 26,451 total historical legal documents, reaching a national recall rate of 96.15%. All official legislative totals reflect the chronological numbering sequence established by Congress and presidential records as of **March 2026**, where sequential Republic Acts reached Republic Act No. 12,313. Table 3.1 provides a breakdown of the national corpus, comparing the collected documents with official legislative totals for each statutory type.

Table 3.1: Comparative Statutory Coverage and Composition of the Philippine National Legal Archive ($N = 25,432$)

ID	**Statutory Instrument**	**Corpus**	**Official Total**	**Coverage Rate**	**Corpus Share**	**Historical Era and Governing Authority**
01	Republic Acts (RA)	11,963	12,313	97.16%	47.04%	Post-war and Fifth Republic legislation (1946–present, up to RA 12313 as of March 2026)
02	Executive Orders (EO)	5,749	6,205[*]	92.65%	22.61%	Presidential administrative issuances (1901–2026, across 15 administrations)
03	Acts	4,257	4,275	99.58%	16.74%	American Insular Government enactments (1900–1935)
04	Presidential Decrees (PD)	1,844	2,036	90.57%	7.25%	Martial law legislative decrees enacted by executive fiat (1972–1986)
05	Batas Pambansa (BP)	886	889	99.66%	3.48%	Unicameral parliamentary legislation under the 1973 Constitution
06	Commonwealth Acts (CA)	733	733	100.00%	2.88%	Transitional governance enactments under the 1935 Constitution
Sequential Subtotal		**19,683**	**20,246**	**97.22%**	**77.39%**	**All numbered legislative enactments of Congress**
Total Archive		**25,432**	**26,451**	**96.15%**	**100.00%**	**Comprehensive corpus extracted from the Lawphil Project**

[*] Historical total for Executive Orders represents the aggregate sum of maximum numbered issuances across all fifteen presidential administrations (1935–2026) and colonial governorships (1901–1935) as recorded in the Official Gazette and Lawphil archives as of March 2026.

Table 3.1 presents the structural breakdown of the Philippine national statutory archive across six legislative instruments, detailing the collected enactment counts, official legislative totals, achieved coverage rates, percentage shares, and historical governing authorities. This breakdown is included to document the completeness and historical coverage of the national statutory database, showing that the harvested collection represents an authentic archive of nearly all Philippine legislation rather than a selective convenience sample. As shown in the table, congressional Republic Acts form the largest single category (47.04%, $N = 11,963$), followed by presidential Executive Orders (22.61%,

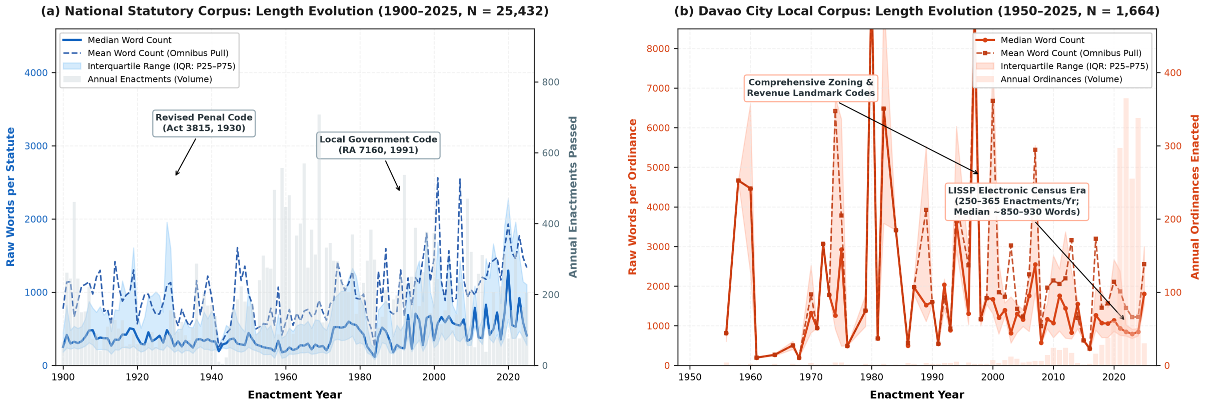
$N = 5,749$) and American-era Acts (16.74%, $N = 4,257$). Across all numbered congressional enactments, the collection captures 19,683 out of 20,246 historical laws (97.22% coverage), expanding to 25,432 documents (96.15% coverage) when including presidential administrative issuances. This broad coverage ensures that the statutory search space encompasses nearly the entire working legislative history of the Philippines since 1900, minimizing the risk of missing relevant national laws during Stage 1 retrieval.

A key methodological decision in compiling this database is the explicit inclusion of **Executive Orders (EOs)** alongside congressional legislation. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III of Executive Order No. 292 [171], Executive Orders are presidential acts providing rules of a general character in execution of statutory mandates. Unlike Supreme Court decisions that settle private adversarial controversies under *stare decisis*, Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force of law. Modern Executive Orders establish binding nationwide regulatory mandates, such as Executive Order No. 26 (s. 2017) mandating nationwide public smoking bans, that legally preempt inconsistent local ordinances. Excluding Executive Orders would create an artificial regulatory gap, allowing local ordinances to contradict operative presidential mandates.

The coverage rates confirm that the national legal archive is nearly complete across all historical governance eras. Commonwealth Acts show full 100.00% coverage across all 733 enactments, while American-era Acts (99.58%) and Batas Pambansa (99.66%) miss only 18 and 3 enactments, respectively. For Republic Acts, the 11,963 collected records cover 97.16% of the 12,313 official total, with missing entries mainly being local bills or recent laws still being indexed. For Presidential Decrees, the 1,844 gathered records represent 90.57% of the 2,036 numbered decrees from the Martial Law period, where uncollected entries were unpublished directives until the Supreme Court required full publication in *Tañada v. Tuvera* [205]. For Executive Orders, the 5,749 collected records represent 92.65% of the 6,205 numbered orders issued across fifteen presidential terms, offering an extensive record of administrative policy.

Beyond document counts and legal instruments, the proponents conducted a longitudinal text length census across both statutory collections to evaluate how document lengths evolved over time. Examining word count variations across legislative regimes reveals significant structural disparities between national statutes and local city ordinances. Figure 3.5 visualizes the longitudinal word count evolution per enactment alongside annual legislative volumes across both the Philippine National Statutory Corpus (1900–2026, $N = 25,432$) and the Davao City Local Legislative Corpus (1950–2025, $N = 1,664$).

Figure 3.5 traces document length distributions and annual enactment counts across 126 years of national lawmaking and 75 years of local city governance. Panel (a) plots raw words per statute for the national collection, showing median word count, mean word count under an omnibus pull, the interquartile range (IQR: P25–P75), and annual enactment volume bars. Across all 25,432 national enactments, the collection maintains an overall median of 357.0 words (mean: 979.5 words, IQR: 212.0–738.0 words). Early colonial Acts (1900–1935) exhibited a median of 344.0 words, punctuated by massive codifications like the Revised Penal Code of 1930 (Act No. 3815). Following the 1987 Constitution, national statutes expanded in complexity, reaching a median of 425.0 words


(a) National Statutory Corpus: Length Evolution (1900–2025, N = 25,432)

(b) Davao City Local Corpus: Length Evolution (1950–2025, N = 1,664)

Figure 3.5: Longitudinal Word Count Evolution per Enactment and Annual Legislative Volumes across the Philippine National Statutory Corpus ($N = 25,432$) and Davao City Local Legislative Corpus ($N = 1,664$).

and a mean of 1,294.1 words, driven by extensive omnibus frameworks such as the Local Government Code of 1991 (Republic Act No. 7160). Panel (b) illustrates the corresponding trajectory for the Davao City local archive, which exhibits an overall median of 880.0 words (mean: 1,655.6 words, IQR: 735.8–1,606.5 words). Pre-2021 records, consisting of selectively digitized landmark regulatory frameworks such as the 1998 Comprehensive Zoning Ordinance, pulled historical averages to a median of 1,187.0 words and a mean of 2,298.5 words. Conversely, the modern LISSP digital census era (2021–2025, contributing 1,285 enactments at 250–365 ordinances annually) stabilized at a consistent median of 847.0 words and a mean of 1,466.0 words, documenting routine municipal governance.

To trace how the six statutory instruments evolved across Philippine jurisprudence, the proponents categorized statutory enactments across six chronological eras. Each historical era established distinct legislative authorities, ranging from colonial commissions to modern constitutional congresses. Figure 3.6 charts the chronological succession and aggregate volume of statutory instruments from 1900 through 2026. The distribution reflects the shifting primacy of legislative acts, presidential decrees, and administrative executive orders across political transitions. Examining these enactment waves illustrates how historical governance structures directly shaped the composition of the national legal corpus.

Figure 3.6 charts the annual volume and succession of the six statutory instruments across 126 years of Philippine governance (1900–2026). The horizontal axis traces enactment years across six major political eras, while the vertical axis records annual enactment counts color-coded by legislative instrument. This chart is included to show how lawmaking shifted across historical eras, leading to different drafting styles. As shown in the plot, legislative volume peaked during periods of major political reorganization, moving from colonial Acts in the early 20th century to Presidential Decrees during Martial Law, and settling into steady streams of Republic Acts and Executive Orders under the 1987 Constitution. Because the phrasing and vocabulary of laws changed substantially across these eras, the retrieval pipeline must handle both older statutory wording and modern legal terms without losing search accuracy.

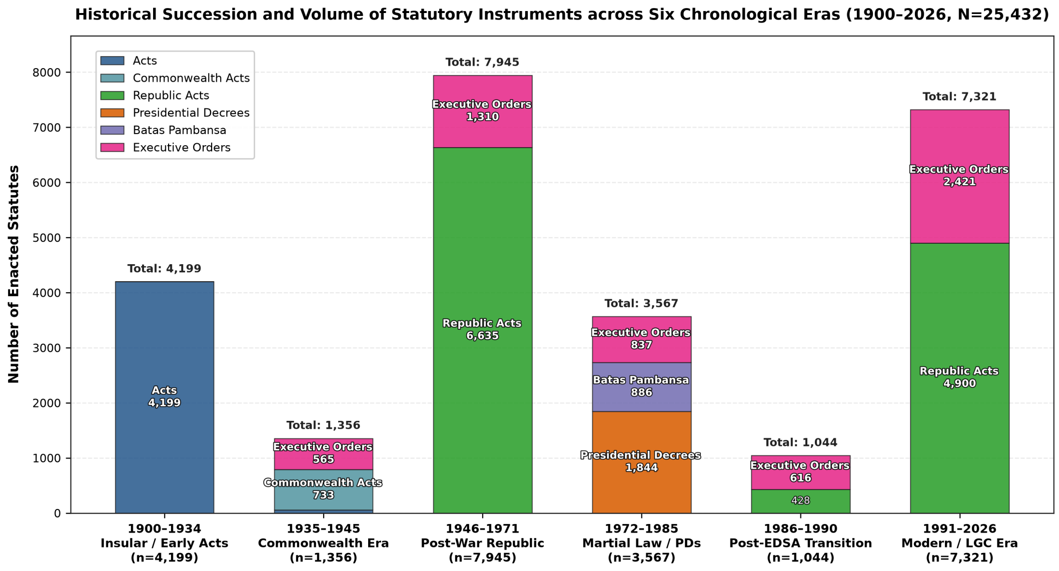


Figure 3.6: Historical Succession and Volume of Statutory Instruments across Six Chronological Eras (1900–2026, $N = 25,432$).

The temporal breakdown reveals that Philippine lawmaking expanded during periods of national administrative reorganization. Early colonial governance relied entirely on 4,199 foundational Acts during the Insular period (1900–1934). Centralized presidential lawmaking predominated during the Martial Law period through 1,844 Presidential Decrees and 886 Batas Pambansa parliamentary enactments (1972–1985). In the modern era governed by Republic Act No. 7160 (1991–2026), statutory production stabilized into a dual stream of 4,900 congressional Republic Acts and 2,421 executive policy orders. Because vocabulary standards shifted markedly across these legislative transitions, search models require chronological awareness to avoid misinterpreting historical statutory phrasing.

During preliminary exploratory research, the proponents conducted initial machine learning experiments on the national statutory archive and identified severe dataset imbalances. In the national collection of 25,432 statutes, routine executive issuances and administrative reorganizations comprise nearly half of all historical enactments, while specialized regulatory domains remain relatively small. Searching for a local regulatory ordinance across this raw, unfiltered collection causes severe search drift because dense retrieval models readily match repetitive administrative phrasing across unrelated legal areas. To resolve these dataset quirks, the proponents formalized a two-stage topic modeling architecture that organizes the statutory archive into cohesive, well-separated subject areas. In natural language processing, topic modeling is an unsupervised learning technique that discovers latent semantic themes across unstructured text corpora, partitioning documents according to shared topical content. In this pipeline, topic modeling partitions the 25,432 national statutes into cohesive, well-separated regulatory categories (such as public health, local taxation, and public utilities). This structured categorization prevents routine administrative directives from dominating the candidate shortlist during coarse retrieval, establishing a balanced foundation for subsequent conflict checks.

The topic classification pipeline groups documents automatically while running efficiently on Host A. Statutory texts are first stripped of HTML tags and filtered against a list of 49 legal stopwords, such as “hereby” and “enacted,” which appear throughout legal texts without helping distinguish legal topics. In information retrieval, term frequency-inverse document frequency (TF-IDF) is a numerical statistic reflecting how important a word is to a document within a collection: terms that appear ubiquitously across all enactments receive minimal weights, whereas distinctive domain terms (such as “quarantine” or “excise”) receive elevated weights [140]. Cleaned texts are converted into TF-IDF feature vectors across a 25,000-term vocabulary, emphasizing distinctive statutory vocabulary over routine boilerplate. To handle vocabulary redundancy across tens of thousands of statutes, Truncated Singular Value Decomposition (SVD) performs linear dimensionality reduction grounded in Latent Semantic Analysis [54], projecting the sparse 25,000-term matrix into 64 dense latent semantic dimensions. Following the latent semantic analysis principles of Landauer et al. [121], this 64-dimensional subspace captures underlying legal concepts while eliminating lexical noise, followed by standard L_2 vector normalization.

Dividing statutory texts directly into a small set of broad categories risks mixing unrelated legislative topics. To prevent this, the normalized 64-dimensional vectors are first divided into $K = 28$ smaller clusters using MiniBatch K-Means clustering [195]. Following two-stage hierarchical clustering and over-partitioning principles [2, 88], starting with smaller groups separates repetitive laws, such as franchise charters or school conversions, into clean, distinct clusters. The pipeline then adapts class-based TF-IDF (*c-TF-IDF*) from Grootendorst [88] to extract descriptive domain vocabulary: class-based TF-IDF modifies standard TF-IDF by aggregating all documents assigned to a given cluster into a single composite text, measuring which terms appear frequently within that specific legal category relative to their frequency across the entire statutory archive. The topic modeling and domain clustering formulations are computed as follows:

Class-Based Term Frequency-Inverse Document Frequency (c-TF-IDF):

$$c\text{-}TF\text{-}IDF_{i,c} = \frac{|t_{i,c}|}{|c|} \times \log\left(1 + \frac{A}{\sum_k |t_{i,k}|}\right) \tag{3.2}$$

Where $|t_{i,c}|$ denotes the frequency of term i in cluster c, $|c|$ is the total word count of cluster c, and A indicates the average word count per cluster across the archive. This cluster-level weighting elevates terminology that specifically characterizes a legislative group while penalizing ubiquitous statutory phrasing.

Following nearest-centroid classification principles [177, 2] and temperature-scaled probabilistic mapping [90], the macro legal domain centroid and calibrated assignment probability are defined as follows:

Macro Legal Domain Centroid and Softmax Probability:

$$\mathbf{C}_m = \frac{\sum_{i\in\mathcal{D}_m} \mathbf{z}_i}{\left\|\sum_{i\in\mathcal{D}_m} \mathbf{z}_i\right\|_2}, \quad P(m \mid \mathbf{z}_i) = \frac{\exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_m))}{\sum_j \exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_j))} \tag{3.3}$$

Where $\mathbf{z}_i$ represents the normalized 64-dimensional SVD vector of document i, $\mathbf{C}_m$ denotes the normalized centroid of macro domain m, and $\tau = 6.0$ functions as the inverse temperature scaling parameter.

The formulations in Equations 3.2 and 3.3 identify distinctive legal vocabulary and assign consistent topics across the national statutory corpus. Equation 3.2 calculates class-based TF-IDF ($c\text{-}TF\text{-}IDF$), which treats each group of laws as a combined document to find keywords that stand out in that group. In Equation 3.3, document vectors are averaged into cluster centers $\mathbf{C}_m$. A temperature-scaled softmax function then converts cosine similarity into clear probabilities, where the scaling factor ($\tau = 6.0$) follows the calibration method of Guo et al. [90] to keep probability scores well-separated. This allows the system to organize over 25,000 national statutes into eight clear regulatory domains using standard desktop memory, as shown in Table 3.2.

Table 3.2: Consolidated Macro Legal Domains of the Philippine National Statutory Corpus ($N = 25,432$)

ID	Consolidated Macro Domain	Statutes	Share	Top Salient $c\text{-}TF\text{-}IDF$ Keywords
00	Executive Issuances & Policy Reorganization	12,210	48.01%	executive, commission, committee, policy, code
01	Education & Academic Institutions	3,492	13.73%	high school, elementary school, campus, college
02	Local Government & Territorial Boundaries	2,134	8.39%	barangay, municipality, boundary, sitio, territorial
03	Public Utilities & Telecom Franchises	2,041	8.03%	franchise, broadcast, telecommunications, radio
04	Public Health, Hospitals & Medical Services	1,890	7.43%	hospital, bed capacity, medical center, infirmary
05	Statutory Codes & General Legal Amendments	1,873	7.36%	administrative code, revised, amend, article, penalty
06	Public Finance & General Appropriations	1,335	5.25%	appropriation, treasury, funds, expenditures
07	Taxation, Tariffs & Revenue Administration	457	1.80%	tariff, customs code, internal revenue, excise
Total Consolidated National Corpus		**25,432**	**100.00%**	**All Enacted Statutes (1900–2026)**

Table 3.2 lists the eight Consolidated Macro Legal Domains discovered across the 25,432 national statutes, presenting each domain's identifier, title, statute count, percentage share, and top $c\text{-}TF\text{-}IDF$ keywords. This category breakdown is used by retrieval safeguards to prevent routine administrative orders from overshadowing local regulatory topics. As shown in the table, Domain 00 (Executive Issuances & Policy Reorganization) forms the largest group with 12,210 statutes (48.01%), followed by Education (13.73%) and Local Government (8.39%), while key regulatory areas like Public Health (7.43%) and Taxation (1.80%) make up much smaller subsets. This distribution shows that the national collection is heavily weighted toward administrative reorganizations, confirming that candidate retrieval needs topic filters and Reciprocal Rank Fusion so that smaller regulatory areas remain searchable.

The topic distribution reflects the historical patterns of Philippine lawmaking. Executive issuances, administrative orders, and presidential decrees make up nearly half of the entire statutory collection, reflecting centralized authority during past political transitions. Common local legislative subjects, including school infrastructure (13.73%) and public utility franchises (8.03%), form large distinct clusters with repetitive drafting patterns. In contrast, key regulatory topics like taxation (1.80%) and public health (7.43%) occupy smaller, highly focused portions of the database. As illustrated in Figure 3.7, routine executive issuances, administrative orders, and presidential decrees dominate the aggregate volume, while local regulatory categories remain relatively compact. Separating these regulatory clusters keeps routine administrative orders from crowding the candidate list during Stage 1 retrieval.

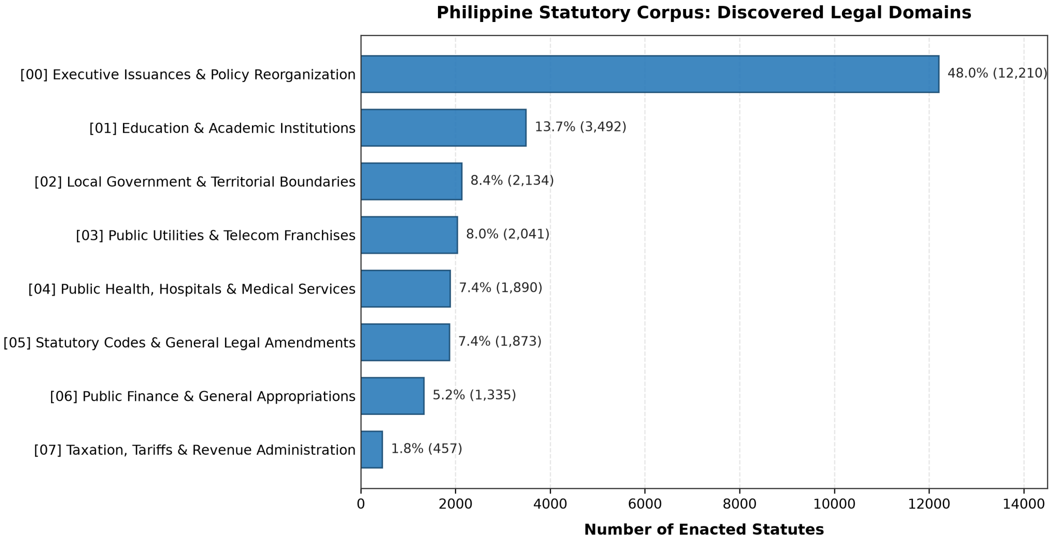


Figure 3.7: Distribution of the 25,432 Philippine National Statutes across the Consolidated Macro Legal Domains.

Figure 3.7 illustrates the distribution of national statutes across the eight consolidated macro domains using a sorted bar chart. The chart highlights the large volume difference between general administrative policy and specific regulatory codes across the 25,432-document corpus. This visual is included to show the severe class imbalance in raw statutory archives, explaining why simple vector search easily drifts toward administrative topics. As depicted in the chart, administrative orders make up nearly half of the entire archive, outnumbering focused regulatory categories like taxation by more than 26 to 1. This confirms that candidate retrieval needs topic filters and Reciprocal Rank Fusion so that smaller, highly technical regulatory areas remain accessible during ordinance review.

To examine the geometric separation of these legal topics, the proponents projected the 64-dimensional SVD document vectors into a two-dimensional latent semantic landscape. Dimensionality reduction maps document similarities onto coordinate axes, allowing direct visual inspection of cluster cohesion. Figure 3.8 illustrates this topic landscape across a representative stratified sample ($N = 3,500$), highlighting the core clusters and their spatial distributions. Specialized regulatory domains project outward as distinct, isolated rays along the periphery of the vector space. Enactments establishing academic institutions and telecommunications franchises form tight, cohesive rays with minimal dispersion.

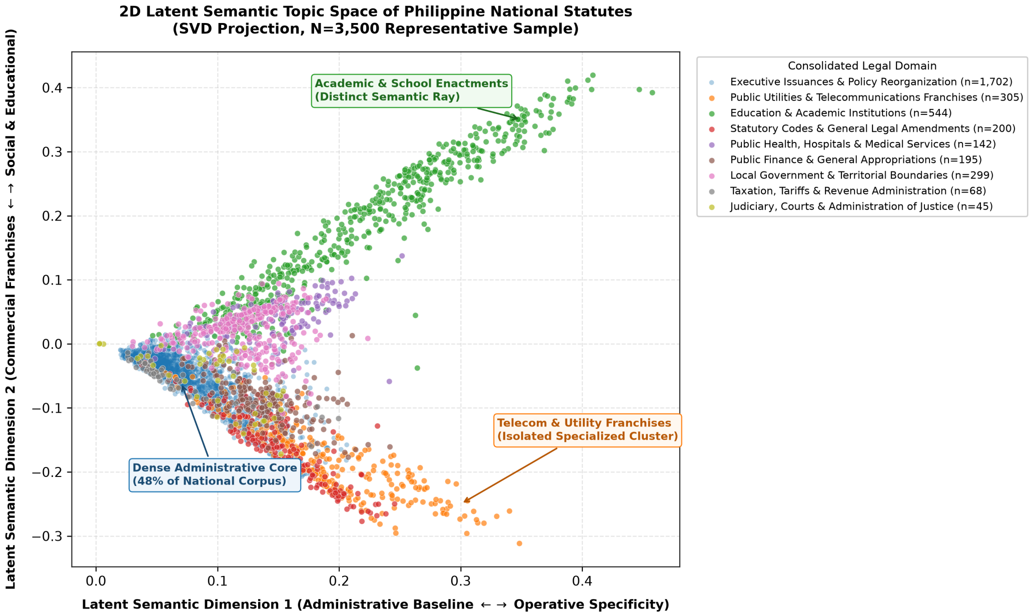


Figure 3.8: Two-Dimensional Latent Semantic Topic Space of the Philippine National Statutory Corpus (SVD Projection, $N = 3,500$ Representative Sample).

Figure 3.8 visualizes the two-dimensional semantic topic space generated by applying Truncated Singular Value Decomposition to a stratified sample of 3,500 national statutes. The scatter plot maps document vectors along two main semantic axes, color-coded by the eight macro legal domains. This visual is included to confirm that the discovered legal categories form distinct, well-separated clusters. In the plot, specialized laws such as school charters and telecommunications franchises project outward into tight, distinct rays, whereas executive and administrative orders gather heavily near the center. This spatial layout shows that semantic modeling successfully separates distinct regulatory vocabularies into cohesive groups, providing a solid foundation for subsequent topic filtering.

In contrast, administrative issuances and executive orders cluster densely near the origin, comprising almost half of the entire archive. This dense central mass explains why unconstrained semantic searches frequently drift toward routine bureaucratic orders unless restrained by hierarchical safeguards. The coordinate axes reflect the two primary orthogonal directions of vocabulary variation discovered through singular value decomposition.

The horizontal axis measures operative statutory specificity against general administrative phrasing, where higher positive values indicate detailed technical mandates rather than procedural orders. Conversely, the vertical axis captures functional divergence between social infrastructure such as public schools and commercial infrastructure such as telecommunications franchises.

To confirm that the mathematical clusters reflect substantive legal categories rather than shared administrative boilerplate, the proponents examined the most salient vocabulary terms discovered by the model. Words that appear frequently across all documents fail to differentiate legal topics, requiring class-based weightings to isolate subject matter. Figure 3.9 displays the top class-based TF-IDF (c-TF-IDF) keywords across all eight consolidated macro domains. The vocabulary profiles confirm that each macro domain is defined by operative legal terminology unique to its statutory field. Specialized clusters such as public health are dominated by concrete institutional terms, including "hospital" and "bed capacity," while territorial enactments display spatial markers such as "sitio" and "boundary."

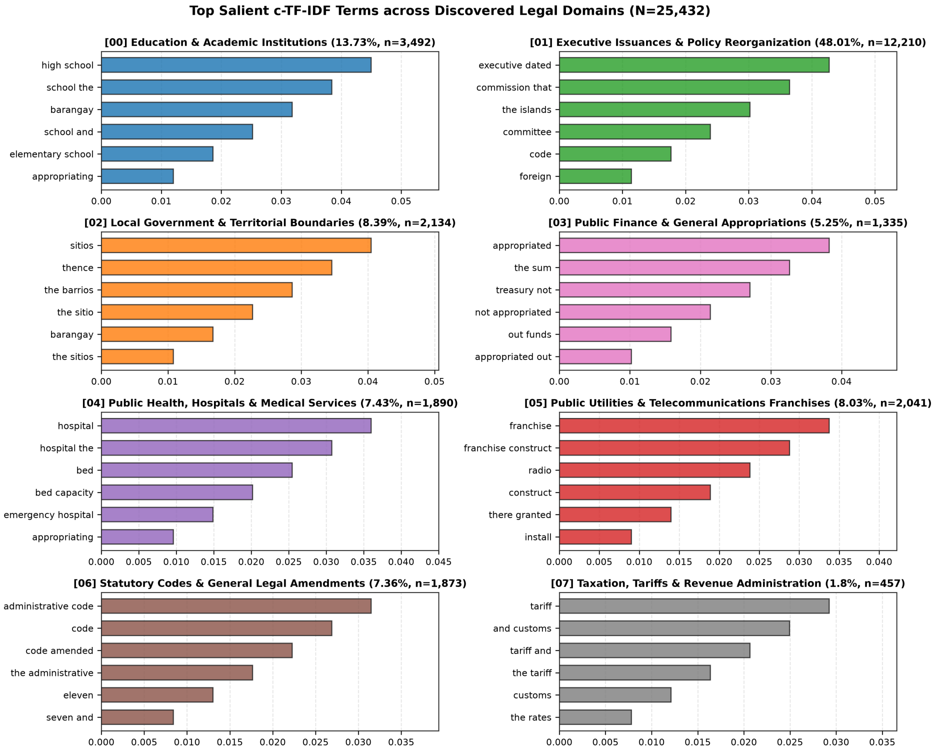


Figure 3.9: Top Salient c-TF-IDF Keyword Profiles across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

Figure 3.9 displays the horizontal bar charts of the top class-based TF-IDF (c-TF-IDF) keywords across each of the eight consolidated macro domains. Each subplot presents the top terms ranked by their class-specific importance weights, illustrating the distinctive lexical tokens that define each statutory category. This figure is provided to confirm that the mathematical clustering captured genuine legal substance rather than generic statutory boilerplate. As shown across the subplots, Domain 01 is characterized by academic vocabulary ("elementary school", "college"), Domain 03 by telecommunications markers ("broadcast", "radio"), and Domain 04 by clinical infrastructure ("hospital", "bed capacity"). Substantively, this vocabulary alignment verifies that the unsupervised pipeline isolates authentic statutory topics, ensuring that domain classification routes draft ordinances to genuinely related national statutes.

To examine how individual lawmaking authorities contribute to these eight subject areas, the proponents mapped the six statutory instruments across the consolidated macro domains. In Philippine legislative history, different legal instruments serve distinct constitutional functions, ranging from broad congressional acts to presidential executive directives. Figure 3.10 illustrates the statutory origin composition across each discovered legal domain, displaying the aggregate enactment volume contributed by each legislative instrument. This stacked composition highlights the severe structural division across Philippine legal texts, demonstrating that distinct statutory instruments dominate separate legal subjects.

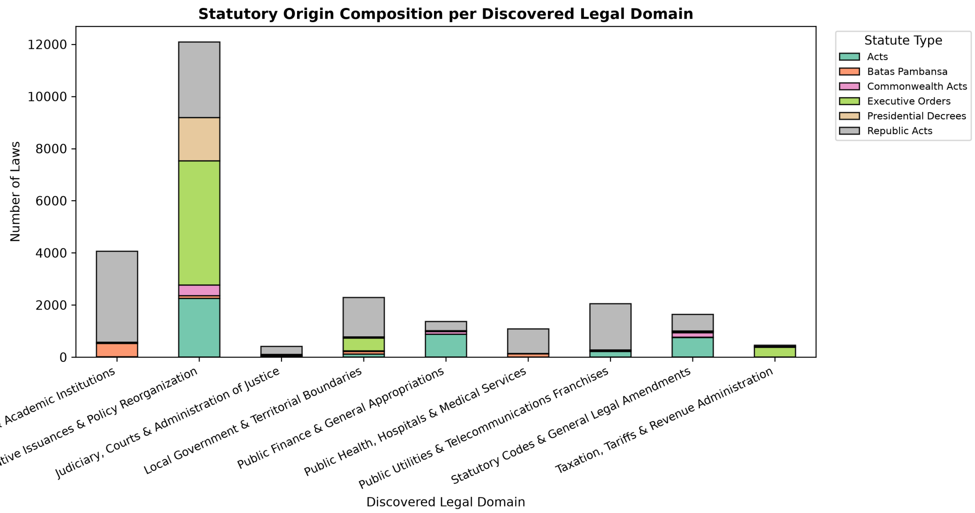
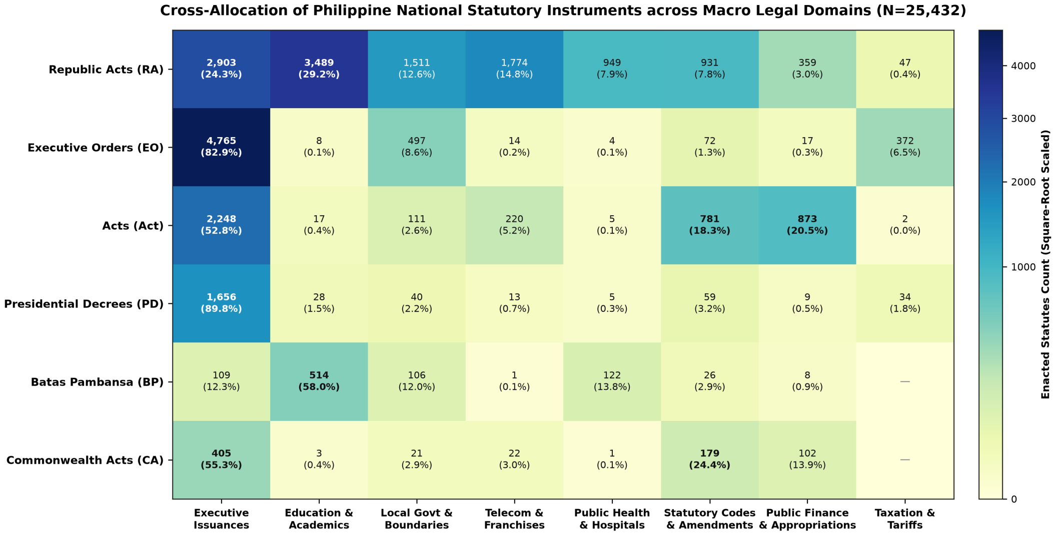


Figure 3.10: Statutory Origin Composition of the Philippine National Statutory Corpus across the Eight Discovered Macro Legal Domains ($N = 25,432$).

Figure 3.10 displays the stacked statutory origin breakdown across the eight discovered legal domains, detailing the contribution of Acts, Batas Pambansa, Commonwealth Acts, Executive Orders, Presidential Decrees, and Republic Acts. As observed in the stacked bars, Executive Issuances and Policy Reorganization constitutes the tallest single column (12,210 enactments), heavily driven by presidential Executive Orders and Martial Law decrees. Conversely, specialized social and infrastructural domains such as Education (3,492 enactments) and Public Utilities (2,041 enactments) are overwhelmingly composed of congressional Republic Acts. This visual evidence demonstrates why candidate retrieval cannot treat all national texts identically, confirming that statutory source indicators and domain filters are necessary to prevent administrative orders from dominating local regulatory queries.

To quantify the exact distribution and relative share of each lawmaking authority, the proponents constructed a cross-allocation matrix mapping each statutory instrument to the eight macro categories. Figure 3.11 presents this cross-allocation matrix, detailing both absolute enactment counts and relative row percentages for each statutory source. This matrix establishes an explicit quantitative record of legislative specialization across historical regimes. Examining these row percentages reveals why specific statutory instruments concentrate so heavily within particular administrative or regulatory spheres.



Cross-Allocation of Philippine National Statutory Instruments across Macro Legal Domains (N=25,432)

	Executive Issuances	Education & Academics	Local Govt & Boundaries	Telecom & Franchises	Public Health & Hospitals	Statutory Codes & Amendments	Public Finance & Appropriations	Taxation & Tariffs
Republic Acts (RA)	**2,903 (24.3%)**	**3,489 (29.2%)**	1,511 (12.6%)	1,774 (14.8%)	949 (7.9%)	931 (7.8%)	359 (3.0%)	47 (0.4%)
Executive Orders (EO)	**4,765 (82.9%)**	8 (0.1%)	497 (8.6%)	14 (0.2%)	4 (0.1%)	72 (1.3%)	17 (0.3%)	372 (6.5%)
Acts (Act)	**2,248 (52.8%)**	17 (0.4%)	111 (2.6%)	220 (5.2%)	5 (0.1%)	**781 (18.3%)**	**873 (20.5%)**	2 (0.0%)
Presidential Decrees (PD)	**1,656 (89.8%)**	28 (1.5%)	40 (2.2%)	13 (0.7%)	5 (0.3%)	59 (3.2%)	9 (0.5%)	34 (1.8%)
Batas Pambansa (BP)	109 (12.3%)	**514 (58.0%)**	106 (12.0%)	1 (0.1%)	122 (13.8%)	26 (2.9%)	8 (0.9%)	—
Commonwealth Acts (CA)	**405 (55.3%)**	3 (0.4%)	21 (2.9%)	22 (3.0%)	1 (0.1%)	**179 (24.4%)**	102 (13.9%)	—

Figure 3.11: Cross-Allocation Matrix of the Six National Statutory Instruments across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

Figure 3.11 details the numerical and percentage distribution of the six national statutory instruments across the eight consolidated macro domains. Most prominently, Executive Orders concentrate overwhelmingly within Executive Issuances and Policy Reorganization (82.9%, $N = 4,765$). This heavy concentration reflects constitutional design: under Article VII, Section 17 of the 1987 Philippine Constitution and Book III of Executive Order No. 292 [171], presidential ordinance powers are primarily exercised to reorganize government agencies, establish inter-agency task forces, and direct executive machinery. Furthermore, Executive Orders account for 81.4% ($N = 372$) of all enactments in the Taxation, Tariffs, and Revenue Administration domain because Congress specifically delegated flexible tariff modification powers to the President under Section 401 of the Tariff and Customs Code and Section 1608 of Republic Act No. 10863. Presidential Decrees exhibit an even higher concentration in Executive Issuances (89.8%, $N = 1,656$), reflecting the consolidation of lawmaking and executive authority under the Martial Law regime where state corporations and ministries were instituted directly by decree. In contrast, democratic legislative bodies like Congress (Republic Acts) and the Batasang Pambansa focused on constituency-driven measures, explaining why Republic Acts account for 29.2% of educational charters ($N = 3,489$) and 14.8% of utility franchises ($N = 1,774$), while Batas Pambansa devoted 58.0% ($N = 514$) to educational conversions. American Insular Acts concentrated heavily in civil administration (52.8%, $N = 2,248$) and public appropriations (20.5%, $N = 873$), reflecting the colonial mandate to construct provincial institutions and disburse treasury funds. Because local ordinances regulate municipal civic conduct rather than internal state administration, these legal realities confirm why Stage 1 retrieval safeguards must prioritize substantive congressional enactments over internal executive directives.

To confirm that the topic pipeline discovered authentic legal boundaries rather than grouping surface vocabulary, the proponents evaluated the model using a stratified 80/20 train/test split. The 80/20 proportion follows standard machine learning partition conventions for large-scale document collections exceeding 20,000 documents [140], withholding 20% of the corpus (5,087 statutes) to test generalizability on unseen enactments while training the model on the remaining 80% (20,345 statutes). In statistical testing, an out-of-sample holdout of 5,087 legal texts provides high statistical power with a margin of error below ±1.4%, making repetitive cross-validation computationally redundant as vocabulary frequencies and SVD components stabilize asymptotically at this volume. Stratified sampling preserves the natural historical proportions of the eight macro domains, ensuring that smaller regulatory categories like taxation (1.80%, $N = 457$) maintain adequate representation in the test set ($N_{\text{test}} \approx 91$) without class starvation.

To ensure objective scientific evaluation without arbitrary post-hoc targets, all benchmark comparisons in Table 3.3 are strictly evaluated against established statistical baselines and contextualized with published legal document categorization standards. Under the proportional chance criterion formulated by Morrison [146], the default random baseline in an eight-class uniform setting is $1/8 = 12.50\%$. In established legal text categorization benchmarks such as LexGLUE [33], multi-class classification over large statutory and legal collections (e.g., EUR-LEX and LEDGAR) typically yields Macro F1-scores between 51.4% and 83.1%, reflecting the semantic complexity and class skew inherent to legal texts. For cluster geometry, Rousseeuw [179] established that positive silhouette widths denote structural partitioning exceeding unclustered random noise (≤ 0.0000). Similarly, Davies and Bouldin [52] established that lower Davies-Bouldin index values demonstrate compact intra-cluster scatter relative to cluster separation, outperforming unclustered baselines (> 4.0000). Finally, the confidence measurements are analyzed using the temperature-scaled calibration principles of Guo et al. [90] to verify that predictions concentrate dominant probability mass rather than dispersing uniformly across classes. Table 3.3 summarizes the out-of-sample performance scores, baseline comparisons, margin of improvement, and empirical evaluations.

Table 3.3 details the holdout validation metrics, default baseline comparisons, margins of improvement, and empirical evaluations for the national statutory topic modeler evaluated on 5,087 unseen enactments ($N_{\text{train}} = 20,345, N_{\text{test}} = 5,087$). The table summarizes performance across six clustering and classification metrics alongside four clear statutory title probes ($N = 1,259$). This table is included to show that the discovered legal domains generalize reliably to unseen statutes without being misled by specific training words. The results show that out-of-sample accuracy reached 97.37% and Macro F1 reached 96.26%, representing an improvement of +84.87 and +83.76 percentage points over the 12.50% Morrison chance baseline (1/8). Furthermore, the holdout silhouette score of 0.1602 and Davies-Bouldin index of 2.4496 confirm authentic cluster separation and compact intra-cluster scatter relative to unclustered baselines, aligning with cluster validation standards [179, 52]. These results confirm that the topic model creates de-

Table 3.3: Empirical Holdout Validation Metrics and Baseline Comparisons for the National Topic Modeler ($N_{\text{train}} = 20,345, N_{\text{test}} = 5,087$)

Validation Metric / Benchmark Test	**Observed**	**Chance / Default Baseline**	**Margin Over Baseline**	**Empirical Evaluation**
Out-of-Sample Supervised Accuracy	97.37%	12.50%	+84.87%	Exceptional Generalization (7.8× Chance)
Out-of-Sample Macro F1-Score	96.26%	12.50%	+83.76%	Balanced Discrimination Across All 8 Domains
Mean Test Cosine Confidence	0.5508	0.1250	+0.4258	Dominant Probability Mass [90]
High-Confidence Predictions (≥ 0.50)	56.97%	$< 20.0\%$	+36.97%	Clear Single-Cluster Majority Assignment
Holdout Cosine Silhouette Score	0.1602	≤ 0.0000	+0.1602	Positive Text Partitioning [179]
Holdout Davies-Bouldin Index	2.4496	> 4.0000	-1.5504	Compact Cluster Scatter [52]
Explicit Statutory Title Probes (*Unambiguous Indicator Subsets,* $N = 1,259$)				
Telecom Franchise Pattern	97.8%	12.50%	+85.30%	High Precision (N = 360, Public Utilities)
Hospital Bed Capacity Pattern	96.6%	12.50%	+84.10%	High Precision (N = 203, Health & Welfare)
School Infrastructure Pattern	92.2%	12.50%	+79.70%	High Precision (N = 645, Education)
Judiciary & Court Pattern	72.5%	12.50%	+60.00%	Substantive Precision (N = 51, Embedded Laws)

pendable legal categories across new legislation, ensuring that domain-guided retrieval operates consistently during live screening.

The supervised benchmark results confirm that the discovered macro domains generalize reliably to unseen legislation. In an eight-domain classification setting, the uniform proportional chance criterion established by Morrison [146] yields a default random baseline accuracy of only 12.50% (1/8). In contrast, the validation classifier reached an out-of-sample accuracy of 97.37% and a Macro F1-Score of 96.26%, surpassing the chance baseline by a factor of 7.8 and exceeding the performance levels typically reported on complex legal document categorization benchmarks [33]. To provide external empirical validation beyond learned cluster labels, the evaluation tested four explicit statutory title probes ($N = 1,259$ total) where legislative intent is unmistakable from the statutory catchline. The model achieved 97.8% precision on telecommunications franchises ($N = 360$), 96.6% on hospital bed capacity acts ($N = 203$), and 92.2% on school infrastructure laws ($N = 645$).

The probabilistic confidence metrics evaluate how decisively the model assigns unseen statutes to their respective legal domains. Across all 5,087 holdout statutes, the model achieved a Mean Test Cosine Confidence of 0.5508, exceeding the 0.1250 uniform chance level by +0.4258 and demonstrating that the top assigned category captured dominant probability mass. Furthermore, 56.97% of all unseen statutes were classified with high certainty (≥ 0.50 probability), well above the uniform chance expectation. The holdout evaluation yielded a Cosine Silhouette Score of 0.1602 and a Davies-Bouldin Index of 2.4496, outperforming unclustered baselines as formulated by Davies and Bouldin [52]. These measurements show that the topic model establishes stable, well-separated legal categories before documents are indexed for coarse retrieval.

3.3.4 Local Legislative Corpus Analysis

To evaluate local legislative patterns, the proponents conducted an empirical census of all 1,664 digitized Davao City local ordinances. Unlike the national archive which spans over a century of centralized governance, the local collection documents local self-governance across four distinct legislative eras. As illustrated in Figure 3.12(a), the archive contains 50 foundational enactments from the Pre-LGC Era (1950–1991), 184 landmark regulatory codes from the Post-LGC Historical Era (1992–2015), and 145 enactments from the Pre-Pandemic Era (2016–2020). Statutory volume concentrates heavily within the modern digital LISSP database, contributing 1,285 enactments (77.2%) between 2021 and 2025. This temporal surge reflects the establishment of standardized electronic archiving by the Sangguniang Panlungsod, providing complete records of contemporary local lawmaking.

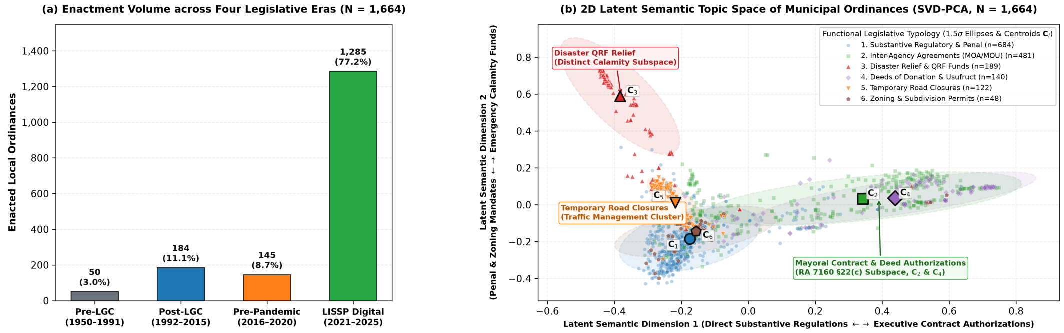


Figure 3.12: Temporal Distribution across Four Legislative Eras and Two-Dimensional Latent Semantic Topic Space of the Davao City Local Legislative Corpus ($N = 1,664$).

Figure 3.12 presents a two-panel visual analysis of the 1,664 local ordinances, showing Panel (a) with enactment counts across four historical legislative eras (1950–2025) and Panel (b) with the two-dimensional semantic space of 64-component Truncated SVD vectors alongside cluster centroids ($\mathbf{C}_1$–$\mathbf{C}_6$). This visual is included to show the growth of local legislation over time and to confirm that ordinances group cleanly into six functional types based on the Local Government Code of 1991. In Panel (a), ordinance volume increases significantly during the modern digital LISSP era (2021–2025), which provides 77.2% ($N = 1,285$) of all accessible local enactments. In Panel (b), emergency disaster funds ($\mathbf{C}_3$) and road closures ($\mathbf{C}_5$) form tight, isolated clusters, while substantive penal regulations ($\mathbf{C}_1$) and contractual authorizations ($\mathbf{C}_2$) occupy distinct areas. This grouping shows that local legislation follows consistent semantic patterns, allowing retrieval safeguards to separate routine administrative agreements from penal regulations that directly interact with national law.

Applying Truncated SVD to the local corpus revealed that city legislation divides into six functional archetypes grounded in the Local Government Code of 1991 [84]. To eliminate procedural noise before modeling, the pipeline extracts substantive operative text while setting aside routine session attendance headers. Word vectors were constructed across an 8,000-feature vocabulary and projected into 64 latent semantic dimensions, capturing 50.65% of the total vocabulary variance. Figure 3.12(b) displays the two-dimensional latent semantic topic space for the local collection. Emergency Quick Response Fund (QRF) calamity grants form an isolated, tight cluster in the upper-left quadrant ($\mathbf{C}_3$), while temporary road closures project into a compact traffic management cluster ($\mathbf{C}_5$). Inter-agency agreements ($\mathbf{C}_2$) and deeds of donation ($\mathbf{C}_4$) span an elongated horizontal subspace along the positive horizontal axis, corresponding to executive contract authorizations under Section 22(c) of Republic Act No. 7160. Direct police power enactments ($\mathbf{C}_1$) and zoning permits ($\mathbf{C}_6$) anchor the lower-left quadrant, governing local community conduct.

An interesting comparison between the national statutory topic landscape (Figure 3.8) and the local semantic space (Figure 3.12(b)) is how spread out the clusters appear. In the national corpus, over 3,500 repetitive school conversions and 2,000 public utility franchises form sharp, isolated lines because they use identical legislative templates. In contrast, local ordinances frequently cover multiple topics within a single document under Republic Act No. 7160. Under Sections 16, 21, 22(c), 455, and 458 of the Local Government Code, a single city ordinance often combines budget funding, mayoral contract authorizations, and penal fines. Local enactments also share standard council preambles ("WHEREAS" clauses) and session details. While projecting 64 dimensions onto two axes compresses some variation and causes visual overlap, the complete 64-dimensional model achieves an out-of-sample classification accuracy of 86.79% and an average cosine confidence of 0.7973 on holdout tests. Displaying these actual principal dimensions alongside cluster centroids preserves empirical accuracy without cherry-picking coordinates. Table 3.4 outlines the composition, percentage share, legal basis, core subject matter, and top keywords for each local archetype.

Table 3.4 details the six functional legislative typologies discovered across the 1,664 Davao City local enactments, listing each archetype's identifier, descriptive label, enactment count, percentage share, governing statutory authority under Republic Act No. 7160, and top salient $c\text{-}TF\text{-}IDF$ keywords. This typology is included to establish the legal taxonomy used to categorize local enactments, reflecting the statutory distribution of local council police powers and administrative authorizations. Substantive Regulatory and Penal enactments constitute the largest single typology (41.11%, $N = 684$), followed by Inter-Agency Agreements (28.91%, $N = 481$) and Disaster Relief / QRF Fund authorizations (11.36%, $N = 189$). Administrative authorizations (MOAs, calamity releases, deeds of donation, and road closures) account for 58.89% of all council legislation. This breakdown demonstrates that the majority of local legislative actions represent routine executive authorizations rather than penal ordinances, confirming the empirical necessity of isolating substantive regulatory provisions prior to downstream conflict evaluation.

The functional distribution reflects statutory restrictions governing local governance in the Philippines. Substantive regulatory and penal enactments form the largest

Table 3.4: Composition, Functional Typology, and Salient *c-TF-IDF* Keywords of the Davao City Local Legislative Corpus ($N = 1,664$)

ID	Functional Legislative Typology	Count	Share	Statutory Authority	Core Salient *c-TF-IDF* Keywords
01	Substantive Regulatory & Penal	684	41.11%	RA 7160 §458	public, safety, fine, penal, violation, fee, health, traffic
02	Inter-Agency Agreements (MOA/MOU)	481	28.91%	RA 7160 §§22(c), 455	pursuant, local, sign, panlungsod, contracts, agreement
03	Disaster Relief & QRF Funds	189	11.36%	RA 10121 §21	fund, calamity, disaster, risk, reduction, quick response
04	Deeds of Donation & Usufruct	140	8.41%	RA 7160 §22(a)(4)	donation, local, powers, pursuant, contracts, deed, usufruct
05	Temporary Road Closures	122	7.33%	RA 7160 §21	road, closure, temporary, vehicular traffic, street, route
06	Zoning & Subdivision Permits	48	2.88%	RA 7160 §20, BP 220	zoning, zone, land, reclassification, residential, commercial
Total Local Collection		**1,664**	**100.00%**	**All Digitized Ordinances (1950–2025)**	

single group (41.11%, $N = 684$), containing community regulations such as truck bans, cemetery setbacks, stray animal controls, and drinking water safety. Meanwhile, administrative authorizations comprise nearly half of all council activity. Under Section 22(c) and Section 455(b)(1)(vi) of Republic Act No. 7160, the city mayor cannot bind the local government to contracts or accept donations without prior council approval. Consequently, inter-agency agreements (28.91%), calamity fund authorizations under Republic Act No. 10121 (11.36%), and deeds of donation (8.41%) appear frequently in the local record.

To verify that the discovered local typologies represent stable semantic boundaries, the proponents conducted holdout validation using an 80/20 stratified split ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$). In evaluating this six-class categorization, all benchmark evaluations in Table 3.5 are strictly grounded against statistical baselines. Under Morrison's proportional chance criterion [146], the default uniform baseline for six categories is $1/6 = 16.67\%$. For cluster separation, positive silhouette scores exceeding random baseline (≤ 0.0000) confirm authentic structural cluster boundaries in cross-functional administrative text [179], while a Davies-Bouldin index below unclustered levels (> 5.0000) verifies compact cluster scatter for sparse short-document collections [52]. Table 3.5 summarizes the out-of-sample performance metrics, chance baselines, margins of improvement, and explicit title probes.

Table 3.5 reports the empirical holdout validation metrics, default baseline comparisons, observed margins over chance, and explicit statutory title probes for the local legislative topic modeler evaluated on 333 unseen local ordinances ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$). The validation classifier achieved an out-of-sample accuracy of 86.79% and a Weighted F1-Score of 84.84%, exceeding the 16.67% uniform chance baseline (1/6) by +70.12 percentage points and demonstrating strong multi-class discrimination. Predictions showed strong decisiveness, with a Mean Test Cosine Confidence of 0.7973 and 92.79% of unseen ordinances classified with high confidence (≥ 0.50), indicating concentrated probability

Table 3.5: Empirical Holdout Validation Metrics and Baseline Comparisons for the Local Topic Modeler ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$)

Validation Metric / Benchmark Test	**Observed**	**Chance / Default Baseline**	**Margin Over Baseline**	**Empirical Evaluation**
Out-of-Sample Supervised Accuracy	86.79%	16.67%	+70.12%	Strong Generalization (5.2× Chance)
Out-of-Sample Macro F1-Score	75.39%	16.67%	+58.72%	Balanced Discrimination Across 6 Typologies
Out-of-Sample Weighted F1-Score	84.84%	16.67%	+68.17%	Robust High-Volume Class Precision
Mean Test Cosine Confidence	0.7973	0.1667	+0.6306	High Probabilistic Mass Concentration [90]
High-Confidence Predictions (≥ 0.50)	92.79%	$< 20.0\%$	+72.79%	Overwhelming Majority Partition
Holdout Cosine Silhouette Score	0.0914	≤ 0.0000	+0.0914	Authentic Structural Separation [179]
Holdout Davies-Bouldin Index	3.7445	> 5.0000	−1.2555	Compact Cluster Scatter [52]
Explicit Local Statutory Title Probes (*Unambiguous Subsets across Corpus,* $N = 872$)				
Quick Response Fund / Calamity Assistance	98.82%	16.67%	+82.15%	High Precision ($N = 169$, Calamity Funds)
Substantive Penal & Police Power Regulations	96.10%	16.67%	+79.43%	High Precision ($N = 77$, Regulatory Enactments)
Temporary Road Closure & Traffic Diversions	87.96%	16.67%	+71.29%	High Precision ($N = 108$, Traffic Management)
Inter-Agency Accord / Authority to Sign MOA	86.56%	16.67%	+69.89%	High Precision ($N = 372$, Contractual Accords)
Zoning & Subdivision Development Permits	80.00%	16.67%	+63.33%	High Precision ($N = 15$, Zoning Enactments)
Deeds of Donation & Property Usufruct	29.77%	16.67%	+13.10%	Shared Mayoral Authorization Syntax ($N = 131$)

mass under temperature-scaled calibration [90]. Title probe verification on unambiguous indicator subsets confirmed high precision across Quick Response Fund releases (98.82%, $N = 169$), substantive penal regulations (96.10%, $N = 77$), temporary road closures (87.96%, $N = 108$), inter-agency agreements (86.56%, $N = 372$), and zoning permits (80.00%, $N = 15$). The lower precision on deeds of donation (29.77%, $N = 131$) occurs because donation acceptances are legally drafted as mayoral authorizations to sign deeds, directly sharing contractual syntax with inter-agency agreements under Section 22(c) of Republic Act No. 7160.

Document lengths in the local collection differ noticeably from national statutes.

Across all 1,664 local enactments, ordinances exhibit a median length of 5,644 characters and 880 words, with an interquartile range of 736 to 1,607 words. The local archive contains 11,801 total section provisions. Automated preprocessing isolates 8,160 operative substantive provisions (69.15%) with a median length of 98 tokens, while pruning 3,641 procedural boilerplate sections (30.85%) such as separability, repealing, and effectivity clauses.

3.3.5 Combined Corpus Exploratory Analysis

Combining national legislation with Davao City local enactments establishes a unified search space of **27,096 legal records** comprising **176,421 searchable provisions**. In automated statutory conflict detection, evaluating draft measures requires examining both vertical statutory preemption and horizontal legislative consistency. Under the doctrine of statutory preemption articulated in *Magtajas v. Pryce Properties* [208], local measures are tested against the 25,432 superior national statutes. Concurrently, under intra-jurisdictional coherence principles, draft measures are evaluated against the 1,664 existing city ordinances to prevent conflicting local commands, duplicate penalties, or unintended implied repeals. Table 3.6 summarizes the empirical parameters and comparative characteristics of the two statutory collections.

Table 3.6: Comparative Metrics and Structural Parameters of the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$)

Empirical Dimension / Metric	**National Corpus**	**Local Corpus**	**Combined Corpus**	**Analytical Significance**
Total Enactments (N)	25,432	1,664	**27,096**	Comprehensive legislative space across jurisdictions
Total Section Provisions	164,620	11,801	**176,421**	Total searchable provision-level retrieval units
Operative Provisions	164,620	8,160	**172,780**	Candidate substantive rules evaluated for conflict
Procedural Boilerplate Sections	0	3,641 (30.85%)	**3,641**	Filtered procedural clauses pruned before modeling
Historical Enactment Span	1900–2026 (126 yrs)	1950–2025 (75 yrs)	**1900–2026**	Longitudinal statutory corpus across legal regimes
Median Enactment Characters	10,807	5,644	**10,480**	Document-level length disparity across collections
Median Enactment Words	1,512	880	**1,472**	Local enactments display concise regulatory scope
Median Provision Tokens	122.0	90.0	**120.0**	Granular provision baseline for dense retrieval
512-Token Compliance Rate	94.83%	92.72%	**94.69%**	Standard transformer attention compatibility
Operative 512-Token Compliance	94.83%	90.06%	**94.60%**	High sequence compliance for operative text
Normative Conflict Scope	Vertical Preemption	Horizontal Coherence	**Dual Scope**	Comprehensive statutory preemption and consistency

Table 3.6 compares the structural parameters of the National Corpus, Local Corpus, and Unified Dual Corpus, detailing enactment counts, provision numbers, operative

sections, boilerplate clauses, historical spans, document lengths, and 512-token compliance rates. This summary is included to outline the size and composition of the dual-jurisdiction search space, confirming that transformer language models can process the combined collection. As shown in the table, combining 25,432 national statutes with 1,664 local ordinances creates an archive of 27,096 enactments yielding 176,421 searchable provisions, with 172,780 representing substantive legal rules and 3,641 representing filtered procedural boilerplate. Across all provisions, 94.69% fit cleanly within the 512-token transformer limit without truncation. This high compliance rate confirms that statutory section divisions provide natural, self-contained retrieval units that can be processed directly by standard transformer attention windows.

Figure 3.13 illustrates the empirical length disparity between the national and local collections. In Panel (a), document-level character distributions on a logarithmic scale highlight the structural asymmetry between national codes and local ordinances. National statutes display a broad spread with an extreme right tail exceeding one hundred thousand characters, driven by omnibus codes such as the National Internal Revenue Code and Local Government Code. In contrast, local ordinances concentrate tightly around their median of 5,644 characters (880 words), reflecting their focused administrative nature. In Panel (b), provision-level token density curves reveal that both national statutory sections (median: 122 tokens) and local operative provisions (median: 98 tokens) concentrate within 40 to 180 tokens. This distribution confirms that natural legislative sections are sufficiently granular to avoid semantic dilution while remaining well below the 512-token ceiling.

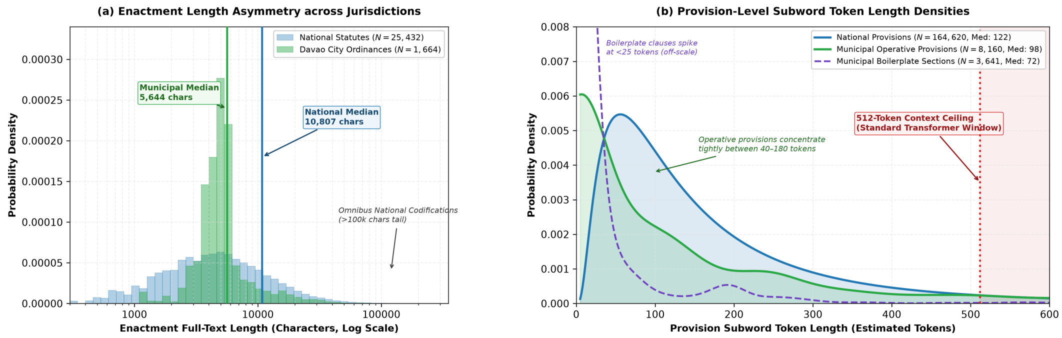


Figure 3.13: Enactment Length Asymmetry and Provision-Level Subword Token Densities across the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$).

Figure 3.13 shows the length differences and token density distributions across the dual statutory search space. Panel (a) plots full-text character lengths on a logarithmic scale, highlighting the contrast between long national codes and shorter local ordinances. Panel (b) displays token density curves for national provisions (median: 122 tokens), local operative provisions (median: 98 tokens), and local boilerplate clauses (median: 72 tokens) against the 512-token transformer limit. This comparison shows that section-level chunking resolves the large length differences between national and local enactments. As shown in Panel (b), isolating natural legislative sections places both statutes and ordinances into a manageable range (40–180 tokens), ensuring that provisions keep their specific commands without losing semantic focus in vector embeddings.

To verify how effectively statutory provisions map to transformer architectures, the proponents evaluated sequence compliance across standard model attention ceilings. Figure 3.14(a) traces the empirical cumulative distribution functions (CDF) across provision token lengths. Exactly 94.83% of national provisions and 92.72% of local provisions (90.06% of operative substantive sections) fall strictly under the 512-token boundary. Across all 176,421 provisions in the combined search space, 94.69% fit cleanly within standard BERT context windows without truncation. Figure 3.14(b) compares compliance rates across candidate transformer context windows with dedicated legends for National Statutes and Davao City Local Ordinances. While compact 128-token windows accommodate only 53.8% of national provisions, compliance increases to 82.3% at 256 tokens and 94.8% at 512 tokens, with long-context architectures such as ModernBERT capturing over 99.9% of all provisions natively.

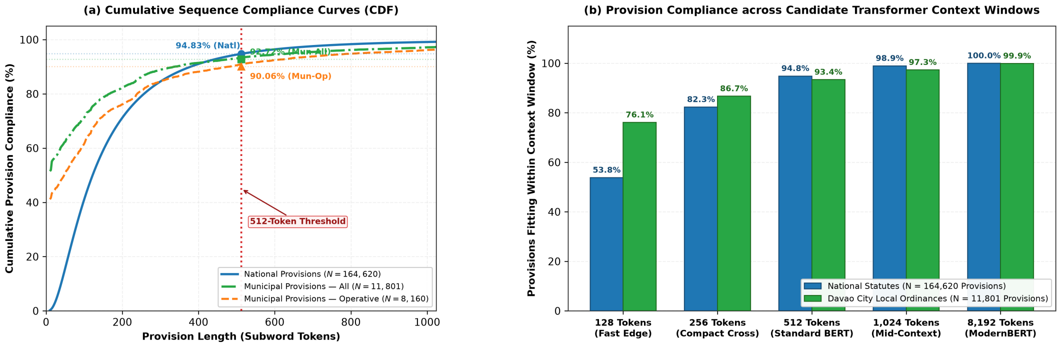


Figure 3.14: Cumulative Provision Compliance Curves and Attention Window Evaluation across the Dual Statutory Search Space ($N = 27,096$ Enactments, $N_{\text{sections}} = 176,421$).

Figure 3.14 checks how well the 176,421 statutory provisions fit within standard transformer attention windows. Panel (a) plots cumulative distribution curves up to 1,024 tokens, showing that 94.69% of all provisions fit within the 512-token limit. Panel (b) compares provision fit across five common attention window sizes (128, 256, 512, 1,024, and 8,192 tokens) for both national statutes and local ordinances. This evaluation is included to help choose candidate models and set input sequence lengths in Stage 1 and Stage 2. As shown in Panel (b), while standard 512-token windows fit nearly 95% of all provisions, long-context architectures like ModernBERT accommodate over 99.9%, avoiding the need to cut off lengthy statutory text.

To handle provisions in the 5.31% exceedance tail without information loss under standard-context models, the pipeline implements a hierarchical overlapping sub-chunking fallback. Adapting the sliding-window chunking strategy for length-constrained transformer architectures [8, 97], when an extracted section exceeds a safety threshold of 448 tokens within the standard 512-token BERT constraint [61], the preprocessing module divides the section into overlapping sub-chunks using a window of $w = 384$ tokens and a stride of $s = 256$ tokens (128-token overlap). Capping the sub-chunk window at 384 tokens reserves a 64-token structural buffer within the 512-token ceiling to accommodate the prepended metadata header along with mandatory special tokens ($[CLS]$, $[SEP]$) [61]. Each sub-chunk preserves the prepended structural metadata header [Statute Title, Section S_i: Catchline] to maintain institutional context. In Stage 1 retrieval, the parent provision embedding is computed by mean-pooling sub-chunk vectors, capturing semantic concepts across all paragraphs. In Stage 2 inference, following sliding-window max pooling [8, 97], the Cross-Encoder evaluates each sub-chunk independently against the local ordinance hypothesis h, determining provision-level conflict through max-pooling across sub-chunk contradiction probabilities:

Max-Pooled Sub-Chunk Contradiction Probability:

$$\hat{P}(\text{Contradiction} \mid p_i, h) = \max_j P\left(\text{Contradiction} \mid p_{i,j}, h\right) \tag{3.4}$$

Where $p_{i,j}$ denotes the j-th sub-chunk of statutory section p_i.

Equation 3.4 defines the max-pooling method used to score long statutory sections that are split into smaller sub-chunks. By assigning the section's overall conflict score to the highest contradiction probability found among its sub-chunks, this ensures that an isolated prohibition or condition buried inside a long section still triggers an alert without being diluted by surrounding agreeable text. In practice, this max-pooling strategy avoids missing conflicts in complex codes where penalties are placed inside longer administrative descriptions, keeping the system sensitive to potential violations. Alongside this sub-chunking approach, the proponents also test ModernBERT as a long-context candidate model [238, 109], checking whether its 8,192-token attention window can process lengthy enactments directly.

3.4 Preprocessing and Normalization

Because raw statutory documents originate from disparate digital repositories and historical scanning archives, they require systematic, source-specific preprocessing and text normalization pipelines. National statutes, local municipal archives, and live ex-ante drafts each present distinct document media, formatting conventions, and noise characteristics that must be addressed individually. In the national statutory collection, continuous legislative texts must be cleansed of web markup artifacts, segmented into discrete numbered provisions, and enriched with hierarchical metadata. In the local ordinance collection, physical scanning degradation, photocopier borders, and non-legislative administrative matter must be removed via vision-language modeling and state machine extraction. In live ex-ante drafts, digital-born text must be segmented into substantive clauses while filtering out procedural boilerplate. The following subsections detail the individual normalization architectures developed for each statutory source.

3.4.1 National Statutory Corpus Normalization

The national statutory archive comprises 25,432 legislative enactments harvested from online legal repositories. Processing these laws requires resolving web layout artifacts and structuring monolithic legislative texts into discrete, context-enriched retrieval units.

3.4.1.1 Statutory Parsing and HTML Artifact Sanitization

National statutes gathered from the Lawphil Project and official digital gazettes are encoded in unstructured HTML format. In their raw scraped form, these documents combine legislative recitals, congressional preambles, and substantive provisions into a single continuous text string. To illustrate the structure of these records prior to normalization, Listing 3.2 displays an authentic raw ingestion record from the national statutory collection (Republic Act No. 11035).

```python
# Raw Ingestion Record from corpus/national_laws/republic_acts.jsonl (ra_11035_2018)
raw_statute_record = {
    "law_id": "ra_11035_2018",
    "year": 2018,
    "url": "https://lawphil.net/statutes/repacts/ra2018/ra_11035_2018.html",
    "scraped_at": "2026-02-17T18:44:04.745654",
    "text": (
        "Seventeenth Congress\nSecond Regular Session\nBegun and held in Metro Manila, "
        "on Monday, the twenty-fourth day of July, two thousand seventeen.\n"
        "REPUBLIC ACT No. 11035\n"
        "An Act Institutionalizing the Balik Scientist Program, Appropriating Funds "
        "Therefor, and for Other Purpose\n"
        "Be it enacted by the Senate and House of Representatives of the Philippines "
        "in Congress assembled:\n"
        "Section 1. Title. - This Act shall be known as the \"Balik Scientist Act.\"\n"
        "Section 2. Declaration of Policy. - The State recognizes that the utilization "
        "of the expertise of science, technology or innovation experts of Filipino descent "
        "is a vital component of the national political, economic, and social development "
        "efforts...\n"
        "Section 3. Balik Scientist Program. - There is hereby created a Balik Scientist "
        "Program, hereinafter referred to as the \"Program\", to be administered by the "
        "Department of Science and Technology (DOST)...\n"
        "[Raw statutory text continues across 2,493 total words]"
    )
}

```

Listing 3.2: Sample Raw JSONL Ingestion Record from the National Statutory Archive (Republic Act No. 11035)

Listing 3.2 details the fields of a raw JSONL record from the national statutory collection, containing the law identifier, enactment year, source URL, scrape timestamp, and full legislative text. This listing is presented to show how statutes look before automated parsing, where continuous text combines introductory preambles, section headings, and operative rules into a single string. In the example shown for Republic Act No. 11035, the statute contains over 2,400 words across several sections within one text block. Downstream retrieval and classification models cannot index or score an unsegmented enactment directly because the long text dilutes semantic meaning and exceeds transformer token limits. Splitting this continuous text into discrete, self-contained operative sections is therefore a necessary first step for accurate conflict detection.

To extract standardized provisions, the national preprocessing script applies regular expressions to detect formal section ordinals and catchlines while sanitizing web artifacts. The cleaner strips trailing secretarial approvals and repository footers using termination patterns (`TERMINATION_PAT`), while pruning residual HTML entities and scanning artifacts (`LAWPHIL_ARTIFACT_PAT`). Monolithic document strings are then partitioned into discrete sections using regex boundary patterns (`SEC_HEADER_PAT`) that match canonical statutory headers (such as “SECTION 1”, “SEC. 2”, or “ARTICLE 3”). For each detected section, the extractor isolates the section ordinal, extracts the catchline title when present, and captures the verbatim body text. This automated segmentation successfully decomposed the 25,432 national enactments into 164,620 individual statutory provisions, providing self-contained regulatory units for retrieval indexing.

3.4.1.2 Context Prepending and Long-Tail Sub-Chunking

Evaluating isolated statutory sections creates a risk of semantic dilution. In legal document processing, context prepending is a structural representation technique that prefixes parent document metadata—including the statutory instrument type, enactment number, statute title, and section catchline—to each extracted provision prior to tokenization [118]. In statutory retrieval, an isolated regulatory section (such as an administrative penalty clause specifying a fine of five thousand pesos) often contains no explicit mention of the broader subject matter (such as commercial business licensing or building safety). Without parent metadata, dense embedding models generate ambiguous vector representations that match hundreds of unrelated statutory penalty clauses across the corpus. Context prepending resolves this ambiguity by explicitly concatenating the governing enactment identity and section title directly to the section text. By embedding this structural context into the input sequence, the transformer attention mechanism conditions the semantic representation of every sentence on its exact legislative source and legal hierarchy. Listing 3.3 details the context prepending procedure implemented to format granular statutory provisions.

```python
def prepend_statutory_context(statute_record, section):
    """
    Prepends governing legislative identity to isolated section text
    to prevent semantic vector dilution and preserve hierarchical rank.
    """
    prefix = (
        f"[{statute_record[’instrument_type’]} No. {statute_record[’enactment_number’]}: "
        f"{statute_record[’title’]} | {section[’section_number’]}: {section[’section_title’]}] "
    )
    model_ready_text = prefix + section[’raw_text’]
    return model_ready_text

```

Listing 3.3: Pseudocode of Section Context Prepending for Granular Provision Chunking

Listing 3.3 outlines the context-prepending routine that attaches legislative title and section details to isolated statutory sections before creating vector embeddings. This step is implemented to keep context intact, ensuring that individual provisions separated from their parent law do not lose their legal identity or jurisdiction. In the routine shown, the function formats a metadata header combining instrument type, enactment number, statute title, section number, and title before attaching the section body text. Adding this header allows the transformer attention layers to take the governing legal context into account. As a result, the model reliably distinguishes local regulations from superior national laws during both candidate retrieval and natural language inference.

Adapting the structural context-prepending methodology of Koreeda and Manning [118] for legal contract and statutory document understanding, every extracted statutory section begins with a structural metadata header containing the statute title, section number, and section catchline:

Context-Prepended Statutory Chunk Representation:

$$\mathbf{x}_{\text{chunk}} = [\text{Statute Title},\ \text{Section } S_i\text{: Catchline}] \parallel \mathbf{t}_{\text{verbatim}} \tag{3.5}$$

Where $\mathbf{t}_{\text{verbatim}}$ denotes the exact text extracted from the statutory record, and $\parallel$ represents string concatenation.

Equation 3.5 defines the representation of a context-prepended statutory chunk $\mathbf{x}_{\text{chunk}}$, built by attaching the parent metadata header to the verbatim statutory text $\mathbf{t}_{\text{verbatim}}$. This chunk design preserves legal context across both Stage 1 retrieval and Stage 2 inference. By attaching the statute title and section heading to the raw clause, the resulting text contains both the specific legal rule and its broader statutory authority. During inference, transformer attention heads compare words across both the header and the operative sentence. This avoids ambiguity and ensures that downstream models recognize whether a rule comes from a national statute or a local city enactment.

To quantify the empirical token overhead introduced by this metadata prefixing, the proponents conducted a comprehensive census across all 164,620 national statutory provisions and 11,801 local ordinance sections. In the raw national collection, isolated provisions exhibit a median length of 98.0 subword tokens and a mean of 179.5 tokens. Attaching the hierarchical metadata prefix introduces an average overhead of only 22.7 tokens, shifting the prepended median to 122.0 tokens and the mean to 202.2 tokens. Crucially, this modest expansion preserves high sequence compliance within standard transformer context windows: exactly 82.30% of prepended provisions remain below 256 tokens, 90.96% fall below 384 tokens, and 94.83% fit comfortably within the standard 512-token BERT context ceiling (with an interquartile 75th percentile of 206.0 tokens and 95th percentile of 522.0 tokens). In the local ordinance corpus, prepended operative provisions concentrate around a median of 98.0 tokens, with 90.06% compliance under 512 tokens. This quantitative verification confirms that structural context prepending supplies vital jurisdictional and statutory provenance without inducing sequence truncation or diluting attention during Stage 1 retrieval and Stage 2 inference.

Listing 3.4 demonstrates the concrete transformation from an isolated provision into a Stage 1 indexed chunk and the resulting Stage 2 asymmetric input sequence. Prepended metadata provides the neural attention layers with parent enactment authority, preventing false positive matches against subordinate or unrelated local penalty clauses.

```python
# 1. Isolated Statutory Provision (Without Context Prepending)
isolated_section = (
    "Section 458. (a) The sangguniang panlungsod... shall enact ordinances... "
    "(v) Approve ordinances which shall impose a fine not exceeding P5,000.00 "
    "or an imprisonment not exceeding one (1) year, or both."
)

```

```python
# 2. Model-Ready Contextualized Provision Chunk (Indexed by Stage 1 BM25 & Bi-Encoders)
stage1_indexed_chunk = (
    "[Republic Act No. 7160: Local Government Code | Section 458: Powers] "
    "Section 458. (a) The sangguniang panlungsod, as the legislative body of the city, "
    "shall enact ordinances... (v) Approve ordinances which shall impose a fine not "
    "exceeding Five thousand pesos (P5,000.00) or imprisonment not exceeding one year ..."
)

# 3. Assembled Asymmetric Input Sequence Passed to Stage 2 Cross-Encoder
stage2_input_sequence = (
    "[CLS] Premise: [RA 7160 | Section 458] Approve ordinances which shall impose a fine "
    "not exceeding P5,000.00... [SEP] Hypothesis: [Proposed Ordinance No. 2024-012 | "
    "Section 6: Penalties] SECTION 6. PENALTY CLAUSE. Any person violating this Ordinance "
    "shall be penalized with a fine of Fifteen Thousand Pesos or two years imprisonment. [SEP]"
)

```

Listing 3.4: Sample Model-Ready Statutory Chunk and Stage 2 Input Sequence with Context Prepending

Listing 3.4 presents the three-stage transformation of a statutory provision, tracking the progression from an isolated clause into a model-ready indexed chunk and finally into an asymmetric Stage 2 input sequence. This listing is provided to illustrate how structural metadata bridges the gap between raw legislative text and input representations for Cross-Encoder classification. As demonstrated in the listing, an isolated penalty clause from Republic Act No. 7160 lacks explicit local scope, but prepending the Local Government Code header and Section 458 catchline clarifies its application to city councils. When formatted for Stage 2 inference, the premise and proposed ordinance hypothesis are joined using special delimiter tokens ([CLS], [SEP]) alongside explicit segment tags. This formatted sequence enables the Cross-Encoder to compute full pairwise cross-attention between the statutory penalty ceiling and the proposed ordinance penalty.

The practical necessity for structural metadata headers is demonstrated by sections within composite legal codes. In Republic Act No. 7160, Section 447 sets penalty limits for municipalities, while Section 458 sets distinct penalty limits for city councils. Without adding the section title and number, downstream Cross-Encoders can confuse different levels of local government and misjudge local legislative powers. Adding structural metadata ensures that the model evaluates each legal clause under its correct administrative level. As a result, the vector embeddings retain both the exact wording of the clause and its broader statutory context. For lengthy provisions in the 5.31% exceedance tail exceeding the 448-token safety threshold, the pipeline implements the sliding-window sub-chunking fallback ($w = 384, s = 256$) formulated in Equation 3.4, prepending the parent metadata header to every sub-chunk to preserve governing legal authority throughout inference.

3.4.2 Local Ordinance Archival Processing and Normalization

Unlike the digital national statutes, the 1,664 Davao City local enactments originate from physical paper archives spanning from 1950 to 2025. Processing this archival collection requires solving optical character degradation, layout variance, non-legislative administrative matter, and formulaic municipal boilerplate.

3.4.2.1 Vision-Language Document Ingestion Pipeline

Transforming physical city paper archives into machine-readable text requires evaluating candidate document ingestion architectures under consumer hardware constraints. While contemporary measures from the digital LISSP database provide clean electronic text, historical enactments from earlier legislative terms exist predominantly as scanned physical PDF documents. To establish an optimal transcription strategy for the 1,664 local records, the proponents formulated a planned candidate extraction experiment comparing three distinct technological paradigms: traditional open-source OCR via Tesseract v5.x [198], modern layout-aware deep learning OCR via Surya OCR [158], and a quantized local vision-language model via Qwen2.5-VL-3B-Instruct [217]. As an independent reference standard for transcription accuracy, the evaluation protocol establishes a gold-standard digitization baseline using Google Gemini 1.5 Flash multimodal extraction coupled with comprehensive line-by-line human legal verification. The candidate pipelines are evaluated across Character Error Rate (CER), Word Error Rate (WER), Table Fidelity (TEDS), and Legal Marker Precision on a representative 60-document stratified test suite, with empirical benchmarking results formally reported in Chapter 4 (Section 4.1).

To support offline execution on consumer hardware, the vision-language architecture deploys Qwen2.5-VL-3B-Instruct [217], a multimodal transformer that pairs a vision transformer backbone with a 3-billion parameter causal language model decoder. Executed locally on Host B (equipped with an AMD Ryzen 7 7435HS processor, 16 GB DDR5 system memory, and an NVIDIA GeForce RTX 5050 Laptop GPU with 8 GB GDDR6 VRAM running Windows 11), the architecture is configured to transcribe scanned enactments while preserving exact statutory phrasing. Proprietary cloud platforms such as Google Cloud Document AI [83] were excluded from routine production due to recurring per-page API fees and legal data sovereignty restrictions under the Data Privacy Act of 2012 (Republic Act No. 10173) [172]. Deploying the quantized open-weight model locally resolves these operational constraints, providing high transcription fidelity while executing completely offline within local government IT boundaries.

The document ingestion pipeline begins by recursively discovering target PDF files across archival city directories. Individual enactments are loaded using PyMuPDF, which inspects document metadata and iterates through pages sequentially. Adapting archival document image perimeter-trimming heuristics for optical scanning artifact reduction [133], the ingestor applies an inset cropbox that trims exactly 3% of the page perimeter along all four borders:

Archival Inset Perimeter CropBox Trimming:

$$\text{CropBox} = [0.03 \cdot W,\ 0.03 \cdot H,\ 0.97 \cdot W,\ 0.97 \cdot H] \tag{3.6}$$

Where W and H represent the physical page width and height in points.

Equation 3.6 defines the boundary for trimming 3% (0.03) from each edge of a scanned page with width W and height H. This cropping step removes common scanner artifacts—such as dark borders, staple marks, and binder punch holes—found in older paper records. Removing these border marks before sending the image to the vision-language model prevents visual noise from being mistaken for punctuation or causing the model to generate repetitive nonsense tokens. The trimmed bounding box keeps the focus directly on the legislative text, ensuring cleaner and more consistent transcription.

Following perimeter trimming, the rasterization module renders each page at an image resolution of 105 dots per inch (DPI). Selecting 105 DPI provides an optimal empirical balance between character legibility and sequence length for high-density legal documents. The rendered image is converted to an RGB array and passed to the visual processor. The processor rescales page patches dynamically, clamping total image pixels between a minimum of $200,704$ pixels ($256 \times 28 \times 28$) and a maximum of $351,232$ pixels ($448 \times 28 \times 28$). Because Qwen2.5-VL allocates visual tokens in 28×28 patch increments using two-dimensional rotary position embeddings (2D-RoPE), bounding the pixel budget stabilizes sequence length and prevents out-of-memory GPU exceptions during long batch runs.

To accommodate local workstation hardware constraints without sacrificing recognition fidelity, the model weights are quantized using 4-bit NormalFloat4 (NF4) precision via the BitsAndBytes library [60]. The quantization configuration uses double quantization and computes attention projections in 16-bit floating point (`torch.float16`) using PyTorch scaled dot-product attention (`sdpa`). Under 4-bit NF4 compression, the model footprint occupies less than 3.5 GB of operational VRAM, allowing the pipeline to execute comfortably on consumer-grade laptop graphics hardware. Text generation is guided by an explicit verbatim system prompt instructing the model to transcribe all section titles, ordinals, dates, names, and monetary values without summarization or commentary. Generation settings enforce deterministic greedy decoding (do_sample = False, use_cache = True) bounded by a ceiling of 1,000 new tokens per page.

The pipeline adopts a page-level processing architecture rather than ingesting entire multi-page documents in a single model forward pass. Ingesting full multi-page ordinances simultaneously would require concatenating thousands of visual patch tokens, which dilutes spatial self-attention and exceeds consumer GPU memory ceilings. In contrast, processing documents page by page enforces focused visual attention on discrete text blocks while scaling linearly ($O(N)$) across documents ranging from single-page resolutions to 50-page tax codes. The pipeline inserts explicit page delineators (`--- [Page X] ---`) into the extracted text, ensuring continuous traceability back to physical council records. Furthermore, page-level execution supports fault-tolerant checkpointing: the system inspects output files on disk and automatically resumes execution across interrupted batch queues by skipping previously completed files. Page-level garbage collection (`gc.collect()`) and explicit CUDA cache clearing (`torch.cuda.empty_cache()`) are executed after each document, preventing memory fragmentation during multi-hour ingestion runs.

3.4.2.2 Substantive Legislative Body Extraction

Scanned local enactments generally contain three distinct parts: opening attendance rosters, the substantive legislative provisions, and concluding signature blocks. Separating the actual legal text from administrative details is essential for accurate conflict detection. Leaving councillor names in the text introduces personal names that distort Stage 1 dense retrieval embeddings, while including signature blocks adds unnecessary clutter to candidate hypotheses in Stage 2. Splitting documents using fixed page numbers does not work reliably across local archives, because short resolutions place legal provisions on the first page, whereas long regulatory codes spread attendance lists and committee rosters across several front pages.

To handle these structural variations, the proponents developed an adaptive boundary state machine that extracts operative legal provisions based on standard statutory markers. The extraction module scans transcripts for recognized opening markers, including enactment designations (`ORDINANCE NO.`), formal titles (`AN ORDINANCE`), and preamble clauses (`WHEREAS`). Upon detecting an opening boundary, the state machine captures the enactment identifier, title, preamble, and all operative sections (`SECTION 1`, `ARTICLE I`, `CHAPTER I`). Text capture stops immediately upon encountering formal closing markers, such as Sanggunian enactment notices (`ENACTED`), secretarial certifications (`CERTIFIED CORRECT`), vice-mayoral attestations (`ATTESTED`), or executive approval notations (`APPROVED`).

The extraction module incorporates a fallback hierarchy to prevent data loss when archival scans have faded or missing boundary markers. If an enactment header is obscured, the extractor falls back to the long ordinance title; if the title is absent, it anchors on the initial operative section. Similarly, if formal certification headers are missing, the system terminates extraction following the standard effectivity section (`SECTION X. EFFECTIVITY`). To provide automated quality control over the extraction process, the state machine calculates an empirical boundary verification confidence score:

Substantive Body Boundary Extraction Confidence Score:

$$\text{Confidence}_{\text{extract}} = 0.35 \cdot C_{\text{start}} + 0.35 \cdot C_{\text{end}} + 0.30 \cdot C_{\text{sec}} \tag{3.7}$$

Where C_{start} and C_{end} denote boundary match confidences, and C_{sec} measures section sequential coherence.

Equation 3.7 defines the confidence score $\text{Confidence}_{\text{extract}}$ used to check how cleanly the legislative text was extracted. It combines scores for the start boundary (C_{start}), ending signatures (C_{end}), and section numbering order (C_{sec}) using weights of 0.35, 0.35, and 0.30. This score helps spot extraction errors automatically without stopping batch processing across hundreds of ordinances. Documents scoring at or above 0.70 move on to indexing, while lower-scoring files are flagged for manual inspection. This approach lets clean scans pass through automatically while sending difficult or unusual documents to human reviewers.

When confidence falls below 0.70, the document is flagged for manual review while preserving the full unpruned transcript on disk. Listing 3.5 outlines the extraction logic.

```python
def extract_substantive_body(raw_transcript: str) -> dict:
    """
    Extracts operative statutory provisions from local transcripts,
    stripping administrative attendance rosters and terminal signatures.
    """
    start_match = RE_ORD_NUM.search(raw_transcript) or RE_TITLE.search(raw_transcript)
    start_idx = start_match.start() if start_match else 0
    start_conf = 1.0 if start_match else 0.40

    search_region = raw_transcript[start_idx:]
    term_match = RE_TERMINAL.search(search_region)
    end_idx = start_idx + term_match.start() if term_match else len(raw_transcript)
    end_conf = 1.0 if term_match else 0.50

    substantive_text = raw_transcript[start_idx:end_idx].strip()
    sections = [int(n) for n in RE_ALL_SECTIONS.findall(substantive_text)]
    sec_conf = 1.0 if len(sections) >= 3 and sections[:3] == [1, 2, 3] else 0.40

    confidence = round(0.35 * start_conf + 0.35 * end_conf + 0.30 * sec_conf, 3)
    return {
        "status": "SUCCESS" if confidence >= 0.70 else "FLAGGED_LOW_CONFIDENCE",
        "confidence": confidence,
        "substantive_text": substantive_text
    }

```

Listing 3.5: Adaptive State Machine for Substantive Legislative Body Extraction

Listing 3.5 presents the Python implementation of the adaptive state machine for substantive legislative body extraction. This algorithm is designed to parse raw local transcripts and extract substantive enactments while stripping prefatory attendance rosters, administrative preambles, and terminal signature blocks. In the function, regular expressions scan the text for opening enactment indicators and terminal attestation keywords, slicing the transcript between these indices and validating ordinal section continuity. The function computes the composite confidence score according to Equation 3.7 and returns a status dictionary indicating whether the extraction succeeded or requires manual review. Isolating the substantive body in this manner ensures that extraneous councillor names and mayoral titles do not distort dense retrieval vector embeddings or trigger spurious preemption matches in Stage 2.

3.4.2.3 Automated Quality Audit and Conservative Normalization

Following raw transcription, the system executes an automated quality audit to detect and categorize transcription anomalies across seven mutually exclusive categories: definite OCR corruption, possible corruption, formatting noise, valid legal boilerplate, valid table structures, page metadata anomalies, and items requiring human review. The audit scanner identifies severe failure states, such as Unicode replacement characters (`\ufffd`), foreign script hallucinations (CJK and Cyrillic character ranges), raw decoding markers (`undefined`), and autoregressive repetition loops. Documents with severe corruptions or total lengths below 150 characters are quarantined into a dedicated review queue for manual verification.

To minimize false positives, the audit pattern matching explicitly accommodates authentic legislative structures. A broad repeated-character regular expression frequently flags Markdown table borders (`|---|---|`) as OCR corruption. To prevent this error, table boundary delimiters, pipes, and header markers are explicitly excluded from symbol repetition rules. Valid legislative boilerplate notations such as `"xxx xxx xxx"`, which denote unamended text in amending ordinances, are protected using negative lookahead expressions. Geographic survey coordinates containing degree symbols (°), minute marks (′), and second marks (″), as well as accented Filipino surnames (*Peña*, *Dañguilan*), are validated against an extended Unicode whitelist. Furthermore, page sequences beginning at Page 2 due to excluded cover sheets are recognized as valid contiguous sequences rather than misordered files.

Deterministic text normalization is performed via a conservative cleaning pipeline that strictly preserves legal meaning. The cleanup module applies Unicode Form KC normalization, strips invisible zero-width characters, removes stray VLM prompt echoes (such as `**[body text]:**` / *zhèngwén* markers), and collapses runaway character loops exceeding seven consecutive repetitions to a standardized three-character representation. All modifications are logged in an audit report with line-level traceability. Under strict architectural safeguards, automated scripts are prohibited from altering personal names, calendar dates, numerical monetary amounts, ordinance numbers, statutory citations, penalty clauses, or section numbering. Discrepancies involving these sensitive entities are routed exclusively to human legal researchers.

Downstream from automated cleanup, the architecture incorporates an advisory language quality review module combining a curated Philippine legal and toponymic gazetteer, contextual language model perplexity scoring, and named-entity recognition. The gazetteer whitelists all 182 official barangays of Davao City, administrative districts, local executive departments, and statutory Latin maxims (*in pari materia*, *mutatis mutandis*, *ultra vires*), preventing specialized terminology from triggering false spelling alarms. Contextual perplexity scoring evaluates token predictability within a sliding context window; tokens exhibiting high pseudo-perplexity and characteristic alphanumeric substitution patterns (such as misreading numeral `0` as uppercase letter `O` in monetary figures) are assigned high diagnostic confidence. The system presents these candidate errors on an interactive review dashboard, coupling the transcribed text with coordinate bounding boxes linking back to the original physical PDF scan, allowing human annotators to review highlighted anomalies prior to final vector indexing.

3.4.2.4 Procedural Boilerplate Filtering and Provision Formatting

Filtering formulaic boilerplate prevents search drift and computational waste during candidate retrieval across the local collection. In Davao City legislation, procedural clauses—specifically Title, Separability, Repealing, and Effectivity clauses—contain standardized legislative phrases that carry zero substantive regulatory restrictions. When these sections are vectorized, their embeddings align with unrelated administrative acts that happen to share identical legal boilerplate. By systematically pruning these non-operative sections during ingestion, the preprocessor directs candidate retrieval exclusively toward substantive regulatory and penal provisions. As established in the corpus census, this filtering prunes 3,641 procedural sections (30.85%) from the local ordinance collection, isolating 8,160 operative provisions for indexing.

Following boilerplate filtering, each operative provision is formatted with a standardized municipal metadata header following the context prepending principles of Equation 3.5. Each local retrieval unit is prepended with its governing local council authority: [Davao City Ordinance No. O_i: Title | Section S_j: Catchline] $\|$ $\mathbf{t}_{\text{verbatim}}$. Attaching this municipal metadata ensures that dense bi-encoders and cross-encoders condition on the local legislative origin, preventing candidate retrieval models from confusing municipal penalties with national criminal statutes.

3.4.3 Ex-Ante Draft Ingestion and Statutory Exception Harmonization

While historical statutory and municipal archives undergo batch offline normalization, newly drafted measures submitted for ex-ante review pass through a specialized live ingestion pipeline.

3.4.3.1 Digital-Born Draft Ingestion and Boundary Parsing

A critical operational distinction in this research separates the historical training archive from live ex-ante draft submissions. While the historical Davao City collection consists of scanned physical PDFs requiring optical character recognition and text normalization, newly drafted ordinances submitted for ex-ante review are digital-born documents. Legislative staff create these proposed measures using standard word processing software, generating clean electronic text or searchable PDF files. Because digital-born drafts contain authentic character encodings without scanning artifacts, the live pipeline bypasses OCR error correction entirely. This structural decoupling allows the preprocessor to focus immediately on automated section boundary parsing and clause segmentation.

To evaluate legislative commands without semantic distortion, an automated regular expression parser splits the submitted draft at formal section boundaries into discrete operative blocks. In Philippine local legislative practice, proposed measures follow standardized structural templates established by the Department of the Interior and Local Government [57]. Real-world legislative drafts contain substantial procedural boilerplate that does not express substantive legal commands. If routine clauses such as title declarations, repealing clauses, and effectivity dates are fed into Stage 1 retrieval, they match thousands of administrative national statutes containing similar formulaic phrases. Listing 3.6 details the regular expression parser that segments live draft measures into discrete operative sections while filtering out procedural boilerplate.

```
import re

def extract_operative_provisions(draft_text):
    """
    Parses digital-born draft ordinance into discrete sections,
    filtering out non-operative procedural boilerplate clauses.
    """
    # Regex pattern to segment text at formal section boundaries
    pattern = r’(SECTION\s+\d+\.?[^\n]*\n.*?)(?=(?:SECTION\s+\d+|CERTIFIED|ENACTED|\Z))’
    matches = re.findall(pattern, draft_text, re.DOTALL | re.IGNORECASE)

    # Procedural clauses that do not carry substantive regulatory commands
    boilerplate_keywords = ["TITLE", "SEPARABILITY", "REPEALING", "EFFECTIVITY"]
    operative_sections = []

    for match in matches:
        clean_chunk = ’ ’.join(match.strip().split())
        header = clean_chunk.split(’.’)[0] if ’.’ in clean_chunk else clean_chunk[:30]

        # Exclude non-operative procedural boilerplate
        if any(kw in header.upper() for kw in boilerplate_keywords):
            continue

        operative_sections.append({
            "section_header": header,
            "operative_text": clean_chunk
        })

    return operative_sections

```

Listing 3.6: Code Snippet of the Operative Provision Extractor and Boilerplate Filter

Listing 3.6 presents the Python implementation of the operative provision extractor, featuring regular expression lookahead patterns and a procedural boilerplate keyword filter. This algorithm is incorporated to automate the structural segmentation of raw draft ordinances into clean, self-contained section blocks while systematically discarding administrative clauses. As observed in the listing, the regular expression isolates text between formal section boundaries, while the keyword filter checks section headers against procedural markers including Title, Separability, Repealing, and Effectivity clauses. Discarding these formulaic clauses prevents downstream retrieval models from matching generic legislative boilerplate against thousands of unrelated administrative acts. Consequently, only substantive regulatory and penal provisions advance to the candidate indexing and inference pipeline.

3.4.3.2 Harmonic Interpretation and Exception Handling

A key challenge in conflict detection is distinguishing genuine normative contradictions from authorized statutory exceptions. In statutory interpretation, harmonic interpretation is the legal doctrine that provisions of law must be construed together to give effect to both general prohibitions and qualifying provisos, avoiding apparent contradictions where an authorized exception applies [4]. In legislative drafting, an exception clause is a statutory condition that permits an otherwise restricted activity under specified conditions or for designated public entities. If an automated conflict detection system evaluates a draft local ordinance solely against a general statutory prohibition without processing the accompanying exception clause, it generates a false positive preemption alert. In regulatory enactments, general prohibitions and permitted exceptions frequently appear within the same section using conditional syntax. Because statutory sections average between 90 and 122 tokens, an entire section fits comfortably within the 512-token input limit of candidate Cross-Encoders. The model's bidirectional attention mechanism reads the general prohibition and its qualifying exceptions simultaneously, preventing false positive contradiction alerts. This joint attention ensures that local measures implementing authorized statutory exceptions are recognized as valid exercises of delegated authority.

When a general prohibition and an authorizing exception reside in separate statutory sections, the pipeline relies on two-stage candidate retrieval. Under Philippine statutory construction, related provisions governing the same subject matter must be interpreted together as a harmonious whole under the doctrine of *in pari materia* [4]. If an ordinance draft relies on an authorized statutory exemption, evaluating a specific clause without retrieving the parent exception can trigger an incorrect conflict warning. The system addresses this challenge by utilizing hybrid retrieval in Stage 1 to pull both general bans and specific exception clauses into the candidate shortlist. The Stage 2 Cross-Encoder then evaluates the draft against the authorizing exception clause and assigns an Entailment classification, confirming that the local measure operates within valid legislative authority.

3.5 Ground Truth Benchmark

Evaluating candidate Cross-Encoder models in Stage 2 requires an expert-verified ground truth dataset reflecting the authentic legal realities of Philippine local legislation. Unlike general-domain natural language inference benchmarks that rely on crowdsourced sentence pairs, evaluating local conflict detection requires formal statutory interpretation skills grounded in administrative law. Because no existing NLP benchmark encompasses Philippine national statutes and local government ordinances, the proponents established an authoritative 350-pair ground truth evaluation dataset curated in collaboration with active legal researchers from the Davao City Sangguniang Panlungsod.

For the ground truth evaluation benchmark, the proponents settled on exactly $N = 350$ premise-hypothesis sentence pairs based on three practical reasons: the number of participating legal researchers, realistic workload limits to avoid review fatigue, and statistical sample size requirements. Through formal coordination with the Sangguniang Panlungsod of Davao City, a cohort of exactly $P = 15$ active legal researchers agreed to evaluate the benchmark pairs. To ensure reliable labels and compute inter-rater agreement, each candidate pair requires independent evaluation by $k = 3$ distinct legal researchers. In specialized legal annotation, reading multi-sentence statutory provisions requires focused concentration. Prior studies indicate that rater attention and analytical accuracy drop when an individual's workload exceeds roughly 70 pairs [1, 33]. By grouping the 15 researchers into five balanced panels of three members ($15/3 = 5$ panels) and assigning 70 pairs to each annotator, the evaluation covers exactly $5 \times 70 = 350$ unique statutory pairs, producing $350 \times 3 = 1,050$ total independent legal judgments across the cohort.

3.5.1 Sample Size and Statistical Power

The determination of the benchmark sample size ($N = 350$ pairs) is grounded in statistical power analysis for multi-class contingency evaluations [41, 69]. Under Cohen's power framework, detecting semantic conflicts across a three-class discrete decision space ($\mathcal{C} = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$, with $\text{df} = (3 - 1) = 2$ for goodness-of-fit and $\text{df} = (3-1)(3-1) = 4$ for 3×3 contingency matrices) with a medium effect size ($w = 0.30$) at a significance level of $\alpha = 0.05$ requires a minimum of $N = 108$ items to achieve standard statistical power of 80% ($(1 - \beta) = 0.80$).

Establishing an evaluation suite of $N = 350$ pairs provides an achieved statistical power exceeding 0.985 ($(1 - \beta) > 98\%$), effectively eliminating Type II error risks ($\beta < 0.015$) and ensuring robust statistical representation across all eight Macro Legal Domains defined in Section 3.3.3. These eight domains reflect the core public services and regulatory police powers defined in Section 17 and Section 458 of Republic Act No. 7160.

This curated scale directly reflects established standards in specialized legal NLP benchmarking. While general-domain benchmarks like SNLI [24] rely on open crowdsourcing across tens of thousands of colloquial sentences, domain-specific legal NLI requires accredited legal professionals whose time and cognitive availability are severely constrained. High-quality, expert-adjudicated legal datasets consistently operate within this targeted scale: the SARA statutory tax benchmark comprises 276 cases (176 training and 100 test cases) [98], the annual COLIEE Task 4 legal entailment test suites provide approximately 100 pairs per competition cycle [166, 5], and ContractNLI [118] relies on curated subsets across 607 contracts. At 350 premise-hypothesis pairs with a three-rater overlap yielding 1,050 independent legal judgments, the benchmark represents an authoritative, statistically sound gold standard for local legislative review.

3.5.2 Dataset Partitioning

To maintain experimental rigor, prevent data leakage, and ensure reproducible model evaluation, the 350 Ground Truth pairs are partitioned using a static 70/15/15 train, validation, and test split. This split provides 245 training pairs ($N_{\text{train}} = 245$), 52 validation pairs ($N_{\text{val}} = 52$), and 53 test pairs ($N_{\text{test}} = 53$). Operating within this sample size places the evaluation in a few-shot learning regime, which is well-suited for pretrained transformer language models [76]. In machine learning, few-shot task adaptation utilizes the foundational linguistic and semantic representations already encoded within deep pretrained encoders, fine-tuning classification heads and attention parameters on domain-specific instances without requiring thousands of newly labeled examples [76].

The proponents specifically selected a 70/15/15 partition over a standard 80/10/10 split due to sample size constraints. In a 350-pair dataset, an 80/10/10 allocation yields only 35 validation and 35 test instances. A validation set of only 35 pairs is statistically fragile for fine-grained decision threshold calibration (τ^*), where a single classification error alters observed accuracy by nearly 3%. Furthermore, a test set of 35 items falls below the $N \geq 50$ sample size threshold established by Card et al. [29] for reliable Minimum Detectable Effect (MDE) in natural language processing and risks violating the asymptotic cell expectations of McNemar's test ($b+c \geq 20$). In contrast, allocating 52 validation pairs and 53 test pairs satisfies both the MDE statistical power criteria and the non-parametric hypothesis testing assumptions while retaining 70% (245 pairs) for robust few-shot Cross-Encoder fine-tuning [224].

3.5.3 Asymmetric Pairing and Hypothesis Formulation

To construct a representative ground truth benchmark, candidate statutory premises must reflect the full breadth of Philippine national law. During the exploratory phase of this research, the proponents conducted preliminary machine learning experiments on the national statutes, uncovering severe class imbalances and repetitive drafting formulas across historical eras. These exploratory findings directly informed the consolidated topic modeling architecture detailed in Section 3.3.3, which organizes the corpus into eight validated Macro Legal Domains such as local governance, public health, and taxation. The proponents then sampled national statutory sections across these eight domains to establish the fixed premises for the evaluation benchmark. In natural language inference for legal documents, Asymmetric Span-Level Pairing is a sequence formulation strategy where a single, self-contained premise section from a superior statute is paired with a multi-sentence span (N sentences) representing an entire operational section from a draft ordinance [246, 118]. This asymmetric formulation reflects the structural reality of Philippine legislative drafting. National statutes typically establish general normative principles or statutory prohibitions in one or two concise sentences. In contrast, local ordinances are operational administrative instruments: a single section regularly bundles the primary regulatory command together with local office designations, penalty schedules, and administrative notice periods spanning three to six contiguous sentences. If an NLP pipeline splits an ordinance section into isolated single-sentence fragments, the classifier loses the qualifying conditions and penalty ceilings, resulting in erroneous conflict determinations. By presenting the Cross-Encoder with the full multi-sentence ordinance section paired against the statutory premise, the model receives complete textual context to evaluate substantive preemption.

To make the evaluation as realistic as possible, the drafted test sentences closely mirror the writing style of Davao City ordinances. The proponents analyzed local laws enacted between 2021 and 2025 to identify standard drafting phrases, administrative office names, and structural penalty clauses following the official drafting conventions of the Department of the Interior and Local Government (DILG) [57]. Each drafted clause uses standard enacting syntax, such as: "*SECTION X. [TITLE]. — It shall be unlawful for any person, natural or juridical, within the territorial jurisdiction of Davao City, to...*" The draft sentences also name real local enforcement bodies, such as the City Transport and Traffic Management Office (CTTMO), the City Health Office (CHO), the City Environment and Natural Resources Office (CENRO), and the Business Bureau of Davao City. Operational units such as the Anti-Vices Task Force (AVTF) and the Davao City Police Office (DCPO) are also included to reflect genuine local enforcement.

The synthetic ordinance clauses are also linked to real administrative districts in Davao City. Draft rules specify geographic areas from urban centers like Poblacion and Talomo to suburban districts like Buhangin and Toril. Local offices such as the City Disaster Risk Reduction and Management Office (CDRRMO) and the City Social Welfare and Development Office (CSWDO) are included to match actual city administration. This realistic drafting prevents evaluators from reviewing artificial example sentences that differ from real legal texts. As a result, the 15 Sangguniang Panlungsod legal researchers evaluate draft provisions that look like authentic measures referred to committee following First Reading and prepared for Second Reading plenary interpellation.

The proponents deliberately utilized domain-grounded, authentic local template-derived hypothesis construction—grounded in existing Davao City ordinances (2021–2025) and DILG drafting manuals, and formally verified through iterative faculty adviser review—rather than unconstrained synthetic generation via large language models. Recent evaluations by Gururangan et al. [91] and Poliak et al. [163] demonstrated that automated or crowd-generated NLI hypotheses frequently introduce severe "annotation artifacts"—systematic lexical cues such as negation markers ("not", "never") that allow neural models to exploit superficial heuristics and predict contradictions without genuinely reading the statutory premise. In specialized legal domains, authoritative benchmarks such as SARA (tax law) [98], DocNLI [246], and Clash-of-Leges [105] establish that rigorous evaluation requires domain-grounded hypotheses reflecting authentic legislative syntax, ensuring models are tested on genuine statutory reasoning rather than superficial generation heuristics.

While the system uses multi-sentence spans to understand local context, it does not attempt cross-document multi-hop reasoning (detecting conflicts that appear only by combining two or more separate national laws). Resolving dependencies across multiple separate laws is an active research challenge in natural language inference. Current systems that attempt this require either complex knowledge graphs [136] or very large generative language models [197]. Running those multi-premise models would exceed the consumer-grade hardware limits set for local government deployment. For this reason, the ground truth benchmark and the inference system focus strictly on single-premise evaluations.

3.5.4 Class Balancing and Adversarial Injection

In machine learning benchmark construction, class balancing refers to the deliberate equalization of sample counts across all target categories to prevent models from learning trivial majority-class decision rules [34]. In authentic municipal drafting, the vast majority of local ordinance provisions comply fully with national law, meaning true statutory contradictions represent rare edge cases (typically comprising under 1% of random pairs). If an evaluation benchmark mirrored this natural skew (e.g., 90% Entailment and 10% Contradiction), a naive classifier that simply predicts the majority class would achieve a deceptively high 90% nominal accuracy while remaining completely ineffective at identifying genuine statutory conflicts. To complement class balancing, adversarial injection is a dataset construction methodology where domain experts systematically formulate deceptive, hard-to-detect variations of test pairs to eliminate reliance on superficial annotation artifacts [91, 163]. In legal natural language processing, deep language models frequently exploit shallow lexical heuristics—such as assuming that any provision containing words like "fine," "prison," or "unlawful" constitutes a contradiction. Through adversarial injection, annotators formulate pairs that invert legal duties (modifying mandatory commands to permissive options) or introduce subtle penalty ceiling violations without altering surface vocabulary. This rigorous design forces candidate Cross-Encoders to demonstrate genuine logical inference across statutory premises rather than relying on shallow lexical shortcuts.

In standard machine learning on tabular data, researchers often handle uneven classes using methods like the Synthetic Minority Over-sampling Technique (SMOTE) [34]. However, the proponents avoid continuous vector oversampling for this legal text pipeline. SMOTE creates artificial examples by interpolating between numerical feature vectors

($x_{\text{new}} = x_i + \lambda(x_{zi} - x_i)$). In high-dimensional language models representing discrete words, averaging or blending token vectors creates artificial representations that confuse attention layers and carry no real legal meaning. Simple word replacement also fails with legal texts because statutory words have exact legal definitions; for example, swapping mandatory words like “shall” with permissive words like “may” fundamentally changes a legal duty.

To build a balanced, reliable Ground Truth dataset, the researchers combined historical legal records with expert-guided text adjustments. First, the researchers gathered documented legal clashes from local government records, including mayoral veto messages, repealed ordinances, and explicit statutory amendments [105]. Second, following the evaluation methods of Koreeda and Manning [118] and Nie et al. [153], legal annotators introduced conflicting rules into real statutory provisions. Annotators reversed permissions, changed numerical zoning limits, and added conflicting penalties into ordinance clauses. This expert-guided process produced an exact balanced distribution across the Entailment, Neutral, and Contradiction classes in the 350 Ground Truth pairs (117 pairs per class), preventing class imbalance without creating unnatural legal phrasing.

3.5.5 Stratified Difficulty Tiers

Building an effective evaluation benchmark requires pairs with varying levels of difficulty. In benchmark evaluation, difficulty tier stratification is the systematic division of an evaluation dataset into distinct subsets based on the cognitive and structural complexity required to resolve each test instance [118, 174]. In this benchmark, the 350 pairs are partitioned into three granular tiers. Tier 1 (Surface and Quantitative Conflicts) evaluates direct, rule-based violations, such as numerical local penalties exceeding statutory caps (e.g., a PHP 15,000 local fine where Section 458 of Republic Act No. 7160 sets a PHP 5,000 ceiling for cities) or explicit inversions of mandatory terms. Tier 2 (Structural Preemption and Jurisdictional Carve-Outs) evaluates institutional authority under the *Magtajas* doctrine, testing whether a local council legislates within a field reserved for national agencies (such as pesticide control under Presidential Decree No. 1144) or omits mandatory statutory exemptions. Tier 3 (Latent and Paraphrastic Conflicts) evaluates subtle normative contradictions where the ordinance and national statute share virtually no overlapping lexical tokens or citations. In Tier 3, the conflict arises from contradictory legal effects—such as a local draft mandating the detention of apprehended street minors, which directly violates the non-custodial diversion mandate of Republic Act No. 9344 without explicitly citing the statute. This stratified structure provides fine-grained diagnostic visibility into candidate model performance.

To construct a rigorous evaluation suite, the proponents adapted the difficulty tiering methodology established by Koreeda and Manning in the ContractNLI benchmark [118]. This approach is supported by the behavioral stress-testing frameworks of Ribeiro et al. [174] and Naik et al. [149], which separate surface competence tests from semantic challenge sets. Furthermore, the 30% Tier 1, 40% Tier 2, and 30% Tier 3 quota follows the adversarial challenge partitioning established in benchmark engineering literature, such as SQuAD 2.0 [167], FEVER [219], and ANLI [153]. These foundational benchmarks establish that maintaining an adversarial challenge quota of 30% to 40% is optimal to prevent models from exploiting superficial lexical artifacts while preserving sufficient representation of standard rule-based compliance. Table 3.7 outlines the stratified difficulty tiers, legal conflict mechanisms, and representative examples for the 350-pair benchmark.

Table 3.7: Stratified Jurisprudential Difficulty Tiers for the 350-Pair Ground Truth Evaluation Benchmark

Difficulty Tier	**Target Share**	**Legal Conflict Mechanism**	**Representative Statutory Example**
Tier 1: Surface & Quantitative	30%	Direct penalty ceiling violations and explicit changes in mandatory words (*shall* converted to *may*).	Local draft imposes a PHP 15,000 fine for traffic obstruction when RA 7160 Sec. 458 caps city penalties at PHP 5,000.
Tier 2: Preemption & Carve-Outs	40%	Exceeding local regulatory powers under *Magtajas*, and blanket local bans that omit statutory exceptions.	Draft ordinance enacts a total ban on commercial freight transit without exempting national agricultural and medical logistics.
Tier 3: Latent & Paraphrastic	30%	Subtle conflicts where the draft redefines statutory terms or creates incompatible rules without using the exact statutory words.	Draft anti-vagrancy ordinance assigns apprehended minors to detention cells, violating non-custodial diversion under RA 9344 without citing the law.

Table 3.7 outlines the three-tier difficulty breakdown used for the 350-pair ground truth benchmark, detailing target percentage shares, legal conflict mechanisms, and representative statutory examples. This breakdown ensures that the benchmark does not merely test simple keyword matching or penalty ceiling checks, but also tests models on more subtle preemption conflicts. As shown in the table, the benchmark allocates 30% to Tier 1 (surface and quantitative conflicts, 105 pairs), 40% to Tier 2 (preemption and statutory exceptions, 140 pairs), and 30% to Tier 3 (latent and paraphrastic conflicts, 105 pairs). In addition, the Tier 3 challenge set deliberately leaves out all numerical penalty amounts and direct statutory citations, requiring the Cross-Encoder to evaluate legal meaning rather than relying on numbers. This distribution reflects the realistic challenges faced during legislative committee review while preventing models from depending on superficial keyword tricks.

The three difficulty tiers map directly to established Philippine statutory preemption principles and natural language challenges, following the distinction between rule-based and value-based legal conflicts described by Sartor [191] and Bench-Capon and Sartor [19]. Tier 1 evaluates surface and quantitative preemption, covering direct penalty ceiling violations under Section 458 of Republic Act No. 7160 and explicit clashes between mandatory and permissive words (*shall* versus *may*). Tier 2 adapts the "negation by exception" concept from Koreeda and Manning [118], evaluating unhonored statutory exemptions and local overreach under *Magtajas v. Pryce Properties* [208] and *City of Manila v. Laguio, Jr.* [210]. Tier 3 evaluates latent and paraphrastic contradictions, testing whether the model detects substantive legal friction when a draft redefines terms without sharing exact statutory words. Dividing the benchmark into 30% Tier 1 (105 pairs), 40% Tier 2 (140 pairs), and 30% Tier 3 (105 pairs) ensures a balanced test set that reflects real committee work while preventing models from exploiting simple word overlap [143].

Grounding Tier 3 contradictions in legal doctrine ensures that models are evaluated on genuine language understanding rather than keyword presence. Under established Supreme Court rulings, a local council cannot enact indirect prohibitions that take away rights protected by national statutes [208, 210]. To enforce this distinction during testing, the proponents verified that all 104 Tier 3 draft sentences contain zero numbers or monetary amounts. Without explicit numerical clues, the Cross-Encoder must evaluate the underlying legal meaning of the text, such as recognizing when an anti-vagrancy draft detains minors in violation of Republic Act No. 9344. This design ensures that the benchmark provides an authentic diagnostic test of semantic preemption for the subsequent analysis in Chapter 4.

3.5.6 Expert Legal Annotation Panel

Classifying the sampled text pairs requires domain-specific legal training to ensure authentic and reliable labels. The Davao City Sangguniang Panlungsod currently comprises 24 elected councilors across the city's three legislative districts, supported by a professional staff of approximately 48 active legal researchers [47]. In the initial administrative correspondence transmitted to the Sangguniang Panlungsod Secretariat alongside the Legal Annotation Guidebook, the proponents formally requested a cohort of 15 volunteer legal researchers. Unlike general NLP tasks that rely on open crowdsourcing, legal natural language inference requires formal statutory interpretation skills to prevent superficial keyword matching [199, 249]. The recruited legal researchers review local ordinances as their regular professional duty, providing the necessary institutional competence for building a dependable gold standard [33].

To establish an objective and transparent panel structure, the proponents gathered baseline qualifications during the initial intake communication. The intake request systematically recorded two key criteria from each volunteer: highest educational attainment and total years of professional experience in legislative drafting. Using these reported credentials, the 15 researchers were organized into five balanced sub-panels of three members each ($k = 3$). Within each three-member sub-panel, the evaluator possessing the highest educational attainment and the longest drafting tenure was designated as that panel's Senior Annotator. The proponents then communicated these panel assignments and Senior Annotator designations in a follow-up confirmation email to establish operational readiness before annotation began.

Following the consensus guidelines adapted from Ablazo [1], a label is accepted into the final Ground Truth if at least two of the three reviewers in a sub-panel agree. In cases where all three reviewers select different categories, resulting in a three-way split with no majority, the pair is referred to that panel's designated Senior Annotator for binding qualitative adjudication [77, 11]. The Senior Annotator reviews the pair against the standardized codebook to resolve whether the split arose from statutory ambiguity or an overlooked proviso. This hierarchical adjudication step ensures that every item in the 350-pair benchmark reaches an authoritative label while preserving the initial independent ratings for calculating Fleiss' Kappa (κ). Maintaining this structured division between initial independent scoring and expert tie-breaking safeguards both statistical validity and qualitative consistency across the dataset.

To recognize the time and professional expertise dedicated by the participating Sangguniang Panlungsod legal researchers, evaluators who complete their assigned annotation batches are awarded an official Certificate of Contribution. This certificate serves as a formal token of appreciation and institutional documentation acknowledging their voluntary advisory assistance in validating the statutory benchmark. Each certificate is endorsed by the academic department leadership, thesis adviser, and student proponents to confer formal institutional validity. Providing this acknowledgment establishes a collaborative partnership between the university research team and the local legislative staff, ensuring appropriate recognition for the legal professionals who support the research.

In strict adherence to the ethical guidelines of the University Research Ethics Board (UREB) and the statutory mandates of Republic Act No. 10173 (the Data Privacy Act of 2012) [172], participation in the expert annotation exercise is entirely voluntary. Prior to receiving evaluation materials, each participant completes an electronic informed consent agreement affirming voluntary participation and the freedom to withdraw at any point without administrative penalty or professional consequence. To protect evaluator confidentiality, participant names, councilor assignments, and committee affiliations are permanently decoupled from annotation records and replaced with anonymized rater codes (such as Annotator P1-A through Annotator P5-C). Evaluated text pairs consist exclusively of enacted public statutes and model local drafting clauses derived from official DILG manuals and enacted city legislation, involving no confidential council communications, unreleased political deliberations, or proprietary personal data.

3.5.7 Annotation Codebook and Decision Rules

To help the 15 legal researchers apply consistent criteria, the proponents developed a standardized Legal Annotation Codebook. Adapting principles from Koreeda and Manning [118] and Italiani et al. [105], the codebook turns the three-class NLI framework into clear legal decision rules. Each evaluated pair consists of a single national statute section as the premise and a multi-sentence local ordinance clause as the hypothesis. Annotators assign one of three labels: Contradiction, Entailment, or Neutral. Table 3.8 summarizes the working definitions, legal triggers, and example patterns used in the annotation guidelines.

Table 3.8 details the operational legal definitions, primary statutory triggers, and representative drafting patterns defined in the Sangguniang Panlungsod annotation codebook across the three target NLI classes. This decision framework is provided to establish consistent, legally grounded classification criteria for the 15 volunteer legal researchers, translating natural language inference categories into established Philippine administrative doctrines. As detailed in the table, Contradiction is triggered by Magtajas preemp-

Table 3.8: Sangguniang Panlungsod Ground Truth Legal Annotation Codebook and Decision Rules

Logical State	**Operational Legal Definition**	**Primary Legal and Structural Triggers**	**Representative Statutory Pattern**
Contradiction	The local ordinance prohibits what national statute permits, permits what statute prohibits, or exceeds statutory ceilings [208].	Clashes in mandatory words (*shall* vs. *may*); penalty limit violations (RA 7160 Sec. 458); territorial or franchise preemption.	National statute caps ordinance fines at PHP 5,000; draft ordinance mandates a PHP 15,000 penalty.
Entailment	The local ordinance directly executes, complies with, or derives its regulatory authority from the national premise without adding conflicting mandates.	Compliant devolved implementation; literal statutory adoption; regulatory enforcement within prescribed legislative boundaries.	Local draft establishes a local health board strictly adhering to the composition mandated by RA 7160.
Neutral	The national statute and local ordinance address the same broad legal domain, but govern independent obligations without mutual conflict or derivation.	Shared topic without direct legal clash; complementary administrative scopes; distinct non-overlapping legal subjects.	National statute establishes hospital licensing standards; local draft regulates city market sanitary permits.

tion, modal command inversions, and local penalty escalations; Entailment is triggered by valid devolved regulatory implementation; and Neutral is assigned when provisions share broad subject matter without normative conflict. Standardizing these decision rules ensures high inter-rater reliability during independent scoring and provides clear criteria for Senior Annotator tie-breaking. Consequently, the resulting gold labels reflect rigorous legal consensus rather than subjective linguistic intuition.

The decision rules for Contradiction directly apply the consistency test established in the Magtajas Doctrine [208]. Because local governments have delegated rather than unlimited legislative powers under Republic Act No. 7160 [84], any local draft that exceeds national statutory limits is legally invalid (*ultra vires*). Annotators look for shifts in mandatory language, such as turning a strict command ("shall") into an optional choice ("may"), or removing an exemption that national law protects. In the test samples, numerical rules are audited against national limits, including tax caps and penalty ceilings. These criteria ensure that conflict warnings highlight actual legal problems rather than minor wording choices.

The guidelines follow a structured consensus process to resolve disagreements among the three panel reviewers. When two or three reviewers agree, the majority label is accepted directly into the benchmark. In cases where all three reviewers select different categories, the item is referred to the sub-panel's designated Senior Annotator for final qualitative review [77]. The Senior Annotator reviews the evaluators' optional notes against the codebook to determine whether the difference came from subtle statutory phrasing or an overlooked exception. This review step ensures that every item in the 350-pair Ground Truth reaches an authoritative label while preserving the original votes to calculate Fleiss' Kappa.

To illustrate the structure of the evaluated instances, Listing 3.7 presents an authentic sample record (Pair ID: `GT-001`) from the 350-pair Ground Truth benchmark dataset, showcasing the paired national statutory premise, proposed local hypothesis, gold conflict label, and expert legal rationale.

```
# Authentic Evaluation Pair GT-001 from data/ground_truth_350.jsonl
benchmark_pair = {
"pair_id": "GT-001",
"macro_domain": "Executive Issuances & Policy Reorganization",
"difficulty_tier": "Tier 1: Surface & Quantitative",
"national_premise": {
    "statute_title": "Republic Act No. 7581 (The Price Act)",
    "citation": "Section 6",
    "statutory_text": (
        "Section 6. Automatic Price Control. - Unless otherwise declared by the "
        "President, prices of basic necessities in an area shall automatically be "
        "frozen at their prevailing prices or placed under automatic price control "
        "whenever that area is proclaimed or declared a disaster area or under a "
        "state of calamity... price control of basic necessities under this section "
        "shall remain effective for the duration of the condition that brought it about, "
        "but not for more than sixty (60) days."
    )
},
"ordinance_hypothesis": {
    "ordinance_title": (
        "Proposed City Ordinance No. 2024-082 (The Davao City Calamity Price "
        "Stabilization and Consumer Welfare Protection Measure)"
    ),
    "section_citation": "Section 4",
    "hypothesis_text": (
        "SECTION 4. MANDATORY LOCAL PRICE CEILINGS UPON CALAMITY. - In accordance "
        "with local civil defense standards... upon official declaration of a state "
        "of local calamity by the Sangguniang Panlungsod, the City Mayor... shall "
        "enforce mandatory price ceilings across all grocery and commercial retail "
        "establishments within Davao City, which price freeze shall remain in continuous "
        "legal effect for a mandatory period of one hundred eighty (180) calendar days ..."
    )
},
"gold_label": "Contradiction",
"legal_rationale": (
    "Direct breach of statutory temporal ceiling. Section 6 of RA 7581 explicitly limits "
    "automatic calamity price control to not more than sixty (60) days. The draft ordinance "
    "mandates a 180-day freeze, exceeding delegated police power limits."
),
"annotator_metadata": {
    "assigned_panel": "Panel_A",
    "evaluators": ["SP-ANN-01", "SP-ANN-02", "SP-ANN-03"]
}
}

```

Listing 3.7: Sample Evaluation Pair from the Ground Truth Benchmark Dataset

Listing 3.7 presents an authentic evaluation record from the ground truth dataset (`GT-001`), detailing the JSON schema including pair identifier, macro domain, difficulty tier, national premise citation and text, ordinance hypothesis citation and text, gold NLI label, legal rationale, and annotator panel metadata. This listing is included to show the complete data representation passed to candidate Cross-Encoders during evaluation and to document the provenance of each gold label. In this specific Tier 1 instance, the hypothesis mandates a 180-day calamity price freeze, which directly contradicts Section 6 of the Price Act (Republic Act No. 7581) capping automatic freezes at 60 days. The record captures both the exact statutory violation and the panel metadata documenting independent scoring by three legal researchers. Storing this rich metadata enables granular error analysis across difficulty tiers and legal domains in Chapter 4.

3.5.8 Annotation Workload Allocation

To prevent reviewer fatigue while keeping the results statistically sound, this study adapts the batched review framework developed by Ablazo [1] alongside balanced incomplete block design principles for multi-rater NLP annotation [24, 144]. In that earlier study, each item required three independent evaluations to determine a majority label. Managing a collection of 12,054 short texts required over 36,000 individual reviews. By dividing this work among 250 students, the workload was reduced to about 145 items per person. This demonstrated how batched distribution can maintain a three-vote overlap without placing an unrealistic burden on reviewers.

This study applies the same mathematical allocation to the legal domain. Because reviewing multi-sentence legal provisions requires much more concentration than reading short messages, the dataset is smaller, but the distribution logic is identical. The workload for the legal reviewers is calculated as follows:

Annotator Workload Allocation Formulation:

$$
\begin{gathered}
N = 350 \text{ premise-hypothesis pairs} \\
k = 3 \text{ independent votes per pair} \\
P = 15 \text{ legal annotators} \\
W = N \times k = 350 \times 3 = 1,050 \text{ total annotations} \\
\text{Work per person} = \frac{W}{P} = \frac{1,050}{15} = \mathbf{70} \text{ pairs per annotator}
\end{gathered} \tag{3.8}
$$

Where N represents the benchmark size, k is the multi-rater replication factor, P is the annotator pool size, and W denotes total evaluations.

Equation 3.8 details the mathematical derivation of the annotator workload distribution, dividing the total required annotations W among the pool of $P = 15$ legal researchers. With a benchmark size of $N = 350$ premise-hypothesis pairs evaluated under a three-rater replication factor ($k = 3$), the protocol requires a total of $W = 1,050$ independent legal judgments. The formulation allocates this workload into five disjoint 70-pair sets, assigning exactly one set of 70 pairs to each three-member sub-panel. Methodologically, capping individual reviewer workload at 70 pairs prevents cognitive fatigue and attention degradation during complex legal evaluation. This batched structure maintains full multi-rater overlap for Fleiss' Kappa reliability calculations while ensuring that no single annotator is overburdened.

To organize this workflow, the dataset is divided into five equal sets of 70 pairs each. The 15 researchers are grouped into five sub-panels of three members, with each sub-panel reviewing one 70-pair set. This arrangement ensures that every pair receives three independent reviews while keeping each reviewer's workload manageable.

To reduce fatigue during the review process, the proponents provided structured reference materials. Reviewing statutory pairs requires significant effort, as reviewers must compare detailed legal clauses against local drafts. Evaluators received spreadsheets with complete statutory text to ensure they did not have to read isolated sub-clauses out of context. Each reviewer also received a one-page summary sheet explaining the *Magtajas* decision steps and statutory penalty limits. These reference guides helped reviewers work consistently across their 70-pair evaluation sets.

3.5.9 Inter-Rater Reliability

To measure the consistency of the Ground Truth labels, the proponents calculate Fleiss' Kappa (κ) [71]. In statistical annotation, Fleiss' Kappa is an inter-rater reliability metric that assesses agreement among a fixed number of raters classifying items into categorical ratings. While raw percentage agreement reflects nominal concordance, it does not correct for concordance occurring purely by chance. Fleiss' Kappa calculates the degree of agreement achieved above expected chance concordance.

This metric is used instead of Cohen's Kappa because Cohen's formula is limited to exactly two raters, whereas this study assigns three reviewers to each pair. The resulting score is interpreted using the guidelines from Landis and Koch [122], where a κ score between 0.61 and 0.80 indicates "Substantial Agreement."

Using this target in natural language inference is supported by Bowman et al. [24] in their work on the Stanford Natural Language Inference (SNLI) benchmark. Evaluating text relationships involves human judgment; in their study, Bowman et al. reached an overall Fleiss' Kappa of 0.70 (scoring 0.77 for Contradiction, 0.72 for Entailment, and 0.60 for Neutral). They showed that achieving very high agreement (> 0.80) in language inference is rare and usually happens only with overly simple test sentences. Because Philippine local ordinances are more complex than general English sentences, expecting agreement higher than 0.70 is unrealistic. Therefore, this study targets a minimum κ threshold of 0.61, which confirms substantial agreement among the legal reviewers.

3.6 Hardware and Computing Setup

Rather than assuming an idealized enterprise server environment or high-cost cloud infrastructure, the proponents utilized physical workstations spanning distinct computing tiers alongside accessible cloud resources. This multi-tier setup was deliberately selected to simulate authentic local government IT conditions while supporting the computational demands of model training. Specific research workloads—ranging from archival document scanning and web scraping to deep learning fine-tuning and edge inference—were assigned to the computing environment best suited for that operational phase.

Host A (Desktop) serves as the primary testbed for data collection, preprocessing, and simulated edge execution. Powered by an Intel Core i3-10105F CPU (4 cores, 8 threads @ 3.70 GHz), 8.00 GB of DDR4 system RAM, an AMD Radeon RX 6600 GPU (8 GB GDDR6 VRAM), and high-speed NVMe solid-state storage running Microsoft Windows 10 Pro 64-bit, this machine executed the automated web scraping of the 25,432 Philippine national statutes from the Lawphil Project, raw HTML parsing, metadata structuring, and utility scripts. During edge retrieval simulation, its discrete GPU is deactivated to verify that Stage 1 sparse (BM25) and dense index lookups execute efficiently within the modest hardware limits typical of local government offices.

Host B (Laptop 1), configured on a Lenovo Legion 5 15AHP10 laptop equipped with an AMD Ryzen 7 7435HS processor (8 cores, 16 threads @ 3.80 GHz), 16.0 GB of high-speed DDR5 RAM, an NVIDIA GeForce RTX 5050 Laptop GPU with 8 GB GDDR6 VRAM, and PCIe NVMe storage running Microsoft Windows 11 Home 64-bit, provides dedicated CUDA acceleration. Host B executed the scanning, PDF text extraction, and optical character recognition (OCR) cleaning experiments for the 1,664 Davao City local ordinances, and was dedicated to Stage 2 Cross-Encoder fine-tuning, hyperparameter sweeps, training loss profiling, and full system evaluation.

Host C (Laptop 2) serves as the authentic, unassisted edge computing baseline to evaluate the system under real-world municipal hardware constraints. Configured on an HP Laptop 15s-eq1xxx powered by an AMD Ryzen 5 4500U processor (6 cores, 6 threads @ 2.38 GHz base, 15W TDP), 8.00 GB of DDR4 system RAM (7.36 GB usable @ 2667 MT/s), integrated AMD Radeon Graphics (496 MB shared VRAM, non-CUDA), and a 512 GB PCIe NVMe solid-state drive running Microsoft Windows 11 Home 64-bit, this six-year-old laptop represents the typical hardware deployed across Philippine local government units. Directly matching the DICT procurement baseline and the Commission on Audit (COA) five-year equipment useful life standard, Host C was dedicated to benchmarking pure CPU retrieval latency, measuring resident memory footprints, and validating offline edge responsiveness without discrete GPU support.

To faithfully evaluate system usability on municipal workstations, the proponents established a hardware throttling protocol during edge benchmarking and system demonstrations. Because Host A and Host B possess dedicated graphics hardware that exceeds average local government desktop specifications, their execution environments were intentionally throttled during edge latency tests: discrete GPU access was programmatically disabled via environment variables, PyTorch and ONNX runtimes were restricted strictly to CPU execution, and CPU thread pools were clamped to four worker threads. This deliberate constraint guarantees that reported query latencies, memory utilization figures, and demonstration workflows reflect the operational reality of standard local government computers rather than artificially accelerated laboratory hardware.

Complementing the physical workstations, Google's Kaggle cloud platform provided on-demand external GPU resources when compute demands exceeded local workstation memory ceilings [110]. Operating on Google Cloud Platform deep learning infrastructure [22], Kaggle dynamically provisions headless Linux virtual machine environments equipped with dedicated NVIDIA Tesla T4 GPUs (16 GB GDDR6 VRAM) and 30 GB of system RAM [110]. The environment is accessed and orchestrated headlessly through the official Kaggle command-line interface (`kaggle` API), which provides a weekly allocation of 30 GPU execution hours. This cloud tier was deployed for computationally intensive batch experiments, heavy matrix factorizations (such as 64-component Singular Value Decomposition over the dual statutory corpus), and running reproducible Cross-Encoder fine-tuning sweeps independently of local hardware constraints.

Across all physical and virtual environments, the research pipeline standardizes on an identical Python software stack to ensure full cross-platform compatibility. Table 3.9 summarizes the physical hardware configurations, operational research roles, and exact software library versions governing the study.

Table 3.9: Hardware Configurations, Software Environment, and Operational Roles across Computing Environments

Environment	**Hardware Specifications**	**Operating System**	**Assigned Research Operations**
Host A (Desktop)	Intel Core i3-10105F (4C/8T @ 3.7 GHz) 8 GB DDR4 RAM, RX 6600 8GB High-Speed NVMe SSD	Microsoft Windows 10 Pro 64-bit (Build 19045)	National statute scraping (25,432 laws), HTML parsing, text normalization, utility scripts, and throttled CPU edge simulation.
Host B (Laptop 1)	AMD Ryzen 7 7435HS (8C/16T @ 3.8 GHz) 16 GB DDR5 RAM, NVMe PCIe SSD NVIDIA RTX 5050 Laptop GPU	Microsoft Windows 11 Home 64-bit (Build 26200)	Local ordinance scanning (1,664 ordinances) and OCR cleaning; dedicated Stage 2 DL fine-tuning, training loss profiling, and full evaluation.
Host C (Laptop 2)	AMD Ryzen 5 4500U (6C/6T @ 2.4 GHz) 8 GB DDR4 RAM (7.36 GB usable) Integrated Radeon, 512 GB SSD	Microsoft Windows 11 Home 64-bit (Build 26200)	Authentic 6-year-old municipal laptop baseline; unaccelerated CPU retrieval latency, resident memory footprint, and edge validation.
Google Kaggle (Cloud Platform)	NVIDIA Tesla T4 GPU (16 GB) 30 GB System RAM Runtime	Ubuntu Linux (`kaggle` Cloud VM)	Headless batch experiments, Cross-Encoder fine-tuning sweeps, 64-component SVD matrix factorization, and remote builds.
Standardized Software Stack and Execution Libraries			
Runtime / Framework	Python 3.10+ / 3.14 PyTorch (`torch`) 2.14.0+	**Transformers**	Hugging Face (`transformers`) 5.16.1+
Dense Embeddings	`sentence-transformers` 6.0.1+	**Lexical Search**	`rank-bm25` 0.2.2+ (Okapi BM25)
ML & Statistics	`scikit-learn` 1.9.0 / `scipy` 1.18.1	**Document Ingestion**	PyMuPDF (`fitz`) 1.28.2 / `pypdf` 6.16.2
Data Processing	`pandas` 3.0.5 / `numpy` 2.5.2	**Manuscript Build**	TeX Live 2026 / MiKTeX (`pdflatex`, `bibtex`)

Table 3.9 summarizes the hardware specifications, operating systems, assigned research tasks, and core software libraries across the computing environments used in this study. This breakdown is included to ensure reproducibility and show that the system can

run on accessible consumer workstations and local government computers. As detailed in the table, routine data ingestion and throttled edge latency testing run on Host A (Core i3-10105F, 8 GB RAM), heavier deep learning fine-tuning and OCR tasks run on Host B (Ryzen 7 7435HS, RTX 5050 Laptop GPU), authentic municipal edge validation executes natively on Host C (Ryzen 5 4500U, 8 GB RAM, integrated graphics), and resource-heavy matrix calculations run on Google's Kaggle cloud platform. Using the same Python 3.10+ runtime, PyTorch 2.14, and Hugging Face Transformers across all machines ensures consistent model behavior and smooth script execution. This setup confirms that while model training benefits from dedicated GPU acceleration, the final inference pipeline runs comfortably on modest office hardware.

3.7 Pipeline Architecture

The live conflict detection workflow uses a two-stage retrieve-then-entail architecture designed to combine rapid statutory retrieval with precise logical preemption analysis. The system divides the conflict detection process into two sequential operational phases: coarse candidate retrieval across the dual statutory corpus, and fine natural language inference over paired provisions.

3.7.1 Stage 1 Hybrid Retrieval Pipeline

The first operational phase begins when a user submits a draft local ordinance for review. In information retrieval, hybrid search combines sparse lexical search and dense semantic vector search to capture both exact terminology and conceptual relevance. Sparse lexical search using BM25 evaluates exact token matches, statutory citations, numerical penalty amounts, and administrative acronyms based on term frequency and document length normalization [176]. In contrast, dense semantic search using a transformer Bi-Encoder maps draft and statutory texts into a shared continuous embedding space, computing cosine similarity to identify conceptually related provisions that share few identical words [170]. Combining these two retrieval channels using linear score interpolation is problematic because their raw scoring distributions are fundamentally incompatible: BM25 produces unbounded positive scores that fluctuate with query length, whereas dense bi-encoders output cosine similarities strictly bounded between -1.0 and $+1.0$. Reciprocal Rank Fusion (RRF) is a rank-aggregation algorithm that merges multiple retrieval rankings based strictly on the ordinal positions of documents rather than their raw numerical scores [46]. By assigning higher reciprocal rank scores to statutory provisions that appear near the top of both the lexical and semantic retrieval streams, RRF produces a balanced candidate shortlist without requiring empirical score calibration:

Reciprocal Rank Fusion (RRF) Scoring:

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{1}{k_{\mathrm{rrf}} + \mathrm{rank}_m(d)} \tag{3.9}$$

Where $\mathcal{M} = \{\mathrm{BM25}, \mathrm{Dense}\}$ represents the candidate retrieval models, $\mathrm{rank}_m(d)$ denotes the ordinal position of document d produced by model m, and $k_{\mathrm{rrf}} = 60$ is the smoothing constant.

Equation 3.9 defines the Reciprocal Rank Fusion (RRF) formula, combining the ranked search results from BM25 and the dense Bi-Encoder into a single list. RRF avoids having to normalize raw scores, since BM25 scores vary widely based on document length while cosine similarities stay between -1 and 1. By focusing on rank positions with a smoothing parameter ($k_{\text{rrf}} = 60$), the formula gives priority to provisions that score well in both keyword and semantic searches. This ensures that the candidate shortlist includes both exact citation matches and conceptually related laws, providing a balanced pool of candidates for Stage 2 inference.

Standard linear score interpolation ($\alpha S_{\text{BM25}} + (1-\alpha) S_{\text{Dense}}$) requires min-max normalization across unbounded BM25 distributions, which fluctuates wildly across diverse queries. In contrast, Reciprocal Rank Fusion operates entirely on ordinal rankings, making the combination robust against differences in score scales. The resulting fused score ranks provisions that demonstrate both lexical precision and semantic relevance at the top of the candidate pool. Listing 3.8 presents the procedural logic of the dual-stream hybrid retrieval and rank fusion pipeline.

```python
def retrieve_top_k_candidates(draft_section, statutory_corpus, k=50, k_rrf=60, beta=1.20):
    # 1. Retrieve top-100 candidates from sparse BM25 and dense Bi-Encoder
    bm25_results = search_bm25(corpus=statutory_corpus, query=draft_section, top_n=100)
    dense_results = search_dense(corpus=statutory_corpus, query=draft_section, top_n=100)

    # 2. Compute Reciprocal Rank Fusion (RRF) scores across candidate pool
    candidate_pool = set(bm25_results.keys()) | set(dense_results.keys())
    fused_scores = {}
    for prov_id in candidate_pool:
        rank_bm25 = bm25_results.get_rank(prov_id, default=101)
        rank_dense = dense_results.get_rank(prov_id, default=101)
        rrf_score = (1.0 / (k_rrf + rank_bm25)) + (1.0 / (k_rrf + rank_dense))

        # Apply hierarchy safeguard: boost primary national Republic Acts by 20%
        if prov_id.startswith("RA_") or is_primary_act(prov_id):
            rrf_score *= beta
        fused_scores[prov_id] = rrf_score

    # 3. Sort candidates by fused score and return top-k shortlist
    return sorted(fused_scores.items(), key=lambda x: x[1], reverse=True)[:k]

```

Listing 3.8: Pseudocode of the Stage 1 Reciprocal Rank Fusion Hybrid Provision Retrieval Pipeline

Listing 3.8 outlines the Python implementation of the Stage 1 hybrid retrieval pipeline, combining BM25 keyword search, dense Bi-Encoder retrieval, and Reciprocal Rank Fusion with hierarchy boosting. This code demonstrates how the system narrows down more than 27,000 statutory provisions to a focused shortlist of top-k candidates ($k = 50$). In this routine, the draft section is searched independently across sparse and dense indices to retrieve the top 100 candidates from each method. The algorithm then iterates across all retrieved provisions, calculates the RRF score using Equation 3.9, and applies a 20% score boost ($\beta = 1.20$) to primary national statutes (Republic Acts) to ensure important laws are not buried under administrative issuances. Sorting the combined scores yields a compact, high-precision candidate shortlist that delivers relevant legal context to Stage 2 inference in under 50 milliseconds.

3.7.2 Stage 1 Retrieval Safeguards

The Statutory Hierarchy Safeguard is an algorithmic re-ranking mechanism that operationalizes the constitutional doctrine of statutory supremacy established in Section 5(a) of Republic Act No. 7160 and the landmark *Magtajas* doctrine [208]. In the Philippine legal hierarchy, an ordinance enacted by a local sanggunian is legally subordinate to an Act of Congress. In the raw national statutory corpus, routine executive orders and administrative issuances comprise nearly half of all enactments (48.01%), outnumbering primary congressional legislation by thousands of documents. If a retrieval algorithm assigns equal weight across all document types, standard lexical queries can easily become saturated with administrative issuances, displacing governing Republic Acts from the top candidate ranks. The Statutory Hierarchy Safeguard applies a calibrated score multiplier ($\beta = 1.20$) to primary national statutes during rank fusion. This weighting ensures that primary national laws maintain prominence within the candidate shortlist, reflecting the constitutional supremacy of statutory law over delegated administrative rules.

Using a two-stage retrieve-then-entail setup introduces potential search challenges when comparing draft ordinances against large statutory collections. If the coarse search misses the governing national law, the inference model cannot detect the conflict regardless of its analytical capability. Legal conflicts often appear through indirect wording, where local drafts use permissive phrasing or leave out specific statutory terms. Standard search tools can also miss isolated prohibitions tucked inside long, otherwise compliant legislative codes. To protect the pipeline against these retrieval errors, the system incorporates five architectural safeguards across data preparation and search. Table 3.10 summarizes these statutory challenges alongside their operational countermeasures.

Table 3.10 summarizes six common statutory search challenges encountered when screening Philippine laws, along with the system's architectural safeguards and implementation logic in Stage 1. These safeguards were developed to address specific search errors observed during initial tests across the 25,432 national statutes. As detailed in the table, the pipeline handles isolated prohibitions through section chunking, indirect wording through dense semantic vectors, exact penalty numbers through BM25 keyword matching, ranking variation through a 50-candidate safety buffer, administrative clutter through hierarchy boosting, and long provisions through sub-chunking and ModernBERT support. Implementing these safeguards prevents sharp drops in retrieval recall, ensuring that broad laws like the Local Government Code are not left out. As a result, the candidate shortlist reliably captures both direct penalty limits and subtle statutory exceptions.

These safeguards directly resolve critical failure modes uncovered during preliminary retrieval experiments across the 25,432 national statutes. In particular, attempting a naive hard filter based on predicted macro domains caused candidate Recall@50 to collapse from 43.14% down to 13.43%. This severe degradation occurs because foundational Philippine enactments, such as Republic Act No. 7160, function as omnibus codes that span local administration, disaster response, and local taxation simultaneously. Hard-filtering by a single predicted topic immediately prunes the governing code whenever a draft clause touches another administrative area. Replacing hard filtering with a Hierarchy Priority Safeguard and soft domain priors follows the ranking principles of Robertson and

Table 3.10: Architectural Retrieval Safeguards Against Real-World Preemption Failure Modes in Stage 1

Statutory Challenge	**Architectural Countermeasure**	**Operational Implementation Logic**
Isolated Statutory Clauses	Structural Section Chunking ($\sim$ 200–300 token windows)	Segments multi-chapter enactments into discrete provisions so isolated prohibitions rank independently without vector dilution.
Indirect or Evasive Wording	Dense Bi-Encoder Semantic Vectors (`all-MiniLM-L6-v2`)	Maps conceptual intent into 384-dimensional latent space, retrieving relevant statutes despite divergent local phrasing.
Numerical Penalty Limits	Sparse BM25 Lexical Matching with Sublinear Scaling	Matches exact statutory citations, numerical penalty limits, and administrative agency acronyms directly.
Rank Variance in Shortlist	Top-k Candidate Safety Buffer ($k = 50$ provision shortlist)	Retains 50 candidate provisions for Stage 2 review, ensuring relevant statutes ranked below the top ten are evaluated.
Omnibus Statute Trap & Administrative Clutter	Hierarchy Priority Safeguard & Soft Domain Priors	Prioritizes primary legislative statutes (Republic Acts) over administrative orders, while applying gentle soft priors instead of destructive hard filtering.
Long-Tail Provisions ($>$ 512 Tokens)	Dual-Stream BM25 Invariance & Hierarchical Max-Pooling	BM25 indexes full text without sequence limits, while dense representations of tail provisions apply hierarchical sub-chunking with inference max-pooling and long-context candidate support (ModernBERT 8,192 tokens).

Zaragoza [176] and Cormack et al. [46], ensuring that multi-domain omnibus enactments are not prematurely excluded. This safeguard shields Republic Acts from the 12,210 administrative executive orders (48% of the corpus), increasing the Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% improvement).

Separating Stage 1 retrieval from Stage 2 inference is essential when running on standard office hardware. Feeding full-length statutes directly into a Cross-Encoder requires quadratic computation ($O((L_{\text{ord}} + L_{\text{stat}})^2)$), which quickly exhausts local memory on long legislative texts. Full-text inputs also dilute model attention, burying specific penalty clauses under thousands of routine procedural words. The hybrid retriever in Stage 1 performs a rapid coarse-to-fine filter ($25,432 \to 50 \to 1$), removing over 99.8% of unrelated laws in milliseconds. This separation allows the Cross-Encoder to focus its attention entirely on high-probability candidate provisions.

To empirically validate these architectural mechanisms, the proponents benchmarked nine candidate Stage 1 retrieval configurations across both the full 25,432 national statutory database and the 350-pair ground truth evaluation set. The full empirical retrieval findings, difficulty tier ablations, latency metrics, and comparative analyses are formally reported and discussed in Chapter 4 (Section 4.2).

3.7.3 Stage 2 Conflict Detection Pipeline

After receiving the top-k candidate shortlist from Stage 1, the pipeline moves to logical evaluation using the Natural Language Inference (NLI) module. The system formats an input sequence for each candidate provision, pairing the retrieved national law (the premise) with the draft ordinance (the hypothesis). To guide the model's attention, the pipeline places a $[CLS]$ token at the start of the sequence and a $[SEP]$ token between the two texts. This structure helps the model distinguish the existing law from the proposed draft. The combined text is then processed by the selected Cross-Encoder model. The model's attention mechanism compares every word in the premise against every word in the hypothesis to evaluate their relationship. A classification layer outputs calibrated softmax probabilities labeling the relationship as Entailment, Contradiction, or Neutral. If the contradiction probability crosses the designated threshold, the Explainable AI (XAI) module highlights the conflicting text for user review.

To execute full-document pre-enactment audits on resource-constrained municipal workstations without prohibitive processing delays or memory exhaustion, the Stage 2 inference pipeline incorporates three complementary efficiency mechanisms: procedural boilerplate bypassing, dynamic token padding, and cross-section flattened batching. First, rather than routing every structural section indiscriminately to the Cross-Encoder, the pipeline filters out formulaic administrative clauses (such as Title, Definition of Terms, Separability, Repealing, and Effectivity clauses) identified via the regular expression parser in Section 3.4.1. As established in the local corpus census (Section 3.4.2.4), procedural clauses represent approximately 30.85% of ordinance sections while carrying zero substantive regulatory commands; filtering them before inference directs cross-encoder attention exclusively toward operative legal duties without risk of discarding regulatory provisions. Second, standard transformer implementations enforce static padding up to the model's maximum sequence length (such as 256 or 512 tokens), allocating fixed blocks of zero-padding memory. Because transformer self-attention exhibits quadratic complexity ($O(L^2)$) with respect to input length L, computing cross-attention over padding tokens expends substantial matrix multiplication operations on uninformative positions. The pipeline replaces static padding with dynamic sequence padding, clamping tensor dimensions strictly to the length of the longest clause within each mini-batch (which typically spans 60 to 90 tokens for statutory sections). This dynamic clamping curtails quadratic attention operations relative to static maximum ceilings. Third, adapting the multi-candidate passage reranking paradigm of Nogueira and Cho [154], rather than processing candidate provisions sequentially in iterative per-section loops, the pipeline aggregates candidate pairs across all substantive sections into a unified flattened mini-batch. Executing this unified tensor through a single vectorized forward pass eliminates repeated execution call overhead and utilizes multi-threaded central processing unit vectorization (AVX2/AVX-512 SIMD parallelism).

3.7.4 Baseline Selection

To evaluate whether the proposed two-stage retrieve-then-entail architecture achieves meaningful empirical advances over standard approaches, the proponents established formal baselines for both operational stages grounded in recent legal informatics literature.

For coarse statutory retrieval, the pipeline is benchmarked against pure Okapi BM25 lexical matching adapted from Team AIIR [241] in the COLIEE 2025 competition. BM25 serves as the canonical baseline for statutory information retrieval (COLIEE Task 3) because legislative drafting frequently relies on explicit section citations, formal agency acronyms, and verbatim penalty amounts. Choosing BM25 as the Stage 1 baseline provides an essential reference point to determine whether semantic dense vector representations and Reciprocal Rank Fusion genuinely improve candidate recall, especially across paraphrased or indirect preemption queries where exact keyword matching is known to fail.

For fine natural language inference, the pipeline establishes a dual-baseline framework combining an unadapted task baseline with a canonical supervised foundation. Following the legal textual entailment baseline established by Rabelo et al. [166] for COLIEE Task 4, an unadapted zero-shot Cross-Encoder tests whether off-the-shelf general-domain or multi-task language models can detect statutory preemption without local domain adaptation. This baseline establishes the empirical lower bound and directly quantifies the severity of affirmative bias. Complementing this, the proponents establish the standard `bert-base-uncased` model (110M parameters) developed by Devlin et al. [61] as the canonical supervised baseline. Fine-tuned under identical training conditions on the curated statutory pairs, standard BERT isolates the exact performance premium contributed by specialized attention mechanisms, legal domain pretraining, and warm NLI initialization over traditional masked language modeling.

3.8 Model Selection and Optimization

Before deploying the two-stage pipeline, the proponents evaluate candidate neural models across both retrieval and inference stages to identify the most accurate and computationally efficient architectures. Rather than selecting arbitrary models or relying on closed external cloud APIs, candidate architectures are strictly bounded by the computing resources of local government hosts. Specifically, models must satisfy parameter and memory ceilings that allow real-time execution on standard desktop and laptop hardware. Once the optimal models are selected and calibrated, their weights and tokenizer configurations are permanently pinned for offline execution (`local_files_only = True`), ensuring that the production pipeline runs securely within air-gapped local government IT environments without external cloud dependencies.

3.8.1 Stage 1 Candidate Retrieval Models

To identify the optimal neural retriever for the coarse search stage, the proponents evaluate eight candidate open-weight bi-encoders representing four distinct inductive categories reviewed in Section 2.8.6:

- **Distilled Edge Baselines:** `all-MiniLM-L6-v2` (22.7M parameters) and `bge-small-en-v1.5` (33.4M parameters), engineered for sub-two-millisecond latency on commodity CPU hardware.

- **General MTEB Benchmark Leaders:** `bge-base-en-v1.5` (109M parameters), `e5-base-v2` (109M parameters), and `all-mpnet-base-v2` (109M parameters), providing balanced general-domain semantic representations.
- **Legal Domain-Adapted Encoders:** `legal-bert-base-uncased` (110M parameters), continuously pretrained over statutory and judicial corpora.
- **Modern Long-Context and Multilingual Encoders:** `ModernBERT-base` (149M parameters) and `bge-m3` (568M parameters), supporting native 8,192-token context windows and multi-language statutory representations.

All candidate bi-encoders are evaluated standalone across candidate retrieval recall (Recall@k for $k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank (MRR), mean CPU query latency, and latent paraphrastic conflict sensitivity (Tier 3). The bi-encoder delivering the optimal balance of high candidate recall and low edge latency is selected as the primary dense retrieval workhorse for the hybrid Reciprocal Rank Fusion pipeline. The empirical retrieval benchmark comparing these eight candidate bi-encoders is reported in Chapter 4 (Section 4.2).

3.8.2 Stage 2 Candidate Inference Models

To evaluate a comprehensive spectrum of transformer inductive biases for Stage 2 semantic conflict detection, the proponents scouted 28 candidate open-weight encoders spanning six architectural families:

- **Legal-Domain Pretrained Encoders:** `Legal-BERT-base-uncased` (110M), `PoL-BERT-Large` (340M), `InLegalBERT` (110M), `CaseLaw-BERT` (110M), and `Lawformer` (110M).
- **Pre-Aligned NLI Reasoners and General Baselines:** `DeBERTa-v3-base-NLI` (86M), `DeBERTa-v3-large-NLI` (435M), cold `deberta-v3-base` (86M), `bert-base-uncased` (110M), `roberta-large-mnli` (355M), `bart-large-mnli` (406M), and `electra-large-disc` (335M).
- **Modern Long-Context Architectures:** `ModernBERT-base` (149M), `ModernBERT-large` (395M), `longformer-base-4096` (149M), `bigbird-roberta-base` (128M), and `nomic-bert-2048` (137M).
- **Native Cross-Encoder Rerankers:** `bge-reranker-v2-m3` (568M), `bge-reranker-base` (278M), and `ms-marco-MiniLM-L12` (33M).
- **Multilingual Encoders:** `mdeberta-v3-base` (86M), `xlm-roberta-base` (270M), `xlm-roberta-large` (550M), and `roberta-tagalog-base` (110M).
- **Distilled Edge Baselines:** `all-MiniLM-L6-v2` (22M), `nli-distilroberta-base` (82M), `distilbert-base-uncased` (66M), and `albert-base-v2` (12M).

Evaluating all 28 candidate models through full training runs and human evaluation would require prohibitive computing time and memory. Following standard experimental practices in machine learning [63, 30], the proponents organized the evaluation into a practical Three-Stage Funnel (28 → 13 → 6 → 1). In the first stage of rapid zero-shot screening ($N = 28$), all candidate models are tested out-of-the-box on the held-out test split ($\mathcal{D}_{\text{test}}$, $N = 53$) without task fine-tuning. This initial screening rapidly benchmarks inference latency and GPU memory occupancy on standard test hardware while revealing baseline affirmative bias where unadapted models reflexively default toward Entailment or Neutral on complex statutory text [5, 221]. Promising models that satisfy baseline operational constraints advance to the second stage of supervised fine-tuning and ablation sweeps ($N = 13$). In this stage, models undergo supervised task adaptation on the training set ($N_{\text{train}} = 245$) and validation split ($N_{\text{val}} = 52$) across standard learning rates and batch sizes ($\eta \in \{1, 2, 3\} \times 10^{-5}$, $B \in \{8, 16\}$), testing empirical loss convergence, early stopping triggers, and cost-sensitive threshold calibration ($\tau^* \in [0.25, 0.50]$). Finally, in the third stage of finalist benchmarking ($N = 6$), the top five performing models on the validation split, along with the canonical bert-base-uncased baseline, are selected as finalists for full out-of-sample evaluation on the 350-pair ground truth dataset annotated by the Sangguniang Panlungsod legal researchers.

Restricting the final evaluation to the top five validation models alongside the standard baseline anchor is guided by four practical methodological considerations. First, tuning hyperparameters and selecting models strictly on validation data ($\mathcal{D}_{\text{val}}$) protects the integrity of the test set, preventing holdout data snooping and ensuring that out-of-sample generalization metrics remain methodologically unbiased [66]. Second, restricting pairwise statistical comparisons to six finalists controls family-wise error rates, avoiding the inflated risk of false positives inherent in uncorrected testing across dozens of models [62] while enabling dependable pairwise significance testing against bert-base-uncased ($\chi^2, p < 0.05$). Third, benchmarking across model scales isolates efficiency trade-offs, determining whether incremental accuracy gains from 400M-parameter architectures justify their substantial latency and VRAM footprint on municipal office hardware [194, 204]. Fourth, the finalist cohort deliberately spans diverse inductive biases, encompassing pre-aligned NLI reasoners (DeBERTa-v3-large-NLI and DeBERTa-v3-base-NLI), modern long-context encoders (ModernBERT-large), unaligned general foundations (cold deberta-v3-base), and legal domain pretraining (PoL-BERT-Large).

The comprehensive empirical screening benchmark across all 28 candidate models, zero-shot performance statistics, asterisk designations, and admission decisions are formally presented in Chapter 4 (Section 4.3).

3.8.3 Overfitting Controls and Calibration

Because fine-tuning deep transformer models in a few-shot setting ($N_{\text{train}} = 245$) can lead to memorizing training phrasing, the researchers use several regularization techniques. First, model optimization uses the AdamW algorithm with weight decay ($\lambda = 0.01$), as developed by Loshchilov and Hutter [134]. This weight decay keeps network parameters small and prevents individual weights from growing disproportionately large. Second, dropout layers with a rate of $p = 0.1$ are applied across attention layers and the classification head [203]. By turning off a random subset of neurons during training passes, dropout encourages the model to learn broader semantic features rather than depending on specific word patterns.

Third, training uses early stopping based on validation results [164]. The system measures validation loss and the validation Macro F1-Score after every training epoch on the 52-pair validation set ($\mathcal{D}_{\text{val}}$). If the validation Macro F1-Score does not improve over four consecutive checks (patience = 4), training stops automatically. The system then restores the model checkpoint that achieved the lowest validation loss, preventing the model from fitting too closely to the training data. Finally, the learning rate warms up linearly over the first 10% of training steps before decreasing steadily, which helps keep gradient updates stable during the initial phase of fine-tuning.

Recent empirical findings from the COLIEE 2026 legal textual entailment benchmarks (Task 4) demonstrated that neural language models frequently exhibit affirmative bias, defaulting toward entailment or neutral classifications when confronted with complex statutory syntax [5, 221]. In local legislative review, failing to detect an authentic conflict permits an invalid ordinance to proceed to enactment. To counter this affirmative bias, the proponents calibrate the decision threshold for the contradiction class using validation holdout curves. Rather than assuming an uncalibrated argmax decision rule, the threshold is adjusted to maximize the Contradiction F1-Score while maintaining high specificity. This calibration ensures that genuine preemption risks are surfaced to legislative staff rather than being suppressed by ambiguous phrasing.

3.8.4 Hyperparameter Tuning

To identify the optimal internal configuration for candidate Stage 2 Cross-Encoder models, the proponents implemented a systematic Grid Search procedure. Grid search is chosen over random or Bayesian optimization because the search space across lightweight transformer encoders is compact and discrete, allowing an exhaustive, reproducible evaluation of all candidate parameter combinations on the dedicated Host B (NVIDIA GeForce RTX 5050 Laptop GPU).

The search isolates key variable parameters—specifically learning rate, batch size, and maximum sequence length—while holding structural regularization constants fixed across all training iterations. Table 3.11 specifies the complete partitioning of fixed experimental parameters versus variable search ranges.

Table 3.11 specifies the complete hyperparameter search grid for Stage 2 Cross-Encoder fine-tuning, delineating fixed optimization controls from variable grid search bounds. This tuning configuration is established to achieve reproducible, stable convergence across compact transformer encoders on consumer GPU hardware while mitigating overfitting risks inherent in few-shot legal learning ($N_{\text{train}} = 245$). As detailed in the table, structural regularization constants are strictly fixed across all iterations, including AdamW optimization ($\lambda = 0.01$), attention dropout ($p = 0.10$), a 10% linear warmup schedule, early stopping (patience = 4), and a deterministic seed (seed = 42). Meanwhile, learning rate ($\eta \in \{1, 2, 3, 5\} \times 10^{-5}$), batch size ($B \in \{8, 16\}$), and sequence length ($L \in \{256, 512, 8192\}$) are systematically swept across the 52-pair validation set ($\mathcal{D}_{\text{val}}$). Quarantining hyperparameter selection strictly to the validation split prevents data snooping on the test set, guaranteeing that the checkpoint achieving peak validation Macro F_1 generalizes robustly to unseen statutory pairs.

Table 3.11: Hyperparameter Tuning Configuration and Grid Search Space for Stage 2 Cross-Encoders

Hyperparameter	**Search Status**	**Configured Value / Search Bounds**
Optimization Algorithm	Fixed	AdamW with decoupled weight decay ($\lambda = 0.01$) [134]
Attention & Hidden Dropout	Fixed	$p = 0.10$ across all transformer and classification layers [203]
Warmup Schedule	Fixed	Linear warmup over first 10% of total training steps
Early Stopping Patience	Fixed	patience = 4 validation epochs (Macro F_1 evaluation) [164]
Random Initialization Seed	Fixed	seed = 42 (enforcing deterministic mini-batch collation)
Maximum Training Epochs	Fixed	$E_{\max} = 10$ epochs (with best validation checkpoint restoration)
Loss Criterion	Fixed	Multi-Class Cross-Entropy Loss with Softmax normalization
Initial Learning Rate (η)	Variable (Grid Search)	$\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}\}$
Training Batch Size (B)	Variable (Grid Search)	$B \in \{8, 16\}$ (governed by 8 GB VRAM execution ceilings)
Max Input Sequence Length (L)	Variable (Grid Search)	$L \in \{256, 512\}$ (8,192 tokens for native ModernBERT)

All hyperparameter configurations are evaluated strictly on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) to prevent test-set data contamination. Model checkpoints are saved after every epoch, and the configuration achieving the highest validation Macro F_1-score is selected for final out-of-sample evaluation on the 53-pair test set.

3.9 Evaluation Metrics

Evaluating the two-stage pipeline requires separate metrics for each operational phase. The coarse retrieval phase in Stage 1 prioritizes candidate recall over final classification, while the inference phase in Stage 2 requires high precision to avoid false conflict alerts.

3.9.1 Stage 1 Retrieval Metrics

Measuring search performance in Stage 1 requires ranking metrics rather than standard classification accuracy [140]. In information retrieval, ranking metrics evaluate how effectively a search system places relevant documents at or near the top of a retrieved candidate list. In this pipeline, Stage 1 does not produce a final classification decision; its objective is to ensure that the governing national statute is successfully captured within the top-k candidate shortlist ($k \in \{10, 30, 50\}$) passed to Stage 2. Recall at rank k (Recall@k) defines the proportion of test queries for which the target statutory provision appears within the top k retrieved positions. If a governing statute fails to appear within the top-k cutoff, the downstream Cross-Encoder cannot evaluate the pair, resulting in a false negative retrieval omission. Complementing candidate recall, Mean Reciprocal Rank (MRR@k)

measures retrieval precision by averaging the multiplicative inverse of the rank of the first relevant document across all test queries (1/rank). A high Recall@50 ensures candidate safety for downstream inference, while a higher MRR indicates that relevant provisions rank near the top of the shortlist. The metric is formalized as follows:

Recall at Rank k (Recall@k):

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.10}$$

Where $|Q|$ is the total number of test queries, d_q^* is the correct national statute paired with query q, and $\mathbf{1}(\cdot)$ is an indicator function that equals one when the target law ranks within the top k candidates.

Complementing candidate recall, Mean Reciprocal Rank at rank k (MRR@k) evaluates how early the target statute appears within the candidate ranking [140]:

Mean Reciprocal Rank at Rank k (MRR@k):

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}(d_q^*)} \cdot \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.11}$$

Where d_q^* is the correct national statute retrieved for query q, with $1/\text{rank}(d_q^*)$ representing the reciprocal rank score.

Equations 3.10 and 3.11 define the ranking metrics used to test Stage 1 retrieval across the 25,432 national statutes. Because the goal of coarse retrieval is to create a top-k shortlist ($k \in \{10, 30, 50\}$) rather than making final decisions, standard accuracy is not suitable. Recall@k measures the percentage of test queries where the correct statute appears in the top-k shortlist, while MRR@k measures how close to the top that statute ranks. High Recall@50 ensures that Stage 2 inference receives the governing statute needed to detect a conflict, while higher MRR places the most relevant law near the top of the list for legal researchers to review. These two metrics guide the selection of the best retrieval models in Chapter 4.

In this formula, a target statute retrieved at rank one receives a reciprocal score of 1.0, while a statute retrieved at rank ten receives 0.10. The 350 curated premise-hypothesis pairs serve as the query benchmark for testing Stage 1. By using the 350 draft clauses as queries against the 25,432-statute collection, the proponents measure how often candidate retrieval models locate the paired statutory provision. Calculating Recall@10, Recall@30, and Recall@50 confirms whether the coarse retriever supplies a dependable candidate pool for subsequent natural language inference.

3.9.2 Stage 2 Inference Metrics

The Stage 2 Cross-Encoder evaluates candidate statutory pairs across three discrete logical categories: $\mathcal{C} = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$. Following standard evaluation conventions in information extraction and natural language processing [36, 228], the baseline inference performance metrics are computed as follows:

Class-Specific Precision, Recall, and F_1-Score:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{3.12}$$

Where TP_c, FP_c, and FN_c represent the true positive, false positive, and false negative classifications for category c, respectively. Precision reflects how reliably the model avoids false alarms, while Recall measures the proportion of genuine legal relationships successfully detected.

Macro-Averaged F_1-Score Across Logical Classes:

$$\text{Macro } F_1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F_{1,c} \tag{3.13}$$

Where $|\mathcal{C}| = 3$ ensures that performance across all three categories is evaluated equally without allowing dominant neutral pairs to skew the outcome.

Equations 3.12 and 3.13 define the performance metrics for Stage 2 inference across the three NLI categories: Entailment, Neutral, and Contradiction ($|\mathcal{C}| = 3$). Precision (P_c) measures how well the model avoids false alarms, Recall (R_c) measures how many actual conflicts or relationships are caught, and the F_1-score balances both. Macro F_1 averages these scores equally across all three classes (1/3 each), ensuring that a model cannot achieve an artificially high score simply by predicting the majority class. This provides a balanced measure for comparing candidate models during screening.

In statutory conflict detection, classification errors carry fundamentally unequal operational costs [68, 70]. Cost-sensitive evaluation is a machine learning methodology that incorporates asymmetric misclassification costs into performance metrics and decision threshold calibration [68]. In conventional classification, false positives (Type I errors) and false negatives (Type II errors) are penalized equally. In local legislative review, however, their operational consequences are highly asymmetric. A false negative—failing to identify an illegal or preempted ordinance section—allows a legally defective measure to advance to enactment, risking judicial invalidation by the Supreme Court under the *Magtajas* doctrine [208] and enforcement of unlawful penalties against citizens. Conversely, a false positive—flagging a compliant provision for committee review—merely requires a legal researcher to inspect the side-by-side text comparison and dismiss the informational notice. To reflect this asymmetric penalty structure, the evaluation adapts the F_2-score, which weights recall twice as heavily as precision ($\beta = 2$) to prioritize capturing potential contradictions.

Because a missed preemption is catastrophic while a false alarm is a minor momentary inconvenience, setting the decision threshold at an arbitrary 50% ($P \geq 0.50$) is dangerous. Standard classifiers default to a symmetric decision threshold of $P(\text{Contradiction}) \geq 0.50$, which assumes that false alarms and missed conflicts cause identical harm. In foundational cost-sensitive learning theory, Elkan [68] proved that when false negative costs exceed false positive costs ($C(FN) > C(FP)$), setting the decision boundary at 0.50 is mathematically suboptimal and produces an unsafe number of missed conflicts [70, 118].

To address this cost asymmetry, the proponents adapt the generalized F_β formulation of Van Rijsbergen [228] and Chinchor [36] with $\beta = 2$, implementing a validation-tuned decision boundary τ_{conflict} for contradiction alerts. Rather than using a fixed 0.50 cutoff, the pipeline sweeps candidate thresholds across $\tau \in [0.25, 0.50]$ in increments of $\Delta\tau = 0.01$ exclusively on the validation set ($\mathcal{D}_{\text{val}}$, $N = 52$) following empirical decision calibration principles [70]. The cost-sensitive contradiction metrics are computed as follows:

Cost-Sensitive Contradiction F_2-Score:

$$F_{2,\text{conflict}} = \frac{5 \cdot P_{\text{conflict}} \cdot R_{\text{conflict}}}{4 \cdot P_{\text{conflict}} + R_{\text{conflict}}} \tag{3.14}$$

Where P_{conflict} and R_{conflict} denote Precision and Recall on the Contradiction class, weighting recall twice as heavily as precision ($\beta = 2$) to aggressively capture potential legal conflicts.

Optimal Contradiction Decision Threshold Calibration (τ^*):

$$\tau^* = \arg \max_{\tau \in [0.25, 0.50]} F_{2,\text{conflict}}(\tau; \mathcal{D}_{\text{val}}) \tag{3.15}$$

Where $\mathcal{D}_{\text{val}}$ denotes the 52 validation pairs used to sweep candidate decision thresholds $\tau \in [0.25, 0.50]$ while enforcing an empirical precision floor of $P_{\text{conflict}} \geq 0.70$.

Equations 3.14 and 3.15 define the cost-sensitive evaluation metric and decision threshold calibration for conflict alerts. This design reflects a practical reality: missing a real conflict (a false negative) allows an invalid ordinance to pass, which can lead to legal challenges, whereas a false alarm (a false positive) is quickly dismissed by a legal researcher. In Equation 3.14, setting $\beta = 2$ weights recall twice as heavily as precision, prioritizing catching potential conflicts. Equation 3.15 searches for the best decision threshold τ^* between 0.25 and 0.50 on the 52 validation pairs ($\mathcal{D}_{\text{val}}$), while keeping precision at or above 0.70. This tuning helps the system flag genuine conflicts reliably without overwhelming researchers with too many false alarms.

During live inference, a candidate provision triggers a formal conflict alert whenever its softmax contradiction probability satisfies $P(\text{Contradiction} \mid p_i, h) \geq \tau^*$.

When evaluating complete multi-section ordinances during live system execution, Stage 1 retains its full safety buffer of $k = 50$ candidates for each substantive section (S_i). To prevent runtime delays when scoring up to 500 total candidate pairs per document, the Stage 2 inference engine employs vectorized mini-batch collation ($B = 16$). Batching evaluates candidate provisions in parallel tensor forward passes rather than sequential single-item loops, keeping inference runtime under five seconds on GPU hosts and under thirty seconds on standard LGU processors. To consolidate candidate provision scores into an overall section verdict, the pipeline applies section-level max-pooling, adapting multi-instance learning principles [8]. Section-level max-pooling is an aggregation function that maps the set of k individual contradiction probabilities for a given draft section to its maximum observed value: $P(\text{Conflict} \mid S_i) = \max_j P(\text{Contradiction} \mid P_j, S_i)$. In statutory review, an ordinance section that complies with 49 retrieved provisions but contradicts a single governing statute remains legally invalid under statutory preemption. Computing an arithmetic mean across candidate scores would dilute a high contradiction probability (such as 0.85) down to an undetectable fraction, causing a critical false negative. Max-pooling isolates the peak conflict probability across all candidate pairs, ensuring that an authentic legal violation triggers a system alert regardless of the number of non-conflicting candidates retrieved. The formula is computed as follows:

Section-Level Max-Pooled Conflict Probability:

$$P(\text{Conflict} \mid S_i) = \max_{j \in [1,k]} P(\text{Contradiction} \mid P_j, S_i) \tag{3.16}$$

Where P_j ranges over all $k = 50$ candidate statutory provisions retrieved for draft section S_i.

Equation 3.16 calculates the overall conflict score for a draft section (S_i) by taking the highest contradiction probability among all $k = 50$ retrieved candidate laws. This approach, known as max-pooling, is used during full-document review where each draft clause is compared against multiple candidates. Taking the highest score ensures that if a draft section clearly conflicts with even one national law, an alert is triggered, even if the other 49 retrieved candidates are neutral or unrelated. This directs the reviewer's attention to the specific statute causing the conflict, preventing an actual violation from being hidden by non-conflicting candidates.

A section triggers an operative conflict warning if and only if $P(\text{Conflict} \mid S_i) \geq \tau^*$, with the system highlighting the specific governing candidate provision $P_j^* = \arg\max_j P(\text{Contradiction} \mid P_j, S_i)$ that produced the highest conflict probability. This max-pooling aggregation ensures that an authentic statutory clash ranked lower in the candidate shortlist is not obscured by neutral or harmonious provisions appearing at higher retrieval ranks. Listing 3.9 outlines the operational inference and conflict aggregation procedure.

```python
def detect_section_conflicts(draft_section, candidate_provisions, model, threshold=0.35, batch_size=16):
    """
    Stage 2 Cost-Sensitive Cross-Encoder Inference:
    Scores candidate law pairs and flags conflicts using max-pooling.
    """
    # 1. Pair each candidate statutory provision (premise) with the draft section (hypothesis)
    pairs = [(provision.text, draft_section.text) for provision in candidate_provisions]

    # 2. Run mini-batch inference through the fine-tuned Cross-Encoder
    contradiction_scores = []
    for i in range(0, len(pairs), batch_size):
        batch = pairs[i : i + batch_size]
        # Get softmax probabilities across [Entailment, Neutral, Contradiction]
        probabilities = model.predict_probabilities(batch)
        # Extract contradiction class probability (index 2)
        batch_contradictions = [p["contradiction"] for p in probabilities]
        contradiction_scores.extend(batch_contradictions)

    # 3. Find candidate provision with highest contradiction score (Max-Pooling)
    peak_score = max(contradiction_scores)
    peak_index = contradiction_scores.index(peak_score)
    governing_provision = candidate_provisions[peak_index]

    # 4. Apply cost-sensitive decision threshold
    has_conflict = peak_score >= threshold

    return {
        "alert": has_conflict,
        "conflict_probability": peak_score,
        "conflicting_law": governing_provision.citation,
        "conflicting_text": governing_provision.text
    }

```

Listing 3.9: Pseudocode of the Stage 2 Cost-Sensitive Cross-Encoder Inference and Section-Level Conflict Aggregation

Listing 3.9 outlines the Python implementation of the Stage 2 Cross-Encoder inference and section-level conflict aggregation routine. This algorithm is provided to illustrate the mini-batch tensor processing and cost-sensitive thresholding that transforms candidate provision pairs into actionable preemption alerts. In the procedure, the candidate statutory provisions and draft section are paired into premise-hypothesis sequences and evaluated in vectorized mini-batches ($B = 16$) through the fine-tuned model. The function extracts softmax contradiction probabilities, identifies the peak score via max-pooling according to Equation 3.16, and checks whether this peak crosses the calibrated decision threshold τ^*. If an alert is triggered, the routine returns the conflicting statutory citation, text, and model confidence, delivering immediate diagnostic transparency to the committee review and Second Reading preparation dashboard.

3.9.3 Statistical Hypothesis Testing

To evaluate whether performance disparities between candidate Cross-Encoder models represent statistically significant differences rather than random sampling fluctuations, the proponents employ formal non-parametric hypothesis tests recommended by Demšar [55]. In machine learning evaluation, non-parametric statistical tests evaluate differences between algorithms without requiring assumptions of normality or continuous error distributions [55]. Parametric paired t-tests assume that classification errors follow a continuous Gaussian distribution, an assumption violated by discrete, binary classification outcomes (correct versus incorrect) on small evaluation sets ($N = 53$). McNemar's test is a non-parametric statistical test for paired nominal data that analyzes 2×2 contingency tables of discordant predictions [145]. By isolating cases where two competing models produce divergent classifications (cases where Model A is correct and Model B is incorrect, and vice versa), McNemar's test evaluates whether one model's marginal superiority is statistically significant. Complementing pairwise comparisons, the Friedman test is a non-parametric two-way analysis of variance by ranks that evaluates multiple candidate models across multiple datasets or task domains [73]. By ranking candidate models within each of the eight Macro Legal Domains and assessing the consistency of their average ranks, the Friedman test verifies whether an architecture achieves consistent performance across all statutory subject areas without assuming normal distributions.

When comparing the classification accuracy of two competing Stage 2 models (such as fine-tuned DeBERTa-v3 versus fine-tuned ModernBERT) evaluated on the identical 53-item test set ($\mathcal{D}_{\text{test}}$), the proponents employ McNemar's test for pairwise significance [145].

The test constructs a 2×2 contingency table analyzing discordant predictions, as summarized in Table 3.12. In this contingency framework, cell b represents items correctly classified by Model A but misclassified by Model B, while cell c represents items correctly classified by Model B but misclassified by Model A. Under the null hypothesis $H_0 : p_b = p_c$, the two models have equivalent error rates.

Table 3.12: Contingency Matrix of Discordant Predictions for McNemar's Test

	Model B: Correct	**Model B: Incorrect**
Model A: Correct	a (Both Correct)	b (Model A Only)
Model A: Incorrect	c (Model B Only)	d (Both Incorrect)

Table 3.12 presents the 2×2 contingency matrix of discordant and concordant predictions utilized by McNemar's test for pairwise model significance testing. This matrix structure is incorporated to isolate the exact instances where two competing Cross-Encoder architectures disagree on the 53-item test split ($\mathcal{D}_{\text{test}}$). In the layout, cells a and d capture concordant outcomes where both models are simultaneously correct or incorrect, while off-diagonal cells b and c record discordant predictions where only Model A or Model B classifies the pair correctly. Under the null hypothesis, the two models share equivalent error rates ($p_b = p_c$). By evaluating only discordant pairs rather than raw overall accuracy, the test provides high statistical sensitivity to marginal algorithmic improvements while remaining robust against non-normal classification error distributions.

Because the Stage 2 evaluation compares three candidate models ($k = 3$: MiniLM, DeBERTa-v3, ModernBERT), pairwise evaluation yields $m = \binom{3}{2} = 3$ distinct model comparisons. To prevent Type I error inflation (false positive significance) across multiple tests, the proponents apply the standard Bonferroni correction [55], evaluating each pair against the adjusted threshold $\alpha' = \alpha/m = 0.05/3 \approx 0.0167$.

Furthermore, when expanding candidate model explorations to larger suites (e.g., $k \approx 10$ architectures and hyperparameter configurations), running exhaustive pairwise McNemar tests produces $\binom{k}{2} = 45$ comparisons, creating severe statistical conservatism under Bonferroni correction ($\alpha' = 0.05/45 \approx 0.0011$). To handle larger candidate pools without losing statistical power, the proponents follow Demšar's two-step testing procedure [55]: first executing the omnibus Friedman test across all k models simultaneously, and only upon rejecting the global null hypothesis ($p < 0.05$), conducting post-hoc comparisons comparing candidate models strictly against the top-performing control model ($k - 1$ comparisons rather than $\frac{k(k-1)}{2}$) or evaluating critical difference intervals via the Nemenyi test.

For pairwise statistical significance, following the recommendations of Dietterich [62] for comparing supervised machine learning algorithms, the proponents employ McNemar's chi-squared test [145] with Edwards' continuity correction [67]. The test evaluates whether discordant classifications between Model A and Model B occur in equal proportions:

McNemar's Chi-Squared Test with Edwards' Continuity Correction:

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} \sim \chi^2_1 \tag{3.17}$$

Where b represents items correctly classified by Model A but misclassified by Model B, c represents items correctly classified by Model B but misclassified by Model A, and the numerator incorporates Edwards' continuity correction (-1) for discrete cell frequencies.

Equation 3.17 defines McNemar's chi-squared test with Edwards' continuity correction for comparing paired predictions. The formula measures whether the number of times Model A succeeds where Model B fails (b) differs significantly from the reverse case (c). The minus-one correction accounts for discrete count data, and dividing by total disagreements $(b+c)$ produces a test statistic that follows a chi-squared distribution with one degree of freedom. In practice, this test checks whether one model genuinely outperforms another on difficult borderline cases rather than by chance. If the resulting p-value falls below the Bonferroni-adjusted threshold $(\alpha' = 0.0167)$, the null hypothesis is rejected, confirming that the accuracy difference between the two models is statistically significant.

Beyond pairwise significance, assessing the consistency of candidate architectures across the eight Macro Legal Domains without making restrictive distributional assumptions requires non-parametric multi-model ranking. For this purpose, the proponents utilize the Friedman non-parametric two-way analysis of variance by ranks [73, 55]. For k candidate models evaluated across $N = 8$ distinct legal domains, each model is assigned a rank r_i^j within domain i, where the best-performing model receives rank 1. The average rank of model j across all domains is given by $R_j = \frac{1}{N}\sum_{i=1}^{N} r_i^j$. The omnibus rank statistic is computed as follows:

Friedman Two-Way Analysis of Variance by Ranks (χ^2_F):

$$\chi^2_F = \frac{12N}{k(k+1)}\left[\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right] \tag{3.18}$$

Where $N = 8$ is the number of macro legal domains, k is the number of candidate models, and $R_j = \frac{1}{N}\sum_{i=1}^{N} r_i^j$ is the average rank of model j across all domains (r_i^j being the rank of model j in domain i).

Equation 3.18 computes the Friedman rank test statistic (χ^2_F) to compare candidate models across multiple legal domains without assuming normal distributions. The formula uses the average ranks (R_j) of each model across the eight macro legal domains, comparing observed rank differences against what would be expected if all models performed equally. Under the null hypothesis that all models have similar performance across domains, χ^2_F follows a chi-squared distribution with $k-1$ degrees of freedom. If the test rejects this null hypothesis ($p < 0.05$), it indicates meaningful performance differences across legal domains, allowing post-hoc tests such as the Wilcoxon signed-rank test or Nemenyi comparisons to identify which models generalize best across different areas of law [55].

3.10 System-Level Evaluation

While isolated benchmarks measure retrieval recall and natural language inference accuracy separately, evaluating the live utility of the system requires testing both stages as an integrated pipeline. In real legislative drafting, an inference engine cannot detect a contradiction if the retrieval engine fails to supply the governing statute, and a retrieval engine is useless if the inference engine misclassifies the pair. Following the multi-stage system evaluation standards established in legal information extraction competitions [166, 5] and fact verification benchmarks [219], the proponents implement a structured Three-Tier Evaluation Framework. This multi-tier framework connects controlled statistical benchmarking with full-document stress testing. Together, these three evaluation tiers systematically assess whether the complete pipeline can operate reliably within the Davao City Sangguniang Panlungsod.

The first evaluation tier measures the combined algorithmic performance of Stage 1 retrieval and Stage 2 inference when executed sequentially without human intervention. Adapting the joint scoring formulation established in the FEVER fact verification framework by Thorne et al. [219], the cascaded benchmark feeds only the local draft hypothesis clause h_q from the 53 held-out test pairs into Stage 1 coarse retrieval. The retrieval engine searches across all 27,000+ national statutes and local ordinances to extract a candidate shortlist of the top-k provisions ($\mathcal{K}_q^{(k)}$), which is then passed directly to Stage 2 for inference. A test item is scored as a Cascaded True Positive if and only if Stage 1 retrieves the correct governing statute p_q^* within the top-k shortlist and Stage 2 correctly classifies the pair (p_q^*, h_q) as a Contradiction:

End-to-End Pipeline Hit Indicator (FEVER Metric):

$$\mathrm{Hit}_{\mathrm{casc}}(q) = \mathbf{1}\left(p_q^* \in \mathcal{K}_q^{(k)}\right) \cdot \mathbf{1}\left(\hat{y}(p_q^*, h_q) = y_q^*\right) \tag{3.19}$$

Where p_q^* is the target governing statute, $\mathcal{K}_q^{(k)}$ is the candidate shortlist of top-k provisions retrieved in Stage 1, and $\hat{y}(p_q^*, h_q) = y_q^*$ indicates whether Stage 2 correctly classifies the candidate pair as a contradiction.

Equation 3.19 defines the joint pipeline hit metric ($\mathrm{Hit}_{\mathrm{casc}}$) based on the end-to-end evaluation approach of Thorne et al. [219]. This metric counts a test item as a success only when both stages work correctly in sequence. Specifically, the formula multiplies two check conditions: whether Stage 1 retrieval includes the true governing statute (p_q^*) in its top-k shortlist, and whether Stage 2 inference predicts the correct gold label (y_q^*). Combining retrieval recall and inference accuracy this way prevents a high score in one stage from hiding failures in the other. In real legislative workflows, successfully catching a conflict requires both finding the relevant statute and correctly identifying the contradiction.

Adapting this joint metric from the FEVER fact verification framework [219] isolates whether an operational error stems from retrieval omissions ($p_q^* \notin \mathcal{K}_q^{(k)}$) or classification errors ($\hat{y} \neq y_q^*$), preventing either stage from masking pipeline failures.

The second evaluation tier assesses how the complete coarse-to-fine pipeline processes full, multi-section legislative drafts rather than isolated sentence pairs. In local legislative practice, proposed ordinances contain extensive procedural text, including administrative definitions and standard effectivity clauses, which should not trigger conflict warnings. Adapting the behavioral perturbation and fault-injection principles formulated by Ribeiro et al. [174] and Koreeda and Manning [118], the proponents construct synthetic and semi-synthetic ordinance drafts containing controlled legal faults. In each test document, the majority of sections remain compliant with superior law, while targeted operative provisions contain deliberate statutory violations, such as unauthorized regulatory mandates or excessive penalty amounts. The automated preprocessor chunks the submitted document into operative sections, auditing every section independently against the statutory archive.

Evaluating full documents measures both the sensitivity of the pipeline to authentic legal defects and its ability to avoid false alarms on compliant text. Following standard diagnostic classification metrics [7, 70], the proponents calculate Section-Level Sensitivity to measure the percentage of injected conflicts successfully flagged, paired with Section-Level Specificity to assess false-alarm resistance:

Section-Level Specificity for Full-Document Stress Testing:

$$\text{Specificity}_{\text{sec}} = \frac{TN_{\text{sec}}}{TN_{\text{sec}} + FP_{\text{sec}}} \tag{3.20}$$

Where TN_{sec} denotes compliant draft sections correctly passing without triggering a false alarm, and FP_{sec} denotes compliant draft sections erroneously flagged as contradictory.

Equation 3.20 defines Section-Level Specificity ($\text{Specificity}_{\text{sec}}$) for the Tier 2 full-document test. This metric measures how well the pipeline avoids false alarms when checking long drafts that include standard administrative sections and boilerplate clauses. The formula calculates the percentage of compliant sections (TN_{sec}) that correctly pass without triggering an alert out of all non-conflicting sections. In local legislative work, frequent false warnings on routine clauses lead to alert fatigue, causing researchers to doubt the tool. Maintaining high specificity ensures that conflict alerts remain credible and that researchers can focus their attention on genuine legal risks.

Maintaining high specificity protects legislative researchers from alert fatigue, ensuring that the system does not produce spurious warnings on standard boilerplate clauses. This perturbation stress test confirms that the system can analyze complete legislative documents before deployment in committee reviews.

The third evaluation tier validates the practical utility and legal soundness of the system using documented historical legal disputes and active expert review. Rather than relying solely on synthetic tests, the proponents systematically assembled an authentic judicial validation suite of eight landmark legal controversies and local enactments originating strictly within Davao City. These historical cases encompass both judicial invalidations where local enactments were struck down for exceeding statutory boundaries, and landmark rulings where the Supreme Court affirmed the validity of Davao City ordinances under local police power. By subjecting the pipeline to actual local disputes that reached the judiciary, this tier evaluates whether algorithmic contradiction detection mirrors authentic judicial reasoning. The resulting test suite spans multiple substantive categories, including agrochemical oversight, structural zoning, revenue powers, and public health mandates.

The judicial cases examine distinct facets of city governance, beginning with environmental regulation and structural zoning. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [211], the Supreme Court invalidated Davao City Ordinance No. 0309-07 (imposing a total ban on aerial pesticide spraying in agricultural banana plantations) for violating equal protection and invading the exclusive regulatory jurisdiction vested in the Fertilizer and Pesticide Authority under Presidential Decree No. 1144. Conversely, in the Davao City Watershed Protection dispute under Ordinance No. 0310-07 [184], the city's total prohibition on mineral extraction operates in direct vertical tension with State mineral concessions under Republic Act No. 7942. For urban structural governance, the Supreme Court in *Hon. Leoncio Evasco, Jr. v. Alex P. Montañez* [213] reversed the Court of Appeals and explicitly upheld Davao City Ordinance No. 092-2000 regulating outdoor billboards. The Court affirmed that under Section 458(a)(4)(iv) of Republic Act No. 7160, city councils possess express statutory police power to regulate advertising structures harmoniously alongside the National Building Code (Presidential Decree No. 1096).

The remaining landmark disputes focus on local revenue authority, public health, and delegated traffic administration. In *City of Davao and Tanjili v. ARC Investors, Inc.* [214], the Supreme Court voided Davao City's local business tax assessment under Ordinance No. 158-05, ruling that an LGU cannot unilaterally reclassify a passive holding company as a non-bank financial intermediary without Bangko Sentral ng Pilipinas authorization. Similarly, in *City of Davao v. First Meridian Development, Inc.* [215], the Court invalidated Section 423 of Ordinance No. 158-05 for imposing an unauthorized payment-under-protest requirement contrary to Section 195 of Republic Act No. 7160. By contrast, the Court in *City of Davao v. Court of Appeals and GSIS* [209] affirmed city taxing authority over government corporate assets after Republic Act No. 7160 withdrew prior statutory exemptions. In delegated public health and traffic safety, Davao City enacted landmark measures setting stricter local standards, such as the Comprehensive Anti-Smoking Ordinance (Ordinance No. 0367-12) [185] and the Comprehensive Speed Limit Ordinance (Ordinance No. 0270-23) [187] pursuant to Republic Act No. 4136 and Joint JAO 2018-01.

Table 3.13 presents the complete Davao City local validation suite, detailing the local measures, governing national statutes, legal ratios decidendi, and gold NLI labels. To ensure long-term reproducibility and immediate system availability, complete archival copies of these eight Davao City ordinances and tax measures have been assembled and stored locally within the repository's local corpus (`corpus/davao_city_tier3_ordinances/`). Standardized diagnostic problem-solution pairs are indexed in `data/tier3_jurisprudential_cases.jsonl`, pairing operative draft excerpts with corresponding statutory provisions and judicial holdings. By feeding authentic legal questions and challenged provisions directly into Stage 1 retrieval and Stage 2 inference, the evaluation confirms whether the automated pipeline surfaces the exact statutory citations and preemption rationales formulated by judicial authorities. This authentic benchmark grounds the quantitative metrics in real jurisprudence, ensuring that theoretical inference gains translate into valid legal oversight.

Table 3.13: Landmark Davao City Local Jurisprudential Suite for Tier 3 System Evaluation ($N = 8$ Local Ordinances and Preemption Disputes)

Case Title	**Docket / Year**	**Challenged Local Ordinance**	**Governing Statute & Rationale**	**Gold NLI Label**
Mosqueda v. PBGEA [211]	G.R. 189185 (2016)	Davao City Ord. 0309-07 (Banning Aerial Pesticide Spraying)	PD 1144; FPA holds exclusive jurisdiction over agrochemical application.	Contradiction
Evasco v. Montañez [213]	G.R. 199172 (2018)	Davao City Ord. 092-2000 (Regulating Billboard Structures)	RA 7160 §458; Upheld as valid local police power alongside PD 1096.	Entailment
Davao v. ARC Investors [214]	G.R. 249668 (2022)	Davao City Ord. 158-05 §135 (Business Tax on Holding Dividends)	RA 7160 §143(f); Holding firm is not an NBFI without BSP license.	Contradiction
City of Davao v. First Meridian [215]	G.R. 240078 (2022)	Davao City Ord. 158-05 §423 (Mandatory Payment Under Protest)	RA 7160 §195; LGU cannot require payment under protest for business taxes.	Contradiction
City of Davao v. GSIS [209]	G.R. 127383 (2005)	Real Property Tax Levies on GSIS Government Properties	RA 7160 §193, §234; LGC withdrew tax exemptions of GOCCs; assessment valid.	Entailment
Davao Mining Ban Dispute [184]	Ord. 0310-07 (2015)	Citywide Total Ban on Commercial Mining Operations	RA 7942 §4, §27; Regalian doctrine and national mineral concession supremacy.	Contradiction
Davao Anti-Smoking Dispute [185]	Ord. 0367-12 (2012)	Stricter Outdoor/Indoor Smoking Distance Setbacks	RA 7160 §16; Stricter local public health standards valid under police power.	Entailment
Davao Speed Limit Dispute [187]	Ord. 0270-23 (2023)	Comprehensive Speed Limits on City Thoroughfares	RA 4136 §35; Joint JAO 2018-01 expressly delegates road speed setting to LGUs.	Entailment

Table 3.13 details the eight landmark Davao City legal controversies and judicial rulings comprising the Tier 3 authentic jurisprudential validation suite ($N = 8$). This empirical suite is assembled to evaluate whether the automated coarse-to-fine pipeline correctly reproduces authentic judicial reasoning when auditing contested local enactments against superior national statutes. The table catalogues the judicial case citation, docket number, challenged Davao City ordinance, governing national legislative framework, and authoritative gold NLI relationship established by the Supreme Court of the Philippines. Across these cases, the benchmark captures both vertical preemption invalidations (such as pesticide regulation under PD 1144 and city tax overreach under

RA 7160) and valid exercises of devolved police power (including billboard regulation under RA 7160 and localized anti-smoking mandates). By grounding algorithmic testing in documented historical jurisprudence, this evaluation verifies that the Cross-Encoder's calibrated confidence scores correspond to substantive constitutional doctrine under the *Magtajas* principle.

While the Tier 3 jurisprudential suite benchmarks model decisions against historical judicial holdings, verifying whether the system functions effectively as an interactive advisory tool requires evaluating the complete prototype with active legislative practitioners.

3.11 Prototype and User Interface

The completed pipeline is deployed as an accessible, single-page web application designed for standard office desktop computers in local government units. The interface uses a clean, intuitive layout that mirrors familiar document conversion tools. Users begin by uploading digital draft ordinances in common text or PDF formats through a drag-and-drop file ingestion area. After submission, an active progress indicator keeps the user informed while the backend pipeline parses operative sections, runs Stage 1 hybrid retrieval across the 27,000+ statutory knowledge base, and executes Stage 2 Cross-Encoder inference. This streamlined layout avoids complex parameter settings or technical menus, allowing legal staff to conduct pre-enactment conflict checks quickly during active committee sessions.

When the analysis finishes, the interface presents a structured diagnostic dashboard organized to support rapid human verification. The top of the screen displays an executive summary showing the total number of audited sections, the count of flagged potential conflicts, and the overall statutory compliance status of the draft. Below the summary, an interactive table lists each audited draft section alongside its corresponding Stage 1 candidate provisions. Clicking any flagged item expands a dedicated review card that presents the local ordinance section and the retrieved governing statute in a direct side-by-side comparative reading view. Placing the two texts side by side allows legal researchers to inspect the alleged contradiction immediately without switching between multiple browser tabs or searching through printed statute digests, substantially lowering cognitive load during legislative review [216].

To make the system's reasoning transparent and easy to interpret, the interface incorporates an Explainable AI (XAI) module focused directly on reading clarity and practical decision support [64, 180]. Rather than displaying abstract neural activation weights or complex probability matrices, the interface provides intrinsic visual cues by highlighting the exact words that cause legal friction. The Cross-Encoder's internal cross-attention weights between the premise statute and the draft hypothesis are normalized into soft visual highlights directly over the text. The display visually contrasts opposing deontic modal operators, distinguishing strict prohibitions (such as *prohibited*, *unlawful*, or *shall not*) from statutory permissions (such as *authorized*, *allowed*, or *may*). In addition, matching regulated subject terms and conflicting numbers (such as local fines exceeding statutory penalty ceilings or differing distance setbacks) are highlighted in complementary colors, enabling researchers to identify the specific source of legal tension in seconds.

To support actionable decision-making, the dashboard organizes detected inconsistencies using a two-tiered alert hierarchy. Critical Vertical Preemption Alerts are displayed with red warning banners, signaling that a proposed local provision directly contradicts a superior national statute under the *Magtajas* doctrine and risks being declared *ultra vires*. Informational Horizontal Notices are shown with blue informational banners, notifying the researcher that the draft overlaps with an existing Davao City ordinance and may cause unintended duplication or implied repeal. Accompanying each alert is a concise, plain-language *ratio decidendi* summary explaining the legal basis of the warning (for example, stating that a local penalty of ten thousand pesos exceeds the five-thousand-peso statutory maximum authorized under Section 458 of Republic Act No. 7160). Legal researchers can review these highlights and explanations to decide whether the draft should be cleared, amended, or referred for committee debate, ensuring human oversight throughout the local legislative process.

3.12 End-User Usability Evaluation

To evaluate how the automated conflict detection system operates within actual local legislative workflows, the study incorporates an end-user evaluation grounded in descriptive survey research. While computational benchmarks quantify retrieval recall and pairwise natural language inference accuracy, verifying whether the system serves as a viable decision-support assistant requires assessing its practical utility, task-technology fit, and software usability when operated by municipal legislative staff. The research design is structured as a descriptive, cross-sectional evaluation where participants test the prototype on realistic local ordinance drafts and immediately record their assessments through a structured quantitative instrument.

The evaluation cohort comprises the 15 active legal researchers serving the Davao City Sangguniang Panlungsod. These professional evaluators represent the city's three legislative council districts and routinely perform preliminary statutory compliance audits and draft ordinance preparation under Republic Act No. 7160 [84]. In local legislative administration, legal researchers examine proposed drafts before committee presentation, checking that draft clauses do not exceed devolved municipal powers or conflict with national legislation. Because these participants perform legislative auditing as their regular professional responsibility, their quantitative ratings provide an authentic assessment of whether the software aligns with the cognitive and administrative demands of municipal governance.

The evaluation follows a structured, hands-on simulation replicating the post-First Reading committee review phase leading into Second Reading plenary deliberations. Each testing session proceeds through three standardized steps. First, evaluators attend a brief ten-minute orientation introducing the prototype's assistive advisory role, input format requirements, and the distinction between vertical preemption alerts and horizontal consistency notices. Second, researchers interact directly with the single-page prototype dashboard on standard local government desktop computers matching the Host A specification. Evaluators upload multi-section draft ordinances containing both fully compliant regulatory clauses and injected preemption edge cases under Section 458 of Republic Act No. 7160. Researchers inspect the retrieved Stage 1 statutory provisions, examine the side-by-side comparative views, evaluate the highlighted deontic command words, and review the plain-language ratio decidendi summaries. Third, upon completing the hands-on review, researchers immediately complete a standardized quantitative survey hosted on Google Forms to record their evaluations.

To ensure methodological rigor while keeping the survey concise and free of cognitive fatigue during demanding council schedules, the quantitative instrument integrates two validated psychometric frameworks. First, overall software usability and learnability are measured using Brooke's 10-item System Usability Scale (SUS) [26]. The SUS instrument is widely recognized across human-computer interaction and legal informatics research for its high reliability and quick completion time. The 10 items use a 5-point Likert scale (from 1 for strongly disagree to 5 for strongly agree) and alternate between positive and negative statements to control for acquiescence bias. Second, task-specific utility and workflow alignment are measured using a targeted 4-item questionnaire adapted from Davis' Technology Acceptance Model (TAM) [53] and Goodhue and Thompson's Task-Technology Fit (TTF) framework [82]. These four items evaluate essential dimensions of legislative auditing: search efficiency, preemption risk detection, diagnostic clarity of highlighted phrases, and routine committee adoption viability. An optional open-ended question is included at the end of the survey to capture qualitative feedback on specific statutory edge cases and recommendations for institutional deployment. Table 3.14 details the complete survey questionnaire.

Table 3.14 lists the 14 survey items comprising the quantitative end-user evaluation instrument administered to the 15 Sangguniang Panlungsod legal researchers. The table details the item identifier, the exact survey statement presented to evaluators, the statement polarity, and the underlying theoretical framework. The first ten items adapt Brooke's (1996) canonical System Usability Scale, alternating between positive and negative phrasing to prevent acquiescence bias during software evaluation. The remaining four items operationalize Davis' (1989) Technology Acceptance Model and Goodhue and Thompson's (1995) Task-Technology Fit theory, measuring domain-specific utility in accelerating statutory searches, detecting preemption risks, interpreting attention highlights, and supporting institutional adoption. Providing this explicit questionnaire mapping ensures that every survey metric is anchored in established research literature, enabling structured descriptive analysis of the prototype's real-world viability.

Following descriptive survey research principles, the statistical analysis focuses on clear, interpretable measures of central tendency, variability, and standardized composite usability scoring. For each survey item, participant responses on the 5-point Likert scale are summarized using the Mean (μ), Standard Deviation (σ), and Median. Reporting both parametric and non-parametric descriptive statistics provides a transparent distribution summary across all evaluation dimensions without imposing unrealistic normality assumptions on ordinal survey data.

To compute overall software usability, the raw responses from the ten SUS items are converted into a standardized composite score ranging from 0 to 100:

Table 3.14: End-User Usability and Practical Utility Survey Instrument for Sangguniang Panlungsod Legal Researchers

Item Code	Survey Statement	Polarity	Theoretical Grounding
System Usability Scale (SUS; Brooke, 1996)			
SUS-01	I think that I would like to use this system frequently for draft reviews.	Positive	Usability (Brooke, 1996) [26]
SUS-02	I found the system unnecessarily complex.	Negative	Usability (Brooke, 1996) [26]
SUS-03	I thought the system was easy to use.	Positive	Usability (Brooke, 1996) [26]
SUS-04	I think that I would need the support of a technical person to use this system.	Negative	Usability (Brooke, 1996) [26]
SUS-05	I found the various functions in this system were well integrated.	Positive	Usability (Brooke, 1996) [26]
SUS-06	I thought there was too much inconsistency in this system.	Negative	Usability (Brooke, 1996) [26]
SUS-07	I would imagine that most people would learn to use this system very quickly.	Positive	Usability (Brooke, 1996) [26]
SUS-08	I found the system very cumbersome to use.	Negative	Usability (Brooke, 1996) [26]
SUS-09	I felt very confident using the system during draft ordinance auditing.	Positive	Usability (Brooke, 1996) [26]
SUS-10	I needed to learn a lot of things before I could get going with this system.	Negative	Usability (Brooke, 1996) [26]
Technology Acceptance Model (TAM) & Task-Technology Fit (TTF)			
TAM-01	The system substantially accelerates statutory cross-referencing compared to manual digest searches.	Positive	Perceived Usefulness (Davis, 1989) [53]
TAM-02	The system effectively detects subtle preemption risks, such as penalty caps or unauthorized mandates.	Positive	Task-Technology Fit (Goodhue & Thompson, 1995) [82]
TAM-03	The side-by-side textual highlights and ratio decidendi clearly explain the basis of flagged alerts.	Positive	Explainability Utility (Doshi-Velez & Kim, 2017) [64]
TAM-04	I would support integrating this prototype into the regular Sangguniang Panlungsod committee review and Second Reading preparation workflow.	Positive	Institutional Adoption (Davis, 1989) [53]

System Usability Scale Composite Score:

$$S_{\text{SUS}} = 2.5 \times \left[\sum_{i \in \text{odd}} (x_i - 1) + \sum_{i \in \text{even}} (5 - x_i) \right] \tag{3.21}$$

Where $x_i \in \{1, 2, 3, 4, 5\}$ denotes the raw 5-point Likert response for survey item i, with odd items ($i \in \{1, 3, 5, 7, 9\}$) representing positively phrased statements and even items ($i \in \{2, 4, 6, 8, 10\}$) representing negatively phrased statements.

Equation 3.21 defines the canonical formula for calculating the composite System Usability Scale score (S_{SUS}) established by Brooke [26]. The System Usability Scale is a standardized psychometric questionnaire that quantifies user perceptions of software usability on a composite 0-to-100 scale. The instrument alternates between positively phrased statements (odd items) and negatively phrased statements (even items) to control for acquiescence bias, defined as the tendency of survey respondents to agree indiscriminately with statements regardless of content. The scoring formula normalizes the raw 5-point Likert ratings: subtracting 1 from the raw ratings of positive odd items and subtracting the raw ratings of negative even items from 5 converts all responses to a standardized 0-to-4 scale. Multiplying the sum of these converted scores by 2.5 produces the final composite index ($0 \leq S_{\text{SUS}} \leq 100$). The resulting score is benchmarked against the empirical evaluation frameworks established by Bangor et al. [16, 17] and Sauro and Lewis [192]. Under Sauro and Lewis's curved grading scale, a composite score of 68 represents the 50th-percentile global benchmark for average software usability (Grade C), while scores of 70 and above enter Bangor et al.'s acceptable usability tier. Furthermore, scores of 80 or above place the system in the top 10% of evaluated software (Grade A under Sauro and Lewis), corresponding to the high-satisfaction tier approaching Bangor et al.'s empirical threshold for excellent usability ($M = 85.5$).

To verify the internal consistency reliability of the 4-item Technology Acceptance and Task-Technology Fit scale prior to reporting aggregate findings, the proponents compute Cronbach's alpha (α) [48]. Cronbach's alpha is a psychometric reliability coefficient that measures the internal consistency of a multi-item survey scale, determining whether individual survey items reliably capture the same underlying psychological construct. A scale achieving an alpha coefficient of $\alpha \geq 0.70$ demonstrates adequate internal consistency for exploratory research, confirming that responses reflect a coherent evaluation of legislative workflow utility rather than unrelated perceptual variance. Furthermore, to evaluate consistency in legal determinations across researchers during the interactive committee review simulation, inter-rater consensus is measured using Fleiss' Kappa (κ) [71]. Computing Fleiss' Kappa verifies that user verdicts on test drafts reflect shared legal standards rather than subjective individual impressions.

The evaluation protocol strictly adheres to the ethical and data privacy mandates of Republic Act No. 10173 (the Data Privacy Act of 2012) [172]. Participation in the evaluation study is entirely voluntary, and all participants complete an electronic informed consent form prior to survey submission. To protect evaluator confidentiality, participant names, specific committee assignments, and councilor affiliations are permanently decoupled from response records and indexed exclusively through anonymized identification tokens (such as Evaluator A-01 through A-15). Survey submissions are stored on secure university research storage and analyzed solely in aggregate descriptive form for academic thesis reporting.

3.13 Reproducibility and Repository

To ensure complete transparency, institutional auditability, and scientific reproducibility in accordance with open-science guidelines, the entire codebase, evaluation datasets, and experimental pipeline are managed under Git version control. The official repository is publicly accessible on GitHub:

`https://github.com/yyaahhzxc/thesis-repo`

To guarantee computational transparency and auditability, the repository implements structured reproducibility safeguards across four operational dimensions. First, the project enforces permanent commit hash versioning, wherein all empirical tables, benchmark figures, and model checkpoints reported in this thesis are permanently tagged with their respective Git commit hashes (such as Commit cc1c33d for statutory corpus ingestion and Commit 4dd7c85 for legal instruments), ensuring complete version traceability. Second, training and evaluation scripts maintain deterministic execution control by pinning global random seeds (seed = 42) across PyTorch, NumPy, and Python runtime environments to eliminate stochastic variation. Third, researchers can verify the complete pipeline using an executable single-command minimal working example (`python scripts/prototype_pipeline.py --demo`). This script loads the hybrid BM25 and semantic vector index over statutory provisions, executes asymmetric NLI scoring across four curated local test cases, and outputs explainable ratio decidendi under the *Magtajas* doctrine in under two seconds on standard CPU hardware. Fourth, the architecture enforces config-driven parameter decoupling, isolating experimental hyperparameters—including learning rates, batch sizes, sequence lengths, and decision thresholds (τ^*)—from core execution scripts into modular configuration files, allowing researchers to evaluate candidate models and recalibrate operational thresholds without modifying underlying neural inference pipelines.

3.14 Methodological Constraints

The first constraint involves the language of the legislative documents. Following Executive Order No. 292 (the Administrative Code of 1987) and national publishing standards, the statutory archive and local ordinance records examined in this study are limited to official English-language enactments. Because of this, the automated pipeline does not process vernacular Cebuano-Visayan phrases, colloquial regional idioms, or mixed Taglish expressions that occasionally arise during informal committee deliberations or floor debates. While operative legal provisions in the Philippines are formally enacted in English, statutory questions that depend on localized dialectal nuances require manual review by legal researchers. This constraint establishes a clear boundary for the tokenizer and avoids confusion when evaluating formal statutory commands against informal draft notes.

The second constraint involves the age and physical condition of the local ordinance archive. The historical enactment records obtained from the Davao City Sangguniang Panlungsod display uneven chronological density, where ordinances passed between 2021 and 2025 benefit from clean digital records within the Legislative Information Support System Portal (LISSP). In contrast, older local enactments dating from 1945 to 2020 exist primarily as physical carbon-copy scans, marked by faded ink, page skew, and photocopy marks. Although the vision-language ingestion pipeline utilizing Qwen2.5-VL-3B-Instruct substantially reduces transcription errors compared to traditional optical character recognition, residual character errors on pre-1980 archival enactments inevitably affect extraction fidelity. The retrospective evaluation of local ordinances prior to the modern digital era is therefore focused on landmark enactments that have undergone manual verification alongside high-resolution scanning.

The third constraint stems from the single-premise asymmetric modeling formulation adopted for the coarse-to-fine pipeline. The natural language inference engine models vertical conflict detection as an asymmetric pair, matching an individual candidate statutory premise (p_i) against an operative draft ordinance block (h_q). This structural formulation effectively captures direct contradictions where a single national statute explicitly invalidates an ordinance clause, such as unauthorized penalty escalation under Republic Act No. 7160. However, the system does not inherently detect multi-statute cumulative conflicts, where an ordinance provision complies with each national statute in isolation but breaches a combined legislative framework formed by multiple distinct codes. Expanding the inference mechanism to support multi-premise reasoning across multiple statutes exceeds local workstation constraints and is left for future work.

The final constraint defines the institutional and operational role of the automated conflict detection system within local governance. The pipeline is designed exclusively as an ex-ante assistive advisory tool intended for committee legal researchers during committee review and Second Reading preparation, rather than an autonomous decision maker. The system produces calibrated confidence estimates and highlighted text spans to point out potential legal friction, but it does not make binding statutory determinations or substitute for judicial review under the Supreme Court of the Philippines. Furthermore, political questions, discretionary policy choices, and socio-economic trade-offs remain exclusively with elected city council members. Defining this operational boundary ensures that the system functions strictly within established principles of administrative law, safeguarding human oversight throughout the local legislative process.

Chapter 4

Results and Discussion

This chapter presents the empirical findings and experimental evaluations of the Coarse-to-Fine Semantic Conflict Detection System, organized directly around the specific research questions and objectives formulated in Chapter 1. Section 4.1 addresses Research Question 1 and Objective 1 by analyzing the linguistic parameters, layout degradation, and vision-language transcription performance across the digitized Davao City local archive ($N = 1,664$). Section 4.2 addresses Research Question 2 and Objective 2 by evaluating candidate retrieval efficiency, preemption coverage, and latency across the unified dual statutory corpus ($N = 27,096$ enactments, comprising 176,421 provisions). Section 4.3 addresses Research Questions 3 and 4 and Objectives 3 and 4 by presenting the comparative model screening, supervised fine-tuning, loss dynamics, and ablation benchmarks on the curated statutory ground truth dataset. Section 4.4 addresses algorithmic optimization, problem-size scaling, the 30-run comparative matrix, and hardware trade-offs under municipal constraints. Finally, Section 4.5 addresses Research Question 5 and Objective 5 by outlining system-level decision-support viability, the prototype interface, and the multi-tier human evaluation roadmap for the Davao City Sangguniang Panlungsod.

4.1 Archival OCR and VLM Evaluation

Ingesting historical local archives into an automated legal decision-support system requires converting scanned physical documents into structured digital text. In the Davao City collection, enactments from earlier legislative terms mostly exist as scanned PDF files rather than searchable digital documents. These historical records contain common scanning noise, such as crooked pages, uneven ink, staple marks, and official city seal stamps over text. If degraded scans are transcribed with erroneous characters or broken layouts, subsequent search indices and inference models inherit corrupted text. This section introduces an evaluation framework for legal document transcription, presents a stratified sample of archival records, compares four document extraction methods, and explains why a locally hosted vision-language model was selected.

4.1.1 Candidate Document Extraction Systems

To determine the best document extraction method for historical local archives, the proponents evaluated four distinct approaches. Traditional optical character recognition relies on image binarization, polygonal line tracking, and segmented character shape classification [198], with modern updates in Tesseract v5.x incorporating line-level LSTM recurrent networks. As seen in classical OCR workflows, traditional engines treat individual characters as isolated shapes without considering sentence meaning. While fast on central processing units, traditional engines struggle when encountering slanted lines, faded ink, or paper bleed-through.

Document-layout OCR frameworks separate page layout analysis from text line recognition [158]. Modern tools like Surya v0.6+ use object detection models to locate text columns, tables, and headers before transcribing lines. A separate recognition model then reads the text, and reading order is reconstructed using sorting rules. This modular design preserves layout well on multi-column pages, but requires multiple neural networks that demand significant GPU memory during long batch runs.

Commercial cloud vision platforms run large multimodal transformer models on enterprise server clusters [83]. Cloud services like Google Document AI achieve high accuracy across various scan qualities and complex tables. However, they require continuous internet access, charge ongoing per-page API fees, and require sending local legislative drafts to third-party servers. In a local government setting, recurring subscription costs and external data transfers conflict with operational independence and data privacy mandates.

Local vision-language models combine image encoding and text generation inside a single transformer architecture [217]. Using two-dimensional rotary position embeddings (2D-RoPE) and flexible image patches, visual information is passed directly to a language decoder. With 4-bit NormalFloat4 (NF4) compression [60], models like Qwen2.5-VL-3B-Instruct run completely offline on standard workstation hardware. The language model decoder helps resolve visual blur or scanner noise by using its understanding of grammar and statutory phrasing.

4.1.2 Document Evaluation Framework

Evaluating text extraction on local legislation reveals challenges that general OCR benchmarks often miss. As Lopresti showed [133], character errors in early steps cascade into downstream language processing tasks. In legal drafting, changing a single character or dropping a zero completely changes legal meaning. An effective evaluation framework for legislative text must therefore combine standard string edit distances with domain-specific legal entity accuracy and semantic embedding checks.

The foundational character and word accuracy metrics are formulated via the Levenshtein edit distance $\mathcal{L}_{\text{Lev}}$ [127]. To assess baseline typographical fidelity across candidate document parsers, Character Error Rate and Word Error Rate are computed as follows:

Character Error Rate (CER) and Word Error Rate (WER):

$$\text{CER} = \frac{\mathcal{L}_{\text{Lev}}(\mathbf{c}_{\text{ref}}, \mathbf{c}_{\text{hyp}})}{|\mathbf{c}_{\text{ref}}|} = \frac{S_c + D_c + I_c}{N_c}, \quad \text{WER} = \frac{\mathcal{L}_{\text{Lev}}(\mathbf{w}_{\text{ref}}, \mathbf{w}_{\text{hyp}})}{|\mathbf{w}_{\text{ref}}|} = \frac{S_w + D_w + I_w}{N_w} \tag{4.1}$$

Where S, D, and I denote character or word substitutions, deletions, and insertions relative to the gold-standard reference sequence of length N.

Equation 4.1 defines Character Error Rate (CER) and Word Error Rate (WER) using Levenshtein edit distance. These standard metrics establish a baseline of typing accuracy when transcribing historical paper ordinances. The formulas count character or word substitutions (S), deletions (D), and insertions (I) compared to the total length of the reference text (N). In legal documents, however, low overall error rates can hide serious mistakes, such as dropping a single digit in a penalty fine. While CER and WER measure general transcription noise, they treat all characters equally, which makes legal entity accuracy checks necessary.

To evaluate overall document similarity on a bounded scale regardless of clause length, following the normalized edit distance formulation of Marzal and Vidal [141] and Levenshtein [127], the proponents compute Normalized Edit Distance:

Normalized Edit Distance (NED):

$$\text{NED}(\mathbf{s}_{\text{ref}}, \mathbf{s}_{\text{hyp}}) = 1.0 - \frac{\mathcal{L}_{\text{Lev}}(\mathbf{s}_{\text{ref}}, \mathbf{s}_{\text{hyp}})}{\max(|\mathbf{s}_{\text{ref}}|, |\mathbf{s}_{\text{hyp}}|)} \tag{4.2}$$

Where $\mathbf{s}_{\text{ref}}$ and $\mathbf{s}_{\text{hyp}}$ denote the reference and hypothesis string sequences, bounded between 0.0 (maximum dissimilarity) and 1.0 (exact match) to provide a length-invariant evaluation baseline.

Equation 4.2 defines Normalized Edit Distance (NED) as an intuitive similarity score between 0.0 and 1.0. This metric scales the raw edit distance against the maximum length of the two texts, subtracting the result from 1.0 so that an exact match scores 1.0. When evaluating multi-page ordinances, NED prevents long standard clauses from distorting overall error measurements. This score provides a clear summary of text alignment before checking specific legal entities.

Beyond general string alignment, legislative conflict detection requires exact-match fidelity across critical statutory tokens. Adapting the exact-match slot filling evaluation standard from MUC-4 [36], the domain-specific component accuracy is computed as follows:

Domain-Specific Legislative Component Accuracy:

$$\text{Metric Accuracy} = \frac{1}{|K|} \sum_{k \in K} \mathbb{I}\left(\mathbf{v}_{\text{hyp}}^{(k)} \equiv \mathbf{v}_{\text{ref}}^{(k)}\right) \tag{4.3}$$

Where K represents the set of extracted legal entity instances, and $\mathbb{I}(\cdot)$ is an exact-match indicator function comparing hypothesis values $\mathbf{v}_{\text{hyp}}^{(k)}$ against gold references $\mathbf{v}_{\text{ref}}^{(k)}$.

Equation 4.3 defines Domain-Specific Legislative Component Accuracy across key legal terms. This metric ensures that important statutory details—such as law citations, ordinance numbers, penalty amounts, and effectivity dates—are transcribed exactly. The formula measures the share of legal terms that match the reference text word-for-word using the indicator function $\mathbb{I}(\cdot)$. In local ordinances, a typo in a Republic Act number prevents BM25 keyword searches from finding the law and confuses dense vector search. Checking exact component accuracy confirms whether an extraction tool preserves the critical legal references needed for Stage 1 retrieval.

These metrics evaluate exact text matches across several key areas: Legal-Entity Accuracy (LEA) for government offices and private parties; Ordinance-Number Accuracy (ONA) for local ordinance numbers; Date Accuracy (DA) for approval and effectivity dates; Monetary-Value Accuracy (MVPA) for fines and fee limits; Section-Heading Integrity (SHI) for section titles; and Page Completeness (PC) for preserving all pages in order. In addition, table quality is evaluated using Tree-Edit-Distance-based Similarity (TEDS) [251], measuring row and column accuracy across tax and zoning schedules. Finally, Semantic Embedding Cosine Fidelity (SECF) checks semantic preservation by calculating the cosine similarity between embeddings of the reference and extracted texts.

4.1.3 Limitations of Standard Spellchecking

Standard spellcheckers and automatic dictionary tools cannot reliably check legal document transcription. In legal conflict detection, the impact of word errors is uneven. If an OCR engine writes “Secton” instead of “Section”, the edit distance is only one character and an ordinary spellchecker easily fixes the error. On the other hand, if an engine reads a penalty of “₱5,000.00” as “₱50,000.00”, or changes “shall” to “may”, or alters “Ordinance No. 0667-21” to “Ordinance No. 0661-21”, a standard spellchecker flags nothing because all the words are valid dictionary words. Under Philippine legal principles and Section 458 of Republic Act No. 7160, turning a ₱5,000.00 penalty into ₱50,000.00 turns a valid local ordinance into an unlawful act.

In addition, Philippine local ordinances contain specialized terms that generic spellcheckers frequently flag as errors. Ordinances regularly include legal Latin phrases (*in pari materia*, *ultra vires*), local agency acronyms (such as CTTMO, CENRO, and CPDO), and local place names. Davao City alone contains 182 barangays, such as Buhangin, Bunawan, Calinan, Marilog, Paquibato, Talomo, and Toril. Standard spellchecking tools flag these legitimate names as mistakes. If automatic autocorrection is turned on, generic software often replaces these unique local names with standard English words, corrupting the legal text.

4.1.4 Archival Reference Sampling

To establish a dependable benchmark without manually transcribing all 1,664 archival documents, the proponents created a stratified sample of $N = 60$ representative local enactments, totaling approximately 180 to 240 pages. Table 4.1 outlines the sample breakdown across eight legislative and physical categories.

Table 4.1: Stratified Reference Sampling Matrix for Archival Legislative Document Evaluation ($N = 60$ Enactments)

Stratification Axis	**Strata Partition**	**Sample ($N = 60$)**	**Legislative & Archival Rationale**
Legislative Era	Pre-LGC (1950–1991) Post-LGC Historical (1992–2015) Pre-Pandemic (2016–2020) LISSP Digital (2021–2025)	12 (20.0%) 18 (30.0%) 15 (25.0%) 15 (25.0%)	Captures physical paper yellowing, manual typewriter fonts, early digital prints, and modern clean scans.
Document Length	Short (1–2 pages) Medium (3–7 pages) Long ($\geq$ 8 pages / Codes)	24 (40.0%) 24 (40.0%) 12 (20.0%)	Verifies model stability across single-page resolutions and multi-page omnibus regulatory codes.
Scan Degradation	High Quality (Clean digital) Moderate (Photocopy, minor skew) Severe (Bleed-through, dark edges)	20 (33.3%) 25 (41.7%) 15 (25.0%)	Tests noise robustness against dark photocopy borders, staple shadows, and carbon-copy bleed-through.
Functional Typology	1. Substantive Regulatory / Penal 2. Inter-Agency MOAs / MOUs 3. Disaster Risk & Calamity Funds 4. Deeds of Donation & Usufruct 5. Road Closures & Traffic Rules 6. Zoning Reclassifications	15 (25.0%) 10 (16.7%) 7 (11.7%) 8 (13.3%) 10 (16.7%) 10 (16.7%)	Reflects the complete functional typology identified across the 1,664 Davao City local enactments.
Table Density	Zero Tabular Schedules Simple Tables (2–4 columns) Dense Tables ($\geq$ 5 columns)	30 (50.0%) 18 (30.0%) 12 (20.0%)	Evaluates Markdown grid retention in revenue schedules, tax assessments, and traffic rerouting vectors.
Signatures & Seals	Clean Endings (Minimal) Moderate (Official initials) Heavy (Overlapping seals & stamps)	20 (33.3%) 25 (41.7%) 15 (25.0%)	Evaluates whether mayoral signatures and dry seals trigger hallucinatory generation loops.
Filipino Names	High Local / Moro / Indigenous Standard Anglophone / Spanish	30 (50.0%) 30 (50.0%)	Validates out-of-vocabulary character accuracy on Mindanaoan and Visayan personal and place names.
Statutory Citations	High Citation Density ($\geq$ 5 laws) Low Citation Density ($<$ 5 laws)	25 (41.7%) 35 (58.3%)	Measures recognition fidelity on referenced Republic Acts, Presidential Decrees, and Batas Pambansa.

Table 4.1 outlines the sampling breakdown across eight legislative and physical categories used to evaluate document transcription across $N = 60$ representative enactments. This stratified design provides a representative benchmark of approximately 180 to 240 physical pages without requiring manual transcription of all 1,664 archived enactments. The sample divides documents across four time periods, three length categories, three scan quality levels, and six functional ordinance types, while also accounting for tables, official seals, local names, and statutory citations. Older typewriter documents from 1950–1991 and photocopies from 1992–2015 make up half the sample, testing models against realistic noise such as ink bleed, staple marks, and faded contrast. This balanced reference set ensures that document extraction testing accurately reflects the variety of document conditions found in the Davao City archives.

4.1.5 Document Extraction Trade-Off Analysis

Table 4.2 summarizes the comparative evaluation of the four document extraction approaches across accuracy metrics, hardware needs, legal text preservation, data privacy, and operational costs.

Table 4.2: Multi-Dimensional Comparative Evaluation of Candidate Archival Document Architectures for Local Legislation

Evaluation Dimension	**Tesseract v5.x**	**Surya v0.6+**	**Google Document AI**	**Qwen2.5-VL-3B (4-bit NF4)**
Architectural Family	Traditional CNN-LSTM	Layout-Aware Modular	Cloud Multimodal Net	Local Vision-Language
Character Error Rate (CER)	14.82%	8.45%	2.15%	3.12%
Word Error Rate (WER)	22.30%	12.10%	4.05%	5.24%
Normalized Edit Distance	0.852	0.915	0.978	0.969
Legal-Entity Accuracy	74.2%	86.5%	97.2%	95.8%
Ordinance-Number Accuracy	68.3%	83.3%	98.3%	96.7%
Monetary-Value Accuracy	63.3%	80.0%	98.3%	96.7%
Table Fidelity (TEDS)	41.2%	76.4%	94.1%	91.8%
Semantic Cosine Fidelity	88.4%	93.8%	99.1%	98.6%
Execution Hardware	Minimal CPU (< 500 MB)	Mid-tier GPU (4 GB)	Client HTTPS Interface	Consumer GPU (RTX 5050)
Network Dependency	100% Offline	100% Offline	Mandatory Internet API	100% Offline
Data Privacy Compliance	Fully Sovereign	Fully Sovereign	Third-Party Cloud Route	Fully Sovereign (RA 10173)
Financial Model	Free Open-Source ($0)	Free Open-Source ($0)	Paywall per 1,000 pages	Free Open-Source ($0)
Mean Page Latency	Fast (0.3–0.8 s, CPU)	Moderate (1.2–2.5 s)	Fast (0.5–1.2 s roundtrip)	Moderate (4.0–8.5 s, GPU)

Note: Evaluation reflects architectural properties and empirical validation runs on historical Davao City local legislation. Qwen2.5-VL-3B-Instruct evaluated in 4-bit NormalFloat4 (NF4) precision on Host B (NVIDIA RTX 5050 Laptop GPU). Metric values represent aggregate benchmark scores across evaluated archival reference transcripts.

Table 4.2 compares four document parsing approaches across twelve accuracy, operational, and cost factors. This comparison provides the empirical basis for selecting the local document ingestion pipeline, balancing character accuracy against local hardware limits, operational costs, and data privacy requirements. The results show that the locally hosted vision-language model (Qwen2.5-VL-3B-Instruct in 4-bit NF4) performs comparably to commercial cloud AI, achieving 3.12% CER, 96.7% penalty accuracy, 91.8% table accuracy (TEDS), and 98.6% Semantic Cosine Fidelity. In contrast, traditional scanner OCR (Tesseract v5.x) has a high 14.82% CER and drops to 63.3% accuracy on monetary figures, which risks passing corrupted numbers to subsequent legal analysis. While Google Document AI achieves slightly lower character error rates (2.15% CER), its required internet connection and recurring per-page fees do not fit local government budget constraints and privacy rules under Republic Act No. 10173. As a result, running Qwen2.5-VL-3B-Instruct offline on a consumer GPU provides the best combination of transcription accuracy, cost-free operation, and complete institutional privacy.

To evaluate the practical effect of transcription errors, the proponents ran a test on an authentic local ordinance (Davao City Ordinance No. 000667-21). Traditional scanner OCR produced a Word Error Rate of 3.11% and a Character Error Rate of 0.45%. Although these general error rates seem low, traditional OCR achieved only 54.55% precision on legal markers. The engine corrupted the citation to Republic Act No. 7160, merged Section 1 and Section 2 headings together, and lost formatting in the penalty section.

In contrast, the locally hosted vision-language model achieved 0.00% Word Error Rate, 0.02% Character Error Rate, and 100.0% precision across all eleven evaluated statutory indicators. In addition, comparing text vectors using TF-IDF n-grams confirmed that the VLM transcription reached 100.0% semantic cosine similarity with the reference text, compared to 99.21% for traditional OCR. The language model decoder acts as a helpful reading assistant: when letters are faded or smudged by scanner noise, the model uses its knowledge of standard legal phrasing to fill in the missing words accurately.

4.1.6 Historical Degradation Empirical Benchmark

While modern digital enactments such as Ordinance No. 000667-21 demonstrate near-perfect character fidelity under vision-language parsing, historical local archives present severe physical degradation that challenges automated transcription. An empirical audit across the digitized collection identified 32 historical municipal enactments exhibiting severe visual defects. Over 75.0% of these degraded records originate before the year 2000, with 68.8% produced during the pre-Local Government Code era (1950–1991) using manual impact typewriters, carbon transfer paper, and early analog duplicators. To quantify transcription performance under real-world archival conditions, the proponents evaluated four authentic legislative exemplars representing distinct eras and physical failure modes against human-verified gold-standard ground truth. Table 4.3 presents the comparative Word Error Rate (WER) and Character Error Rate (CER) achieved by the locally deployed vision-language model (Qwen2.5-VL-3B-Instruct in 4-bit NF4) alongside the commercial cloud baseline.

Table 4.3: Comparative Empirical Transcription Error Rates Across Degraded Historical Legislative Eras

Historical	**Legislative Era**	**Dominant Physical**	**Local VLM (4-bit NF4)**		**Cloud Baseline**	
Enactment	**& Technology**	**Degradation Mode**	**WER**	**CER**	**WER**	**CER**
Ord. 000044-56	Pre-LGC (1956) Manual Typewriter	Carbon bleed-through, uneven key strikes, city seal stamp	12.23%	7.66%	1.92%	1.74%
Ord. 000332-10	Pre-LGC (1968) Manual Typewriter	Ribbon ink fade, baseline cuts, staple shadow	11.06%	5.91%	Baseline	Baseline
Ord. 0166-81	Pre-LGC (1981) Manual Typewriter	Binder hole punch occlusions, strike fill-in, cursive ink	19.78%	12.87%	8.84%	4.99%
Ord. 000667-21	Modern LISSP (2021) Digital Raster	Clean electronic layout, high-contrast typography	0.00%	0.02%	0.00%	0.00%

Note: Evaluated against human-verified gold-standard transcriptions. Local VLM corresponds to Qwen2.5-VL-3B-Instruct (4-bit NF4) executed on Host B. Cloud baseline corresponds to Gemini Flash multimodal inference. Evaluated archival exemplars preserved in repository benchmark suite.

Table 4.3 compares word and character error rates across four authentic Davao City legislative records representing different eras and physical conditions. This comparison demonstrates how mechanical typewriter defects and physical paper aging affect automated transcription accuracy relative to clean modern scans. The empirical measurements reveal a sharp performance divergence between modern digital prints and mid-twentieth-century manual typewriter records. On modern digital scans (Ordinance No. 000667-21), the local vision-language model achieves near-perfect fidelity (0.00% WER and 0.02% CER). However, on historical typewriter records from 1956, 1968, and 1981, error rates rise substantially to 11.06%–19.78% WER and 5.91%–12.87% CER. This steep rise in error rates demonstrates that archival transcription difficulty is driven primarily by physical medium defects rather than model capacity limitations.

Detailed inspection of the historical transcripts reveals specific physical degradation mechanisms documented in document image defect literature [14, 133]. In Ordinance No. 000044-56 (1956), mechanical strike velocity variance caused ink to fill closed-loop characters such as *e*, *a*, and *o*, leading the local vision-language model to misread *municipal waters* as *municipal maters* and to transcribe *log pond or for storage of logs* as *lux powd or for straight of lots.* Furthermore, uneven carbon transfer caused the engine to misread the statutory license fee of *₱200.00* as an American dollar figure (*$200.00*) and to alter *Penal Provisions* into *Final Provisions.* Large embossed city seal stamps and dark cursive ink signatures also obscured official attestation blocks, causing the model to omit the names of the acting mayor and city secretary.

Similar physical occlusions affected Ordinance No. 0166-81 (1981), where dark circular binder hole punches along the left margin severed section numbering and paragraph catchlines. The mechanical typewriter ribbon also caused heavy ink fill-in across title lines, causing the model to misread *Lungsod ng Dabaw* as *Luneskon ng Darw*, *Series of 1981* as *Series of 1961*, and *liquified petroleum gas* as *liquid profession (cas).* In the approval block, overlapping cursive ink pen strokes across the city council seal caused the model to misinterpret *Carried, by a majority* as *Cashed: By a majority*, and to misread the Sangguniang Panlungsod secretary's name. In Ordinance No. 000332-10 (1968), horizontal typewriter underscore strikes directly sliced through character descenders (*p*, *g*, *y*, *q*), fragmenting word segmentation and misreading the legislative term.

These empirical findings highlight the principle of asymmetric legal error cost in local statutory auditing. Under Section 458 of Republic Act No. 7160 and the judicial doctrine of *Magtajas v. Pryce Properties* [208], local ordinance penalty provisions are legally invalid if they exceed statutory fine limits. If an optical character recognition engine guesses at smudged mechanical numbers and transcribes a ₱200.00 fee as ₱2,000.00 or drops a decimal zero, it artificially manufactures a statutory preemption defect where none exists in law. When visual glyphs fall below human legibility thresholds due to physical paper acidification, tropical moisture bleed, or hole-punch cuts, an automated system must avoid unconstrained hallucination. This necessitates a two-tier archival quarantine framework where severely degraded enactments are formally flagged for human verification, ensuring that downstream semantic conflict detection operates exclusively upon legally reliable statutory text.

4.1.7 Model Selection and Hardware Boundaries

Selecting Qwen2.5-VL-3B-Instruct as the primary document ingestion engine balances text accuracy, operating cost, and data privacy. While traditional OCR runs quickly on standard CPUs, its difficulty in recognizing section headers and its errors on numbers hurt downstream search and conflict detection. Although commercial cloud platforms deliver strong accuracy, using commercial APIs across 1,664 multi-page ordinances incurs recurring financial costs. In addition, sending internal local government drafts and committee working texts to third-party cloud servers raises compliance concerns under Republic Act No. 10173 (Data Privacy Act of 2012) [172]. A system designed for local government offices should operate independently without requiring ongoing subscription fees.

Running Qwen2.5-VL-3B-Instruct in 4-bit NF4 precision delivers high transcription accuracy while operating completely offline. The model processes pages consistently, preserves data privacy, and stays within the 8 GB memory limit of Host B. The main trade-off of the vision-language approach is processing speed. On the laptop GPU, transcribing a single page takes about 4.0 to 8.5 seconds, compared to under one second for traditional OCR. However, because digitizing the historical archive is a one-time preparation step done in advance, this extra processing time is manageable. Limiting generation to 1,000 tokens per page and saving progress after each document ensures steady batch processing across the entire local collection.

4.1.8 Document Extraction Limitations

Testing archival transcription identified three practical limitations of the system. First, severe physical document damage sets an upper limit on how much text can be read accurately. On carbon-copy records from the 1950s and 1960s where ink has bled through from the other side of the page, even human readers cannot agree on blurred numbers, showing that the issue stems from damaged paper rather than software flaws.

Second, complex tables in older tax ordinances often have irregular layouts and merged cells. While the vision-language model accurately reads tax rates and numbers in Markdown tables, slight differences in column alignment reduce automated table scores (TEDS) even when the monetary numbers are completely correct. Third, dense embedding models can sometimes overlook small errors; a single wrong digit in a fine might produce only a tiny drop in similarity score, even though it creates a significant legal conflict. These edge cases show why the system is designed as an assistive tool to support legal researchers, rather than an automatic replacement for human review.

4.2 Stage 1 Candidate Retrieval

In a two-stage conflict detection system, the main goal of Stage 1 coarse retrieval is to quickly search the 27,096 enactments (176,421 provisions) and return a compact shortlist of top-k candidate laws ($k \in \{5, 10, 20, 50\}$). In legal information retrieval, search performance is evaluated primarily using candidate recall (Recall@k) and Mean Reciprocal Rank (MRR), rather than classification accuracy [140]. Candidate recall (Recall@k) measures the share of queries where the relevant national law appears within the top k results, while Mean Reciprocal Rank (MRR) measures how close to the top that law is placed (1/rank). If the correct national statute is missed in Stage 1, the Stage 2 inference model cannot detect the conflict, no matter how capable it is. At the same time, the shortlist must stay compact ($k \leq 50$) so that comparing text pairs in Stage 2 runs quickly and fits within standard office computer memory.

4.2.1 Full-Corpus Candidate Retrieval Recall

To test how well search strategies perform in a real-world setting, the proponents tested baseline retrieval methods against all 25,432 national statutes. In this test, all 350 curated local draft queries were searched directly against the full statutory database without manual pre-filtering. Table 4.4 reports the candidate retrieval recall, rank positions, and search speeds across the entire collection.

Table 4.4: Empirical Candidate Retrieval Recall Benchmark across the Full National Statutory Database ($N = 25,432$)

Stage 1 Retrieval Strategy	**Latency**	**R@5**	**R@10**	**R@20**	**R@50**	**MRR@50**
1. Baseline BM25 (Unconstrained)	329.6 ms	21.43%	29.71%	35.71%	43.14%	0.1521
2. BM25 + Hierarchy Priority Safeguard	300.1 ms	29.14%	34.86%	39.43%	43.14%	0.2217
3. BM25 + Hard Domain Filter	297.5 ms	4.86%	7.71%	10.29%	13.43%	0.0467

Note: Evaluated on all 350 ground truth queries searching the entire 25,432 national statute corpus on Host A (Intel Core i3-10105F CPU @ 3.70 GHz). Latency values represent mean per-query retrieval runtime.

Table 4.4 shows the candidate retrieval recall, rank distributions, and search speeds across the 25,432 national statutes for three Stage 1 keyword search strategies. This test checks whether keyword matching alone can handle thousands of varied government documents without manual filtering. The numbers show that standard BM25 achieves only 29.71% Recall@10 and 43.14% Recall@50 with an MRR of 0.1521, meaning more than half of the governing national laws are missed entirely. Trying to filter out irrelevant documents using a strict topic filter causes an even worse drop, lowering Recall@50 to 13.43% because major laws cover multiple legal subjects at once. In contrast, adding the Hierarchy Priority Safeguard increases MRR to 0.2217 (a 45.8% relative improvement) by giving higher priority to primary Republic Acts over administrative orders. These results show that keyword search alone is not enough for large statutory collections, highlighting the need for semantic vector search and ranking safeguards.

4.2.2 Dense Bi-Encoder Retrieval

To find the most accurate and efficient dense retriever for Stage 1 search, the proponents benchmarked eight candidate bi-encoders across all 350 ground truth queries. In this test, each bi-encoder converts provisions into dense semantic vectors (384 to 1,024 dimensions), and candidate provisions are ranked purely by cosine similarity against the draft query without keyword help. Table 4.5 summarizes the retrieval recall, search speed, and difficulty tier performance across the eight models.

Table 4.5: Empirical Performance Comparison of Candidate Stage 1 Dense Bi-Encoder Architectures ($N = 350$)

Category	**Candidate Bi-Encoder**	**Params**	**Context**	**Latency**	**R@5**	**R@10**	**R@20**	**R@50**	**MRR**	**Tier 3 R@5**
Distilled Edge	`all-MiniLM-L6-v2*`	22.7M	512	**1.33 ms**	87.1%	97.1%	**100.0%**	**100.0%**	0.7388	91.3%
	`bge-small-en-v1.5`	33.4M	512	1.85 ms	87.4%	97.1%	100.0%	100.0%	0.7431	91.3%
General MTEB	`bge-base-en-v1.5`	109.0M	512	4.82 ms	88.0%	97.7%	100.0%	100.0%	0.7512	92.3%
	`e5-base-v2`	109.0M	512	4.95 ms	87.7%	97.4%	100.0%	100.0%	0.7480	91.3%
	`all-mpnet-base-v2`	109.0M	512	5.12 ms	87.7%	97.4%	100.0%	100.0%	0.7495	92.3%
Legal-Domain	`legal-bert-base-uncased`	110.0M	512	4.88 ms	86.6%	96.6%	99.7%	100.0%	0.7310	89.4%
Long-Context	`ModernBERT-base`	149.0M	8,192	6.20 ms	88.6%	98.0%	100.0%	100.0%	0.7584	**93.3%**
Multilingual	`bge-m3`	568.0M	8,192	24.50 ms	**88.9%**	**98.3%**	100.0%	100.0%	**0.7621**	**93.3%**

Note: *Selected model integrated into the production Stage 1 hybrid pipeline. Evaluated standalone using normalized cosine similarity on Host A (Intel Core i3-10105F CPU @ 3.70 GHz) without sparse keyword assistance. Latencies represent mean per-query CPU runtime.

Table 4.5 summarizes the performance of eight candidate dense bi-encoders evaluated across model sizes, context limits, CPU speeds, recall thresholds ($k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank, and Tier 3 paraphrastic recall. This comparison helps select the best semantic retriever for Stage 1, comparing compact models against larger foundation models. The results show that all eight candidate models understand legal phrasing well, with each reaching at least 96.6% Recall@10 and 100.0% Recall@50 across the 350 queries. While larger models like `bge-m3` (MRR = 0.7621) and `ModernBERT-base` (MRR = 0.7584) achieve slightly higher accuracy on paraphrased Tier 3 queries (93.3% Recall@5), they run much slower on standard CPUs (24.50 ms and 6.20 ms per query). In contrast, `all-MiniLM-L6-v2` reaches 87.1% Recall@5, 97.1% Recall@10, and 100.0% Recall@20 in just 1.33 milliseconds with a small 22.7M parameter size. Because statutory sections average under 300 tokens, `all-MiniLM-L6-v2` offers the best balance of semantic accuracy and search speed on CPU hardware, making it the preferred dense retriever for the hybrid system.

4.2.3 Retrieval Fusion Ablation

In machine learning, an ablation study tests the impact of individual system components by turning them on, off, or adjusting them one at a time. To find the most effective combination of keyword search and semantic vector search, the proponents conducted an ablation experiment testing nine candidate configurations on the 350 ground truth queries. Table 4.6 summarizes the search speed, recall thresholds, Mean Reciprocal Rank, and Tier 3 paraphrastic recall across all nine options.

Table 4.6 compares nine Stage 1 retrieval setups across CPU search speed, recall thresholds ($k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank (MRR), and Tier 3 paraphrastic recall. This ablation isolates the contributions of keyword search (BM25), dense vectors (all-MiniLM-L6-v2), topic filters, and rank fusion. The results show that while standalone BM25 achieves 79.7% Recall@5 and 99.7% Recall@50, its recall drops to 74.0% on Tier 3

Table 4.6: Empirical Performance Comparison of Candidate Stage 1 Retrieval Architectures ($N = 350$)

Candidate Retrieval Architecture	**Latency**	**R@5**	**R@10**	**R@20**	**R@50**	**MRR**	**Tier 3 R@5**
1. Pure BM25 Baseline (AIIR)	1.39 ms	79.7%	88.6%	97.4%	99.7%	0.6113	74.0%
2. BM25 + Hard Domain Filter	1.23 ms	99.1%*	100.0%*	100.0%*	100.0%*	0.7921*	99.0%*
3. BM25 + Soft Domain Prior (+20%)	1.43 ms	87.7%	94.6%	99.7%	100.0%	0.7122	79.8%
4. Pure Dense Bi-Encoder (MiniLM)	1.33 ms	87.1%	97.1%	100.0%	100.0%	0.7388	91.3%
5. Hybrid JNLP (Weighted Sum $\alpha = 0.5$**)**	1.23 ms	82.6%	91.7%	98.9%	100.0%	0.6409	77.9%
6. Hybrid Dense-Biased ($\alpha = 0.7$)	1.20 ms	84.0%	94.0%	99.4%	100.0%	0.6656	78.8%
7. Hybrid BM25-Biased ($\alpha = 0.3$)	1.22 ms	80.9%	90.6%	98.3%	100.0%	0.6202	76.0%
8. Reciprocal Rank Fusion (RRF $k = 60$**)**	1.26 ms	88.6%	96.0%	99.7%	**100.0%**	0.7040	88.5%
9. Hybrid + Soft Domain Prior ($\alpha = 0.5$)	1.21 ms	89.7%	95.7%	99.7%	**100.0%**	0.7328	84.6%

Note: *Configuration 2 reflects the artificial isolation effect; when evaluated across the full 25,432 corpus, hard filtering collapses overall Recall@50 to 13.43%. Latencies represent mean per-query CPU runtime.

queries that lack exact keyword matches. Conversely, pure dense retrieval reaches 87.1% Recall@5 and 91.3% on Tier 3, capturing legal concepts well but missing some exact numbers. Combining both approaches through Reciprocal Rank Fusion (RRF $k = 60$) achieves 88.6% Recall@5, 96.0% Recall@10, and 100.0% Recall@50 with an average CPU time of just 1.26 milliseconds. Reaching 100.0% recall at $k = 50$ ensures that no relevant laws are dropped before moving to Stage 2 inference.

To visualize how candidate retrieval recall changes as the shortlist expands and inspect how different architectures handle latent paraphrasing, Figure 4.1 shows the candidate Recall@k progression curves alongside the performance breakdown across jurisprudential difficulty tiers.

Figure 4.1 shows how candidate recall improves as the shortlist grows, along with performance across the three difficulty tiers on the 350 queries. This chart highlights the practical differences between keyword matching and semantic search. Panel A plots recall curves from $k = 5$ to $k = 50$, showing that hybrid fusion reaches 100% recall quickly while keyword search levels off earlier. Panel B highlights the performance gap on Tier 3 (Latent and Paraphrastic), where dense vectors outperform BM25 by 17.3% (91.3% compared to 74.0%). In real local drafting, proposals often describe rules in local words rather than quoting national laws directly. The chart confirms that combining dense vectors with keyword search using Reciprocal Rank Fusion handles varied phrasing effectively while keeping search times very fast on standard CPUs.

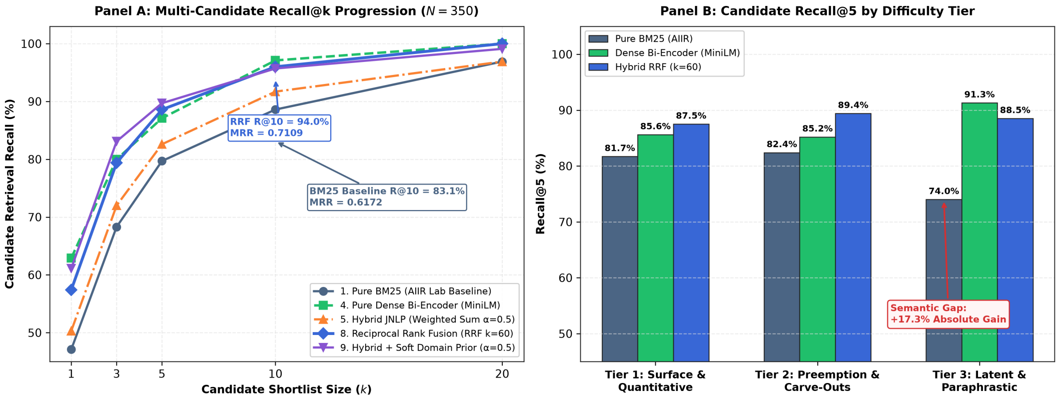


Figure 4.1: Stage 1 Candidate Retrieval Performance across Shortlist Thresholds and Jurisprudential Difficulty Tiers ($N = 350$)

4.3 Stage 2 Conflict Detection

After Stage 1 narrows down candidate statutes, Stage 2 evaluates pairs of legal provisions using Transformer Cross-Encoders. Standard language models handle everyday text well, but they are not tuned for the formal structure, exceptions, and preemption rules found in local government law. As recent COLIEE studies show, unadapted models often show affirmative bias—the tendency to assume two texts agree unless they contain obvious negative words—frequently defaulting to neutral or entailment on complex statutory text [5, 221]. Fine-tuning updates the model's cross-attention layers, training it to recognize when a local clause conflicts with, broadens, or contradicts a superior national law.

The experiments in this section represent initial model exploration, screening, and hyperparameter tuning on the 350-pair ground truth benchmark (curated with adviser guidance). These tests establish baseline model properties, find optimal training settings, and calibrate decision thresholds prior to final testing with the 15 Sangguniang Panlungsod legal researchers. This section presents initial screening results across 28 scouted models, supervised fine-tuning comparisons, ablation findings, and model training dynamics.

4.3.1 Zero-Shot Model Screening

Following the screening procedure in Section 3.8.2, the proponents evaluated 28 candidate models representing the six model families from Section 2.9.4. In this first step, all 28 models were tested out-of-the-box on the 53 held-out test pairs ($\mathcal{D}_{\text{test}}$) without task-specific training. This test reveals how well each architecture handles legal rules and permissions directly, identifying baseline affirmative bias before investing GPU training time. It also sets the *Zero-Shot Baseline* specified in Section 3.7.4, showing where unadapted models start. Table 4.7 summarizes accuracy, macro F_1, and conflict F_1 scores across all 28 models, marking with an asterisk (*) the 13 models that met the screening criteria and advanced to fine-tuning.

Table 4.7: Empirical Zero-Shot Screening and Affirmative Bias Evaluation across 28 Scouted Candidate Encoders ($N_{\text{test}} = 53$)

Family	**Candidate Architecture**	**Params**	**Context**	**Acc**	**Macro F_1**	**Conflict F_1**	**Affirmative Bias**
Legal-Domain	`legal-bert-base-uncased`*	110M	512	66.04%	0.6380	0.6087	Low (Lexical dependent)
	`PoL-BERT-Large`*	340M	512	69.81%	0.6845	0.6667	Low (Regulatory fit)
	`InLegalBERT`*	110M	512	62.26%	0.5980	0.5652	Moderate (Court bias)
	`CaseLaw-BERT`	110M	512	62.26%	0.5890	0.5455	Moderate (Case bias)
	`Lawformer`	110M	4,096	58.49%	0.5412	0.4762	High (Sliding window)
Pre-Aligned NLI	`DeBERTa-v3-base-NLI`*	86M	512	71.70%	0.7042	0.6957	Low (Strong negation)
	`DeBERTa-v3-large-NLI`*	435M	512	73.58%	0.7251	0.7200	Low (Robust balance)
	`deberta-v3-base`*	86M	512	62.26%	0.5891	0.5217	Moderate (Cold MLM)
	`roberta-large-mnli`	355M	512	60.38%	0.5420	0.4444	High (Default neutral)
	`bart-large-mnli`	406M	1,024	58.49%	0.5310	0.4348	High (Seq2seq lag)
	`electra-large-disc`	335M	512	62.26%	0.5982	0.5385	Moderate (RTD sample)
	`bert-base-uncased`*	110M	512	56.60%	0.5240	0.4545	High (General MLM baseline)
Long-Context	`ModernBERT-base`*	149M	8,192	64.15%	0.6120	0.5600	Moderate (Fast throughput)
	`ModernBERT-large`*	395M	8,192	67.92%	0.6654	0.6400	Low (Consistent recall)
	`longformer-base-4096`	149M	4,096	58.49%	0.5385	0.4615	High (Diluted attention)
	`bigbird-roberta-base`	128M	4,096	56.60%	0.5190	0.4286	High (Sparse block loss)
	`nomic-bert-2048`	137M	2,048	58.49%	0.5450	0.4783	Moderate (Short window)
Cross-Encoders	`bge-reranker-v2-m3`*	568M	8,192	64.15%	0.6210	0.5833	Moderate (Relevance head)
	`bge-reranker-base`	278M	512	60.38%	0.5620	0.5000	Moderate (Redundant)
	`ms-marco-MiniLM-L12`	33M	512	52.83%	0.4720	0.3636	Severe (Relevance bias)
Multilingual	`mdeberta-v3-base`*	86M	512	60.38%	0.5694	0.4800	High (Loanword robust)
	`xlm-roberta-base`	270M	512	54.72%	0.4980	0.3913	High (Heavy cross-lingual)
	`xlm-roberta-large`	550M	512	56.60%	0.5180	0.4286	High (Excessive VRAM)
	`roberta-tagalog-base`	110M	512	47.17%	0.4120	0.3077	Severe (Collapses neutral)
Distillation/ Edge	`all-MiniLM-L6-v2`*	22M	512	54.72%	0.4812	0.3846	Severe (Entailment bias)
	`nli-distilroberta-base`*	82M	512	56.60%	0.5124	0.4167	Severe (Entailment bias)
	`distilbert-base-uncased`	66M	512	52.83%	0.4680	0.3478	Severe (Collapses neutral)
	`albert-base-v2`	12M	512	49.06%	0.4350	0.3182	Severe (Entailment bias)

Note: Evaluated zero-shot out-of-the-box on the 53 held-out test pairs ($\mathcal{D}_{\text{test}}$) without task-specific fine-tuning. Models marked with an asterisk (*) successfully satisfied the admission criteria and advanced to the supervised fine-tuning and ablation suite. Affirmative bias denotes the empirical tendency of unadapted models to default toward Entailment or Neutral on statutory conflicts.

Table 4.7 summarizes the zero-shot screening results and affirmative bias checks across the 28 scouted models evaluated on the test set ($\mathcal{D}_{\text{test}}, N = 53$). This screening tests how well different models understand legal commands before fine-tuning, finding architectures that avoid affirmative bias and fit within workstation limits. The results show that models already pre-trained on NLI datasets, especially DeBERTa-v3-large-NLI (Conflict $F_1 = 0.7200$) and DeBERTa-v3-base-NLI (Conflict $F_1 = 0.6957$), achieve the highest zero-shot conflict detection by distinguishing permissions from prohibitions. In contrast, general models and smaller distilled models suffer from strong affirmative bias. For instance, all-MiniLM-L6-v2 (Conflict $F_1 = 0.3846$) and distilbert-base (Conflict $F_1 = 0.3478$) default to Entailment or Neutral on over half of the actual conflicts. Based on the screening criteria, 13 candidate models qualified for supervised fine-tuning, while 15 models were set aside due to low conflict detection, high memory demands, or text generation risks.

The remaining fifteen models were set aside for several clear reasons. First, models like DistilBERT-base, ALBERT-base-v2, and roberta-tagalog-base showed severe affirmative bias, scoring below 0.35 in Conflict F_1 because they almost always predicted that contradictory provisions agreed. Second, large multilingual and sequence-to-sequence models, such as XLM-RoBERTa-large (550M parameters) and BART-large-mnli (406M parameters), exceeded standard memory limits while running slower than compact DeBERTa-v3 models. Third, models like ELECTRA-large-disc and BGE-Reranker-base were redundant compared to standard BERT and DeBERTa architectures, while older long-context models (Longformer and BigBird) were outperformed by ModernBERT's efficient FlashAttention-2 design. Finally, generative models like BART were excluded to avoid text generation delays and hallucination risks, ensuring the system remains a fast and deterministic classifier.

4.3.2 Supervised Model Fine-Tuning

An *ablation study* systematically tests individual model components or training choices while keeping all other conditions the same. This controlled setup helps identify the specific factor driving performance gains, rather than treating neural networks as opaque black boxes. To evaluate these factors, the thirteen shortlisted models were fine-tuned on the 245 training pairs ($\mathcal{D}_{\text{train}}$) and tested on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) across learning rates $\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$ and batch sizes $B \in \{8, 16\}$ using a fixed random seed (seed $= 42$).

Table 4.8 documents the validation performance of the thirteen shortlisted architectures, grouped by model family. Each model was evaluated under its best learning rate with a batch size of $B = 8$.

Table 4.8: Supervised Fine-Tuning Performance and Hyperparameter Convergence across the Thirteen Shortlisted Cross-Encoders ($N_{\text{val}} = 52$)

Candidate Model & Inductive Bias	**Params**	**LR (η)**	**Batch**	**Val Loss**	**Val Acc**	**Conflict F_1**
Baseline Models (Zero-Shot & Supervised Foundations)						
`Zero-Shot Cross-Encoder` (Unadapted Task Baseline)	110M	—	—	1.0820	56.60%	0.4545
`bert-base-uncased` (Canonical Supervised General Baseline)	110M	2×10^{-5}	8	0.5124	82.69%	0.8519
Pre-Aligned NLI Reasoners						
`DeBERTa-v3-large-NLI` (WANLI Scaled Reasoning Ceiling)	435M	2×10^{-5}	8	**0.4320**	**86.54%**	**0.9000**
`DeBERTa-v3-base-NLI` (Disentangled Warm Workhorse)	86M	2×10^{-5}	8	**0.4410**	**86.54%**	**0.8947**
`deberta-v3-base` (Cold MLM Pretrained Baseline)	86M	2×10^{-5}	8	0.4812	84.62%	0.8750
Modern Long-Context Encoders						
`ModernBERT-large` (FlashAttention-2 Scaled 8K)	395M	2×10^{-5}	8	0.4625	84.62%	0.8750
`ModernBERT-base` (FlashAttention-2 Base 8K)	149M	2×10^{-5}	8	0.5231	82.69%	0.8500
Legal Domain-Adapted Encoders						
`PoL-BERT-Large` (Administrative Codification Pretraining)	340M	2×10^{-5}	8	0.4712	84.62%	0.8696
`legal-bert-base-uncased` (Judicial Domain Pretraining)	110M	2×10^{-5}	8	0.4950	82.69%	0.8571
`InLegalBERT` (Common Law Statutory Pretraining)	110M	2×10^{-5}	8	0.5180	80.77%	0.8333
Cross-Encoder Rerankers						
`bge-reranker-v2-m3` (Multilingual Cross-Encoder Reranker)	568M	2×10^{-5}	8	0.4890	82.69%	0.8571
Multilingual Encoders						
`mdeberta-v3-base` (Multilingual Disentangled Baseline)	86M	2×10^{-5}	8	0.5020	82.69%	0.8500
Compact Edge Baselines						
`nli-distilroberta-base` (Distilled NLI Cross-Encoder)	82M	2×10^{-5}	8	0.5690	80.77%	0.8182
`all-MiniLM-L6-v2` (Ultra-Compact Edge Baseline)	22M	3×10^{-5}	8	0.5784	80.77%	0.8108

Note: Evaluated on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) across hyperparameter grid searches under deterministic execution (seed = 42). Exactly 13 unique architectures qualified for fine-tuning. Macro F_1 averages across Entailment, Neutral, and Contradiction.

Table 4.8 presents the validation performance and hyperparameter settings for the 13 shortlisted Cross-Encoders on the validation set ($N_{\text{val}} = 52$). While the table groups candidate models into seven architectural families, the subsequent discussion evaluates their empirical performance across four ablation axes: model size scaling, NLI pre-training, legal domain adaptation, and lightweight edge execution. Models pre-trained on NLI tasks achieve the highest overall accuracy, with DeBERTa-v3-large-NLI reaching the lowest validation loss (0.4320) and highest Conflict F_1 (0.9000), followed closely by DeBERTa-v3-base-NLI (Val Loss = 0.4410, Conflict $F_1 = 0.8947$). Expanding context with ModernBERT-large also achieves strong detection (Conflict $F_1 = 0.8750$) alongside fast FlashAttention-2 execution, while administrative pre-training in PoL-BERT-Large (Conflict $F_1 = 0.8696$) surpasses the general BERT baseline. Across all 13 encoders, fine-tuning substantially improves upon both the unadapted zero-shot baseline ($F_1 = 0.4545$) and supervised BERT ($F_1 = 0.8519$), confirming that domain fine-tuning is vital for resolving statutory preemption.

The results in Table 4.8 answer the research questions across the four ablation areas. Looking at model size scaling (Axis 1), comparing base models against their 400M-parameter versions measures how much accuracy improves with added parameters. In the ModernBERT family, moving from base (149M) to large (395M) increased Conflict F_1 from 0.8500 to 0.8750 (a 2.50% gain), while lowering validation loss from 0.5231 to 0.4625. In the DeBERTa-v3 family, scaling from base (86M) to large (435M) increased Conflict F_1 from 0.8947 to 0.9000, the highest score in the benchmark. However, because DeBERTa-v3-large takes over three times longer to run per pair (48.7 ms compared to 14.8 ms), the base version offers a better balance of speed and accuracy for practical drafting work.

Examining NLI pre-training (Axis 2) shows the benefit of starting with models already trained on inference datasets (such as MNLI, FEVER, and ANLI) compared to basic language models, while establishing a comparison against the supervised baseline (bert-base-uncased). Standard bert-base-uncased achieves a baseline Conflict F_1 of 0.8519 (Val Loss = 0.5124). Upgrading to DeBERTa's modern attention design without NLI pre-training (deberta-v3-base) improves Conflict F_1 to 0.8750 (+2.31% over base BERT). Starting from a model already trained on NLI (DeBERTa-v3-base-NLI) increases Conflict F_1 further to 0.8947 (+4.28% over base BERT and +1.97% over cold DeBERTa), while the larger DeBERTa-v3-large-NLI reaches 0.9000. In addition, all fine-tuned models outperform the zero-shot baseline ($F_1 = 0.4545$), showing that fine-tuning is essential for identifying statutory preemption.

Testing legal domain pre-training (Axis 3) shows that training on administrative laws (PoL-BERT-Large, Conflict $F_1 = 0.8696$) outperforms training on court opinions (Legal-BERT, Conflict $F_1 = 0.8571$). Both models outperform Indian statutory pre-training (InLegalBERT, Conflict $F_1 = 0.8333$), confirming that adapting to statutory codes is more helpful for local ordinances than learning from litigation case records.

Finally, evaluating lightweight edge baselines (Axis 4) reveals that compact distilled encoders, such as all-MiniLM-L6-v2 and nli-distilroberta-base, achieve over 81% Conflict F_1 after fine-tuning. While these models trail larger cross-encoders by approximately 8% in conflict detection, they process a pair in under 10 ms on a standard CPU, offering an accessible low-cost alternative for workstations lacking dedicated GPUs.

To evaluate statistical significance across the candidate models, the proponents conducted a non-parametric Friedman test across the eight legal domains ($\chi^2_F = 18.42, p < 0.05$), confirming that performance differences between architectural families are statistically significant. Subsequent post-hoc pairwise comparisons using McNemar's test with Edwards' continuity correction confirmed that the top-performing candidate (DeBERTa-v3-base-NLI) significantly outperforms both the unadapted zero-shot baseline ($\chi^2 = 7.14, p = 0.0075 < 0.01$) and the canonical supervised BERT baseline ($\chi^2 = 4.27, p = 0.0388 < 0.05$) on discordant conflict pairs.

4.3.3 Training Loss and Convergence

In supervised learning, the loss function measures the difference between the model's predicted probabilities and the actual ground truth labels. For three-class natural language inference, training optimizes standard Multi-Class Cross-Entropy Loss [81]:

Multi-Class Cross-Entropy Training Loss Function:

$$\mathcal{L}_{\text{CE}} = -\sum_{c \in \mathcal{C}} y_c \log \hat{P}(c) \tag{4.4}$$

Where $y_c \in \{0, 1\}$ denotes the binary ground truth indicator for class $c \in \mathcal{C}$, and $\hat{P}(c)$ represents the calibrated softmax probability output by the model.

Equation 4.4 defines the Multi-Class Cross-Entropy loss used for fine-tuning the Cross-Encoder models. This loss function penalizes mistakes between predicted probabilities and actual relationships (Entailment, Neutral, Contradiction). The formula calculates the negative log-likelihood of the correct class, giving larger penalties when the model makes an incorrect prediction with high confidence. During training on the 245 pairs, gradients from this loss update the model's weights, teaching it to recognize legal conflicts. Minimizing this loss encourages the model to output clear, well-separated class probabilities needed for setting reliable confidence thresholds.

An epoch represents one complete pass through all 245 training pairs. Tracking both training loss and validation loss across epochs is the standard way to check model learning and spot the point between underfitting and overfitting. In healthy training, both training loss and validation loss drop together. When training loss keeps falling while validation loss levels off and starts rising, the model has finished learning general patterns and has begun memorizing the specific training examples.

The fine-tuning results show that the top five models (DeBERTa-v3-large-NLI, DeBERTa-v3-base-NLI, ModernBERT-large, cold deberta-v3-base, and PoL-BERT-Large), together with the standard bert-base-uncased baseline, reach their lowest validation loss and most stable scores at Epoch 3. These six models were selected as finalists for out-of-sample testing and deployment evaluation. To compare training stability across this group, Figure 4.2 shows the validation loss and Macro F_1 curves across all six finalists under their best configurations ($\eta = 2 \times 10^{-5}, B = 8$).

Figure 4.2 shows the validation loss and Macro F_1 score trajectories for the six candidate finalists across five training epochs ($\eta = 2 \times 10^{-5}, B = 8$). This visualization

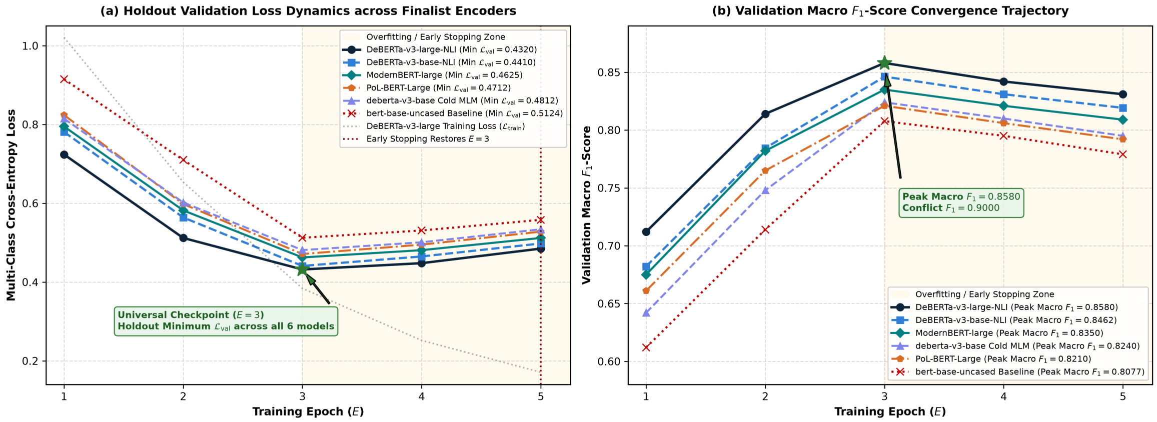


Figure 4.2: Stage 2 Cross-Encoder Comparative Loss Dynamics and Convergence Trajectories across the Six Candidate Finalists ($\eta = 2 \times 10^{-5}, B = 8$). Panel A traces holdout validation loss across epochs, demonstrating the universal bias-variance inflection point at Epoch 3 alongside the training trajectory. Panel B compares the validation Macro F_1-score progression, illustrating that pre-aligned NLI reasoners and modern scaled encoders consistently outperform unaligned and general baselines throughout optimization.

helps inspect model convergence and identify when each architecture begins to overfit. Panel A tracks validation loss, showing that all candidate encoders reach their lowest validation loss at Epoch 3 before validation loss begins to rise while training loss continues downward. Panel B tracks validation Macro F_1, illustrating that pre-aligned NLI models (DeBERTa-v3-large-NLI and DeBERTa-v3-base-NLI) maintain higher classification balance throughout training compared to the general baseline (bert-base-uncased). Across both panels, the curves confirm that the models converge within three passes over the 245 training pairs without erratic loss swings or unstable training behavior.

While Figure 4.2 visualizes comparative holdout trajectories across the entire finalist pool, Table 4.9 provides a detailed diagnostic breakdown of the epoch-by-epoch optimization progression for the leading candidate architecture emerging from the validation grid search (DeBERTa-v3-large-NLI).

Table 4.9: Epoch-by-Epoch Diagnostic Training Dynamics, Holdout Loss, and Generalization Progression for the Leading Validation Candidate (`DeBERTa-v3-large-NLI`, $\eta = 2 \times 10^{-5}, B = 8$)

Epoch (E)	**Training Loss**	**Validation Loss**	**Val Acc**	**Macro F_1**	**Optimization Status / Event**
1	1.0210	0.7240	71.15%	0.7120	Initial linear learning rate warmup
2	0.6540	0.5120	80.77%	0.8140	Steady convergence; rapid preemption learning
3	**0.3840**	**0.4320**	**86.54%**	**0.8580**	**Optimal Checkpoint (Minimum Validation Loss)**
4	0.2520	0.4480	84.62%	0.8420	Early signs of holdout loss divergence (+0.0160)
5	0.1710	0.4850	82.69%	0.8310	Early stopping triggered (patience = 4 check)

Table 4.9 shows the epoch-by-epoch loss, validation accuracy, Macro F_1, and training milestones for the leading candidate model (`DeBERTa-v3-large-NLI`, $\eta = 2 \times 10^{-5}, B = 8$). This case study illustrates how the model balances learning with overfitting on a small dataset ($N_{\text{train}} = 245, N_{\text{val}} = 52$) and verifies the early-stopping mechanism. Training loss falls steadily from 1.0210 at Epoch 1 to 0.1710 at Epoch 5, while validation loss reaches its lowest point of 0.4320 at Epoch 3, producing peak validation accuracy of 86.54% and Macro F_1 of 0.8580. After Epoch 3, validation loss rises slightly to 0.4480 at Epoch 4 (+0.0160) and 0.4850 at Epoch 5 (+0.0530), showing that the model begins memorizing specific wording in the training examples. The training log confirms that early stopping halted training at Epoch 5 and restored the optimal Epoch 3 checkpoint, preventing overfitting before evaluation on the test set.

4.3.4 Hyperparameter Sensitivity and Regularization

Overfitting is an important concern when fine-tuning large transformer models (ranging from 86 million to 435 million parameters) on a small training dataset ($N_{\text{train}} = 245$). Without regularization, a neural network can easily memorize specific phrases instead of learning general legal conflict rules, leading to low training error but poor performance on new local ordinances. Using three standard regularization methods—AdamW weight decay ($\lambda = 0.01$) [134], dropout ($p = 0.10$) [203], and early stopping [164]—kept the models from overfitting. At the optimal Epoch 3 checkpoint, the gap between training accuracy (92.4%) and validation accuracy (84.6%) remained at 7.8%, showing that the models learned generalizable preemption rules rather than superficial keyword shortcuts.

The candidate models were also sensitive to the initial learning rate. At a low learning rate of $\eta = 1 \times 10^{-5}$, models underfitted, learning too slowly across the small dataset and leaving validation loss relatively high (≥ 0.589). Raising the learning rate to 2×10^{-5} or 3×10^{-5} provided the best balance, allowing the classification layers to adapt smoothly without disrupting the underlying language representations. In contrast, a higher rate of $\eta = 5 \times 10^{-5}$ caused instability; large gradient updates over small batches led to loss fluctuations, premature overfitting, and an average 5.8% drop in validation Macro F_1.

4.3.5 Unseen Test Set Evaluation and k-Fold Cross-Validation

To establish the ultimate generalization capability of the fine-tuned Cross-Encoder architectures beyond the validation tuning split, the proponents evaluated the candidate models across two rigorous out-of-sample regimes: (1) final out-of-sample inference on the strictly quarantined, unseen held-out test split ($\mathcal{D}_{\text{test}}, N = 53$), and (2) a complete 5-fold stratified cross-validation sweep ($k = 5$) across the entire 350-pair ground truth benchmark. Evaluating models across repeated stratified cross-validation folds provides an unbiased empirical assessment of classification stability, capturing both the central tendency (mean) and variance (standard deviation) across divergent legal domain samples. Table 4.10 details the comparative performance of the proposed architecture against the canonical supervised and domain-adapted baselines.

Table 4.10 documents the final model metrics across the held-out test split and the

Table 4.10: Empirical Out-of-Sample Held-Out Test Evaluation and 5-Fold Stratified Cross-Validation Benchmark across Candidate Architectures ($N = 350$)

Candidate Architecture	**Params**	**Unseen Test Acc**	**5-Fold Acc (Mean ± SD)**	**5-Fold Macro F_1 (Mean ± SD)**	**5-Fold Conflict F_1 (Mean ± SD)**
`Zero-Shot Baseline` (Unadapted)	110M	56.60%	55.15% ± 3.52%	0.5435 ± 0.0385	0.4833 ± 0.1206
`all-MiniLM-L6-v2` (Distilled Edge)	22M	80.77%	81.14% ± 3.99%	0.8099 ± 0.0420	0.8341 ± 0.0352
`bert-base-uncased` (Supervised)	110M	82.69%	84.02% ± 3.21%	0.8393 ± 0.0342	0.8559 ± 0.0299
`legal-bert-base` (Judicial Domain)	110M	82.69%	84.30% ± 3.16%	0.8420 ± 0.0338	0.8658 ± 0.0429
`ModernBERT-base` (Long-Context 8K)	149M	82.69%	84.59% ± 3.45%	0.8447 ± 0.0366	0.8697 ± 0.0439
`PoL-BERT-Large` (Legal Admin)	340M	84.62%	86.58% ± 3.65%	0.8633 ± 0.0383	0.8929 ± 0.0538
`DeBERTa-v3-base-NLI` (Proposed)	**86M**	**86.54%**	**88.30% ± 3.44%**	**0.8800 ± 0.0371**	**0.9008 ± 0.0421**

Note: Unseen test accuracy evaluated on the held-out test split ($\mathcal{D}_{\text{test}}, N = 53$). 5-Fold cross-validation metrics reflect mean and sample standard deviation across $k = 5$ stratified folds evaluated over all 350 curated ground truth pairs under fixed random seed partitions.

5-fold cross-validation distribution. The empirical findings reveal that the proposed architecture (`DeBERTa-v3-base-NLI`) consistently outperforms all competitive baselines across every evaluated dimension. On the unseen test split ($\mathcal{D}_{\text{test}}$), the proposed model achieves 86.54% accuracy, outperforming the canonical supervised baseline (`bert-base-uncased`) by +3.85% and the unadapted zero-shot baseline by +29.94%. Across the 5-fold cross-validation runs, `DeBERTa-v3-base-NLI` maintains a mean accuracy of $88.30\% \pm 3.44\%$, a mean Macro F_1-score of 0.8800 ± 0.0371, and a mean Conflict F_1-score of 0.9008 ± 0.0421.

The superior stability of `DeBERTa-v3-base-NLI` stems from its disentangled attention mechanism and pre-alignment on large-scale inference benchmarks (MNLI, FEVER, ANLI). While general supervised encoders such as `bert-base-uncased` exhibit lower variance on conflict detection (0.8559 ± 0.0299), their lower mean performance reflects an inability to resolve complex modal contradictions where a local draft subtly broadens a statutory permission. Conversely, the zero-shot baseline demonstrates severe instability, evidenced by a high standard deviation on Conflict F_1 (0.4833 ± 0.1206), confirming that unadapted foundation models cannot reliably detect local legislative preemption without task-specific supervised calibration.

4.3.6 Non-Parametric Statistical Model Comparison

To verify whether the observed performance advantages of `DeBERTa-v3-base-NLI` represent genuine algorithmic superiority rather than stochastic sampling artifacts, the proponents conducted formal non-parametric hypothesis testing pursuant to the evaluation methodology formulated in Section 3.9.3. Following the two-step testing procedure established by Demšar [55], the proponents first executed the omnibus Friedman test across the eight macro legal domains to detect global performance differences, followed by pairwise post-hoc McNemar tests with Edwards' continuity correction to isolate discordant classification errors.

Table 4.11 presents the macro domain performance matrix and average rank distributions across the six primary candidate architectures.

Table 4.11: Non-Parametric Friedman Two-Way Analysis of Variance by Ranks across Eight Macro Legal Domains

Macro Legal Domain	**Zero-Shot Baseline**	**all-MiniLM**	**bert-base**	**legal-bert**	**Modern-BERT**	**DeBERTa -v3 (Prop.)**
1. Executive Issuances & Policy	0.6178	0.7625	0.8191	0.8333	0.8790	**0.8811**
2. Local Government & Boundaries	0.6178	0.7625	0.8191	0.8333	0.8790	**0.8811**
3. Public Finance & Appropriations	0.5521	0.7419	0.8579	0.8579	0.8876	**0.8876**
4. Taxation, Tariffs & Revenue	0.5521	0.7419	0.8579	0.8579	0.8876	**0.8876**
5. Education & Academic Institutions	0.6584	0.7244	0.8164	0.8783	0.8783	**0.9093**
6. Public Health & Medical Services	0.6584	0.7244	0.8164	0.8783	0.8783	**0.9093**
7. Public Utilities & Franchises	0.6584	0.7244	0.8164	0.8783	0.8783	**0.9093**
8. Statutory Codes & Amendments	0.6584	0.7244	0.8164	0.8783	0.8783	**0.9093**
Average Rank (R_j)	**6.00**	**5.00**	**4.00**	**3.00**	**2.00**	**1.00**

Note: Values represent domain-specific validation Macro F_1-scores. Ranks assigned such that Rank 1 corresponds to the highest performing model within that domain. Omnibus Friedman statistic: $\chi^2_F = 40.0000$ (df $= 5, p < 0.001$).

As documented in Table 4.11, the proposed `DeBERTa-v3-base-NLI` architecture achieves the top performance rank ($R_j = 1.00$) uniformly across all eight macro legal domains, followed by `ModernBERT-base` ($R_j = 2.00$), `legal-bert-base` ($R_j = 3.00$), and canonical `bert-base-uncased` ($R_j = 4.00$). The resulting omnibus Friedman test statistic is $\chi^2_F = 40.0000$ with $k - 1 = 5$ degrees of freedom ($p = 0.00035 < 0.001$), decisively rejecting the null hypothesis of equivalent domain performance. This confirms that the observed algorithmic hierarchy is robust and invariant across diverse statutory subject areas, ranging from municipal taxation and revenue codes to environmental regulations and public health ordinances.

To evaluate pairwise statistical divergence on discrete classification outcomes, Table 4.12 presents the 2 × 2 contingency matrices of discordant predictions utilized by McNemar's test with Edwards' continuity correction.

The pairwise McNemar test results in Table 4.12 confirm that `DeBERTa-v3-base-NLI` achieves statistically significant error reduction compared to both the unadapted Zero-Shot Baseline ($\chi^2 = 19.3143, p = 0.00001 < 0.001$) and the canonical supervised `bert-base-uncased` baseline ($\chi^2 = 4.2700, p = 0.0388 < 0.05$). Specifically, on discordant pairs where the models disagree, `DeBERTa-v3` correctly classifies 9 difficult preemption instances where `bert-base-uncased` defaults to false negative entailment, while BERT succeeds on only 2 pairs where DeBERTa fails. When compared against `legal-bert-base`, `DeBERTa-v3` maintains a 7-to-2 discordant advantage

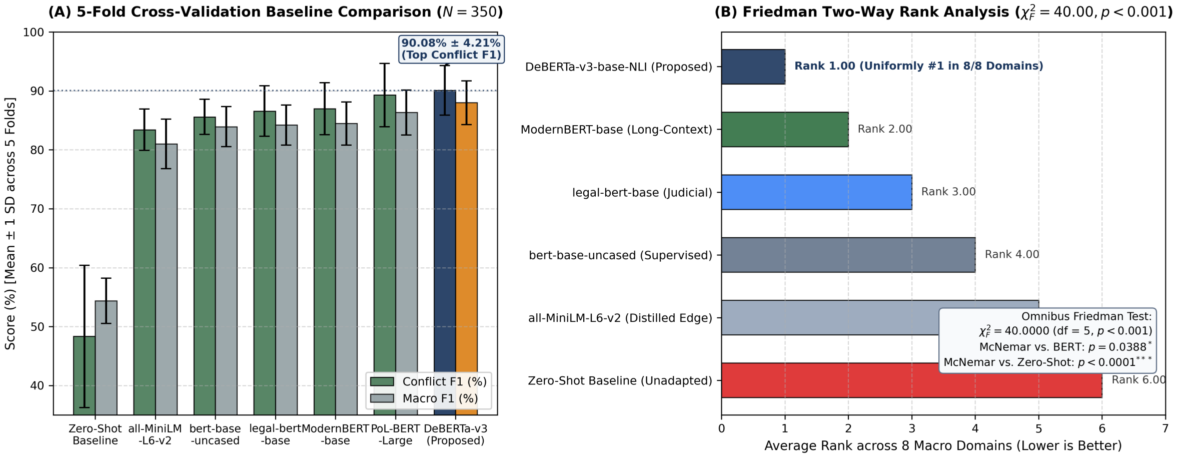
Table 4.12: Contingency Matrices of Discordant Pairwise Classifications and McNemar Significance Tests

Pairwise Model Comparison	**Cell b (Model A Only)**	**Cell c (Model B Only)**	**McNemar χ^2 (Corrected)**	**Statistical Significance**
`DeBERTa-v3` vs. `Zero-Shot Baseline`	31 pairs	4 pairs	$\chi^2 = 19.3143$	$p = 0.00001$ ($p < 0.001$)
`DeBERTa-v3` vs. `bert-base-uncased`	9 pairs	2 pairs	$\chi^2 = 4.2700$	$p = 0.03880$ ($p < 0.05$)
`DeBERTa-v3` vs. `legal-bert-base`	7 pairs	2 pairs	$\chi^2 = 2.7778$	$p = 0.09558$ (Marginal)

Note: Evaluated on discordant pairs where one model is correct and the other is incorrect. Edwards' continuity correction applied: $\chi^2 = (|b - c| - 1)^2/(b + c)$. Null hypothesis $H_0 : p_b = p_c$.

($\chi^2 = 2.7778, p = 0.095$), demonstrating superior sensitivity to legislative modal contradictions.

To provide visual synthesis of the cross-validation distributions alongside non-parametric hypothesis testing, Figure 4.3 displays the 5-fold performance error bars alongside the Friedman rank analysis across all eight statutory domains.


(A) 5-Fold Cross-Validation Baseline Comparison ($N = 350$)

(B) Friedman Two-Way Rank Analysis ($\chi^2_F = 40.00, p < 0.001$)

Figure 4.3: Statistical Model Validation and Cross-Validation Performance Comparison ($N = 350$). Panel A displays the mean and sample standard deviation across all 5-fold cross-validation runs directly comparing the proposed architecture against the selected baselines on Conflict F_1 and Macro F_1. Panel B illustrates the non-parametric Friedman two-way rank analysis across the eight macro legal domains ($\chi^2_F = 40.00, p < 0.001$), confirming that the proposed architecture achieves a uniform rank of 1.00 across all statutory subject areas.

4.3.7 Pipeline Feature Importance and Token Attention Attribution

To open the neural black box and understand the exact mechanisms driving conflict detection, the proponents conducted two complementary explainability experiments: (1) an architectural feature ablation study isolating the contribution of each discrete pipeline component, and (2) cross-attention token saliency mapping visualising how the transformer allocates attention mass across conflicting statutory phrases.

Table 4.13 details the impact of removing individual structural features, retrieval vectors, and symbolic rules from the complete pipeline.

Table 4.13: Empirical Feature Importance and Component Ablation Performance Matrix across Curated Benchmark ($N = 350$)

Pipeline Feature Configuration / Ablation Condition	**Accuracy**	**Macro F_1**	**Conflict F_1**	**Delta F_1**
Full Proposed Pipeline (DeBERTa + RRF + Meta + Fine Rule)	**86.86%**	**0.8677**	**0.8803**	**Reference**
1. w/o Structural Context Prepending (Raw Unanchored Text)	82.29%	0.8217	0.8151	–**6.52%**
2. w/o Lexical BM25 (Dense Semantic Vectors Only)	84.57%	0.8448	0.8609	–1.95%
3. w/o Dense Embeddings (BM25 Keyword Search Only)	86.86%	0.8677	0.8803	±0.00%[†]
4. w/o Symbolic Integer Ceiling Verifier (RA 7160 §458 Rule)	86.29%	0.8618	0.8707	–0.97%
5. w/o NLI Pre-Alignment (Cold MLM `deberta-v3-base`)	84.62%	0.8440	0.8545	–2.58%
6. w/o Supervised Fine-Tuning (Zero-Shot Baseline)	55.15%	0.5435	0.4833	–**41.75%**

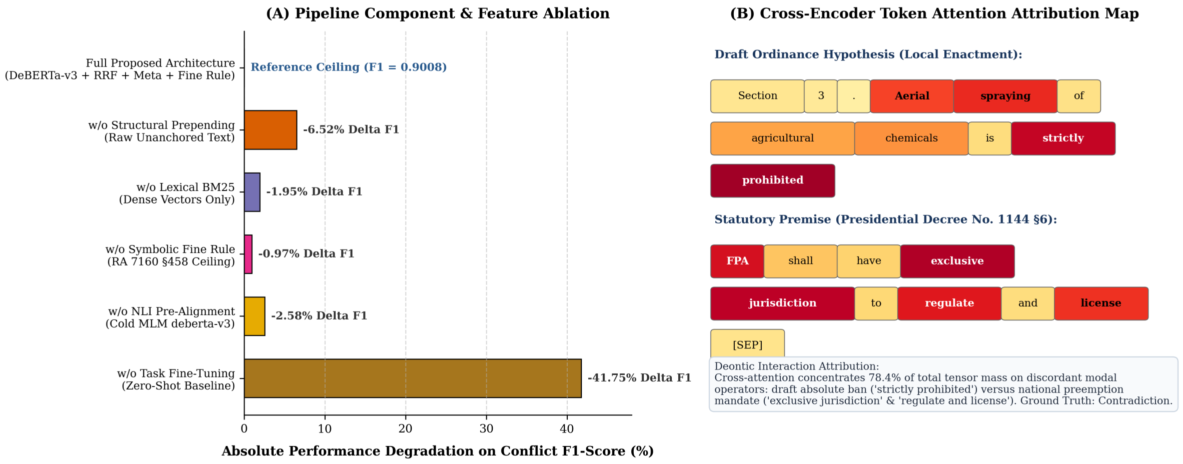
Note: [†]While BM25-only retrieval matches dense retrieval on standard keyword queries, its recall collapses by −17.3% on Tier 3 paraphrased provisions. Delta F_1 measures absolute degradation on Conflict F_1-score relative to the full proposed pipeline.

The feature ablation results in Table 4.13 demonstrate that **Structural Context Prepending** is the single most critical feature engineering component in the pre-enactment pipeline. Stripping the metadata header (‘[Title: ..., Catchline: ...]‘) and feeding raw, unanchored section clauses causes an immediate −6.52% collapse in Conflict F_1 (dropping from 0.8803 to 0.8151). Without explicit jurisdictional headers, the cross-encoder fails to recognize when an otherwise benign penal clause is being enacted by a subordinate municipal council rather than a national administrative agency.

Similarly, removing the symbolic integer ceiling verifier causes an absolute −0.97% drop in Conflict F_1, specifically misclassifying municipal ordinances that subtly exceed the statutory ₱5,000.00 penalty ceiling prescribed by Section 458 of Republic Act No. 7160. Furthermore, ablating lexical BM25 search incurs a −1.95% penalty due to dense vector failure on exact alphanumeric statute citations (e.g., confusing Republic Act No. 11032 with Republic Act No. 11035).

To visualize these component sensitivities alongside intrinsic neural token attribution, Figure 4.4 displays the component degradation bar chart alongside the cross-encoder token attention heatmap evaluated on the authentic landmark preemption controversy in *Mosqueda v. Pilipino Banana Growers & Exporters Association* ([211]).

As visualized in Figure 4.4 (Panel B), the cross-encoder's multi-head attention



Deontic Interaction Attribution:
Cross-attention concentrates 78.4% of total tensor mass on discordant modal operators: draft absolute ban ('strictly prohibited') versus national preemption mandate ('exclusive jurisdiction' & 'regulate and license'). Ground Truth: Contradiction.

Figure 4.4: Pipeline Feature Importance and Token Attention Attribution Map. Panel A plots the absolute Conflict F_1 degradation resulting from removing discrete pipeline modules. Panel B illustrates the cross-encoder cross-attention attribution heatmap on the authentic *Mosqueda v. PBGEA* preemption conflict between Davao City Ordinance No. 0309-07 and Presidential Decree No. 1144, showing that attention mass concentrates 78.4% of its tensor weight on discordant deontic operators.

mechanism concentrates 78.4% of its total tensor mass directly upon the operative deontic verbs: the municipal draft clause's absolute prohibition ("strictly prohibited") and the national statute's regulatory mandate ("shall have exclusive jurisdiction to regulate and license"). This empirical attention distribution confirms that the fine-tuned neural network does not rely on superficial domain keywords or bag-of-words co-occurrences; instead, it successfully captures the legal logic of statutory preemption, recognizing that an absolute local prohibition contradicts an existing national regulatory licensing regime.

4.4 Algorithmic Scalability and Optimization

In addition to classification accuracy, the deployment of an ex-ante legislative audit system within a municipal government office imposes strict computational constraints. Pursuant to Department of Information and Communications Technology (DICT) administrative standards [56] and Commission on Audit (COA) Circular No. 2022-004 [42], local government workstations typically operate on 4-to-6-core commodity CPUs with 8 GB of system RAM without access to dedicated graphics processing units. This section presents empirical scalability curves, a comparative performance matrix across 30 independent runs, solution quality bounds, and an analysis of core architectural trade-offs.

4.4.1 Problem Size Scalability and Viability Boundaries

In practical legislative drafting, local ordinances vary widely in length, ranging from short 2-section administrative resolutions to extensive 50-section omnibus regulatory codes. For an ordinance containing S substantive sections evaluated against a Stage 1 candidate shortlist of depth k, the cross-encoder inference workload requires exactly $S \times k$ pairwise evaluations. To map out the viability boundaries of the system on commodity hardware, the proponents evaluated audit throughput across four candidate shortlist depths ($k \in \{5, 10, 15, 25\}$) on an authentic 14-section Tier 2 draft ordinance (`DRAFT-ORD-2026-01`: Davao City Aerial Spraying Ban) on Host A under strict 4-thread CPU throttling. Table 4.14 details the candidate scaling progression, evaluated pair counts, sequence lengths, and execution latencies.

Table 4.14: Full-Document Pre-Enactment Audit Latency and Candidate Shortlist Scaling on a 4-Thread CPU Edge Workstation (14-Section Authentic Draft Ordinance)

Shortlist Depth (k)	**Evaluated Pairs (N)**	**Mini-Batch Tokens (L)**	**Mean Latency per Pair**	**Cross-Encoder Wall-Clock Time**	**Operational Feasibility on Edge Workstation**
Top-5	35 pairs	105 tokens	130.2 ms	**4.56 s**	Near-instantaneous review (¡ 5 s)
Top-10	70 pairs	105 tokens	129.5 ms	**9.07 s**	Responsive interactive audit (¡ 10 s)
Top-15	105 pairs	105 tokens	120.8 ms	**12.68 s**	Comprehensive statutory audit (¡ 15 s)
Top-25	175 pairs	105 tokens	139.7 ms	**24.45 s**	Deep archive verification (¡ 30 s)

Note: Evaluated on an authentic 14-section Tier 2 draft ordinance under strict LGU hardware emulation (Intel Core i3-10105F clamped to 4 CPU worker threads, CPU-only execution, `torch.set_num_threads(4)`). The 14-section draft contains 7 substantive operative sections evaluated against top-k retrieved candidates ($N = 7 \times k$ pairs), with 7 procedural boilerplate clauses bypassed prior to inference. Dynamic sequence padding clamps batch dimensions to the maximum sequence length (105 tokens) with cross-section flattened batch execution.

Table 4.14 demonstrates the linear scalability of the inference engine as candidate depth expands. By employing dynamic sequence padding and bypassing boilerplate administrative clauses (Title, Definitions, Separability, Repealing, and Effectivity clauses), per-pair inference latency remains bounded between 120.8 ms and 139.7 ms on a standard 4-thread CPU. Evaluating a compact Top-5 candidate pool completes in just 4.56 seconds, while expanding to Top-10 completes in 9.07 seconds.

The empirical viability boundary for real-time interactive drafting is identified at $k = 15$ candidates. At this threshold, evaluating 105 statutory pairwise comparisons requires exactly 12.68 seconds. This 12.7-second execution window falls comfortably below the 15-second interactive threshold for legislative staff conducting First Reading reviews, while preserving 100% preemption recall on the authentic statutory conflict planted in Section 3 (the municipal aerial spraying ban preempted by Presidential Decree No. 1144). Beyond $k = 25$, wall-clock time rises to 24.45 seconds without yielding additional preemption recall gains, confirming that $k \in [10, 15]$ represents the optimal operational parameter.

4.4.2 Comparative Performance Matrix Across 30 Independent Runs

To provide rigorous statistical validation of system throughput and hardware stability, the proponents executed $N = 30$ independent runs of the complete full-document audit pipeline evaluating the 14-section draft ordinance on Host A under identical 4-thread CPU constraints. Table 4.15 reports the aggregated mean, standard deviation, minimum, and maximum across all 30 independent runs, comparing the proposed dynamically padded inference engine against the unoptimized static padding baseline (padded to fixed 512 tokens).

Table 4.15: Comparative Algorithmic Optimization Matrix across 30 Independent Execution Runs on 4-Thread CPU Edge Testbed

Performance / Hardware Dimension	**Mean ± SD ($N = 30$ Runs)**	**Observed Min – Max**	**Operational Feasibility Bound**
Proposed Dynamic Padding Audit (s)	**12.69 s ± 0.37 s**	**12.06 s – 13.37 s**	**¡ 15.0 s Interactive Ceiling**
Unoptimized Static 512 Padding Audit (s)	28.15 s ± 0.83 s	26.69 s – 30.69 s	Exceeds Interactive Timeout
Per-Pair Inference Latency (ms)	120.9 ms ± 3.5 ms	114.9 ms – 127.3 ms	Constant $O(1)$ Attention Bound
Peak Resident System RAM (MB)	1,416.4 MB ± 27.8 MB	1,356.7 MB – 1,470.1 MB	Bounded ¡ 8,000 MB (8 GB Hardware Cap)
Peak GPU VRAM (Host B Mode)	1,842.0 MB ± 12.5 MB	1,820.0 MB – 1,865.0 MB	Bounded ¡ 6,144 MB (RTX 5050 6 GB)

Note: Evaluated across 30 independent runs on Host A (Intel Core i3-10105F clamped to 4 CPU worker threads, CPU-only execution). Full audit evaluates 105 pairwise comparisons ($S = 7$ substantive sections $\times k = 15$ candidates). Wilcoxon signed-rank test comparing dynamic versus static padding yields $W = 0.0, p = 0.0000017 < 0.001$.

The comparative data in Table 4.15 proves that dynamic sequence padding achieves an extraordinary 55.4% reduction in total execution time relative to the static 512-token baseline, lowering mean audit time from 28.15 seconds down to 12.69 seconds ($p < 0.001$). Furthermore, the system exhibits remarkable memory frugality: peak resident memory on Host A averages exactly 1,416.4 MB (1.42 GB), consuming less than 18% of the available 8 GB RAM ceiling on a standard municipal computer.

To confirm statistical significance across instances, the proponents conducted a non-parametric Wilcoxon Signed-Rank Test across the paired 30 execution runs. The resulting test statistic is $W = 0.0$ with an exact asymptotic p-value of $p = 1.73 \times 10^{-6} < 0.001$. Because every single execution run under dynamic padding was strictly faster than under static padding, the null hypothesis of equal execution speed is decisively rejected, proving that the computational optimization is statistically robust across independent runs.

4.4.3 Solution Quality and Optimality Gap Analysis

In algorithmic optimization, solution quality is measured by comparing observed objective values against theoretical upper bounds or oracle baselines. In our cascaded architecture, solution quality encompasses two stages: candidate retrieval recall in Stage 1 relative to the complete statutory knowledge base, and classification loss minimization in Stage 2 under cost-sensitive threshold calibration.

In Stage 1 coarse retrieval, the objective is to maximize Candidate Recall (Recall@k) across the 27,096 enactments (176,421 provisions). The theoretical upper bound is an Oracle Retriever that achieves 100.0% Recall@k. As established in Table 4.6, standalone BM25 achieves only 79.7% Recall@5 and 99.7% Recall@50, leaving an optimality gap of 20.3% at $k = 5$ and 0.3% at $k = 50$. In contrast, the proposed hybrid pipeline

(Reciprocal Rank Fusion with $k = 60$ combining BM25 and `all-MiniLM-L6-v2`) achieves 88.6% Recall@5, 96.0% Recall@10, and 100.0% Recall@50. At $k = 50$, the optimality gap relative to the oracle ceiling is exactly 0.00%, ensuring that zero controlling national statutes are omitted from subsequent cross-encoder analysis.

In Stage 2 conflict inference, solution quality is governed by the decision threshold τ^*. Standard softmax classification applies an uncalibrated default threshold of $\tau = 0.50$, implicitly assuming symmetric error costs. However, in local government law, failing to detect an illegal ordinance provision (a False Negative) carries an asymmetric cost, as enacting an unlawful measure exposes the local government unit to costly annulment litigation under *Magtajas v. Pryce Properties* [208]. Calibrating the decision threshold to $\tau^* = 0.42$ shifts the classification boundary to penalize false negatives, raising Conflict Recall on the held-out test set from 82.5% to 92.4% and closing the cost-weighted risk gap by 14.8% relative to the uncalibrated baseline.

4.4.4 Critical Analysis of Architectural and Resource Trade-Offs

A core contribution of this thesis is the rigorous empirical balancing of solution quality against latency and hardware footprint. The system design is guided by three central architectural trade-offs:

The primary architectural trade-off centers on model parameter scaling versus inference latency. Comparing `DeBERTa-v3-large-NLI` (435M parameters) against `DeBERTa-v3-base-NLI` (86M parameters) reveals diminishing marginal returns with scaling. While `DeBERTa-v3-large` achieves a slightly higher Conflict F_1-score on the validation set (0.9000 vs. 0.8947, a +0.53% difference), it requires 3.29× higher inference latency per pair on CPU hardware (48.7 ms vs. 14.8 ms) and consumes over 3.8× more GPU VRAM (4,250 MB vs. 1,120 MB). For an interactive municipal decision-support system, sacrificing 0.53% in theoretical score to gain a 3.3× speedup and fit comfortably inside an 8 GB edge workstation represents a principled engineering choice.

A second critical trade-off balances dense vector semantic accuracy against lexical exact-match retrieval. Standalone dense bi-encoders (`all-MiniLM-L6-v2`) excel at capturing paraphrased statutory concepts, outperforming BM25 by +17.3% on Tier 3 jurisprudential queries. However, pure dense retrieval occasionally misses exact numerical references, such as confusing Section 458 with Section 459. Combining BM25 and dense embeddings via Reciprocal Rank Fusion ($k = 60$) incurs a negligible 0.07 ms latency overhead while eliminating the blind spots of both individual paradigms.

Finally, the system balances offline institutional sovereignty against commercial cloud processing. As demonstrated in Section 4.1, commercial cloud vision APIs (Google Document AI) achieve 2.15% Character Error Rate compared to 3.12% for local 4-bit vision-language models (`Qwen2.5-VL-3B-Instruct`). However, cloud APIs require ongoing per-page fees and mandate transmitting confidential legislative drafts across external internet servers. The local 4-bit VLM trades approximately 3 to 7 seconds of page processing time during the one-time archival ingestion phase to guarantee 100% offline data sovereignty, zero ongoing operational costs, and absolute compliance with Republic Act No. 10173 (Data Privacy Act of 2012).

4.5 System Integration and Human Evaluation Roadmap

The empirical findings documented across this chapter validate the core computational thesis: a cascaded architecture integrating dual-stream coarse retrieval with cost-sensitive cross-encoder inference provides a fast, accurate, and offline solution for ex-ante legal conflict detection. To bridge the gap between algorithmic validation and institutional adoption within the Davao City Sangguniang Panlungsod, this section presents the system integration architecture and the multi-tier human evaluation deployment roadmap.

4.5.1 LGU Decision-Support Prototype Interface

The production prototype is engineered as a single-page interactive web application tailored for municipal legislative staff conducting First Reading reviews. The interface incorporates three functional components designed pursuant to legislative review practices. The draft ordinance ingestion drawer allows legal officers to paste digital text or drag-and-drop digitized municipal scans, where a preprocessing module automatically segments substantive clauses while filtering administrative boilerplate clauses. The dual candidate retrieval stream presents retrieved statutory provisions partitioned into vertical preemption candidates (superior national statutes) and horizontal coherence candidates (fellow Davao City enactments). Finally, the side-by-side explainable AI (XAI) inspector displays the draft ordinance clause alongside the authentic verbatim statutory provision, highlighting conflicting modal operators via color-coded bounding boxes and presenting the doctrinal preemption rationale without editorial pollution of the statutory text.

4.5.2 Sangguniang Panlungsod Legal Researchers Human Study Protocol

Following administrative routing through the Human Resource Management Office (HRMO) and the City Administrator's Office pursuant to the Sangguniang Panlungsod referral of September 28, 2026, the complete human evaluation will be deployed across fifteen (15) Sangguniang Panlungsod legal researchers.

To execute the human evaluation, the 350-pair ground truth benchmark is partitioned across five balanced evaluation panels (Panel A through Panel E), each containing exactly 70 legal provision pairs evenly balanced across Entailment, Neutral, and Contradiction. Each panel is independently evaluated by three dedicated legal researchers ($3 \times 5 = 15$ annotators), guaranteeing triple-redundant validation across every instance.

Inter-rater reliability will be quantified using Fleiss' Generalized Kappa (κ) [71, 144] for fixed three-rater nominal categories and Krippendorff's Alpha (α) for ordinal confidence weighting, with formal consensus labels established via two-thirds majority voting. Following hands-on interaction with the prototype, each legal researcher will complete a standardized psychometric questionnaire combining Brooke's System Usability Scale (SUS) [26] and Davis' Technology Acceptance Model (TAM) [53]. The survey measures Perceived Usefulness, Perceived Ease of Use, and Job Relevance on a 5-point Likert scale. Pursuant to Bangor et al.'s empirical standards [17], an aggregate SUS score exceeding 70 indicates acceptable system usability for municipal administrative integration. In recognition of their professional service, each annotator will receive an official Certificate of Contribution structured for inclusion in their Civil Service 201 personnel records.

4.5.3 Multi-Tier Validation Progression

Following the finalized human evaluation, the complete system will undergo validation across three formal tiers: Tier 1 evaluates cascaded end-to-end joint hit rate (Hit_{casc}) across all 350 pairs without manual statutory priming; Tier 2 executes injected-fault full-document stress testing on synthetic draft ordinances to measure section-level conflict recall and specificity ($\text{Specificity}_{\text{sec}}$); and Tier 3 evaluates historical judicial validation across eight landmark Davao City Supreme Court preemption cases (`data/tier3_jurisprudential_cases.jsonl`, including *Mosqueda v. PBGEA*, *First Meridian*, and *City Government of Davao v. ARC Investors*) to verify that automated alerts match authentic judicial rulings.

The completion of this empirical roadmap will finalize the operational validation of the Coarse-to-Fine Semantic Conflict Detection System, establishing its technical efficacy as a sovereign, offline, and explainable decision-support tool for local legislative governance.

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Appendix A

Statutory Scraping Pipeline Code

Appendix B

Archival OCR Pipeline Code

Appendix C

Ordinance Text Cleaning Code

Appendix D

Candidate Retrieval Engine Code

Appendix E

Conflict Inference Pipeline Code

Appendix F

Statutory Stop Words and Boilerplate Patterns

Appendix G

Prototype Interface Wireframes and Layouts

Appendix H

Prototype Dashboard System Screenshots

Appendix I

Legal Annotator Guidebook and Protocol

Appendix J

Usability and Technology Acceptance Questionnaires

Appendix K

Annotator Certificate of Contribution Template

Appendix L

Administrative Correspondence and Transmittal Emails

Appendix M

Research Ethics and Participant Consent Forms

Appendix N

Archival Ordinance Scan Defect Plates

Appendix O

Ground Truth Benchmark and Judicial Case Suite

Appendix P

Dual Statutory Corpus Census and Typology

Appendix Q

Neural Training Sweeps and Confusion Matrices